

Customer Release Guide

**CRG-684
March 2013**

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About This Publication

This publication supports **TotalPlant** Solution (TPS) system Network Release. TPS is the evolution of TDC 3000^X.

The Release 684 TPS Network Customer Release Guide contains procedures that will guide you through the migration from Release R43x, R5xx or R6xx to Release 684. It begins with a “Read Me First” section that explains how to use the remainder of the document and a description of the overall steps you must take to perform this migration.

It also contains minimum hardware and firmware revisions for this release, along with other special considerations that can impact your migration.

Section 1.13 contains the minimum skills a person performing a migration should have. This document is written to that skill level.

Caution must be taken when performing this migration, especially when migrating on a live system, where no "down time" is intended for the migration.

In cases where a statement includes more than one release, within a group of releases, the letter “x” is used to represent any possible release in the grouping. For example, R5xx would represent R500 through R530. Rxxx would indicate any R5xx through R6xx release.

There are three documents that provide installation and configuration information for the Global User Station (GUS).

TPS Network Release 684 Customer Release Guide—provides information and procedures to install a Global User Station on your network.

TPS System Administration Guide

TPS System Implementation Guide

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NOTE

Refer to the online support for the latest version of the user documents. In our effort to provide the latest information any updates made to the user documents after the release of TPS/TPN Knowledge Builder or PDF Collection will be available on the online support site.

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1. Read Me First!

1.1 What's In This Document

Overview

This “Read Me First!” section will provide an insight of the entire process of migration, including

- how to use this guide,
- R600 enhancements,
- R610 enhancements,
- R611 enhancements,
- R620 enhancements,
- R630 enhancements,
- R640 enhancements,
- R650 enhancements,
- R660 enhancements,
- R670 enhancements,
- R680 enhancements,
- R681 enhancements,
- R682 enhancements,
- R683 enhancements,
- R684 enhancements
- An overview of this migration process,
- hardware revision requirements,
- memory size checks,
- how to get started,
- how to know that you’re finished, and
- where to go to get help.

For in-depth explanations of the new features in R684, refer to the R684 Differences Summary, DF02-683.

1.2 How To Use This Guide

This manual is organized into the following sections of information. Where possible, the information contained in this manual is in chronological order, so it is there when you need it. A brief explanation of each section is provided in the following table.

Section	Section Name	Description
1	Read Me first!	Provides an overview of the Release Numbering scheme, the migration process, and leads you through the use of this manual.
2	What To Do Before Migrating Your Software	Informs you of the prework that must take place prior to starting the software migration steps.
3	Hardware Preparation	Supplies lists of all hardware and its board revision, firmware revision, and/or memory size requirements in order to successfully implement and run this software release.
4	HM Volume Sizing	Contains size estimation formulas and volume size figures for each node type and each system and user volume that will occupy space on your HM(s).
5	Special Considerations	Critical information that you must be aware of prior to start of a migration, and during the ongoing use of your system. Be sure to read the parts that pertain to the components in your system.
6	Distribution Media	Provides a list and a brief explanation of zip disks, DAT tapes, and CD-ROMs supplied with the migration package.
7	Setup For Migration From R43x	Contains the steps to be taken when migrating from R43x.
8	Setup For Migration From R5xx	Contains the steps to be taken when migrating from R5xx.
9	Loading R6xx Universal Station	Contains the R6xx translations and NCF Configuration changes required for R6xx.
10	Migration from R43x, R53x to R6xx	Contains the steps to be taken to complete the R6xx migration.
11	Backing Out of R6xx	Contains an explanation of the steps required to regress back to a previous release of software.
12	Migration from R6xx to R683	Provides the steps to migrate from R6xx to R683.
13	Backing out of R683	Provides a method to regress to a former software release.
14	Troubleshooting Migration Problems	Provides a methodology used by the experts in troubleshooting common problems experienced when installing migration software.

Table 1-1 Customer Release Guide Contents

1.3 Release Numbering Scheme

R680 and prior to R680	Release numbers consist of four fields, structured as: ABC.D. The range and significance of each field in the release number is as follows: A - Range: 1 - 9 Significance: Designates the release series. This type of a release is known as a Series Release. B - Range: 0 - 9 Significance: A non-zero number designates a release within the release series. This type of a release is known as a Functional Release. C - Range: 0 - 9 Significance: A non-zero number designates a maintenance release that is scheduled on an external roadmap. This type of a release is known as a Roadmap (or Scheduled) Maintenance Release. D - Range: 1 to 99 Significance: A number of 1 indicates a first release of one of the above release types. This type of a release is known as a Point Release. Example: R660.xx, R670.xx
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1.3. Release Numbering Scheme

R681 and later¹	The revised release numbering scheme consist of five fields, structured as: ABB.CD. A - Range: 6 Significance: Designates the release series. This type of a release is known as a Series Release. BB - Range: 80-99 Significance: A non-zero number designates a release within the release series. This type of a release is known as a Functional Release. Example: R681, R682 C - Range: 2 - 9 Significance: A non-zero number designates a release within the release series. This type of release is known as a Roadmap (or Scheduled) Maintenance Release. Example: R681.2, R682.2, R682.3. D - Range: 1 to 9 Significance: A number of 1 indicates the first release of one of the above release types. This type of release is known as a Point Release. Example: R681.21, R682.21, R682.31
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¹The revised release numbering scheme (ABB.CD) will be continued until we reach R700. From R700 onwards the release numbering scheme ABC.D will be continued.

1.4 R5xx Enhancements

Description of R5xx enhancements To see a description of the R5xx enhancements, refer to the “TPS Network, Release 530, *Customer Release Guide*.”

1.5 R60x Enhancements

Introduction TPS network software Release 60x enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from north America, Europe, the Far East, and Australia. The following lists the enhancements that were made as a result of their recommendations:

- Display Parameter
- AM Load Balance Tools
- HG Cable Swap ENABLE/DISABLE Message
- Load Data Warning
- Retain Organizational Summary Selections
- PE Variable Properties and Break Key
- Replace Subpicture Enhancements
- Simplify AM and US NCF Node Changes
- Bad Value Check in Targets
- System Status Summary
- Event History Modifications
- Suppress HG Chattering Digitals
- HG Setpoint/Output Clamp
- Expand HM Journal Sizes

Display Parameters	The Display Parameter is a PSDP, which contains the eight-character name of the schematic currently being displayed on the Universal Station or native window. The user is responsible for the collection of data, e.g., AM/CL routine to collect and store parameter.
AM Load Balance Tools	This is an analysis tool that will provide a composite set of results in graphical format for the schedule loading of all points in a single AM.
HG Cable ENABLE/DISABLE swap message	When automatic cable swapping gets disabled on the HG, either manually or through the Hiway Swap Algorithm, a Real Time Journal entry is recorded. Also, the Hiway Status Display will always show the status of automatic and periodic cable swap (enable or disable).
Load Data warning	When the LOAD DATA target is selected on the Hiway Status Display, a warning message will appear that says the Load Data action will put all outputs in manual mode.
Retain Organizational Summary selections	This function will allow operators to print the Organizational Summary information requested for display without having to re-enter the same Organizational Summary information just to print. It will also retain the summary information for use the next time the organization summary is installed.
PE variable properties and Break key	This package allows the “set collection” properties (collection rate and group ID) of variables in objects or subpictures to be inherited by the destination picture when a Picture Editor command is executed (add/replace subpicture, paste, etc.). This package also allows the Picture Editor to occasionally check for BREAK requests from the keyboard during the period of time that a picture is compiling or writing, and abort the operation if the BREAK key is hit.

Replace subpicture enhancements

This command will be modeled after the Picture Editor Multiple Compile command in that a text file similar to the Multiple Compile text file will be read in allowing replacement of multiple subpictures in multiple schematic files.

An error file will be produced indicating errors if the number and types of the parameters of the subpictures don't match or if the subpicture or file names are invalid.

The original source file will be replaced with the modified source file only if no errors occur on the replace subpicture commands. If no errors occur on the replace, allow compile (if indicated in the text file).

No parameter form will be provided for the user to modify in the automatic mode. A manual mode indicator, in the input text file, will bring up the parameter form for the subpicture and wait for the user to enter any changes to the subpicture parameters. The subpicture replace mode will remain in manual until the input file indicates to change the mode to automatic.

The Picture Editor Replace Subpicture Command will be modified to replace only the selected instances of the indicated subpicture. Note that if only some instances of the old subpicture are selected, the new subpicture must have a different name than the subpicture to be replaced, because subpicture names within the same schematic must be unique.

**Simplify AM and US
NCF node changes**

The NCF configuration of LCN nodes has been modified to allow modifying node attributes without having to first delete and then re-add the node to the NCF. This is being done using the existing MODIFY NODE target. By selecting the MODIFY NODE target, the following attributes can be modified.

For the AM

- Units
- Startup Mode

For the US

- Cartridge disc numbers
- Floppy disc numbers
- Printer numbers
- Trend pen numbers
- Station default access
- Station default personality
- Operator's keyboard (Y/N)
- Engineer's keyboard (Y/N)
- Touchscreen (Y/N)
- Default Area

For the modifications to take, affected nodes will need to be reloaded.

**Bad Value check in
targets**

Two new actors will allow users to check variables. The first will be called BAD_VAL. This actor will return TRUE if the value is bad and FALSE if the value is good.

The input parameter will be considered a bad value if:

- the data owner is down
- the parameter is NULL
- the value is NAN
- the point has been deleted
- the point is invalid
- under any other condition that prevents a good value from being retrieved by data access.

The other actor will be called EXISTS. This actor will return TRUE if the variable name exists and FALSE if it does not exist. A variable is considered non-existent if it cannot be fully resolved by data access.

**System Status
summary**

The actor, SYS_JRNL, was created for use in the Picture Editor and Button Configurator functions. The actor invokes the System Status journal, based upon the value of the actor's arguments. The new System Status journal actor has the following format:

SYS_JRNL(t, a, b, c, d)

t = integer	(default value=1); where: t - represents the number of hours to retrieve data (from current time).
a = 1-5	(default value=5); where: 1-Network Boxes, 2-Network, 3-ALL Boxes 4-Node, 5-ALL.
b = 1-20	(default value = n/a); where: b=Network Number (applicable when a=1 or 2).
c = 1-65	(default value = n/a); where: c=Box Number (applicable when a=1).
d = 1-96	(default value = n/a); where: d=Node Number (applicable when a=4)

**Event History
modifications**

Event History Modifications

This package accomplishes the following objectives:

- It preserves Event History Retrieval (EHR) setup data until an operator takes an overt action to alter, clear, or delete the data. The emphasis is on reuse of the setup data rather than having to enter all of the data each time the EHR setup display appears on the screen.
- It provides an AREA UNITS pick on the setup display for the purpose of requesting retrieval of all of those events referencing one of the units assigned to the local area. This new pick is applicable to all EHR retrievals that are unit related.
- In support of the stated objectives, this package provides a new EHR setup display, better suited for presentation of the preserved setup data.

The new display also provides a measure of consistency in the usage of AREA UNITS across the different kinds of journal retrievals.

**Suppress HG
Chattering Digitals**

The parameter DLYTIME has been added to HG digital inputs and composites. The parameter has a value of 0..60 seconds and specifies the amount of time that a digital alarm is guaranteed to remain in alarm even if the alarm condition ceases.

**HG Setpoint/Output
Clamp**

This function forces high-level control devices, such as AM/CL programs or AM Regulatory Control points, to honor the limits defined by the HG secondary point for its setpoint and output.

The option is selected through the System Wide Values in the Network Configurator.

When the clamp option is selected, any attempt to write a value that is outside the setpoint or output limits results in the targeted parameter being clamped at the appropriate limit, SPLOLM, SPHILM, OPLOLM, or OPHILM.

In the case where an AM/CL program is attempting to violate the limits, the CL program will show a limits violation.

Expand HM Journal sizes

The maximum number of journal events increases which can be historized in the System Status Changes, System Errors, System Maintenance, Area SOE, Process Alarms, Process Changes and Operator Messages (see table below).

Also, the capacity of the current process variable history within the context of the current HM history collection architecture is increased, i.e., provides additional snapshots and/or averages without requiring significant HM or file manager modification.

Journal Type	Maximum Number of Events
System Status	12,500
System Errors	15,000
System Maintenance	15,000
Area Sequence of Events	15,000
Process Alarms	15,000
Process Changes	10,000
Operator Messages	5,000

Table 1-2 Maximum Number of Journal Events

In addition, the maximum number of pre-archive hours has increased for snapshots and user averages as shown in the following table:

Snapshot Save Rate (Seconds)	Maximum Number of Pre-Archive Hours
5	240
10	480
20	960
60	2880
User Average Save Rate (Minutes)	Maximum Number of Pre-Archive Hours
3	746
4	996
5	1246
6	1496
10	2496
12	2996
15	3746
20	4996
30	7496

Table 1-3 Maximum Number of Pre-Archive Hours

The number of events per journal type and the number of pre-archive hours are configured under the VOLUME CONFIGURATION target of the NCF. This function does not change the method of retrieving and displaying events. It only changes the quantity of events that can be retrieved and displayed.

1.6 R610 Enhancements

Introduction

TPS network software Release 610 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee; a worldwide group with representatives from North America, Europe, the Far East and Australia. The following lists the enhancements that were made as a result of their recommendations:

- Extend List Entities Command
- Add Date to Alarm Summary Displays
- Picture Editor Variant Branching to Values
- TPN-Wide US Keylevel Access
- Doc Tool Filter Enhancements
- Period Pre-Defined Doc Tool Queries
- Serial Interface Support for I/O Simulator

In addition, changes and additions are provided to support Fieldbus.

Extend List Entities Command

This function extends the function of the existing DEB EC file 'LS' and 'LI' commands. Currently these commands are limited to listing all of the entities (AKA points) in a module, box or IDF. These commands are extended to enable restricting the entities listed to those that meet "point realm" or "point realm and point type" criteria.

Point realms and their associated point types and format changes are listed in the following tables.

POINT REALM	ASSOCIATED POINT TYPES ²			
AM	REGLATRY	NUMERIC	COUNTER	CUSTOM
	FLAG	TIMER	SWITCH	
HG	COUNTER	DIGIN	DIGOUT	DIGCMP
	ANALOGIN	ANALOGOUT	ANALOGCOMP	FLAG
	TIMER	NUMERIC	LOGBLK	REGLATRY
	CTRLCNTR	BATCHASSN	PROCESSMOD	
CIU	ACIDP	CRDP		
NIM	ANALOGIN	ANALOGOUT	REGPV	REGCTL
	DIGIN	DIGOUT	DIGCOMP	LOGIC
	PROCESSMOD	FLAG	TIMER	NUMERIC
	ARRAY	DEVICECONTROL	CONTROLMODULE	

Table 1-4 LS and LI Command Point Realms and Associated Point Types

EXISTING 'LS' AND 'LI' COMMAND FORMATS				FORMAT ADDITIONS ³
LS	<LIST path>	<MOD # or BOX #>	<hwy# or ucn# > ⁴	<POINT REALM and optional POINT TYPES ⁵ >
LI	<IDF Path>	<LIST PATH>		<POINT REALM and optional POINT TYPES ⁴ >

Table 1-5 LS and LI Command Format Changes

² These point types match those seen in the DEB interactive mode when executing a LIST ENTITIES IN MODULE or LIST ENTITIES IN IDF command, except that any imbedded spaces have been removed.

³ This is the portion of the command format added by this function. It is optional in nature and is omitted when “all” of the point types are to be listed.

⁴ Note: Use “&B or &b” for this field when listing entities in a module on the LCN. Use HWY # or UCN # for this field when listing entities in a box, on a hiway or UCN.

⁵ One or more point types restricted only by the available space on the 80-character command line.

Add Date to Alarm Summary Display

Currently, only the time of occurrence is displayed for each alarm on the Alarm Summary displays (Area Alarm Summary, Unit Alarm Summary and Alarm Annunciator).

This R610 enhancement adds the date of occurrence to each alarm displayed on the Alarm Summary Displays.

A new target appears in the header of the displays, which is used to toggle between time, date and both time and date of the alarm occurrence. The current time field for each alarm line changes accordingly (based on the target indication) to show the time, date and both time and date of the alarm occurrence. In the “both time and date” mode the time and date of the alarm occurrence is displayed in half-height text.

PE Variant Branching to Values

The Add Variant command in the picture editor has been modified to accept a value as its object. The Add Variant command previously handled a string or a subpicture by looking for a quote or SUBPICTURE as a keyword. In R610 the command now also recognizes the VALUE keyword to signal that a value will be used as the object. The picture editor object file created during compilation is compatible with previous releases. The picture editor source file using this new function is not backwards compatible (can not be edited) with a previous version of the picture editor.

TPN-Wide US Keylevel Access

Provides the ability to read and write to the keylevel access parameter of any GUS or (US or EST). This parameter is available to all GUSs or (USs or ESTs) and AMs connected to the same LCN independent of the console.

The parameter is one of the \$PRSTSxx parameters defined by the node number. All changes are journalized to the HM and to the RTJ.

In addition, 100 \$PRSTSxx parameters are added for each of the following parameter types:

- Real
- Integer
- Boolean
- Entity

Fifty \$PRSTSxx parameters are added for the following parameter type:

- String

**Doc-Tool Filter
Enhancement**

This functionality enhances the Documentation Tool filter capability provided by the “Conditions Box” for both the Query and Filter commands as follows:

- The right-hand side of a conditional expression for a HWY/UCN/Unit/Node Query command will be able to be specified by either a parameter name or a value.
- The right-hand side of a conditional expression for a Journal Query or a Filter command will be able to be specified by either a field name or a value.

In order to differentiate a parameter or field name from a value on the right-hand side of the conditional expression, the special prefix character ‘%’ must be used.

For a HWY/UCN/Unit/Node Query command the right-hand side of the conditional expression can now be a parameter name.

Example 1:

Old Functionality: MODE = MAN

New Functionality: MODE = %NMODE

(This compares the value of the parameter MODE to the value of the parameter NMODE)

Example 2:

Old: PV > 60.0

New: PV > %PVHITP

(This compares the value of the parameter PV to the value of the parameter PVHITP.)

**Periodic Pre-Defined
DOC TOOL Queries**

This functionality provides the user the ability to trigger & cancel a pre-defined Documentation Tool query via CL “SEND” messages or Command Processor commands and have the query result output to a file or printer automatically. Additionally, this function enables the user to sort the result by Entity name or by user specified first field. User specified first field is defined as the first field in the “Parameter(s) To Show” list in the pre-built Query. The sort capability is to be extended in a manner such that the Schematic Query Actors may also utilize it.

CL SEND Statement

Operating Instructions:

SEND: "\$QFILE <Node Number> <Descriptor> <Pathname>
<XX>"

SEND: "\$QPRINT <Node Number> <Descriptor> \$P<n> <XX>"

SEND: "\$QCANCEL <Node Number> <YY>"

Where,

\$QFILE is the keyword for specifying a query whose result should be output to a file.

\$QPRINT is the keyword for specifying a query whose result should be output to a printer.

\$QCANCEL is the keyword for canceling the running query being executed by Doc Tool background task.

<Node Number> is a required field, and specifies the LCN US node, which is to process the request.

<Descriptor> is a required field, and specifies the name of the pre-defined query (max of 16 characters with no spaces).

<Pathname> is a required field, and specifies the path to the file to which the query results should be output.

\$P<n> is a required field, and specifies the printer ID on which the query results are to be printed.

<XX> and <YY> are optional fields which can take the values given below:

XX = [\$BYPASS | \$SORT | \$BYSORT | \$SORTF | \$BYSORTF
| \$NSORT | \$BYNSORT]

YY = [\$BYPASS]

The optional fields provided for sorting are further explained below:

Sort Directive	Sorted By
\$BYPASS	Allows the user to specify that the CL Send message itself is not to be output to the Real Time Journal or put into the Operator Message Summary Display (the message will still go into event history on the HM).
\$SORT	Sort the Query Result by the Entity Name.
\$BYSORT	Facilitates both \$SORT & \$BYPASS operations (\$BYPASS + \$SORT).
\$SORTF	Sort the Query Result by the user specified first field.
\$BYSORTF	Facilitates both \$SORTF & \$BYPASS operations (\$BYPASS + \$SORTF).
\$NSORT	Explicitly does not sort the Query Result.
\$BYNSORT	Facilitates both \$NSORT & \$BYPASS operations (\$BYPASS + \$NSORT).

Example:

To sort the Query Result of the pre-built query “HGPOINTS” on US node 35 by user specified first field and store the results into the file “NET>TEMP>HGPOINTS.XX”, issue the following SEND message:

SEND: “\$QFILE 35 HGPOINTS NET>TEMP>HGPOINTS.XX \$SORTF”

Command Processor Commands

The newly added CP commands can be executed in Scheduled ECs both in the foreground and background mode.

Operating Instructions:

QFILE <Descriptor> <Pathname> <XX>

QPRINT <Descriptor> \$P<n> <XX>

QCANCEL

Where,

QFILE is the keyword for specifying a query whose result should be output to a file.

QPRINT is the keyword for specifying a query whose result should be output to a printer.

QCANCEL is the keyword for canceling the running query being executed by Doc Tool background task.

<Descriptor> is a required field, and specifies the name of the pre-defined query.

<Pathname> is a required field, and specifies the path to the file in which to put the query results.

\$P<n> is a required field, and specifies the printer ID on which the query results are to be printed.

<XX> is an optional field, which can take the values as XX = [-SORT | -SORTF | -NSORT]

The optional fields provided for sorting are further explained below:

Sort Directive	Sorted By
-SORT	Sort the Query Result by the Entity Name.
-SORTF	Sort the Query Result by the user specified first field.
-NSORT	Explicitly does not sort the Query Result.

Example:

To sort the results of the pre-built query “HGPOINTS” by entity and print the results to \$P8, issue the following command:

QPRINT HGPOINTS \$P8 -SORT

QRYSORT Parameter (PSDP)

The sort functionality being provided for the CL/CP query is extended to support the schematics, thus allowing schematics that use the existing Schematic Query Actors to make use of the sort capabilities. To support this feature for the Schematic actors, a PSDP **QRYSORT** parameter is added on all US/GUS/EST nodes. The query results will be sorted, or not sorted, depending on the value of the QRYSORT parameter explained in the table below. The QRYSORT parameter is initialized to 0 at node startup. The possible modes are:

TYPE	QRYSORT Value	Definition
Temporary	0	“No Sort”. The Sort Directive in the CL/CP query will override this option.
	1	“Sort by Entity Name”. Automatically reset to ‘No Sort’ at completion of Query. The Sort Directive in the CL/CP will override this option.
	2	“Sort by user specified first field”. Automatically reset to ‘No Sort’ at completion of Query. The Sort Directive in the CL/CP will override this option.
Node Global	3	“No Sort”. This option over rides the CL/CP Sort Directive. This value is not automatically reset. Once set this option can only be changed to another Node Global type or Reset.
	4	“Sort by Entity Name”. This option overrides the CL/CP Sort Directive. This value is not automatically reset. Once set this option can only be changed to another Node Global type or Reset.
	5	“Sort by first user specified field”. This option overrides the CL/CP Sort Directive. This value is not automatically reset. Once set this option can only be changed to another Node Global type or Reset.
Reset	6	Used to Reset the Node Global options. Once a Node Global option is set, it can be changed to a temporary type only by setting this option. Setting this option resets the PSDP value to 0.

**HPM IO Simulator –
SI IOP Support**

SI IOPs can be supported in real mode in the HPM IO Simulator when the IOPs are physically present in the HPM. Under this mode SI data is accessed directly from the SI IOP Card. SI IOP, with no physical hardware present in the configured slot, is simulated as previously done. The support mode of the SI IOP, real or simulation is updated only when there is IOP configuration change in the HPM. Insertion or removal of physical SI IOP during normal operation does not change the support mode of the SI IOP.

SI IOP operating in real mode will have the text string "*** IOP in real mode ***" shown on the module detail display in place of "*** IOP in simulation ***".

- **NOTE:**
The SI is the only IOP that can be supported in this manner in the HPM I/O Simulator. All other IOP types continue to be supported in the simulation mode only.
- The HPM simulator does not support an external array of 32 elements. It supports only an array of 16 elements.

No	Setup	HPM Simulator communication mode (Simulated / Actual data)
1	Load the HPM IO Simulator personality Insert an SI Card into slot #7 only Configure SI Module 5 (Slot #7) and SI Module 6 (slot #8)	SI module 5 is operating in real mode and SI module 6 is operating in simulation mode.
2	Remove the SI Card from slot #7 and insert it into slot #8	SI module 5 shows "NR?" (No Response) and SI module 6 is still simulated.
3	Reconstitute and reload the box configuration	SI module 5 is now simulated and SI module 6 is operating in real mode.

**HPM IO Simulator –
CFG Error on FBUS
IOP Configuration**

Configuration and simulation of FBUS IOP is not supported in the HPM I/O Simulator. A CFG error is posted in the FBUS module slot on the HPM node display after the IOP configuration is being loaded.

**Changes and
additions to
support Fieldbus**

The following changes and additions are provided in R60x to support fieldbus.

1. A new CMTYPE field has been added to the following list of displays to provide type information for fieldbus point types:
 - LCN Node Point Summary
 - UCN Device Point Summary
 - UCN Device Module Point Summary
 - UCN Device IOP Slot Summary
2. Detail displays have been added to support the fieldbus point types. The Group Display will now support the FBCM fieldbus point type.
3. The HPM status display will now support the fieldbus IOP.
4. Support displays for the fieldbus network and devices have been added as follows:
 - IOM Detail Status Display for Fieldbus IOP
 - Fieldbus Network Status Display
 - Fieldbus Device Slot Summary
 - Fieldbus Network Statistics Display

**Fieldbus HPM/CL
functionality**

HPM/CL sequence can access CM point type parameters.

**Fieldbus HPM
functionality**

Continuous control functions in the HPM have been tested accessing fieldbus function AI, AO, and PID blocks parameters for data acquisition and cascade control.

Fieldbus data scanning eliminates the pre-fetch constraint to certain frequently accessed fieldbus parameters in HPM/CL programs and HPM control points.

Logic and CL access to fieldbus parameters allows access to fieldbus parameters from CL programs and Logic points in the HPM.

The HPM also supports FB-IOP software upgrade.

UIS xPM functionality

Expanded Regulatory Control Pre-fetch Table allows pre-fetching up to seven I/O connections to accommodate the configuration of fan-out to multiple AO output connections. This enhancement applies to the PM, APM, and HPM.

Externally Visible bias value is now available for the Ratio Control algorithm of xPM Regulatory Control point. This maintains consistency in output bias behavior and user interface as compared with other Regulatory Control algorithms such as AutoMan, MulDiv, and Summer.

The operator is now able to add a bias term to the calculation if it is needed for the control strategy. Internal ramping bias value can be reset, which previously was not possible. This enhancement applies to the PM, APM, and HPM.

**Fieldbus IOP
functionality**

Fieldbus IOP functionality provides the following:

- Supports up to four independent H1 wire fieldbus networks comprised of a single segment that can be individually loaded, commissioned, and de-commissioned.
- Supplies power for the fieldbus networks.
- Provides fieldbus Link Master capability for each fieldbus network.
- Permits loading to two link schedule domains for primary schedule and “new” schedule for each fieldbus network.
- Provides Application Time Scheduler capability for each fieldbus network.
- Supports access via fieldbus to FB-IOP fieldbus network configuration data.
- Permits up to 63 independent virtual communication relationships per fieldbus network.
- Supports access to fieldbus devices utilized for data acquisition, local control, or supervisory control.
- Interoperates with other vendor devices that have been certified by the fieldbus to be interoperable.
- Provides up to 119 configurable software slots, which may represent:
 - (1) a fieldbus Application Process (FBAP) Virtual Field Device;
 - (2) a fieldbus foundation function block;
 - (3) a fieldbus foundation transducer block; or
 - (4) a combination of a single fieldbus Foundation function block and its associated transducer block, on any of the four independent fieldbus networks to which the FB-IOP is attached.

**Fieldbus IOP
functionality,
continued**

Fieldbus IOP functionality provides the following:

- Supports field device alert and alarm reporting and alarm recovery.
- Provides value added user notification beyond standard fieldbus specifications such as advising the user when simulation mode has been enabled.
- Provides the means for integrating field device data into the TPS system.
- Provides the means for integrating field device status into the TPS system.
- Provides the means for integrating fieldbus defined block status into the TPS system.
- Supports field device configuration, commissioning, and de-commissioning, including validation of “device level” compatibility with associated TPS device configuration.
- Supports interfaces to enable field device save/restore.
- Provides up to 32 software slots for the HPM to fast scan analog information.
- Provides up to 16 software slots for the HPM to fast scan digital information.
- Provides software upgrade capability.

1.7 R611 Enhancements

Introduction

TPS network software Release 611 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee; a worldwide group with representatives from North America, Europe, the Far East and Australia. The following lists the enhancements that were made as a result of their recommendations:

- On-line Migration of Redundant Application Modules
- On-line Migration of AM External Load Module Upgrades
- Allow GUS to Invoke Operator Displays over Engineering Main Menu

On-line Migration of Redundant Application Modules

After upgrading the secondary (backup) AM, the system status display shows the primary as 'OK', and the secondary state as 'WARNING,' with a status detail of "THE SECONDARY IS AN UPGRADE". After shutting down the primary AM, the system status display shows the old primary AM node in 'QUALIF' state and the upgraded primary AM node in 'WARNING' state. The status detail shows "THE PRIMARY HAS NO BACKUP". A reload of the secondary with the same release of software running on the primary now results in the AM database loading from the primary AM through the redundancy channel. The system status display shows the primary as 'OK' and secondary as 'BACKUP'.

On-line Migration of AM External Load Module Upgrades

The secondary AM may be upgraded with a different release of external load modules even though the AM software is not upgraded. If the AM software is being upgraded then the preceding scenarios apply. If the AM software is not being upgraded, then a procedure supplied with the external load modules must be followed. The upgraded external load modules are activated by a failover to the upgraded AM secondary.

Important note for redundant AM (CLM) nodes using the "FURTHER EXTERNAL DIRECTIVE" option (third page of the NCF node configuration): Each node of the redundant pair must have its own APCFGnn.X file. (where nn = node number). Prior to R611, only the primary AM (CLM) of a redundant pair needed this file.

Allow GUS to Invoke Operator Displays over Engineering Main Menu

This package provides the ability to invoke the Point Detail (or an Operator display) in the Native Window from a script, when the Engineering Main Menu is displayed in the Native Window, or when the Native Window was open before the display was called. Prior to this enhancement, the request would merely re-display the Engineering Main Menu.

The table below gives the actions for these functions.

Source of Request for Operator display	Action when target node is at an Operator Display	Action when target node is at an Engineering function	Action when target node running "GUS/U^XS" is at the Engineering Main Menu	Action when target node running "UNP" is at the Engineering Main Menu
US or Native Window picture	Operator display is called	No action at target node	Operator display is called	Engineering Main Menu is redisplayed
GUS picture or script	Operator display is called	No action at target node	Operator display is called	Engineering Main Menu is redisplayed
Multiple Schematics picture (UXS)	Operator display is called	No action at target node	Operator display is called	Engineering Main Menu is redisplayed

1.8 R620 Enhancements

Introduction

TPS network software Release 620 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee; a worldwide group with representatives from North America, Europe, the Far East and Australia. The following lists the enhancements that were made as a result of their recommendations:

- Keylock NCF Access
- Trending Unassigned Unit Points
- Reverse Output Indicator
- Activate/Deactivate Alarm Annunciation during Engineering Functions
- Update Keyword Descriptor on Group Display
- Increase Hairline Trend Value Decimal Places
- On-Delay / Off-Delay –NIM & HG
- Configurable Access Levels
- Enhanced Change Zone
- Change Alarm Priority on Active AM Points
- Custom Detail Display
- Additional Diagnostic Parameters
- Doc Tool Queries with Selection list
- Increase File Size limits for EB Files
- CL SET statement with WHEN ERROR clause
- Do Not Fail HPM CL and APM CL programs when accessing data from a Failed IOP
- Expand DEB Back Build Capabilities
- Logic Point Configuration Diagram
- Backbuild Subpicture from Schematic Source
- New Actors for Console, Station, and Node Number

A summary of the TPS Network R620 enhancements follows:

**Keylock NCF
Access**

Functionality Modifications

- Provides configurable keylevel access for selection of NCF configuration pages. Allows access by operator/supervisor of NCF configuration pages to be blocked.
- Configuration option is located in NCF “System Wide Values/Console Data” configuration section. Default access is “Operator” to retain current operation.
- If operator or supervisor access is allowed, a temporary “View Only” file is created for the session. The temporary file is deleted when the session is ended.
- Multiple “View” sessions are supported.
- Prevents the erroneous creation of an NCF.WF (work file).
- Engineer access will still create a work file as done currently.

**Trending
Unassigned
Unit Points**

Previously an operator/supervisor could not assign “Off Area” points to trend displays.

Functionality Modifications

- Enable operator/supervisor assignment of “Off Area” points to trend displays.
- Keylevel will no longer be checked to trend or invoke the detail display from the group display for “Off Area” points.

Comment

Operator/Supervisor modifications to “Off Area” point.parameters are still not allowed.

“Off Area” within this context refers to a point which belongs to a Unit which is ‘not configured’ for the local Area Data Base or ‘not assigned’ in the Unit Assignment Display.

**Reverse
Output
Indicator**

Functionality Modifications

- A new parameter, OUTIND, is provided to allow engineers to reverse the output indication on the Analog Output and Regulatory Control points of all the xPM modules on the NIM. Implementation is similar to the one already existing in the HG.
- This parameter can be configured as DIRECT or REVERSE and affects the OP% & OP bar graph display in the Detail and Group displays of xPM Regulatory Control & Analog Output points.
 - OUTIND = Direct
4 - 20ma signal drives OP 0 - 100%
 - OUTIND = Reverse
4 - 20ma signal drives OP 100 - 0%

**Activate/
Deactivate
Alarm
Annunciation
during
Engineering
Functions**

Currently, the alarm annunciator horn will not sound if all US or GUS or EST stations within a console have an engineering function displayed in the native window. With US's this may not be a problem because it is impossible to monitor the plant with only the Engineering Menu displayed. However with GUS/ EST, it is possible to have only an engineering function displayed on all Native Windows and still monitor and control the plants via GUS/ EST graphics. The Native Windows can be buried under other windows leading to an unsafe situation.

Functionality Modifications

- Provision has been made to configure alarm horn annunciation behavior on a per node basis in the NCF – to cover those times that an engineering function is on display.
- Configuration provided per US on the NCF “US Configuration” page:
 - “No” – operate as previously done, not sounding the annunciator when an engineering function is active in the native window.
 - “Yes” – sound the annunciator even when an engineering function is active in the native window.

**Update
Keyword
Descriptor on
Group Display**

Background

Currently if the keyword descriptor (PARAMETER KEYWORD) is changed for a point and the point is loaded into the system, the new keyword descriptor appears on the point's Detail Display but not on the Group Display.

Functionality Modifications

- The keyword descriptor shown in the group display will reflect the actual KEYWORD for the point.

**Increase
Hairline Trend
Value Decimal
Places**

The number of places to the right of the decimal point in a hairline trend value shall be determined by the Point.PVFormat parameter.

Functionality Modifications

- Determine the number of places to the right of the decimal point based on the value of Point.PVFormat.

On-Delay/Off-Delay – NIM & HG

Functionality Modifications

- Point parameters have been added to configure the time delays and filtering of alarms on a per point basis.
- \$ALDLOPT – defines the delay type (OnDelay/OffDelay) and the alarm filtering. Selection of the following determines which alarm levels (LOW, HIGH, EMERGENCY) are affected.

On–Delay – suppresses alarm condition reporting for the configured amount of time after the triggering alarm condition. If an alarm condition exists when the time delay expires, the alarm will be reported. If an alarm condition does not exist when the time delay expires, even though the point may have transitioned in and out of alarm, then no alarm will be reported. Selection of the following determines which alarm levels (LOW, HIGH, EMERGENCY) are affected.

ONDLYLO – Suppresses alarms and RTNs with priority <= LOW

ONDLYHI – Suppresses alarms and RTNs with priority <= HIGH

ONDLYEM – Suppresses alarms and RTNs with priority <= EMERGENCY

Off–Delay – suppresses additional alarm condition reporting for the configured amount of time after the reporting of the initial (triggering) alarm condition. If an alarm condition does not exist when the time delay expires, then the return to normal event will be reported. If an alarm condition exists, even though the point may have transitioned in and out of alarm, nothing will be reported. Selection of the following determines which alarm levels (LOW, HIGH, EMERGENCY) are affected.

OFFDLYLO – Suppresses RTNs and repeated alarms with priority <= LOW

OFFDLYHI – Suppresses RTNs and repeated alarms with priority <= HIGH

OFFDLYEM – Suppresses RTNs and repeated alarms with priority <= EMERGENCY

- DLYTIME – defines the effective delay time for the selected option. Reset for all return–to–normal transitions of Off–Delay timing. Reset for all alarm transitions for On–Delay timing.
- Change of DLYTIME/\$ALDLOPT/alarm priority terminates the current timer. The new configuration will be in effect for the next applicable event.

ATTENTION

Alarm events delayed because of alarm delay functionality are reported in the journal without holding/displaying the current PV value.

**Configurable
Access Levels**

This UIS package will allow the engineer to set the access level on a per parameter basis for different types of parameters. This package was originally implemented in UIS Release 530, by grouping parameters into four different categories and then allowing the engineer to configure access to these parameters on a category basis. The four categories provided were Alarm Limits, Control Limits, Range Limits and Tuning Parameters. Instead of having only four different categories to choose from to set access level, a file can be created with a list of parameters and the access level can be chosen for each individual parameter.

- Access level specified in a user defined file (KEYLEVEL.KL)
File size limited to 500 parameters for performance and memory allocation considerations.
File format: ParameterName AccessLevel(P/E/S/O)
Where P = Program, E = Engineer, S = Supervisor, and O = Operator
Located in &ASY directory.
- File is read into memory during US load or area change
File status reported via Aux Node Status message.
File Found/Not Found messages set "OK" node status.
File Found - Errors Detected messages set "WARNING" node status and function is active for valid entries.
- Parameter Access hierarchy
The parameter access levels defined in the KEYLEVEL.KL file will override the majority of other access definitions.
- Configured access will provide a single access override even when the default is different for different point types.
- The operator initiated Alter Parameters DEB function will not use the KEYLEVEL.KL file. It will use the data owner default access levels.

Enhanced Change Zone

Background

The Change Zone sub-picture in TDC is used to view the details of points and to accept key parameter values. This sub-picture may form part of a schematic, thus allowing the user to view/change parameter values without having to leave the schematic. The following functionality is provided.

Functionality Modifications

The purpose of this proposed functionality, called the Enhanced Change Zone (ECZ), is to provide the functionality of the Standard Change Zone (SCZ) and add Point status, Alarm Status, System Alarm, PV, SP, OP Status, and a Trend graph of PV/SP/OP.

The Enhanced Change Zone will support the same point types and contain all the same data as the current standard Change Zone, with the exception of the PTEXECST parameter. In addition to the SCZ, the ECZ will provide the following:

- Indicators for the System Alarm Status, Point Alarm Status, PV Status, OP Status, SP Status and Point Status.
- A Mini-Trend graph automatically scaled with the OVERVAL parameter, with color-coded tick marks of alarm limits reflecting their alarm priorities, with targets to change the time base and targets to scroll.

Pertinent Information:

- User can choose whether to use the SCZ or the ECZ in a specific schematic.
- ECZ features appear for specific point types, i.e. regulatory control, AO, AI, DI, DO, Device Control, etc. SCZ features appear for non-ECZ point types.

NOTE: U^XS Multiple Schematics users should be aware that Multiple Schematics 300.1 (200.6) does not recognize the new Enhanced Change Zone phantoms added to TPN R620. Therefore, any attempt to use an Enhanced Change Zone in a U^XS Multiple Schematics Window will be unsuccessful. Nothing will be drawn to the change zone area on the schematic.

Schematics that use the OVERLAY or MULT_OV method of calling change zone can easily upgrade from standard change zone to the Enhanced Change Zone by renaming file ENHAN_CZ.DO to CHG_ZONE.DO. It is not necessary to edit or recompile the picture source. However, this method should not be used if you want the change zone to work in a U^XS Multiple Schematics window.

**Change Alarm
Priority on
Active AM
Points**

Background

Currently, alarm priorities on AM points cannot be changed unless the point is inactive. NIM points do not have to be inactive in order for alarm priorities to be changed. HG points do not have a ‘point active’ status.

Functionality Modifications

- Allows the changing of alarm priority for AM point types without setting the point active status (PTEXECST) parameter inactive.
- Alarm condition is re-evaluated after the priority is changed.
If value is in alarm, the point is re-alarmed to update the US alarm list—even if the point was already in alarm prior to the change.
A change to NoAction will cause a point in alarm to be removed from the alarm list. A change to Journal, JnlPrint, or Printer will also remove the point from the alarm list but the alarm will also be journaled or printed as configured.
- Includes fix to make the AM Flag point alarming consistent with NIM Flag point.

**Custom Detail
Display**

Allow the user to configure, on a point-by-point basis, what custom display is invoked when the standard Detail Display would normally be invoked. A new parameter (\$CDETAIL) has been added to HG/AM/NIM points. \$CDETAIL holds the name of the custom schematic that defines the user-supplied Custom Detail Display. When the “Detail” of the point is requested, the schematic identified by \$CDETAIL is used instead of the standard Detail Display.

Functionality Modifications

- When the “Detail” of a point is requested, the schematic identified as the Custom Detail Display is used instead of the standard Detail Display.
- If a Custom Detail Display is not specified, the standard Detail Display call is made.
- If a Custom Detail display is specified but cannot be invoked, then the standard Detail Display is invoked. An error message is displayed.
- Applies to all standard Detail Display invocation requests and the DETAIL actor requests. The existing DET_PAGE actor will continue to invoke the Standard Detail Display. This provides the user with a mechanism for invoking Standard Detail when Custom detail is configured.

There is a new PERMENU display (CDET_CLR) that allows the user to modify/clear \$CDETAIL.

**Additional
Diagnostic
Parameters**

Background

History Collection Enable/Disable is currently viewed and/or changed only from the HM node status display. Auto Checkpointing Enable/Disable is currently viewed and/or changed only from the respective data owner's node status display (AM, HG, NIM and CG).

Functionality Modifications

- Add parameters for HM History collection Enable/Disable and LCN node Auto Checkpointing Enable/Disable, which can be viewed and/or modified from custom schematics.
New PSDP parameters.
 - Auto Checkpointing Enable/Disable for the NIM, HG, AM and CG.
 - History Collection Enable/Disable for the HM.
- Keylock access will be consistent with NCF configured levels.

**DOC Tool
Queries with
Selection List**

Background

Currently when building a query specification in the Doc Tool , the “Entity” port allows the entry of one or more entity names, limited by the number that will fit in the port, perhaps 10 or so.

Functionality Modifications

- Provides for entry of a selection list pathname in the “Entity” port.
- “NET>” or “\$Fxx>” identifies a pathname in the “Entity” port.
- More than one pathname can be specified in the “Entity” port.
- A combination of a file pathname and entity name separated by a comma or space is also supported.
Example: NET>SEL>ENTITIES.XX, TAG1 TAG2.
- Supports removable media file pathnames.

**Increase File
Size Limits for
EB Files**

Background

Prior to R620, Exception Build Files were limited in size to 32,767 records.

Functionality Modifications

Increase file size limits for Exception Build files to 65,535 records.

This file size increase was necessary to support the R620 enhancement “Expand DEB Back Build Capabilities”, which captures additional parameters when creating Exception Build Files.

**CL SET
Statement
with WHEN
ERROR
Clause**

Background

Currently CL SET statements require a huge IF/THEN statement to test that the point is in the correct state to accept a store. If the point is not in the correct state then the program fails. This applies to both APM and HPM environments.

Functionality Modifications

- IF an error occurs on a SET statement that has the new WHEN ERROR clause, the WHEN ERROR clause will be executed; the CL sequence will not be stopped.
- When a WHEN ERROR clause exists on a SET statement, the maximum number of consecutive WHEN clauses will be limited to one hundred (100).
- Compliments WAIT and IF statements with WHEN ERROR (Also New Functionality).

NOTE

If the WHEN ERROR clause specifies the execution of a PAUSE statement when an error is detected, execution of the program will NOT stop if the program is in fully automatic mode.

**Do Not Fail
HPM and APM
CL Programs
when
Accessing
Data from a
Failed IOP**

Background

CL in the HPM and APM fails with an I/O link error when it is referencing a point on an analog input board that has failed. CL should detect a bad analog input board without failing the program.

Functionality Modifications

- Abnormal handler option FAIL_IF_ERROR works like present version (FAIL and stop CL).
- Abnormal handler option IGNORE_IF_ERROR directs handler to not execute on error.
- Abnormal handler option INVOKE_IF_ERROR directs handler to run on error.
- IF and WAIT statements WHEN ERROR clause allows user CL to take action on IOP failure.
- Applies to APM and HPM, not PM, AM, MC.
- SET statement WHEN ERROR clause allows user CL to take action on IOP failure. Note that if the WHEN ERROR clause specifies the execution of a PAUSE statement when a failed IOP is detected, execution of the program will NOT stop if the program is in fully automatic mode.

SPECIAL NOTE

If a WHEN ERROR clause specifies the execution of a PAUSE statement when an error is detected, execution of the program will NOT stop if the program is in fully automatic mode. Note that error clauses can be used with SET, READ WRITE, STATE CHANGE, IF THEN ELSE, WAIT, and INITIATE statements.

**Expand DEB
Back Build
Capabilities**

Background

Currently NIM Array and ProcMod parameter data is not captured in an Exception Build (EB) file for parameters of type Flag, Numeric, String and Time.

Likewise Custom Data Segment (CDS) parameters of type Time and which are non-build visible are not captured in Exception Build files.

Functionality Modifications

- Provides the capability to back build Numeric, Flag, String, & Time parameters of NIM ProcMod and Array point types into an EB file. Serial I/O parameters are excluded from this. Only those parameters containing non-default values will be back built to EB files, in order to save space.
- Provides the capability to back build non-build visible CDS parameters of type Time into an EB file. Only those parameters containing non-default values will be back built to EB files, in order to save space.

NOTE

The number of ports seen on a DEB display may be greater than the number actually used when gathering data for the parameters identified in this package. An example of this is where 14 Numerics are specified. The display will show 16 ports because there are 4 ports per line, but only 14 are actually used. If a user enters data in the two extra ports it is ignored, but this may not be obvious to the user.

Comment

These previously missing parameters are now in EB files and loaded to the system by the DEB. It should be noted that both EB files and Intermediate Data Files (IDFs) will increase in size if they contain any of the parameters referred to in this package.

Logic Point Configuration Diagram

Background

The request is to make Logic Point configuration diagram on page 2 of the Detail Display similar to the Device Control Point configuration diagram.

Functionality Modifications

Currently, the Standard Detail Display for APM/HPM Logic Points does not show the logical sequence (interconnections) between logic blocks in a graphical manner. It was decided to produce a custom schematic solution since the graphical capabilities of the Native Window are limited. The Custom Detail Display function will allow this schematic to be configured as the custom detail display for logic points. Following are the functional modifications:

- Enable the user to view the logic connections of the inputs, outputs, and logic gates of a single APM/HPM logic point in a graphical “1-wide” layout. Navigation begins by selecting one of the output connection targets. Selecting one of that block’s inputs makes the next upstream block and its associated inputs visible to its left. The user can continue navigating “upstream” in this manner.
- A chain of six logic blocks is visible at one time.
- Selecting an input to the leftmost gate creates a new page. The leftmost gate on page 1 becomes the rightmost gate on the new page.
- A maximum of 5 pages can be traversed which allows a logic chain of 24 gates.
- Selecting a new input in the chain causes the upstream portion of the graphical chain to be broken at that point.
- Navigation properties are persistent. If the user leaves the display and returns, the display will retain its scrolled position.
- A second “view” of the display will show an overview of every gate associated with the logic point in a non-graphical view along with its immediate inputs and outputs.
- Targets are associated with each gate that allow the user to select any gate as the starting point of a new logic chain.
- This view is useful not only as an overview, but also when the user desires to view a block that has no physical output associated with it.

NOTE:

The functionality related to the Logic Point Custom Detail Display applies only to APM and HPM logic points. PM logic points cannot be used in conjunction with \$LOGIC and \$LOG_OVR; therefore, the user should avoid assigning either of these schematics to the \$CDETAIL parameter for PM logic points.

**Backbuild
Sub-Picture
from
Schematic
Source**

Background

- Schematic maintenance sometimes involves analyzing schematics that hold subpictures that somebody created a long time ago. The source code is not always available.

Functionality Modifications

- New command RECREATE to recreate a single named subpicture source file automatically from a Picture Editor schematic source file. Stores as .DS file.
- New command RECREATALL, to recreate all “top level” subpictures automatically from a PE schematic source file.
- Applies to any PE built subpictures.
- Recreates only “top level” subpicture.
- Nested subpictures can be recreated from previously recreated higher level subpicture source.
- Available as a new batch mode command, MULTRECR.

**New Actors
for Console,
Station, and
Node Number**

Background

Actors do not exist to get the console number, station number, or node number from which the actor is called.

1.8.1 Functionality Modifications

The following three new actors have been added:

- Get Console Number—This actor returns the console number from which the actor is called.
- Get Node Number—This actor returns the node number from which the actor is called.
- Get Station Number—This actor returns the station number from which the actor is called.

The values that are returned by these actors are integers. No input parameters are returned by these actors.

1.9 R630 Enhancements

Introduction

TPS network software Release 630 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee; a worldwide group with representatives from North America, Europe, the Far East and Australia. The following lists the enhancements that were made as a result of their recommendations:

- GUS/APP LCNP Auto Restart
- Engineering Change Journal
- Prevent Mode Selection of Non-Applicable Controller Modes
- Re-Range OP Parameter on Trends
- Doc Tool Efficiency Improvements
- Add Selection List to Find Names
- Feedback on Point Activation / Inactivation Command
- HPM DigComp Initialization Enhancement
- TPS Fieldbus System Alarm Enhancement

GUS/APP LCNP Auto Restart

- Provides automatic loading of TPN personality on the LCNP(4) of a GUS or APP node any time the LCNP(4) goes into the PWR_ON state.
- Option is configurable via check boxes added to the existing GUS & APP node Configurator Utility
- Personality and NCF source paths are defaulted to W and N. They are not configurable.
- Function will continue to retry requests for NCF, allowing time for HMs to restart.

This function can be implemented only if the following software is installed: GUS R31x and/or APP20x.

**Engineering
Change
Journal**

- Provides an audit trail report of prescribed engineering changes. Will include changes accomplished by functions on the Engineering Main Menu, Builder commands, and through ECs. For example:
 - Changes to system configuration files, i.e. NCF
 - Changes to process data bases
 - Compilation of user written applications
- Engineering change events will be written to the System Status Change Journal and Real Time Journal using the 'Off Process System Change Event'. Events are labeled with Event Type code "41" in columns 131 & 132
- Data retrieval will be via
 - New Engr Changes target on the Event History Retrieval Display
 - New Engr Change Messages target on the Doc Tool Journal Page
 - Event Initiated Reports can request report type "ENG"
- User has the ability to sort on the Event Type code using Doc Tool or other third party applications.

NOTE: **All** stations in a given console need to be running R630 (or newer) software for this functionality to work correctly.

**Prevent Mode
Selection of
Non-
Applicable
Controller
Modes**

- New parameter “\$MODESEL” is added for NIM, AM, and HG point types which have a control mode parameter. The new parameter specifies acceptable modes on a per point basis.
- The selection list of MODE and MODEATTR values presented to an operator on the Standard displays will be determined by the point algorithm and the value of \$MODESEL.
- The following standard displays are affected: Detail, Group, Change Zone, Enhanced Change Zone, GUS Faceplate, GUS Group, GUS Change Zone.
- The behavior of the ‘MAN’, ‘AUTO’, and ‘NORM’ keys is also being modified based on \$MODESEL. The GUS mode menu is also affected.
- The new parameter value defaults to allow current (pre-R630) operation.
- \$MODESEL is not visible on any of the standard displays. Its value may be changed via the DEB, TPS Builder, AM/CL, or user schematic whose key-lock is set to ENGINEER.

NOTE: Multiple Schematics 300.1 (200.6) change zone on U^XS will continue to operate as it did with pre-TPN R630. It will not reflect the R630 or later changes made to the native window change zone

**Re-Range OP
Parameter on
Trends**

- Provides re-ranging of the OP parameter on the Group Trend display.
- Existing target is modified to toggle between PV/SP and OP entry.
- The new OP range will be saved via the existing group save function.
- TR_RANGE actor already provides for re-ranging OP parameter.

The “Re-Range OP parameter on Trends” enhancement requires a new file in the HM Area Data Base Directory. Here are some migration issues.

- **Background:**

A TPN R530 UIS enhancement allowed users to save re-ranged PV and SP trend values to an HM-resident file named NET>&Dxx>TRNDARxx.SV, where xx denotes the area data base number being used by the US. For area 01, the data would be saved in NET>&D01>TRNDAR01.SV. (We will refer to this as the “R620 file”.)

Starting with TPN R630, re-ranged OP trend values may also be saved to an HM-resident file. However, the format of the R620 file is no longer valid and a new file must be introduced. The new file will be named NET>&Dxx>TRNDARyy.SV, where xx denotes the area data base number being used by the US and yy = xx + 10. For area 01, the data would be saved in NET>&D01>TRNDAR11.SV. (We will refer to this as the “R630 file”.)

- **Migration Scenarios:**

The new TRNDARyy.SV file is created when a Group Trend Save action is taken. If an old TRNDARxx.SV file exists, it is automatically translated to the new format.

TRNDARyy.SV File (New Exists? (R630))	TRNDARxx.SV File (Old Exists? (R530 – R620))	Action Taken by Group Trend SAVE
No	No	New .SV file will be created in the R630 format.
No	Yes	R630 version of .SV File would be created, translating the R620 version of .SV file into the new one. The new .SV file will preserve all of the contents of the old .SV file and will have provision to store OP range values. So, at this point in time, both the old .SV file (R620 version) and the new .SV file (R630 version) would exist. Subsequent SAVE /RESTORE operations will find the R630 file and then only one read/write is necessary.
Yes	No	The SAVE/RESTORE request would just write/read to/from the R630 file as necessary.
Yes	Yes	The SAVE/RESTORE request would just write/read to/from the R630 file as necessary. The R620 file will not be updated.

NOTES:

1. For each of the Areas configured in the system, a new TRNDARyy.SV file will be created in the NET>&Dxx. The new file will be approximately 60% larger than the old file.
2. To optimize space on the HM, after migration to R630 is complete, force a SAVE request on each of the Area's .SV files. Then manually delete the old R620 files from the HM.
3. Once a R630 file is created using the data from the corresponding R620 file, any further modifications done to the R620 file (from a R620 station) will not be visible to the user in an R630 station.
4. The system might take an extra few seconds when creating a new TRNDARyy.SV file.

**DOC Tool
Efficiency
Improvements**

The System Documentation Tool (DOC Tool) is modified to enable a user to specify which units to query during an event history retrieval. This yields significant efficiency improvement for those cases where a user is interested in only a subset of the total number of units in the system.

- A two line “Unit List” port will be added to the journal query menu with the following characteristics:
 - User may enter one or more specific units to be retrieved and/or the keyword AREA. The keyword AREA requests retrieval from all of the local area units.
 - User may enter the keyword ALL, requesting retrieval from all of the process units in the system.

**Add Selection
List to Find
Names**

- Provides for the specification of a selection list in the Entity Argument port of the Find Names window.
- Entry form would be NET>SEL>ENTITIES.XX
- Output of the FN operation
 - Appears in the order of the searched files, rather than the order of the entities in the selection list. This approach minimizes the execution time hence improves the performance of the new functionality.
- Existing entity search alternatives (one - entity and wildcard entities) can be used inside the selection list, except search alternative ‘all entities’ (*) will not be supported inside the selection list.

**Feedback on
Point
Activation/Ina
ctivation
command**

R630 now provide a new option switch for the ACTIVATE and INACTIVATE commands.

An optional –C switch is added to the Command Processor’s ACTIVATE and INACTIVATE commands.

Presence of the –C switch would result in a confirmation message.

For example:

```
INACTIVATE PSF001 –C
```

```
PSF001 SET INACTIVE
```

```
ACTIVATE PSF001 –C
```

```
PSF001 SET ACTIVE
```

Absence of the –C switch will result in no confirmation message (pre-R620 functionality).

The error message functionality remains unchanged (i.e., an error condition always results in an error message whether the –C switch is present or not present).

**HPM DigComp
Initialization
Enhancement**

The DigComp initialization enhancement function will provide an option to hold the output (HOLDOP option) at the previous state during the initialization processing in the DigComp and DevCtl points of the HPM.

In support of this function a new parameter \$INITOPT has been added for the DigComp and DevCtl algorithms. \$INITOPT can be set to one of 2 values “Normal” or HoldOP”. The setting of \$INITOPT determines whether the output is held at the previous value during initialization. If \$INITOPT is set to “NORMAL”, there is no change in the current behavior of the points. The initialization executes as currently described in the Digital Composite section of the HPM control Theory manual for R620 and prior HPM releases. When \$INITOPT is set to “HOLDOP” the output value (OP) is held at the last value and there is no output store during initialization (unless one of the override interlock inputs are active).

The default value of \$INITOPT is “NORMAL”. The setting of this parameter is configured during the point build. After loading the point configuration, the setting of \$INITOPT can only be changed if the universal station access key is in the Engineering position and the point is inactive.

Currently the discrete control (DigComp and DevCtl) points in the HPM perform an initialization of their output on recovery from a communications failure. During initialization, the output OP of the point is set to a state based on a set of predefined rules and the output value is stored to the final control device. The new initialization enhancement function will provide a feature by which the output can be optionally held at the previous state and the store to the final control device can be disabled by the user. That is, if the hold output option is enabled, the output of the DigComp point will be held at the last value unless one of the override interlocks is active. If any of the override interlocks are active, OP will be set to a state corresponding to the asserted override interlock and an output store will still be performed. An override interlock is active if any of override interlock inputs I0,I1,I2,I3 of the DigComp point is true and the BYPASS parameter is FALSE or if the Safety interlock SIO input is true. If more than one interlock is active, OP will be set to the state of the highest priority interlock. In absence of an active interlock, the value of OP will be preserved and there will be no output stores.

TPS Fieldbus System Alarm Enhancement

The following paragraphs describe the modified FBIOP functionality being released in FBIOP version 605.5 and newer.

Problem Being Solved:

- When a Fieldbus device fails, an FBIOP soft failure event (Device Missing) is reported that causes the OPR/GUS (LCN US) station system alarm LED on the SYST_STAT key to be lit until the given Fieldbus device is repaired. Also, instead of failing completely, a device could have its transducer portion fail making it useless. This condition also generates a soft failure. In fact, any soft failure condition against the device (including errors on function blocks in the device) will cause the SYST_STAT LED to be turned on.
- Since it may take some amount of time (days) for the faulty device to be repaired/replaced, the SYST_STAT LED continues to be lit during this time, potentially masking other more important system level alarms.
- Once that device is known to be failed, capability is given to the user to disable/clear these soft failure alarms so that the SYST_STAT LED can be extinguished.

Summary Of Solution:

The functional solution by which a user can disable and clear out the existing system soft failures associated with a non-operational FBus device is to set the device's corresponding PECM's point execution status (PTEXECST) to inactive. The following summarizes what actions then occur:

- The FBIOP generates return-to-normal events for all of the outstanding soft failures associated with that device. These include soft failures from both its related PECM and all FBCMs. If these were the only soft failures present in the LCN-US area then the System LED on that US keyboard would be extinguished.
- The generation of any soft failure events on that FBus device would remain disabled until its PECM's point execution status (PTEXECST) is returned to active.
- This functional improvement is only done in the TPS FBIOP downloadable personality image. No changes occur in the personality image of any LCN node, the HPM Ctrl/Comm, or in any Fbus device.

1.10 R640 Enhancements

Introduction

TPS network software Release 640 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Far East and Australia.

The following lists the enhancements that were made as a result of their recommendations.

- Update Digital PV on US Alarm Summary
- XPM Sequence Loading from Command Processor
- Modify HG and NIM Point Parameters while Active
- Add ALENBST to HG EB Files
- AM Rolling Average Algorithm
- Reload of Memory Resident Schematics and FFLs
- Add HG OP Alarming
- Doc Tool Performance Improvement
- Automatic TPN System Restart
- Area Database Builder ABORT Function
- Open Line Detection Disable

The following enhancements are implemented in TPS software and are not dependent upon TPN R640 software. They are part of the UIS Release 8 project.

TPS Software Improvements

- Synchronize TPN time to time on TPS nodes

For more information about this enhancement, refer to the following documents:

Network Form Instructions
TPS System Configuration Utility User's Guide
TPS System Planning Guide

- GUS Graphics – Bulk Replacement

For more information about this enhancement, refer to the following document:

GUS Display Builder User's Guide

- Version Management Tool – Net

For more information about this enhancement, refer to the following document:

TPS System Configuration Utility User's Guide

- Increase GUS Emulated Disk Size

This function ensures that 500MByte emulated disks can be used by the GUS. For more information about this enhancement, refer to the following document:

GUS Display Builder User's Guide

Update Digital PV on US Alarm Summary

The current PV of a digital point is now displayed in field 9 of the Unit/Area Alarm Summary and the Alarm Annunciator displays. This value is updated every four seconds (same as the analog points).

**xPM Sequence
Loading from
Command
Processor**

xPM Sequence Program operations can now be performed from the Command Processor. Appropriate events are generated to indicate execution of the requested operations. The following functions are supported:

- LOAD the sequence program onto xPM PROCMOD point
- STOP the sequence program
- STOPWO – Stop the sequence program (with override) which provides the same function as stop but if the operation fails, it will not abort the command and the command will be processed for further options
- START the sequence program
- Change the OPER status to OFF
- Change the MODE of the sequence program

For more information, refer to the following documents:

Command Processor Operation
Engineer's Digest
PM/CL Data Entry
APM/CL Data Entry
HPM/CL Data Entry

**Modify HG and
NIM Point
Parameters while
Active**

Many non-control parameters of HG and NIM points can now be modified without first making the point INACTIVE or the node/box IDLE. The modifications can originate from:

- Detail displays
- Custom schematics
- AM/CL programs
- The Alter Parameters function

The requested modification must still pass the appropriate access level checks.

For more information, refer to the following documents:

Application Module Parameter Reference Dictionary

Hiway Gateway Parameter Reference Dictionary

Logic Manager Parameter Reference Dictionary

PM Family Parameter Reference Dictionary

Process Operations

Safety Manager Parameter Reference Dictionary

**Add ALENBST
Parameters to HG
EB Files**

Printing an HG point to a DEB EB file now includes the parameter ALENBST (Alarm Enable Status), when the parameter exists on the point.

For more information, refer to the following documents:

Hiway Gateway Control Functions

Hiway Gateway Parameter Reference Dictionary

**AM Rolling
Average
Algorithm**

Any scheduled Application Module (AM) point whose PV parameter is of type REAL, can now have an optional rolling average calculation.

The Rolling Average algorithm is based on collecting values at a user-specified frequency, saving a user-specified number of these values in a buffer, and then averaging the values that it has collected to generate the rolling average. The rolling average is calculated on a configured multiple periodic execution of the AM regulatory point (1sec, 2sec, 5sec, 10sec, 15sec, 30sec, 1min, 2min, 5min, 10min, 15min, 30min, 1hr, 8hr, 12hr and 24hr). Having a 'bad' value input to the 'PV' rolling average calculation will not affect the rolling average calculation.

**AM Rolling
Average
Algorithm**
(continued)

Several new parameters have been added to support the Rolling Average Algorithm:

- \$ROLLBUF/\$ROLLBBS: Used to specify the size of the Rolling Average buffer.
- \$ROLLFRQ/\$ROLLFBS: Used to specify the frequency of data collection.
- \$ROLLRST: Boolean parameter used to reset the rolling average on the AM regulatory point, set the number of samples to zero and cause the rolling average calculation to restart at the next scheduled execution.
- \$ROLLAVG: Used to read the size of the Rolling Average buffer in terms of sample number.
- \$ROLLMLT: Used to read the frequency of data collection as an integer multiple of the point execution period.
- \$ROLLPV: Used to read the current rolling average calculated
- \$ROLLOK: Number of good samples in the current rolling average calculation
- \$ROLLBAD: Number of bad samples in the current rolling average period

The Rolling Average point parameters are saved with an AM checkpoint but the sample buffer is not saved.

For more information, refer to the following documents:

Application Module Algorithm Engineering Data
Application Module Implementation Guidelines
Application Module Parameter Reference Dictionary
Engineer's Reference Manual

Reload of Memory Resident Schematics and FFLs

The user can now reload memory resident schematics and Free Format Logs (FFLs) by selecting the “RELOAD SCHM/FFL” target in the Console Status Display.

The user can select the US node(s) to reload in three ways:

- Select a single NODE
- Select all nodes in the console using AREA yy
- Select all nodes in the console

The memory resident schematics and FFLs specified in the pathname catalog of the area database currently in use by the affected node shall be reloaded. After reload, the memory cache (where the recently invoked schematics are stored) of the affected node(s) shall not be cleared out.

The access level of “CHANGE AREA” as specified in the NCF shall be considered the required access level for this functionality. Events will be generated to indicate the status/results of this operation.

For more information, refer to the following documents:

Area Form Instructions
Operator's Digest
Process Operations

Add HG OP Alarming

R640 provides the ability for HG regulatory points to generate alarms based on the value of the Output (OP) parameter.

Several new parameters have been added to HG regulatory control points in order to support the function:

- OPHITP: OP High Trip point parameter
- OPLOTP: OP Low Trip point parameter
- OPHIPR : OP High Alarm priority parameter
- OPLOPR: OP Low Alarm priority parameter
- OPALDB : OP Alarm Deadband parameter
- OPHAFL: Status of the OP High alarm
- OPLAFL: Status of the OP Low alarm
- \$OPALEN: OP Alarm Enable parameter. Determines if the point to be scanned for OP alarming

The parameter PTINAL is set TRUE if the point has an OP alarm.

A new Data Hiway parameter is added:

- \$OPSNPRD: HG OP Alarm Scan Period in seconds (0 to 60). All HG OP Alarm scanning is disabled by setting \$OPSNPRD to zero. Changes to this parameter are logged in the appropriate HM journals.

Several new PSDP parameters are added to the HG node to monitor OP Alarm scan performance:

- \$OPOVRUN: Total number of OP Alarm scan overruns since the HG was loaded
- \$OPOVRC: Total number of OP Alarm scan overruns during this current hour
- \$OPOVRP: Total number of OP Alarm scan overruns during the previous hour

**Add HG OP
Alarming
(Continued)**

HG OP Alarming and its Interaction with Existing Alarm Concepts and Functions:

- Event Initiated Processing (EIP)

HG OP Alarming is performed by a cyclic task that is independent of the tasks that handle HG PV alarms. Therefore, if EIP is configured for an HG regulatory control point, you will get separate EIP actions for the PV alarm and the OP alarm. This must be considered in your overall EIP strategy.

- Contact Cutout

An HG OP alarm will initiate contact cutout of secondary point alarms in the same manner as an HG PV alarm.

- Alarm Delay (\$ALDLOPT)

HG OP alarms are not affected by the alarm delay function. OP alarms will be reported at the assigned scan period (i.e., the Data Hiway parameter \$OPSNPRD).

For more information, refer to the following documents:

Hiway Gateway Control Functions

Hiway Gateway Parameter Reference Dictionary

Operator's Digest

Process Operations

**Migration
Considerations
for HG OP
Alarming**

If an HG point is using the new HG Output Alarm feature (i.e., its \$OPALEN parameter is set to ENABLE), the value entered for OPHILM / OPLOLM will now be checked against the OPHITP / OPLOTP values for validity.

If an HG point is not using the new HG Output Alarm feature (i.e., its \$OPALEN parameter is set to DISABLE), the validity checking for OPHILM / OPLOLM will be the same as R630 (and earlier).

When migrating to R640 (or newer), the default value of \$OPALEN is DISABLE.

If \$OPALEN = DISABLE, the point detail display will not show the OPHITP / OPLOTP parameters and values

When changing \$OPALEN from DISABLE to ENABLE, the HG checks the OPHITP / OPLOTP and OPHILM / OPLOLM parameters for the proper value relationship, else an error is returned.

**Doc Tool
Performance
Improvement**

The performance of System Documentation Tool Journal queries is improved in R640.

Journal query software has been redesigned to reduce the number of File accesses for a given query, thus reducing query time.

A new query option field specifies whether or not you want the journal query results to be sorted. No sort will return results in less time.

Special symbols in the date-time field provide short-cut methods of specifying current time and date:

- “@” shall be accepted in the starting and ending time port to represent the current time.
- “~” shall be accepted in the starting and ending time port to represent the current date time rounded off to the nearest whole hour

Journal Queries can be pre-built and saved with these new features.

For more information, refer to the following documents:

Documentation Tool Manual

**Automatic TPN
System Restart**

The Automatic TPN System Restart removes the manual intervention previously necessary to restart a TPN system that had failed due to power failure. TPN-resident nodes can be automatically loaded upon power recovery. (Hiway-resident boxes and UCN-resident devices are not automatically reloaded.) Automatic TPN System Restart builds upon the R500 “Easy Reload” function.

To properly implement Automatic TPN System Restart, the “Easy Reload” function must first be configured. “Easy Reload” defines the nodes to be loaded from a US when the LOAD CONFIG target is selected from the System Status display.

Next, a new configuration file (NET>&ASY>RESTART.LD) is created to define the US nodes that will become “Restart Supervisor Nodes” (RSNs). An RSN is a US that can be automatically started from the System HM and will automatically initiate an “Easy Reload” via the LOAD CONFIG target, with no operator intervention necessary. Up to six US nodes can be designated as Restart Supervisor Nodes (RSN). When each RSN completes its “Easy Reload”, the entire TPN system will be restarted automatically. This assumes that the “Easy Reload” is properly configured for each RSN.

The NET>&ASY>RESTART.LD file is created using the Command Processor Editor. A new Auxiliary Node Status object for the System HM has an associated display that can be used to check the validity of the file contents and to enable or disable this function.

When the System HM first starts up, and periodically as TPN node status changes are detected, a check is made to see if any US nodes are running. If no US nodes are running, the System HM attempts to load the defined RSN nodes. A US started as an RSN will then automatically initiate an “Easy Reload”. (Please recall that “Easy Reload” skips nodes that are already running.)

**Automatic TPN
System Restart
(Continued)**

If you want to exclude certain US nodes from the System HM's check for running US nodes, these can be specified in the NET>&ASY>RESTART.LD file. For example, you may have two GUS nodes that are configured for automatic reload. You can exclude these two nodes since the System HM would most likely always view them as "running" after a power outage/recovery. This will allow the System HM to go ahead with an automatic start of the RSN nodes.

The System Status display can be used to monitor the progress of the load and can be used to stop the "Easy Reload" currently in progress, if necessary.

For more information, refer to the following documents:

TPS System Configuration Utility User's Guide
Process Operations

**Automatic TPN
System Restart on
an all-GUS
system.**

If your system uses all GUS nodes (no standard US nodes), here is how you can configure Automatic TPN System Restart:

1. Turn off the Auto LCNP Restart on one or more GUS nodes.
2. Make these GUS nodes your RSNs. This HM functionality will only start up RSN nodes.
3. Do not put these GUS nodes into the exclusion list.
4. Exclude all the GUS nodes that will be using the Auto LCNP Restart functionality. They cannot be RSNs.

**Area DB Builder
ABORT Function**

There is a new ABORT function added to the Area Database Builder. The ABORT function (F3 key) deletes the selected Area work file, AREAxx.WA, and terminates the Area Database Builder.

For more information, refer to the following documents:

Data Entity Builder Manual
Engineer's Digest

**Open Line
Detection Disable**

You can now disable the annunciation of open line detection events by using a NIM-resident, point-specific parameter (\$OLDEVDS) associated with UCN Analog Inputs and Analog Outputs which are subject to open line detection functionality.

- \$OLDEVDS is used to Enable/Disable OUTPUTFL soft failure event generated by the AO IOP.
- \$OLDEVDS is used to Enable/Disable BADPV alarms generated by LLAI/LLMUX IOPs when configured for the detection of open thermocouple or RTD inputs.

The parameter \$OLDEVDS has the following states:

ENABLE

DISABLE

NOOTDETC (The AI point is not affected by this functionality)

For more information, refer to the following documents:

Advanced Process Manager Control Functions & Algorithms

Advanced Process Manager Parameters Pocket Guide

HPM Control Theory

HPM Parameters Pocket Guide

Process Manager Control Functions & Algorithms

Process Manager Parameters Pocket Guide

Process Manager Family Parameter Reference Dictionary

1.11 R650 Enhancements

Introduction

TPS network software Release 650 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Far East and Australia.

The following lists the enhancements that were made as a result of their recommendations.

- Print Date in Real Time Journal Events
- New HPM Switch Algorithm – OPSELECT
- Logic Source Parameter Target Navigation
- Strict Enforcement of View-Only Mode
- Journal AM/CL-Initiated Process Changes
- Increase the Number of Alarmable HPM Flag Points
- APP Node LCNP4 Loading without the System HM
- Increase the Resolution of the Ratio Low Limit in the HPM/APM PID Algorithm
- Standard X-Y Plot for US Schematics
- Triconex Validation with TPN R650
- System AUDIT Tool – Command Processor Overlay

The following enhancements are implemented in TPS software and are not dependent upon TPN R650 software. They are part of the UIS Release 9 project.

**TPS Software
Improvements**

- Allow disabling of the “Print Trend” key on a GUS/ EST node.

For more information about this enhancement, refer to the following documents:

TPS System Configuration Utility User’s Guide

- GUS Graphics Validation – Cancel Operation

For more information about this enhancement, refer to the following document:

GUS Display Builder User’s Guide

- GUS Alarm Summary Enhancement

Display the current PV value for points on the GUS Alarm Summary. For more information about this enhancement, refer to the GUS documentation

The following enhancements are implemented in PlantScape software and are not dependent upon TPN R650 software. They are part of the UIS Release 9 project.

**PlantScape
Software
Improvements**

- PlantScape Bulk Edit Parameters Tool

For more information about this enhancement, refer to PlantScape documentation

**Print Date in Real
Time Journal
Events**

This function ensures that a date is printed on every page where RTJ events are printed.

RTJ header is printed whenever the print task changes from a non-RTJ print task to an RTJ print task.

The date and time in the RTJ header will be the date and time of occurrence of the first event/alarm. Currently it is the print time.

New functionality does not apply to a form feed whenever there is a switch over to/from RTJ print task from/to any other print task.

**New HPM Switch
Algorithm –
OPSELECT**

Provides a new OPSELECT algorithm for the HPM which is the reverse of the present switch control algorithm. The new algorithm takes one input and switches it to selectable outputs with an option to provide a value for the non-selected outputs.

The new algorithm will allow simplified switching strategies to be implemented, strategies that would require fewer HPM P.U.s.

The Control Output connections will be connected in sequential order as defined by NOCOPTS and a new parameter named \$NOPCONN.

\$NOPCONN defines the total output connections existing on this point.

NOCOPTS defines the number of active output connections.

–e.g.: if \$NOPCONN = 3, NOCOPTS = 2, first 2 outputs would be of PUSH type and 3rd would be FETCH type)

**Logic Source
Parameter Target
Navigation**

Targets will be added to the detail display for digital composite and device control point LOGICSRC parameters.

Activating the target(s) will cause the source point to be ‘selected’.

The selected point can then be used to invoke standard displays, i.e. Detail Display, Associated Display, etc.

**Strict
Enforcement of
View-Only Mode**

A station in View-Only mode will now strictly enforce this mode. In the past, there were still non-view actions it could perform.

There is no option in the NCF to return to the pre-R650 behavior for a station in View-Only mode. Modified View-Only behavior for R650 includes the following:

- Alarms will only be silenced at the local station via the Silence key or Alarm Summary invocation.
- Control/Change actions will be blocked in nine areas.
 1. UCN Cable Commands
 2. IOL Cable Commands
 3. Hiway Commands
 4. Reset LCN Cable Diagnostic counters*
 5. Alter parameters via Builder Commands
 6. Modify Default Volume Paths
 7. Change printer assignment of a scheduled/demanded report, log, trend or journal
 8. Save modified trend settings.
 9. Clear acknowledged operator messages at the Message Summary Display.

*Engineer access level will be required to reset cable diagnostic times and their counters.

**Journal AM/CL-
Initiated Process
Changes**

This package provides for the modification of AM/CL to record point.parameter changes made from AM/CL programs. The output can be used to record an event similar to operator process changes.

The AM/CL parameter changes output can be optionally directed to the printer (RTJ) and/or unit journal history (HM).

The output can be used as a debug tool. The AM/CL parameter changes can optionally be output to an AM memory buffer or user-defined file. A parameter fetch or file print can be used to view the data.

The Event History Retrieval (EHR) and documentation tool (SDT) will recall AM/CL changes along with operator process changes. EHR function is modified to provide a unique retrieval mechanism for the AM/CL events.

To reduce the volume of AM/CL point.parameter change journals, a filter mechanism is provided to facilitate defining which specific changes should be journaled.

The function provides the user with on/off activation and filter selection lists of AM point and/or CL block names and destination point.parameter names to journal.

Function enable/disable and configuration is accomplished on a per AM basis.

To prevent AM performance impact, an internal buffer will be added for 100 value changes before 'overflow' occurs.

AM parameters are added to monitor package performance.

**Increase the
Number of
Alarmable HPM
Flag Points**

The number of alarmable flag points in the HPM is increased from the first 128 to the first 1024.

When migrating to R650, the alarm priority (OFFNRMPR) for slots 129 – 1024 are set to NO ACTION. (This will prevent any burst of alarms upon migration.)

If desired, the user may wish to move existing flag points in slots 129 – 1024 that do not need an alarm to a slot that is not alarmed.

**APP Node LCNP4
Loading without
the System HM**

Prior to TPN R650, the load of an APP node's LCNP4 card required access to the System HM (unless you were using the APP node only as an AM replacement). This function changes the AM Node Status Display so that it can load an APP completely without the System HM.

The APP "fast_load" disk needs to contain the files normally accessed from the System HM (NCF, checkpoint, load modules, etc.). A new .EC file (FAST_APP.EC) helps users to create the necessary "fast_load" disk for an APP node.

**GUS Node LCNP4
Loading without
the System HM**

Prior to TPN R650, a remote load of a GUS node's LCNP4 card required access to the System HM (unless you were using the GUS node only as a US replacement). This function changes the US Node Status Display so that it can load a GUS completely without the System HM.

The GUS "fast_load" disk needs to contain the files normally accessed from the System HM (NCF, Area Data Base, &DSY, load modules, etc.). EC file FST_VOLZ.EC has been modified to create the necessary "fast_load" disk for a GUS node.

**Increase the
Resolution of the
Ratio Low Limit in
the HPM/APM PID
Algorithm**

This function provides higher resolution for the Ratio low limit (RTLLOLM) parameter of the HPM/APM PID Algorithm.

- Pre-R650 resolution was limited to two decimal places (0.01).
- In R650, the resolution increases to four decimal places (0.0001).

The increased resolution will benefit customers who use the autoratio feature of the PID algorithm in control strategies where flows are extremely small and precise.

**Standard X-Y Plot
for US
Schematics**

This package provides a standard X-Y plotting capability for US schematics in a manner similar to the existing trend capabilities.

- All of the applicable trend actors will be duplicated as standard X-Y plot actors.
- Allows the plotting of one to three data sets (Y-axis) against one other data set that is used as the basis for the X-axis.
- Three types of plots will be available and a new actor is provided to allow dynamic toggling between the three plot types.
 1. Scatter Plots
 2. Scatter Plots with lines
 3. Operation Point Plot
- Existing trend data sources will be valid: real time, HM and HG trend memory.

The trend collectors such as TR_TIME and TR_COLOR will work with the standard X-Y plots.

Note: This new standard X-Y Plot function does not replace nor obsolete the existing Repeatable Special named “X-Y Plot”, model number ES-SWXY01.

**Triconex
Validation with
TPN R650**

Provides system validation testing to ensure Triconex compatibility with the TPN 650 release.

**System AUDIT
Tool**

AUDIT is a new standard TPN overlay based on the “LCN Node Version Revision Logging tool” (LVRLOG). The AUDIT overlay is called from the Command Processor and produces output identical to LVRLOG. (Refer to Section 34 of the Engineer’s Reference Manual for notes on LVRLOG.)

Output:

AUDIT.RD – Pure text file.

AUDIT.RE – Comma-separated values file.

Syntax:

audit [*keyword-or-option*] ...

Where *keyword-or-option* may be zero or more of the following:

Keywords:

lcn audit LCN nodes

dh audit Data Hiway nodes

ucn audit UCN nodes

Options:

-s *site-id* set the site ID on the report to "site-id"

-n *lcn-id* set the LCN ID on the report to "lcn-id"

-h omit report headings

-o *out-path* set the output path (dev>dir) to "out-path"

-? print the overlay’s help message

1.12 R660 Enhancements

Introduction

TPS network software Release 660 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Far East, and Australia.

The following lists the enhancements that were made as a result of their recommendations.

- Add NIM/HG Load Information to PERFMENU Displays
- Picture Editor Schematic Set Collection Sorting
- Rebroadcast of Process Alarms. (GUS Alarm Summary requires new TPS software as well as TPN R660.)
- AM/UCN Point Interconnections Display
- xPM – Set Deleted Digital Points Inactive
- Alarm Delay for HPM Digital Input Points
- xPM Numeric Point Alarming
- TPN Node Backup/Restore Utility. (Requires TPS nodes and new TPS software.)

The following enhancements to TPN/TPS are in addition to the UIS Release 10 project.

Additional Enhancements

- Enhance HPM I/O Link Error Handling
- GUS Support for Enhanced Flat Panel Displays
- Improvements to the Version Management Tool
- GUS Scripting Trigger of Keyboard Audible Annunciators

For more information about these enhancements, refer to the appropriate Software Change Notices.

The following enhancements are implemented in TPS software and do not depend on TPN R660 software. They are part of the UIS Release 10 project.

**TPS Software
Improvements**

- TPS Node Backup/Restore Utility
- Network Installs of TPS Software
- GUS Display Builder “Find and Replace” Function
- GUS Faceplate Consistent Operation
- GUS Standard X-Y Plot

For more information about these enhancements, refer to the following documents:

GUS Display Builder User’s Guide

TPS System Administrator’s Guide

GUS R350 Software Change Notice

The following enhancements are implemented in Knowledge Builder Documentation for TPS/TPN and do not depend on TPN R660 software. They are part of the UIS Release 10 project.

**Knowledge
Builder
Documentation
for TPN/TPS**

- System Message User Documentation
- Customer Resource Manual for Knowledge Builder

For more information about this enhancement, refer to Knowledge Builder for TPS/TPN.

**Add NIM/HG Load
Information to
PERFMENU
Displays**

Provides a diagnostic capability in PERFMENU to view the loading on the HG and the NIM.

- New PSDP parameters are added to the HG and NIM that can indicate the loading on these nodes.
- The HG and NIM nodes collect and process data related to the new parameters (like Event Processing, Parameter Access, Checkpointing, and others) to keep the PSDP parameter values current.
- New system schematics (HGDETAIL and NMDETAIL) are added to PERFMENU for the display of HG and NIM PSDP information.
- PERFMENU schematic is updated to facilitate access to these schematics.

**Picture Editor
Schematic Set
Collection Sorting**

Allow the user to organize the Set Collection list of parameters in the Picture Editor. The PE Set Collection command is modified to include sort options and search criteria. The help display is modified to include the syntax of set collection command with sort option and search criteria.

The five sort options are as follows:

1. By Symbol – Symbols displayed in an alphanumeric order
2. By Parameter Name – Sorted first by alphanumeric Parameter name followed by alphanumeric sort of the “Symbol” name. The Symbols, which do not have the form entity.parameter, are organized alphanumerically before the Symbols of the form entity.parameter
3. By Collection Rate – Sorted first by numeric order of the Collection Rate followed by alphanumeric sort of the Symbol.
4. By Group ID – Sorted first by numeric order of the “Group ID” followed by alphanumeric sort of the Symbols. After the numeric ordering of the group ids from 0 to 245, the symbols with group id of FST will be displayed followed by symbols with CZ.
5. NOSORT – The Symbols shall be displayed in the unsorted order like the existing functionality. If the search criterion is to be used with the existing functionality then the sort option NOSORT should be specified with the Set Collection command. When used with a NOSORT option, the search criteria shall be searched only in the symbol names of the symbol.

An option is provided to navigate directly to a particular page of symbols in the Set collection display. The Set Collection command is enhanced to accept optional search criterion with the sort-option. The set collection command displays the page of symbols having the first occurrence of the search criterion in the attribute specified as the sort option.

If the search criterion is specified with a symbol or parameter-based sort, the page displayed has the symbol with symbol-name/parameter-name starting with the specified search criterion (an exact match is not required).

When the specified sort option is Rate or Group then the search criterion should have an exact match with the corresponding value, rate or group id, in the symbol with which it is compared.

These functions also work for TPN Free Format logs.

**Rebroadcast of
Process Alarms**

Provides the capability for process alarms that have a “normal” condition to be rebroadcast as new alarms if they have not returned to normal within a specified period.

New parameter \$REALARM is added to alarmable point types in the AM, HG, and NIM to specify the number of minutes allowed to elapse before alarms on the specified point are rebroadcast. The range is 0 to 9999 minutes. A value of zero minutes disables the re-broadcast function for that point. There is only one \$REALARM parameter per point; therefore, the user-configured re-broadcast interval is the same for all alarms associated with that point. For points with multiple active alarms, the alarms will be rebroadcast on the same cycle.

At the Alarm Summary displays, rebroadcast items look and behave exactly like new alarms except that their updated timestamps are displayed in yellow, rather than green. The alarm is annunciated and the alarm priority field on the alarm displays returns to its unacknowledged, blinking state. The rebroadcast alarm is also placed into the new alarm window of the alarm display if the user is currently viewing the first page of the Area/Unit Alarm Summary; otherwise the alarm is shown at the top of the list.

Rebroadcast alarms are also be sent to the History Module, with a new timestamp, and reported to the RTJ and Process Alarm Journal, where they appear as alarms (ALM).

The rebroadcast of a given alarm continues indefinitely, as described above, until the alarm limit is no longer violated (alarm has returned to normal) or the user disables the rebroadcasting of alarms for that point by setting \$REALARM to 0.

The \$REALARM parameter is not supported for reported alarms in the APM and HPM I/O Simulator.

**AM/UCN Point
Interconnections
Display**

This is a new, operator-accessible tool, which finds the usage of a NIM or AM point by other NIM/AM points in the system. The point interconnection search function is similar to the existing Find Names function except it can be called by a US Operator Station and will run in the background so the station can perform other operator-related tasks during the point interconnection search.

The user can display the results, send the results to a user specified file, print the results to a user specified printer, and can cancel the request prior to completion.

A new schematic, \$INTRCON, allows the user to initiate/cancel/monitor the search for point interconnections.

Engineers can create pre-defined queries using the Documentation Tool to simplify operator actions and reduce search times. The predefined query is used by the \$INTRCON schematic to run the point interconnection search. A variety of query files can be created to match typical operator usage.

The point interconnection search feature requires an external load module to be added to the station's Network Configuration File (NCF).

(The current plan is to enhance this function to include HG points in the on-process beta release of R660.)

**xPM – Set Deleted
Digital Points
Inactive**

When a DI or DO point is deleted, the IOP firmware sets the point execution state of the slot to ACTIVE. Starting with R660 NIM, the NIM will send an INACTIVE command to the IOP slot after deleting the DI or DO point. The INACTIVE command overrides the ACTIVE command sent by IOP firmware.

There is no option provided to the user to prevent the NIM from sending the INACTIVE command.

**Alarm Delay for
HPM Digital Input
Points**

A NIM-based ON and OFF delay function for Digital Input points, similar to existing analog alarm delay options.

New parameter \$DLYOPT has the following values:

PMOFFDLY – existing (i.e., pre-R660) IOP-based OFF-delay function is in use.

NMOFFDLY – new NIM OFF-delay function is in use. When the point returns to the normal state from an alarm state, the return to normal state is not reported to the US, RTJ, and HM journal until the time period specified in DLYTIME has elapsed. After the time period has elapsed, if the point continues to be in the normal condition, the return to normal is reported.

NMONDLY – new NIM ON-delay function is in use. When the point goes into alarm, the alarm is not reported to the US, RTJ, and HM journal until the time period specified in DLYTIME has elapsed. After the time period has elapsed, if the point continues to be in alarm, the alarm is reported.

NONE – there is no alarm delay.

The \$DLYOPT options are mutually exclusive and can be modified at runtime.

If the value of \$DLYOPT is NMONDLY or NMOFFDLY, only the reporting of the alarm and the return to normal is suppressed during the time period. Detection and indication of alarm state through alarms flags like HIGHAL or HIGHALPR is not affected

HPM Numeric Point Alarming

Adds PV trip point alarming capability to HPM Box Numeric data points that have tag names.

The user configures the number of alarmable HPM Box Numeric by assigning a value to \$NNUMALM. Slots with SLOTNUM less than \$NNUMALM can be configured for PV alarming.

This new function requires additional HPM Processing Units and Memory Units.

The behavior of the HPM Box Numeric PV trip point alarm is similar to the PV trip point alarm of the Regulatory PV point type.

The HPM Box Numeric data point has to have a tag name, and the point form configured as FULL for the point to be alarmable.

Note:

Attempting to load an HPM Box Numeric point using a pre-R660 EB file, without increasing the value of \$NNUMALM, will report error messages. These error messages are reported when the value of \$NNUMALM is less than the value of the slot number. Only points that have slot numbers less than or equal to value of \$NNUMALM can be built as FULL points.

After successfully upgrading the software to R660, you should create or revise your EB files to change the value of the HPM Box Numeric point's PNTFORM parameter to COMPONNT.

Increasing the value of \$NNUMALM

For all the HPM Box Numeric points whose slot numbers are between the old \$NNUMALM value and new increased \$NNUMALM value, the PNTFORM parameter will be displayed and the value of PNTFORM must be set to COMPONNT, indicating that there is no PV alarming.

To enable the HPM Box Numeric points whose slot numbers are between the old \$NNUMALM value and the new \$NNUMALM value to be processed for PV trip point alarming, the user has to rebuild the points and set the PNTFORM values to FULL.

**HPM Numeric
Point Alarming
(continued)**

Decreasing the value of \$NNUMALM

Before decreasing the value of \$NNUMALM, all the points whose slot numbers are between the currentt value of \$NNUMALM and the new reduced value of \$NNUMALM, must be rebuilt and have the PNTFORM set to COMPONNT. An error message “PNT EXIST IN NIM” is reported by the Parameter Entry Display (PED) if the user tries to decrease the value of \$NNUMALM without changing the PNTFORM to COMPONNT.

**TPN Node
Backup/Restore
Utility**

A new TPS PC tool for performing automatic backups of one or more History Module hard drives to a PC-based storage area. The tool includes the ability to restore one or more HM hard drives from a saved backup.

The user can automatically backup the HM data (files) on a user-defined schedule or initiate a backup on a demand basis. The TPN Backup/Restore also optionally requests database checkpoints prior to transferring the HM data to the PC. (Need TPN R660 software for the checkpoint request to work.) The backup files can be used later to restore an HM disk drive, volume, directory, or file.

The TPN Backup/Restore functionality is TPS based and leverages the Honeywell File Transfer (TPSxfer) as the means to move the files between the HMs and the PC. Therefore, File Transfer must be active and configured for all appropriate HMs and file types. (Must use TPSxfer version R350 or newer.)

TPN Backup/Restore allows viewing of the contents of the backup. Because these files are stored on the PC, the files may be accessed from TPS, but will not be accessible from TPN unless the files are also saved as a .lcn file and the .lcn file is mounted.

The backup executes at a lower priority to minimize impact to operator actions, control, and data presentation. Scheduled backups run in background mode. The status of the backup and checkpoint are available in log files for reference.

Note: This function cannot backup History Collection Data files or Journals.

1.13 R670 Enhancements

Introduction

TPS network software Release 670 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Far East, and Australia. The following lists the enhancements that were made as a result of their recommendations:

- Default alarm priority is changed from LOW to NoAction.
- Logic points can be viewed using the Logic Tool. (This requires new TPS software as well as TPN R670)
- xPM output indications and alarm parameters are reversed in accordance with the OUTIND parameter value. (The changes in the alarm parameters are also implemented in HG)
- LVRLOG can be initiated from TPS. (This requires new TPS software as well as TPN R670)⁶
- System alarms can be disabled at network/node/box/IOP level.
- PTEXECST parameter is added to SI array point in HPM (enabling start/stop of communication).
- New commands are introduced in TPN for mounting/dismounting GUS emulated disks. (This requires TPS R400 software as well as TPN R670)⁷
- Alarm summary provides an NCF option for displaying OP values in output alarms instead of PV values. (This requires TPS R400 software as well as TPN R670)⁶
- In the Picture Editor, original values of generic parameters are displayed when modification is attempted.

⁶ This function will be validated for ES-T in a later release (post R300) of Experion.

⁷ This function is not available for ES-T in R300 of Experion. It will be implemented in a later release.

The following enhancements are implemented in Knowledge Builder Documentation for TPS/TPN and do not depend on TPN R670 software. They are part of the UIS Release 11 project.

**Knowledge
Builder
Documentation
for TPN/TPS**

- *xPM IO Simulator User's Guide* is added to Knowledge Builder,

For more information about this enhancement, refer to Knowledge Builder for TPS/TPN.

**Default alarm
priority is changed
from LOW to
NoAction**

Changes the default value of alarm priority parameters from LOW to NoAction.

- PEDs of all data points, HG, NIM, and AM are modified to have NoAction as the default value for all alarm priority parameters.
- TPS Builder is also modified to have NoAction as the default value for all alarm priority parameters.

**Logic points can be
viewed using the
Logic Tool**

Provides a graphical interface to view HPM/APM/PM Logic points using TPS GUS nodes without using Native Window.

- A display depicts the logic implemented in a Logic point of xPM.
- The interface is implemented as a GUS Picture Builder display with an ActiveX control embedded in it.
- The Logic point display is dynamically updated and can also be printed.
- The tool supports monitors with 1280 x 1024 and 1024 x 768 resolution.

xPM output indications and alarm parameters are reversed in accordance with the OUTIND parameter value

Reverses the output and output alarm parameters of the Regulatory Control and Analog Output points of NIM if OUTIND parameter is set to REVERSE.

- Reversal of the OP parameter is retained.
- NIM parameters OPHILM, OPLOLM, OPHITP, OPLOTP, OPHIPR, OPLOPR, OPHIFL, OPLOFL, OPHAFL and OPLAFL are reversed if OUTIND is set to REVERSE.
- ARWOP remains unaffected. It is not reversed if OUTIND is set to REVERSE.
- OPHITP, OPLOTP, OPHIPR, OPLOPR, OPHAFL, and OPLAFL parameters of HG are reversed if OUTIND is set to REVERSE. OPHILM and OPLOLM are already reversed in HG. This makes HG consistent with NIM.

LVRLOG can be initiated from TPS

Provides an interface to initiate LVRLOG generation from GUS and ES-T nodes in the TPS domain.

- A Windows executable command is added on GUS to initiate the generation of LVRLOG with all the options available in the LVRLOG schematics.
- LVRLOG can be initiated from scripts and scheduler and you can also track the completion status.
- Native Window operations are not blocked because LVRLOG executes as a background operation.

**System alarms
can be disabled
at
network/node/box/IOP level**

Provides an interface to disable/enable system alarms generated from LCN Node/Network/Box/IOP.

- Allows redefining the online annunciation policy implemented through the area database. If an area change is made, the online annunciation policy is reset and must be re-established.
- New schematics are provided to disable/enable system alarms of LCN node, Network (HIWAY/UCN), box/UCN node, IOP.
- The schematics can be accessed from US/GUS/EST or Native Window.
- Changes made in one station can be propagated to the others in the console.
- Annunciation policy details are available as PSDP parameters. It also accommodates scripts to change the online Annunciation policy.
- All changes and propagation are logged in the Engineering Change journal.

**PTEXECST
parameter is
added to SI array
point in HPM**

Adds PTEXECST parameter to SI Array point of HPM:

- The state of SI Array point can be changed to Active/Inactive using this parameter.
- SI Array point does not report any soft failures while in Inactive state.
- Parameter access, change logs, alarm behavior, and so on are identical to the existing PTEXECST parameters available in other HPM points.
- SI Array point does not need reconfiguration of serial communication parameters while activating from Inactive state. The preserved values are restored on activation.

New commands are introduced in TPN for mounting/dismounting GUS emulated disks

Adds commands in TPN Command Processor to mount or dismount an emulated disk on a GUS station.

- Commands are provided for mounting, dismounting, creating, or deleting emulated disk on a GUS station from the TPN command processor of US/Native Window of GUS or EST.
- Commands are provided to lock or unlock the mounted emulated drive on TPN Command Processor.
- Emulated-disk commands can be added to EC files and schedulers to support automation of emulated disk handling. EC files can display the completion/error status of the emulated-disk commands.

Alarm summary provides an option for displaying OP values in output alarms

Indicates OP value for OP alarms annunciated in the US/Native Window Alarm Summary, Unit Alarm Summary, and the GUS Alarm Summary.

- NCF option is provided to choose PV or OP values for OP alarms of regulatory points residing in PM, APM, HPM, CB, EC, and MC.
- OP values in GUS alarm summary are indicated in yellow.

In the Picture Editor, original values of generic parameters are displayed when modification is attempted

Displays the parameters to the sub-pictures referenced by the variant in the “MODIFY VARIANT” command of the Picture Editor.

- The user may change or retain the original value of the parameters that are displayed.

1.14 R680 Enhancements

Introduction

TPN network software Release 680 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Far East, and Australia.

The following lists the TPN 680 enhancements that are made as a result of their recommendations:

- The xPM device utilization is displayed through a schematic.
- The disabled Process Alarms will be identified with the single character 'D' in journals.
- The Recall Last Query option recalls the previously executed queries of the system Documentation Tool on a per station basis.
- Hiway Box Alarms can be inhibited when a parameter changes or a point is loaded.
- Provides an on-line mechanism to calculate the total LCN load and the information is presented in a schematic.
- Adds behavior option in HPM, following the IOL/IOP Communication problem.
- Enhanced guidelines for TPS Domain Organization.
- TPS on XP and Windows 2003 server. (This requires new TPS software.)

The xPM device utilization is displayed through a schematic

- Based on the formula, the device utilization is calculated as follows:
 - The PUs, MUs and the IOL bandwidth utilized by the xPM nodes.
 - The PUs, and MUs utilized by the LM/SM nodes.
- The calculated values are displayed through the UCN Device Resource Utilization schematic.
- Targets to invoke the schematic are provided in the PERFMENU and in the Configuration Display of the xPM and LM/SM node.

The disabled Process Alarms will be identified with the single character 'D' in journals

- Identifies DISABLED ALARMS in the Process Alarm Journal entry.
- This indication is applicable for alarms generated by AM, NIM and HG with event number 21 and HG SOE events with event number 27.
- The new indication is represented by the single character 'D'.
- This indication will appear in the Process alarm journal, which can be retrieved through mechanisms such as Event History Retrieval, System Documentation Tool and so on.
- In the NCF, an option (YES/NO) is provided for the optional selection of this functionality. The option is to be specified in the System Wide Values/ Console Data portion of the NCF.

The Recall Last Query option recalls the previously executed queries of the system Documentation Tool on a per station basis

- Provides an option to recall the executed query of the system Documentation Tool on a per station basis.
- When "Recall Last Query" option is selected, the executed query screen is displayed with the user entered values. The user can modify or enter some more additional parameters and execute the query again.
- The query is automatically saved to the SYSQRY.DC file for the recall.
- There are five types of queries and there will be only one recall option for all the queries. The recall option is provided only to the last submitted query type and not for each one of them.
- It is not applicable to the queries that are executed in the background. For example, the recall option is not applicable to QPRINT and QFILE.

Hiway Box Alarms can be inhibited when a parameter changes or a point is loaded

- Provides an option to inhibit the alarm that occurs when a DHP/PIU is reset due to a DEB point load.
- Provides an option to inhibit the alarm that occurs when a DHP/PIU is reset due to a change in any of the following parameters:
 - PVCHAR, PCADDRxy, SPECIFxy, PCBITxy.
- A PSDP parameter \$BXRSTRP is provided to choose between the options, ENABLE and INHIBIT. The default option is “ENABLE”.
- The changes that occur due to other reasons such as manual RESET and Box failure will continue to be annunciated.
- In the NCF, an option (Enable/Inhibit) is provided for the optional selection of this functionality.
- The changes to the NCF parameter and the PSDP parameter are journaled in the Engineering Change Journal.

Provides an on-line mechanism to calculate the total LCN load and the information is presented in a schematic

- Provides an online mechanism to calculate the total LCN load and other LCN load related parameters of the selected LCN node(s).
- A system schematic ‘LCNPERF’ is added to the ‘PERFMENU’ for displaying the performance indices of the LCN.
- The ‘LCNPERF’ schematic displays the following details for the LCN:
 - Total LCN parameters per second.
 - Total LCN messages per second.
 - Average parameters per message.
 - Estimated % LCN load.
 - LCN load status (Low, Moderate, Heavy, and Extremely Heavy).The calculation can be started / stopped by the user and the status is displayed on the ‘LCNPERF’ schematic.
- Configurable alarm limits and dead band:
 - When the parameter value crosses the configured limits, the color of the respective parameter changes.
 - System alarms will be reported if the LCN load crosses the trip limits. There is an option to suppress the system alarm.

Adds behavior option in HPM, following the IOL/IOP Communication problem

- When a point experiences an IOL or IOP communication error, by using the new parameter \$BADOCMD, the User can determine whether to retain the mode at the current value or to switch to MANual immediately or after a 15 second or 60 second delay.
- This new parameter is available in every HPM Regulatory Control point. However, the parameter will have no effect, when the output is connected to the soft points.
- This functionality will not have any effect on a regulatory control point which writes the OP to a cascaded regulatory point.
- User can change the mode of the point even when the new option is configured and active. This new functionality will not prevent the User from setting the mode to MANual.

Enhanced guidelines for TPS Domain Organization

- Provides a detailed description about the design, implementation and changes to the TPS Domain Setup, applicable for Windows 2000, Windows 2003 Server and Windows XP.
- Enhances user documentation of the TPS user accounts, domain security, access privileges, and trust relationships.

TPS on XP and Windows 2003 server

- TPS software components of this release are GUS400, APP 400 and TPSB400.
- They are part of a Windows based TPS domain with the domain controller being Windows 2003 Server.
- GUS 400 and TPSB 400 are supported in Windows XP.
- TPS 400 supports APP 400 on Windows 2003 Server.
- Domain controller and Nodes will have Honeywell high security package.
- Refer to the following documents for migrating to TPS 400.

TPS System Planning Guide for Windows 2003/XP

TPS System Implementation Guide for Windows 2003/XP

TPS System Administration Guide for Windows 2003/XP

1.15 R681 Enhancements

Introduction	TPN network software Release 681 enhancements are a result of user-identified needs. These enhancements were recommended by the User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Far East, and Australia. The following lists the TPN 681 enhancements that are made as a result of their recommendations: • An alarm is generated in TPN when an HCI component fails on GUS/APP/ES-T/ESVT/ACE-T nodes. • OFF delay option is added in BADPV and BADCTL alarms of NIM and BADPV alarms of HG. • Error blocks of a failed node are retrieved in system HM on receiving the “FAIL” status change notification. • An option is provided to allow specific alarms of selected points in NIM, AM and HG to pass through, though the contact cutout is enabled. • An option is provided to print the NCF settings displayed in various NCF forms to a text file. • An option is provided for an early indication of LCN Cable problems. A tool is provided for Diagnosis.
Send an alarm to the TPN operator when the state of an HCI component state changes/fails on GUS/APP/ES-T/ESVT/ACE-T nodes	• The Node Composite state is propagated by TPN Communication Service to TPN. In the Devices/Services page, the option TPN Communication Service is provided to enable the service. By default, this service is selected in the Devices/Services page. This requires TPS R410 in addition to TPN R681. • An AUX Node status for GUS/APP/ES-T/ESVT/ACE-T node is added to the TPN to represent the overall status of an HCI Component. • A system alarm is generated, whenever a Honeywell software component running on a Windows environment fails in any one of the nodes.

1.15. R681 Enhancements

	• The status changes to WARNING when any of the HCI components in the node goes to WARNING state and changes to SEVERE when any of the HCI components fails. • The annunciation of alarms corresponding to the change in the auxiliary status of the node is recorded in the System Status Journal.
OFF delay option in HG/NIM to minimize the chatter of BADPV and BADCTL alarms	• The OFF delay feature is available for BADPV (HG & NIM) and BADCTL (NIM) alarms through the \$DLBADAL parameter. This parameter allows the user to delay the return to normal condition (RTN) for a time as specified by the DLYTIME parameter. • This feature is supported in the following point types: - o NIM: Analog Input, Regulatory Control, Regulatory PV - o HG: EC Regulatory Control point • During the delay period, if any of the BADPV or BADCTL alarm returns to normal, the timer is restarted from its initial configured value. • During delay period, if any of the BADPV or BADCTL alarm returns to normal, the timer is restarted from its initial configured value. • When the timer expires, if the point is not in alarm, the return to normal is reported. The RTN event is reported with the time stamp when the timer expires and not the actual RTN time. • While in delay, the following instances can cause the RTN to get reported before the expiry of the timer: • Change in \$DLBADAL parameter value, and change in DLYTIME • Change in priority of the BADPV and BADCTL alarms

System HM Node to retrieve error blocks of a failed node on receiving the “FAIL” status change notification from the failed node	• When a node fails, the System HM retrieves and saves the information to a file in the system HM. • Valid error blocks, register information and node attributes are retrieved in system HM. • The new feature supports the failure of the following LCN node types: - o AM ,Non-System HM, HG, NIM, CG, NG, US, GUS, EST, and APP • The error block information is saved in ERRBK.XX file in the NET>&ASY directory of HM. The text file can accommodate the error block information for a maximum of 10 failed nodes. • If the ERRBK.XX file is not accessible, the error block information is retrieved and saved in a temporary file. • The System HM can save the error block information for multiple failed nodes.
For selected points in NIM, AM and HG an option is provided to allow specific alarms of a point to pass through, though the contact cutout is enabled	• The Selective Cutout feature allows the alarms of a point to pass through, though the secondary point is in the cutout state. • This feature is supported in the following point types: - o NIM: Flag, Numeric, Regulatory Control, Regulatory PV, Digital Input, Analog Input, Digital Composite, and Device Control Points. - o AM: Regulatory, Counter, and flag points. - o HG: Analog Input, Regulatory, Digital Input, Analog Composite, Digital Composite • Nine different alarms can be allowed to pass through by configuring the parameters. The parameters are \$HISELCT, \$HHSELCT, \$LOSELCT, \$LLSELCT, \$DHSELCT, \$DLSELCT, \$RPSELCT, \$RNSLCT, and \$OFSELCT. The user requires ENG access to configure these parameters. The changes are recorded in journal.

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An option is provided to print the NCF settings displayed in various NCF forms to a text file	• The NCF will be printed as a text file in NET>&ASY directory with the file name as NCF.XX. • Path can be specified by the user and the existing file can be overwritten with ENG access. • Printing to a text file is initiated by the F4 function key, same as that of printing to a printer. • Pressing CTL + Break will abort printing. • Parameter names and values are separated by “=”. Parameter with multiple values is printed continuously separated by a space. If the number of characters exceeds 80, the next line begins with &\.
Provides an early indication of LCN Cable Problems and Diagnosis	• Captures additional LCN Cable diagnostic information through PSDP parameters. • Detects Marginal Node state as it relates to cable communication. Indicates the marginal node state in System Status display. Journalizes the marginal state in system error journal. • Detects Marginal Cable state as it relates to cable communication. Indicates the marginal cable state in System Status display. Journalizes the marginal state in system error journal. • PSDP parameters are provided to specify the threshold for Nodes and Cables to enter into the marginal state. • A suite of displays is available to analyse the cable problems. • A tool is added to check the health of the communication between any two nodes.
	ATTENTION If you are migrating from R680 to R681 or later, do not use this functionality until the migration is completed.

1.16 R682 Enhancements

Introduction	TPN network software Release 682 enhancements are a result of user-identified needs. The User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Middle East, Africa and Australia, recommended these enhancements. The following lists the TPN 682 enhancements that are made as a result of their recommendations: • An option is provided to change the Clock Master and slave nodes without rebooting any of the LCN Nodes. • Native Window is enabled to display appropriate access levels (ENG, SUP, OPR, or VIEW) when a user logs in using either a Windows login or through SignOn Manager. • An option is provided to the user to take a backup of box parameters in xPM, LM and SM to a file and helps to restore the values later. This option is applicable to APM and HPM simulator box parameters also. • An option is provided to process HPM Process module points twice a second. User has an option to configure the number of Process Module Slots that need to be processed twice in a second. Additional PU is consumed for this process.
Switching the Clock Master and Slave nodes online without rebooting any of the LCN nodes	• In all the LCN node personalities, an option is available to change the Clock Master and Slave nodes without rebooting any of the Nodes. This option consists of the following functionalities: • Switching the clock mastership and Slave to User specified nodes • Switch only the Master node and keep the existing Slave with out any change • Swap the Master and Slave nodes • Switch only the Slave node and keep the existing Master with out any change • The switching/swapping will happen online and does not require reboot of any node in the network. During the clock mastership transition, the nodes shall go to Local mode. When the Master / Slave transition is

1.16. R682 Enhancements

	complete all the nodes shall become Listener to the new Master. • The enhancement will allow the override of the NCF selection for clock sources to switch the mastership. The NCF status will be changed to "No Impact" when the clock source node changes. "No Impact" is an NCF state that indicates that the NCF has changed, but has no impact on the operations of the LCN nodes and the system does not require the LCN nodes to be rebooted to update their local NCF information. • To synchronize the clock source information between the user directed changes and the NCF configuration, the user will have to update the NCF using the existing NCF configurator mechanism. As the changes are made online, the user need not restart the nodes. • If the NCF is not synchronized with the new clock source information and a Master and Slave node exists in the network, then reboot of any node that is defined in NCF as Clock source shall try to assign itself as Slave.
Native window access level and current default access level are synchronized with sign on access level	• When a user logs in using either a Windows login or through SignOn Manager, the Native Window will display the following on its status bar, irrespective of the setting in the CURRENT DEFAULT value of ACCESS CHG target. • With "View Only" access, the status bar will display the status as VIEW and • With "Operator" access, the status bar will display the status as OPR • This new option will work with the SignOn Manager Override. On loading/reloading the Native Window, the LCN access level shall match the access level of the currently logged on user. • There shall be consistency of behavior and visual representation of access levels between Native Window and SOM.

User has an option to backup the box parameters in xPM, LM and SM	• An option is available to the user to take a backup of box parameters in xPM, LM and SM to a file and to restore the values later. • This option is applicable to APM and HPM simulator box parameters also. • The user shall be able to select the following: ○ UCN number ○ Box number ○ Starting subscript ○ Ending subscript ○ Parameter type (numerics/flags/times/strings) ○ File name for Backup/Restore • The Restore feature will allow the user to restore the values of the box parameters when required. • This option is accessible through DEB with two options -- one for 'backup' and one for 'restore'. • The user can either select a combination of box parameters for backup/restore or select ALL. The location where the backup file is to be stored shall be selectable by the user. For restore, the user can specify the pathname of the file from which the parameters are to be restored. • This option is available as a DEBEC command also. • xPM and LM can be in any run state (IDLE, OK) for the backup and restore option to work. However SM should be in OK state.
Option to configure the number of Process Module points to be processed twice in a second	• User has an option to configure the number of HPM Process Module Points that need to be processed twice in a second. • Additional PUs is consumed for executing this operation. • The points that need to be processed twice should be placed in the slots of lower value. • The points which are configured on slots after the \$NSFPMS value will be processed once a second.

1.17 R683 Enhancements

Introduction	TPN network software Release 683 enhancements are a result of user-identified needs. The User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Middle East, Africa and Australia, recommended these enhancements. The following lists the TPN R683 enhancements that are made as a result of their recommendations: • The number of characters in the CALCEXP parameter of HPM Regulatory PV is increased from 40 to 60 characters. The functionality is applicable only for the CALCULTR algorithm of HPM. • A deviation high alarm is introduced in HPM Regulatory PV points with HiLoAvg and MidOf3 algorithms if the absolute difference between any of the inputs exceeds the deviation alarm trip point specified by the user. • PED and EB files are modified to display and print the priority parameters and other alarm parameters, irrespective of the value of the corresponding alarming trip point parameter. • The file names of the TPS/TPN documents are changed to have the document name in addition to the regular document number/model number. The renamed documents will be delivered in Knowledge Builder, as well as in a DVD as a PDF collection.
The maximum number of characters in the CALCEXP parameter of HPM Regulatory PV CALCULTR algorithm is increased from 40 to 60 characters	• This functionality is applicable only for CALCULTR algorithm of HPM Regulatory PV points. • The data type of the CALCEXP parameter is changed to String_60. The number of equations, the number of intermediate storage constants and the number of inputs remain the same. Only the expression length is modified to allow up to 60 characters.
A deviation high alarm is introduced in HPM Regulatory PV points with HiLoAvg and MidOf3 algorithms	• A deviation high alarm is introduced in HPM Regulatory PV points with HiLoAvg and MidOf3 algorithms if the absolute difference between any of the inputs exceeds the deviation alarm trip point specified by the user.

	• Inputs which are active and for which the status is Normal or Uncertain are considered in the deviation alarming. The deviation alarm trip point is stored by the user in the DEVHITP parameter. • DEVHIFL indicates the status of input PV deviation. The difference between inputs will be continuously monitored and the DEVHIFL parameter is set if the input PV deviation exceeds the limit specified by the user (DEVHITP). • Selective cut-out, re-alarm and alarm delay options are applicable for deviation alarming on Regulatory PV points using MidOf3 and HiLoAvg algorithms. • By default the deviation alarm functionality remains disabled. • Deviation alarm is applicable only for HPM Regulatory PV points with HiLoAvg and MidOf3 algorithms.
PED and EB files are modified to display and print the priority and other related parameters, irrespective of the value of the corresponding alarming trip point parameter	• PED is modified to display and modify EB file to print the priority parameters and other related parameters such as xxSELECT, PVHHTP, PVLLTP and SVHHTP irrespective of the value of their corresponding alarming trip point parameter. • An option is added in the NCF System Wide Values to enable or disable this functionality. By default the option is disabled. • When this functionality is enabled, the priority parameters and other alarm parameters are displayed/restored/printed at the following instances irrespective of the value of its trip point parameter. ➤ When a point is reconstituted ➤ When a point is backed up to an EB file ➤ When a point is restored to the NIM via an EB file ➤ When a point is backed up to an IDF file ➤ When a point is restored to the NIM via an IDF file

1.17. R683 Enhancements

	• As the parameters PVSGCHTP, SVALDB, MASKTIM and AVDEV2TP cannot be stored and have no value when its trip point parameter is set to NaN, they are not affected by this functionality. These parameters are hidden if their corresponding trip point parameters are set to NaN. • An option is provided under TPS Builder Parameter configuration. Users need to select this option to print the NIM priorities in the PCF and EB.
The file names of the TPS/TPN documents are changed to have a text name in addition to the regular document number/model number	• The file names of the TPS/TPN documents are changed to have the document name in addition to the regular document number/model number. The renamed documents will be delivered in Knowledge Builder, as well as in a DVD as a PDF collection. With R683, the PDF collection DVD is introduced as an additional deliverable for the user documentation. • The Knowledge Builder will include: ➤ The folders and the documents with the revised names i.e. – The document name is followed by the document number. Example: Engineer's_Reference_SW09-601.pdf. ➤ The existing directory structure. • The PDF collection DVD will include: ➤ The renamed documents to display the document name followed by the document number. Example: Engineer's_Reference_SW09-601.pdf. ➤ The option to have the book set of the previous release on the same machine. ➤ An installer for the PDF Collection. • The Document attributes will indicate the Document number followed by Document Name in the “Title” field and the JTA Category in the “Subject” field of the document.

1.18 R684 Enhancements

Introduction	TPN network software Release 684 enhancements are a result of user-identified needs. The User Input Subcommittee, a worldwide group with representatives from North America, Europe, the Middle East, Africa and Australia, recommended these enhancements. The following lists the TPN R684 enhancements that are made as a result of their recommendations: • An option is introduced to automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals. • Alarm delay option is introduced on AM Regulatory type points, similar to NIM and HG. This option allows the user to configure either an On Delay or an Off Delay, to either suppress the alarms which are generated as a result of unstable process (On Delay), or minimize the Transient alarms (Off Delay). • The number of HPM Regulatory PV slots is increased from 125 slots to 400 slots. User shall have the option to configure the number of HPM Regulatory PV slots and the number of fast Regulatory PV slots to the extended range. This functionality is applicable only for HPM. • From R684, version revision supports 3 digits in TPN personalities and software. • Along with these UIS enhancements the Enhanced Universal Control Network (EUCN) is also delivered with TPN R684. An option is introduced to replace the traditional coaxial cable system Fault Tolerant Ethernet (FTE) based communication network, with Enhanced Network Interface Module (ENIM) replacing NIM, Enhanced High Performance Process Manager (EHPM) replacing HPM and the network being named Enhanced Universal Control Network (EUCN). The UCN status displays are updated for EUCN. Displays that are important for fault scenarios or maintenance of the EUCN FTE network and their elements are modified to ENIM or EHPM.
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An option is introduced to automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals	• This functionality is applicable only for HPM. • This option is applicable only for the TOTALIZR algorithm of HPM Regulatory PV point. • Two new parameters are added to reset the PV and restart the accumulation for the Totalizer RegPV points. 1. Automatic Reset Option: (\$AUTROPT) This parameter is an enumeration with the following values: NONE, HOURLY, 8HOURS, 12HOURS, DAILY, WEEKLY, MONTHLY, and YEARLY. The PV value is reset based on the option selected. 2. Automatic Reset Time: (\$AUTRTIM) This parameter allows the user to enter the offset time within the interval determined by the reset option chosen. With 0 00:00:00 being the lower limit for all the options, the automatic reset functionality is active at 0 00:00:00 if the \$AUTROPT is set to a value other than NONE.
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Alarm delay option is introduced on AM Regulatory type points, similar to NIM and HG	• This option allows the user to configure either an On Delay or an Off Delay, to either suppress the alarms which are generated as a result of unstable process (On Delay), or minimize the Transient alarms (Off Delay). • Similar to NIM and HG, the following two alarm delay options are available: 1. \$ALDLOPT: For PVHH, PVHI, PVLO, PVLL, PVROCN, PVROCP, ADVDEV, DEVHI and DEVLO alarm types 2. \$DLBADAL: For BADPV and BADCTL alarm types • This functionality also provides the option to configure the time duration, during which the On Delay or Off Delay can be effective. • The user can select (per point) a time window for \$ALDLOPT and \$DLBADAL, similar to the DLYTIME parameter which is used in NIM and HG. The Delay Time parameter specifies the duration of the time during which the alarm reporting (in the case of On Delay) and the RTN reporting (in the case of Off Delay) are suppressed. • Alarms that RTN during the time delay, and have not gone back to the alarm state before the expiration of the time delay will not be reported. The timer restarts either on every transition to the alarm state (for On Delay) or for every transition out of alarm state (for Off Delay).
The number of HPM Regulatory PV slots is increased from 125 slots to 400 slots	• This functionality is applicable only for HPM. • User shall have the option to configure the number of HPM Regulatory PV slots and the number of fast Regulatory PV slots to the extended range. • The range of NPVSLOT parameter is increased from 0-125 to 0-400. • The range of NFASTPV parameter is increased from 0-125 to 0-200. • The HPM will consume additional PUs and MUs for the increased HPM Regulatory PV slots and Fast Regulatory PV slots.

1.18. R684 Enhancements

Version revision supports 3 digits	For the following, the version revision is displayed in 3 digits. • HPMM Version/Revision display • HPMM Communications Error Block Display • Control Error Block display • SMCC – Rev / Config status • Audit tool • LVRLOG tool • Perfmenu – UCNVERSN schematic • Perfmenu – IOPMDATA schematic • CBREV schematic • LCNP debug tool • CBREV(n) – PSDP parameters on all nodes except NIM PSDP
Enhanced Universal Control Network (EUCN)	• The Enhanced Universal Control Network (EUCN) introduces an Enhanced Network Interface Module (ENIM) and Enhanced High Performance Process Manager Module (EHPM), wherein the traditional co-axial cable system is replaced by the Fault Tolerant Ethernet (FTE). The user functional difference is the configuration of the FTE based EUCN Ethernet network. A new parameter \$DEVIDX is available for : - o the ENIM to hold the value for the Authentication Group - o the EHPM to configure the FTE device index. The \$DEVIDX shall be zero for legacy nodes and non-zero for EUCN nodes having an FTE connection. • The FTE status can be traced using the two new FTE status schematics \$FTESTS1 and \$FTESTS2 available on the PERFMENU. The UCN status displays, where there are specific physical hardware differences that are necessary to be identified to support the maintenance of the EUCN system hardware are modified. Note: EUCN and UCN nodes cannot communicate with each other. If they are configured in a remote UCN configuration, then both NIM and HPM must be upgraded to their respective EUCN nodes. Upgrading UCN nodes to EUCN nodes must be done off-line.

1.19 Effects and Requirements of this Release

Take note of R43x to R6xx requirements The following table shows requirements and changes in support strategies that must be considered and any required adjustments that must be made before start of a migration from any R43x release to R6xx.

Subject	Description
AM memory	All installed Application Modules require 1 Mw of additional memory over and above what they previously accessed up to 8 MB. (Refer to section 3.4 for more complete memory requirement information.)
A ^X M memory	All Application Modules ^X now require 6 Mw of memory. This is 2 Mw higher than required for R4xx.
68000 processors	LCN nodes containing 68000-based processors (MCPU and EMPU boards) are no longer supported, and must be upgraded to a 68020-based processor (K2LCN, HPK2, or HMPU) or to a 68040-based processor (K4LCN).
WREN I and WREN II drives	HMs containing WREN I or WREN II disk drives are no longer supported. SCSI drives are the only disk drives supported by R6xx or later.
Zip drives	Release 6xx Software will be distributed on Zip disks or CD-ROM.
NIM & UCN Checkpoints	NIM and all UCN Node type checkpoints will be on Bernoulli cartridges or Zip drives. Floppies will not be supported. NIM checkpoint size increased from R43x to R6xx.
Separate Engineering Personality	There is no longer a separate Engineering Personality.
K4LCN upgrades	For the best performance improvements, the first LCN nodes to be converted to K4LCN must be the NIM, HG, PLCG, AM, and CG. AM Redundancy using the K4LCN processor also requires the use of EAMR Redundancy boards.
HPM	It is recommended that an HPM on the UCN be driven by a NIM containing a K4LCN board.
HM	New maximum journal event sizes have changed for: Process Alarms to 15,000 Process Changes to 10,000 SOE events to 15,000 System Status to 12,500 System Errors to 15,000 System Maintenance to 15,000 Operator Messages to 5,000
US	If a process alarm type is configured for color, all printers on the LCN that will receive that alarm type must support ASPI color escape sequences.

Table 1-6 R6xx Requirements

1.20 Overview of the Migration Process

Migration - R43x and R5xx to R6xx The steps to migrate from R43x and R5xx to R6xx are provided in this manual. A "one-step" method is used for R5xx to R6xx, and a "two-step" method is used for R43x to R6xx.

If your system has any FSC Safety Managers, they must be the last devices migrated to R6xx. Once they are on R6xx there is no on-line method available to go back to R510/R52x/R53x.

MANDATORY - Update K4LCN board hardware revision The K4LCN board hardware revision has been updated in the NCF to a "C" to conform to the mandatory minimum revision level.

Responsibilities When performing migrations, even if the migration is being performed by a Honeywell specialist:

- the customer is expected to take the lead,
- the customer will continue to maintain primary responsibility for the integrity and safety of the process operation, and
- all process-related activity required during the migration process must be performed by the customer.

Contents of release media Listings containing the Honeywell-provided media for Release 6xx can be found in the following sections of this document.

Media name...	Can be found in ...
ZIP contents	Section 6.2
CD-ROM Contents	Section 6.3
DAT Tape/CD-ROM Contents	Section 6.4

ATTENTION

If you are migrating from R680 to R681 or later, do not use the LCN Cable Diagnostic functionality until the migration is completed.

Expertise level of person performing migration

At minimum, the person(s) performing this migration is expected to have the following skills.

Prior to migration:

- Know how to prepare node (from US) for removal of power
- Know how to power down nodes (or each half of a dual node), and can remove and replace circuit boards without damage, and with human and equipment safety in mind
- Know how to check hardware and firmware revision status of boards (both through the system and physically on the circuit boards)
- Understand the negative affects of ESD (Electrostatic discharge) on hardware components, and know what to do to prevent it
- Know how to remove and replace fragile integrated circuits (chips) on boards
- Know how to check and set switches on circuit cards
- Know how to initialize media

During migration:

- Proficient in manipulation through all standard TPS screens on a Universal Station, Universal Work Station, Universal Station^X or GUS.
- Proficient in using Command Processor
- Proficient in using Data Entity Builder
- Know how to initialize disk drives on HMs
- Can correctly calculate HM Sizing Estimates
- Know how to “back out” of a migration that cannot be completed.

Migration steps in this manual are written to the level of a person with the above skills. Do not attempt to perform this migration if you do not have these skills, or have someone available who has these skills.

1.20. Overview of the Migration Process

Migration expert assistance available

To best serve our customers' needs in this area, Honeywell has developed and field-tested a detailed migration process involving the use of skilled personnel.

In the USA, contact your local Honeywell service branch for a quote on an installation service that includes the services shown in the following table.

Service	Result
Site audit	Identifies all required items and tasks as they apply to your site.
Assist with migration plan and schedule	Developed by local Honeywell Migration Coordinators in conjunction with the customer, and are based upon proven procedures, which are tailored to the customer's needs.
Parts procurement assistance	Honeywell provides assistance in obtaining the upgraded circuit boards, firmware chips, memory chips, disk drives, etc. that may be needed prior to the start of a software migration.
Expert assistance in the actual migration	In the form of trained technical teams.

Table 1-7 Available Honeywell Installation Services

Suggestions for most efficient HM backups

Backing up HM data is time consuming.

The following suggestions can speed up the process:

- You can create a very large volume on zip or cartridge with up to 32618 files, but the time to backup HMs to such volumes can be excessive. It is suggested that you always specify a maximum directory block size (1700) when using a large number of files. Limit the number of files on the volume to what you actually need for your HM.
- Specification of File Descriptors on zip or cartridge volume severely limits the maximum block size you can specify. Use File Descriptors SPARINGLY. Only specify File Descriptors for volumes that truly need them.

CAUTION

If File Descriptors are in use, do not format Bernoulli cartridges for block sizes greater than 600. Doing so may result in corruption of data on the cartridge. Conversely, do not use File Descriptors if block sizes are in excess of 600.

These volumes should be copied individually to cartridges with volumes that support File Descriptors. Then BACKUP the total HM to cartridges without File Descriptors. After the normal RESTORE, re-RESTORE from the -FD cartridge. When using the -FD option, you cannot specify a Block Size greater than 600.

For HMs without File Descriptors (no -FD), the following forms of the CREATE VOLUME call provide a means for minimizing the number of cartridges, and the actual time to perform the backup.

Cartridge Size	Create Volume Input (If not using File Descriptors)
35 MB	CR \$Fd>xxxx> -BS 1700 -MF 3400
150 MB	CR \$Fd>xxxx> -BS 1700 -MF 6800
ZIP100 MB	CR \$Fd>xxxx> -BS 1700 -MF 6800

Table 1-8 Create Volume Input—No File Descriptors

**Suggestions for
most efficient HM
backups
continued**

By contrast, if you use File Descriptors on the HM, the best usage of the CREATE VOLUME call is shown in the following table.

Cartridge Size	Create Volume Input (Using File Descriptors)
35 MB	CR \$Fd>xxxx> -FD -BS 300 -MF 3300 (If migrating from R43x) CR \$Fd>xxxx> -FD -BS 600 -MF 3300 (If migrating from R500, R510 or R53x) (Ten times slower than without File Descriptors)
150 MB	CR \$Fd>xxxx> -FD -BS 300 -MF 6600 (If migrating from R43x) CR \$Fd>xxxx> -FD -BS 600 -MF 6600 (If migrating from R500, R510 or R53x)) (Twenty times slower than without File Descriptors)
ZIP 100 MB	CR \$Fd>xxxx> -FD -BS 300 -MF 6600 (If migrating from R43x) CR \$Fd>xxxx> -FD -BS 600 -MF 6600 (If migrating from R500, R510 or R53x)) (Twenty times slower than without File Descriptors)

Table 1-9 Create Volume Input–Using File Descriptors

CAUTION

Use of the -MD (Memory-resident Directory) option is discouraged. If you use the same number for the -BS (Block Size) and the -MF (Maximum Files) parameters, you actually cause all of the directory information to remain in memory, without the need for the -MD option.

While the new Extended Directory Support function provided in R510 and later supports up to 2046 directories, for migration/translation purposes the limit of 1 volume and 63 directories still applies.

Migrating your system from R43x or R5xx to R6xx

The following two pictures show the process of migrating your system from either R43x or R5xx to R6xx.

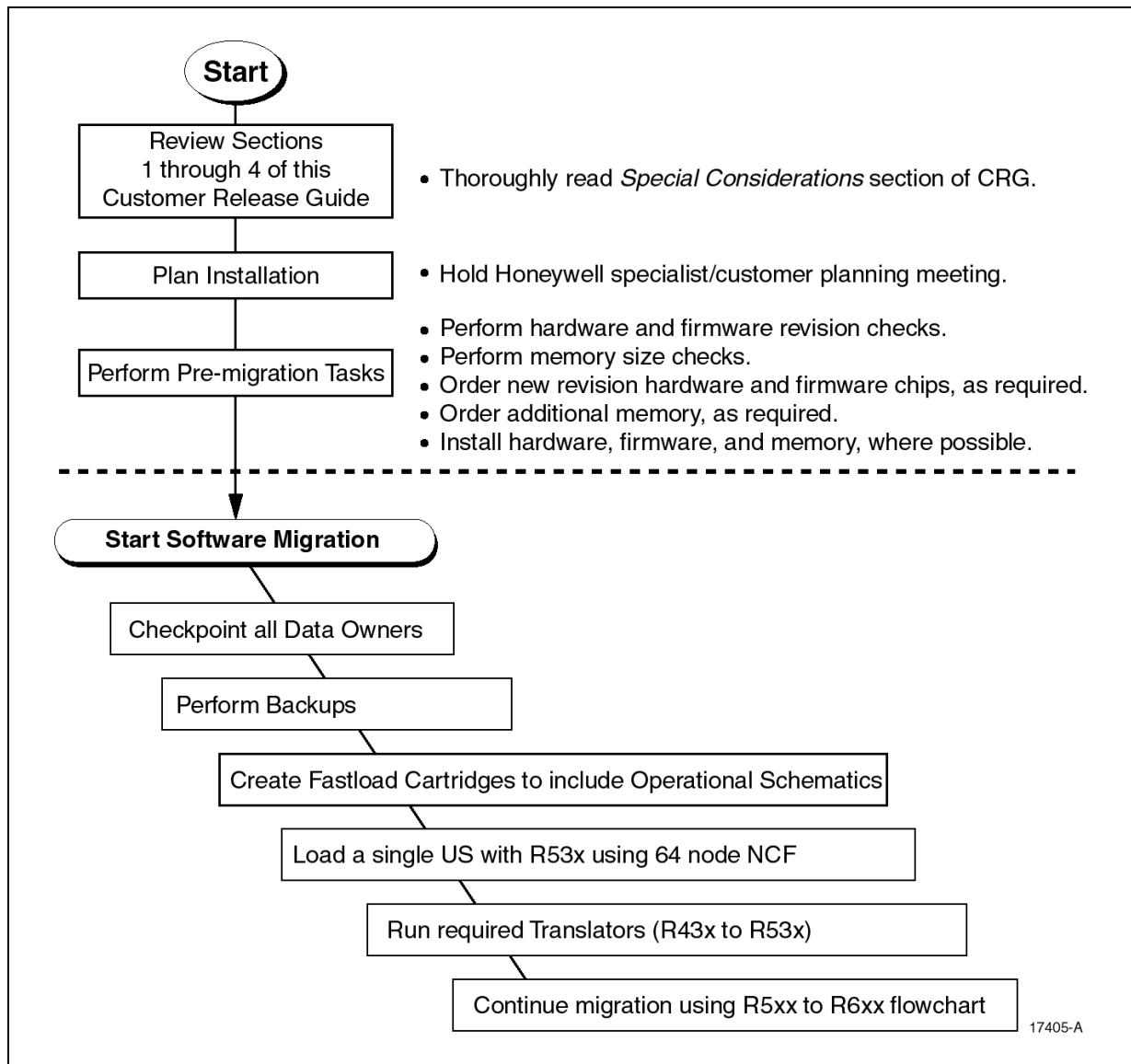


Figure 1-1 Big Picture of Migration from R43x to R6xx

Migrating from R43x or R5xx to R6xx

The following two pictures show the process of migrating your system from either R43x or R5xx to R6xx.

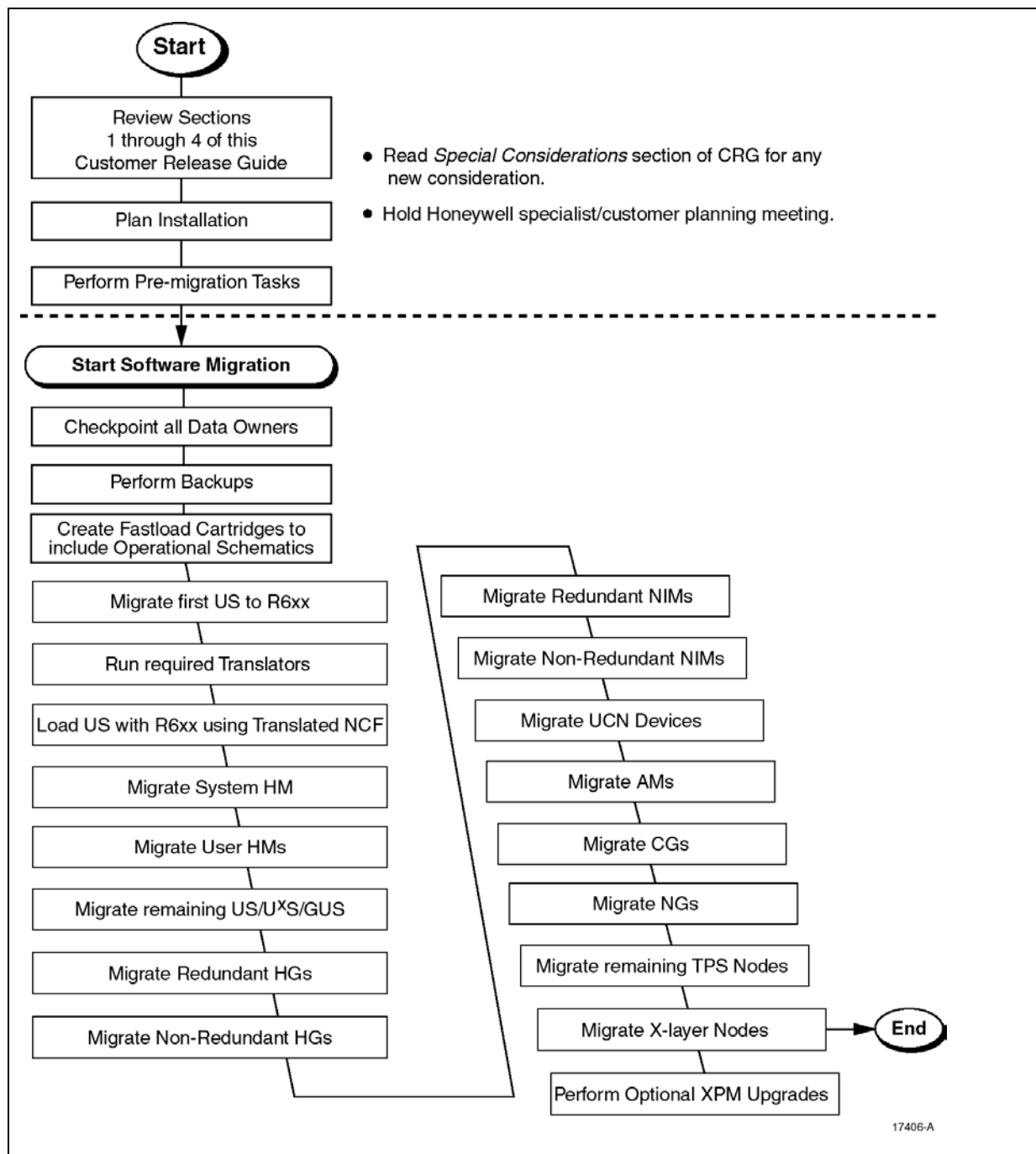


Figure 1-2 Overview of Migration from R5xx to R6xx

1.21 Hardware Revision Requirements

Hardware revision checks

Whether you're migrating from software release 43x or release 5xx to 6xx, the hardware must be at (or above) a given board revision, firmware revision, and memory size, as applicable.

Before starting the software migration, all hardware should have been checked and should be ready for migration, as needed.

Those revisions and memory sizes are listed in Section 3, *Hardware Preparation*.

If migrating from an R4xx release, and you are upgrading your

- PM to an APM,
- PM to an HPM, or
- APM to an HPM,

that upgrade must be performed **AFTER** the R6xx software migration is completed.

1.22 Where To Go From Here

Where to start

If you have:

- read all of the applicable information in the Special Considerations section,
- performed all prework (applicable hardware upgrades, firmware upgrades, memory size adjustments, file cleanup, etc.),
- checked to see you have enough blank ZIP cartridges (using 100 meg ZIP cartridges two (2) would be a minimum and more would be needed if you have a larger system),
- checked to see you have received all of the required R6xx software disks,
- ensured that you have all the necessary custom software for R6xx (Reference: *Special Application Software User Manual*, and *Installation Manuals*), and
- read and performed all applicable preparation items in this Read Me First section,

You are ready to start the R6xx software migration..

How to know when you're finished

The total migration from R43x includes:
Setup for migration from R43x; Section 7.
Overview from Setup for Migration from R5xx in Section 8.
R6xx Migration NCF Configuration Changes; Section 9 and
R6xx Migration; Section 10

The total migration from R5xx includes:
Setup for migration from R5xx; Section 8.
Migration NCF Configuration Changes; Section 9 and
Migration; Section 10.

2. What To Do Before Migrating Your Software

2.1 Overview

Overview

This section describes the work that must be performed prior to the start of the actual software migration.

Taking these actions ahead of the software migration will help ensure a smooth and timely software migration.

You should not be using this document to bring up the software on a newly installed **TotalPlant** Solution (TPS) system. This document is for migrating a system currently running on R43x or R5xx to R6xx.

CAUTION

During migration, you will lose the ability to get to your System HM (or any HMs) until your system is up on Release 6xx software. This could be a long period of time. You should, therefore, provide alternate access to system operational schematics through the use of removable media containing such schematics. These are referenced from the Area Database pathnames.

Added LM information

Starting with R410, certain Ladder Logic lines are identified that now show several error conditions that were not previously available. They support Enhanced Serial I/O System Diagnostics. Until these lines are included in your program, those "PARTFAIL" error conditions are not available to your system.

To enable the new Enhanced Serial I/O System Diagnostics, the Ladder Logic changes must be loaded. Save the Ladder Logic to the HM following the software migration.

Also, if any external shut down timers are in use in conjunction with the LM, be sure to check the timer function and make appropriate adjustments. The time will vary as a function of the amount of ladder logic in the LM. As you increase your ladder logic, you should adjust the timer function as necessary.

Performing a swap of LM primaries requires the external timers to be set at a minimum 6.5 seconds. Set external timers to a time of 6.5 seconds or greater to eliminate problems in this area.

External shutdown timers on Serial Interface modules

If external shutdown timers are in use in conjunction with the UCN devices, be sure to check the timer function to ensure the timers are set at a minimum of 3 seconds.

Performing a swap of the UCN devices requires the external timers to be set to 3 seconds or longer.

Added Option kit Info.

If installing options, refer to the instructions that come with the Option kit.

Tool Kit displays

Honeywell provides a set of displays that help you monitor the performance of your system. These are referred to as the Tool Kit displays and are distributed on the Zip &Z1, under the directories TLK1 and DIA1.

You can call up the Tool Kit displays by invoking the PERFMENU schematic. Throughout this document, we will be referring to these displays as the Tool Kit displays.

**LCN I/O board
revision**

The LCN I/O board must be at hardware Rev. T or later if the node has a K2LCN or K4LCN processor. Refer to “Hardware Preparation” section for details (section 3.3).

**Additional memory
for External Load
Modules**

When determining the minimum memory requirements for each of your nodes, be sure to consider any External Load Modules resident in the node, and the possible increased memory requirements each of the External Load Modules may have. Refer to the documentation for each External Load Module to determine its minimum memory requirements.

Keep ‘em in order!

For the best results, follow the preparation steps in the rest of Section 2 in the order shown.

Section Number	Major Task
2.2	Ensure System Stability • Check for LCN communications problems • Obtain SMCC maintenance information • Check for UCN communications problems • Ensure all Hiway boxes are OK
2.3	Check LCN Load & NCF Status • Check LCN load using Tool Kit displays • Ensure there is no LCN RECONNECT in progress • Ensure all nodes are in-synch with current NCF
2.4	Print Contents of NCF & Check Area Data Base • Print entire NCF • Print HM Volume Configuration • Check Area Data Base pathnames

2.2 Ensure System Stability

Check for LCN communications problems

If the master node has a K2LCN board with firmware revision P or later, or a K4LCN board, the \$LNMENU “Crossed Cable Information” display \$LNACCT will provide a count of single cable transmissions received by each node.

The \$LNACCT display provides a count for each node that can be used to determine the location of crossed LCN cables. The master node sends a count message on only the active cable. Any node with crossed cables will not see the message, so its count should be off, by two or more.

Prior to starting the software migration, perform the following steps to check the system for LCN communications problems, including crossed cable situations.

Use \$LNMENU Tool Kit displays to check the following:

Step	Action	Done (√)
1	Check the NET Media Dead (For help, see the <i>LCN Guidelines Manual</i>)	
2	Use the NETWORK view to check the errors per 10,000 for CABLE A&B rates.	
3	Use the Node specific view for CABLE A&B Rates.	
4	Correct all problems before continuing.	

Table 2-1 Check for LCN Communications Problems

Perform the following steps to check the Clock Status Display(s) (CLOKSTAT).

Step	Action	Done (√)
1	Use the Tool Kit displays to check the Clock Status Display(s).	
2	CORRECT all problems before continuing.	

Table 2-2 Check the Clock Status Display

Check your application(s) for special requirements

Since applications running on the system may have special requirements, check your application(s) for specific System Stability requirements. Some control packages require the final controllers be placed in “manual” mode prior to migration.

Perform Custom Software check

Many customers have purchased special Honeywell-provided software for the TPS System. The program object files for this custom software is found on the NET under directory &CUS. The standard migration procedures in this document do not cover the migration of special software packages.

You should identify all custom software packages currently in use and identify which LCN nodes are configured (in the NCF) that requires this custom software.

When migrating to a new series release (i.e., R4xx to R5xx), it is often necessary to order newer versions of these software packages as the older version will not work with the new series release. You should ensure you have all the necessary “certified” custom software before you begin this migration.

Because the &CUS directory is for optional custom software, the directory is not created during a History Module initialization procedure. Honeywell recommends that the &CUS directory be added to a USER VOLUME on the HM, which can be manually created or by running the ECs.

With R500 and later, several “optional” custom software packages are provided on the standard personality media set. They may or may not be required on your system. The standard personality load “EC” files will copy these custom software program object files to NET if you use PRS2VOLZ.EC and/or PRS4VOLZ.EC for Zip disk and CD-ROM.

However, if you have additional custom software packages, you must be sure to copy the necessary files to NET. Nodes configured (in the NCF) to require custom software will fail to load successfully if these files cannot be located at load time.

2.2. Ensure System Stability**Obtain SMCC maintenance info**

Using the SMCC MAINTENANCE target from the Engineering Main menu, obtain the following information. Correct any problems that are shown in these printouts.

Step	Action	Done (√)
1	Using the SMCC MAINTENANCE target from the Engineering Personality Main menu, Select the SYSTEM MAINT JOURNAL target, then select the PRINT ENTIRE SMJ target.	
2	Do a CTRL/PAGE BACK to get back to the SMCC main menu. Select ACTIVE MAINT JOURNAL target, Select DISPLAY ACTIVE MAINT RECOMMENDATIONS AND ENTER CORRECTIVE ACTION target, Press <ENTER>, Press PRINT DISP key, (This will print the first page), Then, if there are more pages, Press PAGE FWD key, and print the next page, etc., Press CTL/PAGE BACK twice.	
3	After ENTER CA Text: type some letters (xx...) and select a maintenance line (the maintenance line is <u>not</u> highlighted when selected) then press ENTER to clear it. Clear all maintenance recommendations in this same way.	
4	Select INITIATE ON-PROCESS ANALYSIS CYCLE. Select PRINT ALL ERROR AGGREGATES. If any board indicates INCREASING ERRORS, corrective action is required. If the board is still under warranty, replace the board. If the board is out of warranty, • either clean the board's contacts (using appropriate methods) then re-install the board; or • move the board to another node with the same board type and print the error aggregates the next day. This will determine if the board still has the same problem. If it does, replace the board immediately.	
5	CORRECT all problems before continuing.	

Table 2-3 Obtain SMCC Maintenance Information

Check for UCN communication problems

Perform the following steps to check the UCN devices and cables for failures.

Step	Action	Done (√)
1	CHECK the UCN statistics for all NIM, PMM, APM, HPM, SM, and LMMs.	
2	CHECK the IOPs for failures.	
3	CORRECT all problems before continuing.	

Table 2-4 Check for UCN Communication Problems**Check for EUCN communication problems**

Perform the following steps to check the EUCN devices and cables for failures.

Step	Action	Done (√)
1	CHECK the FTE communication problems for ENIM and EHPM. This can be performed using the two new schematics FTESTS1 and FTESTS2 under PERFMENU.	
2	CHECK the IOPs for failures.	
3	FIX all the problems before continuing.	

Table 2-4a Check for EUCN Communication Problems

2.2. Ensure System Stability

Ensure all Hiway boxes are OK**(Skip to the next procedure if you do not have a Data Hiway.)**

Perform the following steps to check the status of all Hiway boxes.

Step	Action	Done (✓)
1	If migrating from R43x, from the System Status Display, CALL UP the PROCESS NETWORK STATUS for each Data Hiway. If migrating from R5xx, from the System Status Display, CALL UP the NTWK/HWY STATUS for each Data Hiway. ENSURE all Hiway boxes are OK.	
2	CORRECT all problems before continuing.	

Table 2-5 Ensure All Hiway Boxes Are OK

2.3 Check LCN Load & NCF Status

Check LCN load

Perform steps 1 through 5 in the following table to check the LCN load.

Step	Action	Done (√)
	Using the Tool Kit displays, you will do a Performance BENCHMARK. You will use this benchmark as a comparison to the system performance benchmark done after the migration.	
1	PRINT NODEPERF schematic for all nodes.	
2	PRINT AMDETAIL schematic for all AMs.	
3	PRINT NGDETAIL schematic for all NGs.	
4	PRINT HMDETAIL schematic for all HMs.	
5	PRINT UCNSUMM, NODESTA1, and NODESTA2 schematics for all NIMs.	

Table 2-6 Check LCN Load

Ensure that no LCN RECONNECT is in Progress

Call the System Status Display. Make sure a valid LCN Revision is shown (e.g., R600 or R500).

Do not continue with migration if “INIT LCN RECONNECT” is shown.

Ensure that all nodes are in-synch with current NCF

Call the System Menu Display. Select NCF Status. Verify that all nodes are in-synch (i.e., backlit in green). If not, shut down and reload the node so it will be in-synch.

2.4 Print Contents of NCF & Check Area Data Base

Print entire NCF Print out the entire contents of the NCF, to include:

Components of NCF to be Printed	Printed (√)
UNIT NAMES	
AREA NAMES	
CONSOLE NAMES	
LCN NODES	
SYSTEM WIDE VALUES • SYSTEM ID • CLOCK SOURCE • USER AVERAGE PERIOD • SHIFT DATA • CONSOLE DATA • NG LOCAL SYSTEM ID • NG REMOTE SYSTEMS • NG SECURITY DATABASE (Print out for each remote NG PIN) • NG MODEM DEFINITION • TAG NAME OPTIONS • SOFTWARE OPTIONS (mark those that are cyan in color) • ALARM ACKNOWLEDGE	 _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____
HM VOLUME CONFIGURATION (for each node pair)	

Check Area Data Base pathnames Check to see that all Area Database pathnames contain removable media pathnames (\$Fx>dir) to access schematics and FFLs while the HM is not available.

2.5 Additional Manuals

Upgrading or adding special nodes

If you are adding or upgrading a GUS node, an FSC-Safety Manager or an AMW/APP, the following documents are needed:

- *Software Change Notice (SCN)*
- *TPS System Administration Guide*
- *TPS System Implementation Guide*
- *FSC Software Manual*

3. Hardware Preparation

3.1 Overview

Overview

The migration from R43x to R6xx requires that hardware on the **TotalPlant** Solution (TPS) system:

- be equal to or higher than a minimum revision level
- have circuit boards that contain firmware chips loaded with firmware equal to or higher than minimum firmware revisions
- consists of LCN Nodes equipped with a memory size equal to or greater than the memory sizes listed in this section.

The following Hardware/Firmware revision lists the affected circuit boards and their respective revision levels, followed by the minimum memory size requirements for each LCN node. The circuit board lists are grouped in the following manner:

- LCN Node circuit board minimum hardware/firmware revisions
- LCN Node minimum memory requirements
- PM/APM/HPM and IOP and FTA circuit board minimums
- Logic Manager (LM) requirements



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

3.1. Overview

68000 processors not supported

R500 and later does not support 68000 processor-based LCN nodes. If 68000 processor boards exist in your system, they must be upgraded to 68020 or 68040, as appropriate.

Hardware/Firmware minimums, vs. enhancements

The following table draws a correlation between the required hardware or firmware upgrade and the enhancement it will affect (or enable). The revision shown is a minimum revision, so later revisions are also allowed.

You need to determine the hardware and firmware you need to upgrade, depending on the R6xx-related enhancements or features you intend to implement now, or in the future.

To Use This Enhancement or Feature...	This Revision of Hardware or Firmware Must Be in Place...
HMPU or HPK2 processor	3/4 M MEM hardware rev. C
HMPU or HPK2 processor	1 M EMEM hardware rev. E
HMPU or HPK2 processor	1 M MEM hardware rev. C
K2LCN or K4LCN processor	LCN I/O hardware rev. T
Fiber optic modems (CD Networks obsolete) Network Gateway Fiber Optic Modem CE 51306959-200 (replaced by Honeywell – Active)	NGI firmware rev. 4
SOE Time Synchronization	EPNI hardware rev. F
Change of State Alarming	DI IOP firmware rev. D (5.0)
RTD Open Wire Softfail Alarming	LLAI IOP firmware rev. G (5.0)
Zip Drives	EPDG firmware Rev. M EPDG2 firmware Rev. P TPDG/TPDGX firmware Rev. G

3.2 Finding Hardware/Firmware Information - On-line

Check revision levels on UCN Perform the following steps to check the hardware and firmware revision levels of all PMs, APMs, HPMs, LMs, and SMs on each UCN.

Step	Action	Done (✓)
1	Use the UCNVERSN Tool Kit display to check the version/rev. of each PMM, APM, HPM, LMM, and SMs. Using this section as a reference (sections 3.3 through 3.6), ensure that all hardware and firmware are at the correct minimum revision level.	
2	Using Section 3 of this manual as a reference, CHECK each IOP module for correct minimum hardware and firmware revision levels. NOTE: Under PERFMENU, a schematic called IOPMDATA can be used to do this.	
3	UPGRADE all UCN boards to minimum revisions, as needed.	
4	Use LVRLOG schematic to print entire system revisions.	

Use REV/CONFIG for LCN nodes Perform the following steps, using SMCC REVISION/CONFIGURATION STATUS to perform the checks.

Step	Action	Done (√)
1	Using the SMCC MAINTENANCE target from the Engineering Main menu, PRINT all REV/CONFIG displays for all LCN nodes.	
2	Using Section 3 of this manual as a reference, CHECK the boards in each module for correct minimum hardware and firmware revision levels.	
3	UPGRADE all LCN boards to minimum revisions, as needed. NOTE: Revisions on all I/O boards must be checked manually.	

The REV/CONFIG display shows the following:

- Device
- Status
- The version and revisions for hardware and software (firmware)

NOTE:

K4LCN boards and EAMR boards **cannot** be installed at this time unless you are migrating from R5xx.

3.3 LCN Node Hardware/Firmware Requirements

LCN NODE - Hardware/firmware revision list

When migrating to R6xx, the hardware shown in the following table must be at (or above) the listed board revision and firmware revision (where firmware is included).

The symbol Δ indicates there is a change from R4xx.

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
AMR	51401070-100	E (Note 0)	
AMR I/O	51304159-100	A	
CLCN A/B (1)	51305072-100	D	N/A
CLCN A/B (2)	51305072-400	D (Note 1)	N/A
CLCN A/B (3)	51305072-700	D	N/A
CLCN -A (4)	51305072-200	D	N/A
CLCN -A (5)	51305072-500	D (Note 1)	N/A
CLCN -A (6)	51305072-800	D	N/A
CLCN -B (4)	51305072-300	D	N/A
CLCN -B (7)	51305072-600	D (Note 1)	N/A
CLCN -B (6)	51305072-900	D	N/A
CLI (BiSync Protocol)	80360206-001	E	D
CLI (HDLC Protocol)	80360206-001	G	F
CLI I/O RS232	80360209-001	C	
CLI I/O RS422	80360230-001	E	

Note 0 Use AMRs with HMPUs only and EAMRs with K4LCNs only.

Table 3-1 LCN Node Hardware/Firmware Requirements

3.3. LCN Node Hardware/Firmware Requirements**LCN NODE -
Hardware/firmware
revision list, continued**

The symbol Δ indicates there is a change from R4xx.

	Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
	PWA, PCSR, Non-EC	51304286-100PWA, PCSR, Non-EC	A (Note 2)	
Δ See note	CNI - Downloaded F/W Flashprom	51401088-100	C	F (Note 3)
	CNI I/O	51304537-100	B	
	DHI (displayed as HWY on SMCC)	51400700-100	E	J
	DHI I/O	51108088-100	A	
	DUAL NODE Pkg.	51401498-100	B	
	DUAL NODE Pwr.	51401497-100	A	
Δ New Board	EAMR	51401996-100	A (Note 0)	A
	ENG KEYBD	51400751-100	B	
Δ See note	EPDG	51401286-100	D (Note 4)	C (Note 4)
Δ See note	EPDG2 (Note 9)	51402089-100	P (Note 5)	K (Note 5)

Note 0 Use AMRs with HMPUs only and EAMRs with K4LCNs only.

Note 1 CLCN A or B or A/B boards cannot be mixed with KLCN A or B board in the same dual node chassis.

Note 2 This board has been replaced by the Precision Clock Source Repeater 51304286-100. 51109919-100 and 51304286-100 Precision Clock Source Repeaters **cannot** be used in conjunction with K2LCN or K4LCN boards.

Note 3 Use firmware Rev.F for CM50S. For any other application, contact Honeywell TAC.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

**LCN NODE -
Hardware/firmware
revision list, continued**

The symbol Δ indicates there is a change from R4xx.

	Board Name	Part Number	Min. Board Revision	Min. Firmware Revision
See note Δ New Board	EPDG I/O	51304270-100	C	
	EPDGP I/O (COLOR PALETTE)	51304584-100	C (Note 6)	
	EPLCI	51400997-200	B	J
	EPLC I/O	51304812-100	Alpha	
	EPLC I/O CE compliant	51304812-200	C	
Δ See note	EPNI	51401583-100	A (Note 7)	B
See note	FDC	51400669-100	G	J
	FDC I/O	51109336-100	A	
	FIBER OPTIC LINK	51108899-100	A (Note 8)	
	F/O CLOCK RECEIVER	51304161-200	A	
	F/O CLOCK TRANSMITTER	51304161-100	A	

Note 4 To support Trackball on Micro TDC 3000^X, this must be at minimum hardware rev. F and firmware rev. E.

For Color Palette, hardware must be at minimum rev. H.

If using Logitech Mouse, Part. No. 51196713-200, firmware must be at a minimum of rev. L. For Zip Drive support, firmware must be at a minimum Rev. M.

Note 5 If using Logitech Mouse, Part. No. 51196713-200, firmware must be at a minimum of rev. N. For Zip Drive support, firmware must be at a minimum Rev. P.

Note 6 Only required if you want to set your own CRT default background colors upon initialization.
Revision "G" or above is required if the following is true:
a) The station has Bernoulli cartridge or Zip drives and
b) The station is installed in a dual node chassis or
c) The cartridge or Zip drives do not use the node power supply.

Note 7 Rev. F hardware recommended if using SOE time synchronization.

Note 8 Board must be Rev. F to work with the K2LCN or K4LCN board.

Note 9 On the SMCC display REV/CONFIG STATUS this board is displayed as an EPDG board with a board hardware revision of P or greater.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

3.3. LCN Node Hardware/Firmware Requirements

**ENIM NODE –
Hardware/firmware
revision list,**

	Board Name	Part Number	Min. Board Revision	Min. Firmware Revision
See note	EPNI2, Enhanced Process Network Interface (CE Compliant)	51454196-100	C (Note 9A)	6F
See note	EPNI2 I/O, Enhanced Process Network Interface I/O (Non-CE compliant)	51306816-100	C (Note 9A)	NA
See note	EPNI2 I/O, Enhanced Process Network Interface I/O (CE Compliant)	51306816-200	C (Note 9A)	NA

Note: 9A: EPNI2 is the Enhanced version of EPNI card to support the FTE based EUCN

Supported from TPN R684 and above. EPNI2 supports both CE Compliant and Non-CE compliant EPNI2 I/O cards. For more details, refer to NIM to ENIM Upgrade Kit Instructions.

ATTENTION

The minimum supported LCN board for ENIM is K2LCN-3.
Also note that ENIM does not support LCN boards of HMPU and HPK2 types.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

**LCN NODE -
Hardware/firmware
revision list,
continued**

The symbol Δ indicates there is a change from R4xx.

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
GGSI	51401914-100	B	A
GUS Integrated Keyboard Keyboard EPROM U17 Keyboard EPROM U12 IKB12 EPROM		See Software Change Notice (SCN)	
GUS Power PC EPROM Firmware HAL		See Software Change Notice (SCN)	
HDDT I/O	51304913-100	A	N/A
HMPU	51400978-100	A (Note 10)	A
HPK2-2	51401288-100	A (Note 11)	A
HPK2-3	51401288-200	A (Note 11)	A

Note 10 If an additional EMEM is required or if an AMR board is required in the configuration, then the HMPU hardware must be rev. E.

Note 11 The HPK2-2 and HPK2-3 boards must be at revision G, or greater, to work with any of the memory boards, or EPDG, or EPDG2 boards. When an EPNI is in the same node as a Rev L HPK2, you cannot run TEST2 of the Hardware Verification Test System.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

LCN NODE - Hardware/firmware revision list, continued

For K2LCN boards only—If you are migrating from a software release prior to R430, SMCC REV / CONFIG STATUS does not display the correct revisions. You must visually inspect each board.

ATTENTION

If you are migrating from R43x software to R6xx, use Figure 3-1 as a guide to locating K2LCN board, kernel firmware, and LCN firmware revisions when using SMCC, and as a physical locator on the board.

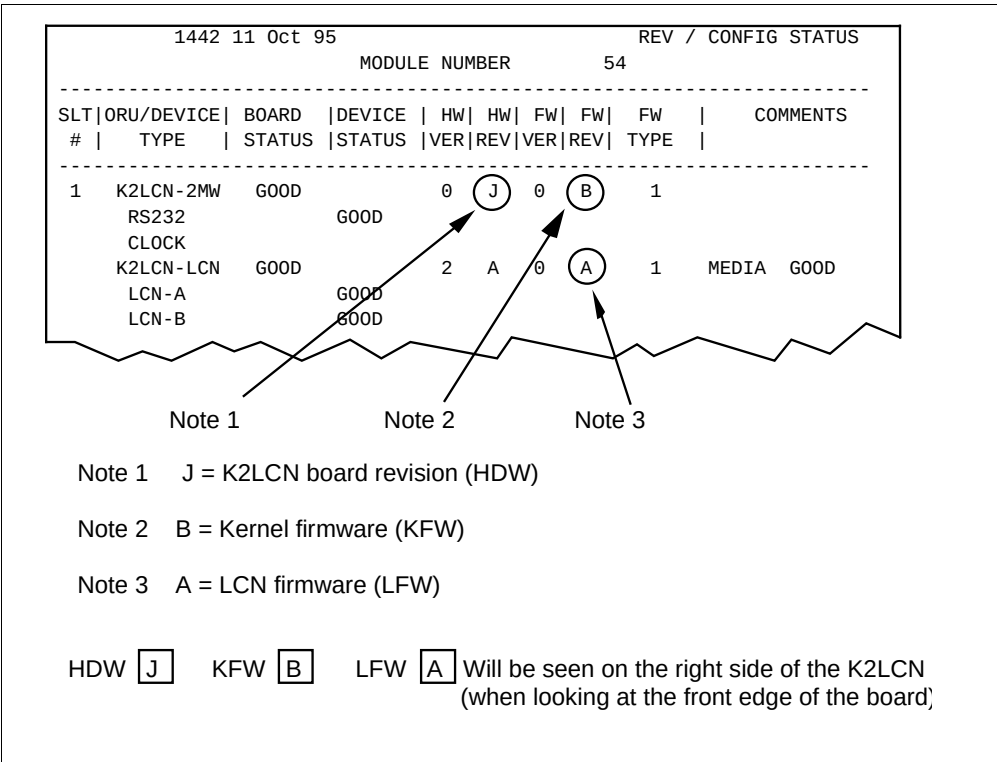


Figure 3-1 Defining Revisions of K2LCN with R430 and later SMCC

**LCN NODE -
Hardware/firmware
revision list,
continued****K2LCN and K4LCN – Remove Clock Source Repeater board(s)—**

If upgrading a node from HMPU or HPK2 processors to K2LCN or K4LCN processors, be sure to remove the CS/R Clock Source Repeater board from that node. If, after the upgrade, your system still has a combination of K2/K4 and non-K2/K4, you must reinstall the CS/R boards in the non-K2/K4 nodes and make the necessary NCF System Wide Value change. Instructions are in the K2LCN or K4LCN upgrade kit.

In the past, instructions for upgrading or configuring 5 and 10 slot chassis with newer processors (K2LCN and above) have specified that the K2LCN or K4LCN be placed in slot 2 rather than slot 1. These configurations may exist in field installations. This was done because some nodes might have a CSR (Clock Source Repeater) I/O board on the back of slot 1. The CSR board needed to be in the system to ground the LCN cable shield. But if a K2LCN or K4LCN were plugged into slot 1 with a CSR on the back, the KxLCN board would be damaged. By placing the KxLCN board in slot 2 with an LCN I/O card behind it, this could be avoided. These type of configurations are functional, but the KxLCN board can not detect the 'Fan Bad' signal which is only available in slot 1. Thus the system displays the incorrect fan status.

With the advent of the dual node chassis and the 51305492 grounded terminator, other means of grounding the shield of the LCN cable are available. The CSR I/O board is not needed to perform this function. Therefore, the preferred configuration is that the K2LCN or K4LCN be placed in slot 1 (so it can properly detect the 'Fan Bad' signal) and there be no more than 2 empty slots between boards.

3.3. LCN Node Hardware/Firmware Requirements**LCN NODE -
Hardware/firmware
revision list, continued**

The symbol Δ indicates there is a change from R4xx.

	Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
Δ See note	K2LCN (original)	51401551-x00 (Note 12)	B (Note 17)	
	- Kernel Firmware	51120624-100		A
	- LCN Firmware	51120625-10x		A
See note	K2LCN (with Crossed Cables Detect. functionality)	51401551-x01 (Note 12)	P (Note 17)	
	- Kernel Firmware	51120624-101		A (Note 13)
	- LCN Firmware	51120834-10x		P (Note 14)
Δ New Board	K2LCN (SMT)(with Crossed Cables Detect. functionality)	51402615-x00	U (Note 17)	
	- Kernel Firmware	51120908-101		U (Note 15)
	- LCN Firmware	51120908-101		P (Note 14)
Δ New Board	K4LCN (no memory)	51401946-100	C (Note 17)	
	- Kernel Firmware (Note 16)	51120809-100		A
	- LCN Firmware	51120809-100		A
	K4LCN second generation	51402755-100	(Note 17)	
	KLCNA	51304542-100	B (Note 17A)	
	KLCNB	51304544-100	B (Note 17A)	

3.3. LCN Node Hardware/Firmware Requirements

	Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
Note 12	-x00/-x01 indicates amount of memory, as follows: -20x = 2mw; -40x = 4mw, etc.			
Note 13	Must be Rev. B to work in U ^X S node.			
Note 14	To support the Crossed LCN Cables display (\$LNACCT), the K2LCN in the Master Node (US with lowest address) must have LCN Firmware at Rev. P or greater. Crossed LCN Cables display is not supported by 51401551-x00 board.			
Note 15	The hardware revision on the K2LCN SMT assembly is contained in the FLASH PROM 51120908, and the revision of this device is always the same as the Kernel Hardware revision. The LCN firmware and Memory Controller gate array program contents are also contained in this device.			
Note 16	The minimum required Firmware revision for K4LCN Kennel Firmware is really B but SMCC and LVRLOG will display it as A.			
Note 17	Nodes with K2LCN or K4LCN processors require Rev. T or later LCN I/O board (51305072-400 I/O Card, CLCN A/B, non-EC).			
Note 17A	KLCN A or B boards cannot be mixed with CLCN A or B boards in the same dual node chassis.			

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)**LCN NODE -
Hardware/firmware
revision list, continued**The symbol Δ indicates there is a change from R4xx.

	Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
	PWA, LCNE II, non-CE	51304540-100	A	
	LCNE2 (replaces LCNE)	51304540-100	C	
	LCNFL I/O	51108899-100	A (Note 18)	
	PWA, Enhanced LCN	51401291-100	F	C
	LCNI2		A	A
	LCNI4		A	A
Δ See note	I/O Card, CLCN A/B, non-EC	51305072-400	L (Note 19)	

3. Hardware Preparation

3.3. LCN Node Hardware/Firmware Requirements

	Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
	LCNP (GUS)	51402797-100	Software Change Notice (SCN)	
	LCNP Board	51305348-100		
	MAU - LCN	51120915-101		
	915-101 NVRAM	51120916-100		
	916 OTP			
	LCNP4 GPS Assy w/o Card Guide	51403299-100		
	LCNP4M	51403776-100	F	F
	LLCN	51401291-100	H	C
	5S LOGIC PS	51400712-100	E	
	Pwr Supply, 5 Slot	51201436-400	E	

Note 18 Must be Rev. F or later to work with the K2LCN or K4LCN board.

Note 19 Must be Rev. T or later to work with the K2LCN or K4LCN board.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

3.3. LCN Node Hardware/Firmware Requirements**LCN NODE -
Hardware/firmware
revision list, continued**

The symbol Δ indicates there is a change from R4xx.



New Bd. Rev.



New Bd. Rev.



New Bd. Rev.



New Bd. Rev.

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
1/4 M MEM	51400659-100	D	
3/4 M MEM	80360211-200	C	
PWA, QMEM 2MW	51401072-200	E	
1 M MEM	80360211-100	C	
2 M PMEM	51400903-100	A	
2 M QMEM	51401072-200	A	
3 M QMEM	51401072-300	A	
4 M QMEM	51401072-400	A	
NETWORK GATEWAY – NGI	51401583-200	B	B (Note 20)
- NGIO	51304472-100	A (Note 21)	
- NG FOM	51306959-200	B (Note 21A)	
PWA, NIM MODEM	51305896-100	C	

Note 20 Must be Rev. C or later if using fiber optic modems or carrier band modem communications.

Note 21 The fiber optic modem from CD Networks must be Rev. 4 (silk screened on bd.). If fiber optic modems are used, the I/O card from the CD Networks™ with the date codes 9304-9407 and the CD Network modem must be upgraded (part number CD-2005A).

Note 21A The Fiber optic modem (HPN-38002188-100) was manufactured by CD Networks. This part is now replaced and supplied by Honeywell (HPN-51306959-200).

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

3.3. LCN Node Hardware/Firmware Requirements

**LCN NODE -
Hardware/firmware
revision list, continued**

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision (*)
OP KEYBD	51400771-100	C	C
PWA, KEYBOARD INTERFACE	51401570-100	B	A
PWA, Enh PDG	51401286-100	B	B (Note 22)
PDG RS170 I/O	51109394-100	A	
PDG TTL I/O	51195058-100	A	
PIC	51400666-100	H	H
PIC I/O	51108020-100	A	
PLC I/O	51195096-100	A	
I/O Card, PLCG Relay, EC	51304421-200	A (Note 23)	
PLCI (Gateway)	51400997-100	B	J
PNI	51400955-100	A	E (8.1)
PNI I/O	51303944-100	A	
Replacement Kit, NIM Modem	51195760-100	A	
PNM I/O	51303947-100	A	
PS CONTROL	51303294-100	C	
QWERTY Keyboard	51401560-100	B	

(*) First value is the label rev. Value in parentheses is the displayed rev.
(as applicable)

Note 22 If the PDG is in a US that has a printer, the firmware must be Rev. F or later.

Note 23 Can be used in all applications, **including** Allen-Bradley with redundant PLC protocol.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

**LCN NODE -
Hardware/firmware
revision list, continued**

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
SIO	51400855-100	B	
SIO I/O	51108843-100	B	
SPC (Smart Periph Ctlr)	51401052-100	B	B (Note 25)
SPC I/O	51304156-100	A	
SPC3 I/O	51305088-100	A	N/A
SPCII I/O	51304907-100	A	N/A
TP485	51304776-100	A	N/A
TP485	51304776-200	B	N/A
TP485-3	51304776-300	D	N/A
TP485-4	51304776-400	D	N/A
TPDG 60 Hz (U ^X S)	51402000-100	A	A (Note 26)
TPDG 72 Hz (U ^X S)	51402000-200	A	A (Note 26)
TPDG I/O (U ^X S)	51304831-100	B	
TPDGX	51402610-200 51402645-200	R S	B B (Note 26)
TRACKBALL CONSOLE	51401748-100	A	
TREND TERM SIO	51108845-100	B	
TREND TERM SIO/EXT	51108845-200	B	

Note 25 If new Seagate disk drive is to be used, the firmware must be Rev. G or later. If the new Dual Logical drives are to be used, the firmware must be Rev J or later. (Rev H will work but performance can be degraded.)

Note 26 For Zip drive support, firmware must be at a minimum Rev. G.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

3.3. LCN Node Hardware/Firmware Requirements

**LCN NODE -
Hardware/firmware
revision list, continued**

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
U ^X S Operator Keyboard	51401570-100	A	D
VDG	51400665-100	D	
VDG I/O	51107756-100	B	
WDI I/O	51304903-100	A	N/A
WSI2 Assembly (U ^X S, A ^X M)	51402083-xxx	A (Note 27)	
WSI2 Circuit Board (U ^X S, A ^X M)	51402081-100	A (Note 28)	
WSI2 I/O (U ^X S, A ^X M)	51304924-200		
WSI2R I/O	51304947-100	A	N/A

Note 27 The WSI2 Assembly (51402083-xxx) hardware revision is not visible through SMCC. The revision letter is located next to the right extractor lever.

Note 28 The WSI2 Circuit Board (51402081-100) hardware revision is read through SMCC and must be Revision "D" for nodes with K2LCN and K4LCN boards. Any rev of the WSI2 circuit board can be made revision "D" by removing jumpers. Jumper TS1 needs jumpers 1, 2, and 8 removed. Removal of 1 and 2 changes the hardware revision to "D," and removal of jumper 8 sets the board to read and write 16 bits. Setting the board to read and write 16 bits is the change associated with revision "D." This was documented in a CPN dated November 19, 1995.

Table 3-1 LCN Node Hardware/Firmware Requirements (continued)

3.4 LCN Node Memory Requirements

LCN node memory requirements

The HMs, NIMs, Universal Stations with Operator personalities, and most AMs, require an additional 1 Mw of memory over what was needed for R4xx. Universal Stations with Universal personalities require an additional 2 Mw of memory. This is true since LCN nodes, once they are loaded, essentially run from their own memory. Therefore, an increase in software size translates directly to an increase in physical memory requirements.

The K2LCN is available in 2, 4, 6, and 8 Mw versions, and the K4LCN is available in 4, 8, and 16 Mw sizes.

Larger memory sizes (such as 8 Mw) are available for the Universal Stations to accommodate, for example, more memory resident schematics.

The following table contains the minimum memory size requirements for LCN nodes.

LCN Node	R6xx Minimum Memory Size	Changed from R4xx (√)
Application Module (AM), Application Module Redundancy (AMR), and CLM	3.0 Mw	√ (+1 Mw) (Note 1)
Application Module ^X (A ^X M)	6.0 Mw	√ (+2 Mw)
Computer Gateway (CG) (68020 processor)	2.0 Mw	
Enhanced Programmable Logic Controller Gateway (EPLCG)	2.0 Mw	
GUS	8 Mw	
History Module (HM)	3.0 Mw	√ (+1 Mw)
Hiway Gateway (HG)	2.0 Mw	(Note 2)
Network Gateway (NG)	2.0 Mw	
Network Interface Module (NIM)	3.0 Mw	√ (+1 Mw) (Note 2)
Plant Network (PLNM)	2.0 Mw	
Programmable Logic Controller Gateway (PLCG)	2.0 Mw	
Universal Station (US) or UWS - Opr	4.0 Mw	√ (+1 Mw)
Universal Station (US) or UWS - UNP	6.0 Mw	√ (+2 Mw)

Table 3-2 Minimum Memory Size Requirements for LCN Nodes

3.4. LCN Node Memory Requirements

LCN Node		R6xx Minimum Memory Size	Changed from R4xx (√)
Universal Station ^X (U ^X S)		8.0 Mw	
Note 1	On all AMs, AMRs, and CLMs, add an additional 1 Mw of memory over R4xx levels.		
Note 2	HGs and NIMs, as data owners must have K4LCN processor boards if any other nodes have K4LCN processor boards or LCNP4 based boards.		

Table 3-2 Minimum Memory Size Requirements for LCN Nodes (continued)**Available User Memory****NOTE:**

The A^XM available user memory for the: 6 Mw A^XM corresponds to the 3 Mw AM; 7 Mw A^XM corresponds to the 4 Mw AM; 8 Mw A^XM corresponds to the 5 Mw AM; and 16 Mw A^XM is 11,058,224 words.

The following table shows how the AM memory available to the user is related to the total AM memory size.

Total AM Memory Size	Available AM User Memory (R500)
3.0 Mw	1,016,358 words
4.0 Mw	2,020,545 words (2,009,435 for K4LCN)
5.0 Mw	3,024,731 words
6.0 Mw	4,028,918 words
7.0 Mw (HPK2/HMPU maximum)	5,033,105 words
7.75 Mw (K2LCN maximum)	5,786,245 words
8.0 Mw	6,037,291 words (6,020,211 for K4LCN)
16.0 Mw	14,070,784 words (14,045,430 for K4LCN)

Table 3-3 Available AM User Memory

3.5 Memory Board/Processor Type Compatibility

Compatibility table

Certain memory boards and processor types are not compatible with each other. The following table describes the memory board types and the processor types, and whether or not they are compatible.

Is the memory board below compatible with this processor? →	HMPU (Note 1)	HPK2-2 & HPK2-3 (Note 2)	K2LCN (Note 3)	K4LCN (Note 3)
MEM - 1/4 Mw (51400659-100)	Yes (Note 4)	Yes (Note 4)	No	No
MMEM - 3/4 Mw (80360211-200)	Yes (Notes 4, 5)	Yes (Notes 4, 5)	No	No
MMEM - 1 Mw (80360211-100)	Yes (Notes 4, 5)	Yes (Notes 4, 5)	No	No
EMEM - 1 Mw (51400910-100)	Yes (Notes 6, 7)	Yes (Notes 6, 7)	No	No
PMEM - 2 Mw (51400903-100) (Notes 8 & 9)	Yes	Yes (Note 4)	No	No
QMEM (51401072-x00) (Notes 10, 11 & 12)	Yes (Note 6)	Yes (Note 6)	No	No

3.5. Memory Board/Processor Type Compatibility

All memory size is in 16-bit words.

The HMPU board contains 2 Mw of on-board memory. The HPK2-2 board contains 2 Mw of on-board memory. The HPK2-3 board contains 3 Mw of on-board memory.

Because of limited space, only HMPU and HPK2 can be used in Multinode Modules.

- Note 1 HMPU, plus external memory, cannot exceed 7 Mw.
- Note 2 The HPK2-2 and HPK2-3 boards must be Rev. G or later to work with external memory boards.
- Note 3 Cannot have any external memory.
- Note 4 Cannot be mixed with EMEM or QMEM.
- Note 5 MMEMs must be at Rev. C.
- Note 6 Cannot be mixed with MEM, MMEM, or PMEM.
- Note 7 EMEMs must be at Rev. E.
- Note 8 Parity only memory, no EDAC functionality. CAUTION: **PMEMs are NOT to be used in any node that is used for control.**
- Note 9 PMEMs shouldn't be mixed with any other memory.
- Note 10 QMEM Tab -200 = 2 Mw, -300 = 3 Mw, etc. (2 Mw is the minimum size)
- Note 11 When combining QMEM and other memory boards, QMEM must be in lowest slot.
- Note 12 Existing memory boards may be utilized when upgrading a node to R530 memory requirements. However, when making new purchases, it is now more cost-effective to purchase R530-ready processors, instead of additional memory.

Table 3-4 Memory Board and Processor Type Compatibility

3.6 PM/APM/HPM Hardware/Firmware Requirements

PM NODE - Hdwr/ Fmwr revision list

Board Name	Part Number	Minimum Board Revision (*)	Minimum Firmware Revision (*)
EXCHPWA, PM COMM R300 CC	51304685-250	A (00)	A (3.1)
PMM Comm	51304685-200	D (03)	A (3.1)
PMM CTRL Daughter Bd.	51304045-100	A (00)	
PMM CTRL	51303982-400 51304518-200	A (00) K (2A) (Note b)	A (3.1) B (3.2)
PWA, R300 CONTROLLER CC	51304518-250	A (00)	A (3.1)
PMM CTRL R230 w/o redundancy	51303982-800	A (00)	A (3.1)
PMM I/O Link I/F	51303979-400	A (00)	A (3.2)
PMM I/O Link I/F (conformally coated)	51303979-450	U (04)	A (3.2)
PMM MODEM	51304163-300	A (00)	
PMM MODEM (surface mount)	51304493-100	A (20) (Note a)	
PMM Redundancy Drvr	51304501-100	A (00)	

(*) First value is the label rev. Value in parentheses is the displayed rev. (as applicable)

Note a The integrated PM Modem, 51304493-100, is a direct replacement for modem 51304163-300. Both modems are release independent, functionally equivalent, and interchangeable. The 51304493-100 modem is used in new PM/APM production and for spares.

Note b The PMM CTRL, 51304518-200, is a direct replacement for 51303982-400 and is only used in PM production. The APMM CTRL 51304518-100 is only used in APM production.

Table 3-5 PM Node Hardware/Firmware Requirements

3.6. PM/APM/HPM Hardware/Firmware Requirements**APM NODE – Hdwr /
Firmware revision
list**

Board Name	Part Number	Minimum Board Revision (*)	Minimum Firmware Revision (*)
APMM COM	51304685-100	D (23)	D(4.2)
APMM CTL	51304518-100	C (22)	C (4.3)
APMM I/O LINK I/F	51303979-500	N (0D)	B (4.1)
APMM Filler Card	51304190-200	A (00)	
APMM MODEM	51304163-300	A (00)	
APMM MODEM (surface mount)	51304493-100	A (20) (Note a)	

(*) First value is the label rev. Value in parentheses is the displayed rev. (as applicable)

Note a The integrated PM Modem, 51304493-100, is a direct replacement for modem 51304163-300. Both modems are release independent, functionally equivalent, and interchangeable. The 51304493-100 modem is used in new PM/APM production and for spares.

Table 3-6 APM Node Hardware/Firmware Requirements**HPM NODE - Hdwr/
Fmwr revision list**

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
PWA, HPM COMM CONTROL (conformally coated)	51403988-150	C	Comm F/W = C
PWA, HPM I/O LINK PWA, HPM I/O LINK (conformally coated)	51309276-150 51309276-150	C C	
HPM UCN Interface	51402573-100 (obsolete)	A	
UCNIF Module Assembly (conformally coated)	51402573-250	E	

Note The HPM COMM firmware gets upgraded automatically upon software download if the embedded firmware in the board is not the latest revision. Therefore, no other action is required to upgrade firmware.

Table 3-7 HPM Node Hardware/Firmware Requirements

3.6. PM/APM/HPM Hardware/Firmware Requirements**EHPM NODE - Hdwr/
Fmwr revision list**

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision
EHPM Communication/Control Card	51454192-150	F	C
PWA, HPM I/O LINK (conformally coated)	51309276-150	C	B/A
FTE Interface Module Assembly	51454307-100	A	A
100-Base-T Crossover cable with white boots, 2Mtr Length For FTE Redundancy	51305786-502		
Control Firewall Ethernet Switch (8 ports + 1 uplink)	51405047-175	(Note A)	
Control Firewall IOTA	51308301-125	(Note A)	NA
Carrier Adaptor Assembly	51454413-100	(Note A)	NA
DC power cable	51307006-103	(Note A)	NA
Industrial 8-Port Ethernet switch	NE-SW508S	(Note B)	NA
8-port Copper Expansion Module	NE-SW58TM	(Note B)	NA
8-port Fiber Expansion Module	NE-SW58FM	(Note B)	NA
AC Power Adapter	NE-SW51P1	(Note B)	NA
AC Power Cord	51307101-200	Note B	NA
Fiber Optic AC Power Cord (120V)	51305409-500	Note B	NA
Fiber Optic AC Power Cord (240V)	51305410-500	Note B	NA

Note The HPM COMM firmware gets upgraded automatically upon software download if the embedded firmware in the board is not the latest revision. Therefore, no other action is required to upgrade firmware.

For more information on EHPM node refer to HPM to EHPM upgrade kit Instructions

Note A The eight-port, Control Firewall (CF9) serves as the Fault Tolerant Ethernet (FTE) communication center for a given EHPM network. Supported from TPN R684 and above. For more details, refer to the CF9_Switch_UpgradeKit Instructions

Note B Supported from TPN R684 and above. For more details, refer to the FTE_Switch_Usedwith_LCNCabinet_Upgrade Instructions

Table 3-7a EHPM Node Hardware/Firmware Requirements

3.7 IOP Hardware/Firmware Requirements

IOP - Hardware/ firmware revision list

The symbol Δ indicates there is a change from R4xx.

	Board Name	Part Number	Min. Board Revision (*)	Min. Firmware Revision (*)
	IOP, A/O CC	51309152-175	A (20)	D (3.4) (Note c)
	AO ₂ IOP	51304672-100	C (52)	D(3.4)
	IOP, A/O CC	51309152-175	A (00)	D (3.4) (Note c)
		51309152-175	A (50)	D (3.4) (Note c)
Δ New IOP	AO IOP (16)	80363969-100 & -150	G (56)	C (1.2)
Δ	IOP, DI, CC	51304485-150	A (00)	C (3.3) (Note d)
	IOP, DI, CC	51304485-150	A (00)	C (3.3)
	IOP, Redundant DISOE2, CC	51402625-175	B (51)	F (4.4)
Δ New IOP	IOP, Redundant DISOE2, CC	51402625-175	A (70)	G (4.5)
Δ New IOP	DI IOP, 24VDC (32)	80363972-100 & -150	G (56) (Note f)	C (1.2)
	DO IOP	51304487-100	A (00)	F (3.6)
	IOP, D/O	51304487-100	A (00)	F (3.6)
Δ New IOP	DO IOP (32)	80363975-100 & -150	H (57) (Note g)	D (1.3)
	IOP, HLAI CC	51304754-150	A (20)	B (3.3)
	IOP, HLAI CC	51304754-150	E (54)	D (5.0) (Note e)
	IOP, HLAI CC	51304754-150	A (00)	B (3.3)

(*) The alpha character is the paper label revision found on the board. The numeric value is the equivalent revision displayed on the IOP "Detail Status" screen

Note c AO Rev. E (3.9) supports 0 mA operation.

Note d DI Rev. D (5.0) firmware is required for DI change of state alarming

Note e HLAI2 IOPs in a redundant configuration must have a minimum of Rev. D (version 5.0).

Note f For Redundant DI IOP (32) operation, minimum board revision is G.

Table 3-8 IOP Hardware/Firmware Requirements

**IOP - Hardware/
firmware revision
list, continued**

The symbol Δ indicates there is a change from R4xx.

Δ
See note

Δ
See note

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision (*)
IOP, LLAI, CC	51304481-150	A (20)	D (3.4) (Note h, i, j, k)
IOP, LLAI, CC	51304481-150	A (00)	D (3.4) (Note h, i, j, k)
IOP, LLAI MUX CC	51304362-150	A (00)	D (3.7)
IOP, Pulse Input, CC	51304386-150	A (20)	B (3.3)
SDI IOP	51304362-200	F (45)	B (4.1)
IOP, Serial I/F, CC	51304362-300	F (05)	F (4.5)

(*) First value is the label rev. Value in parentheses is the displayed rev. (as applicable)

Note g For Redundant DO IOP (32) operation, minimum board revision is H.

Note h The IOP LLAI firmware revision level is shown as D (3.4). This revision is fine if you are using LLAI points with the parameter of PVTEMP = RANKINE. Then you should use the EPROM Rev. E (3.9). Be aware that the checkpoint for those points may not restore with the R4xx/R500/R510 software installed. The LLAI points will have to be rebuilt using the IDFs or the .EB files. No changes to the points are required.

Note i Customers running on 50 Hz MUST use Rev. F 3.7 or later.

Note j LLAI Rev G (5.0) is required for RTD open wire softfail alarming.

Note k A minimum of rev. H firmware is required when used in conjunction with LLAI FTA 51309209-XXX.

Table 3-8 IOP Hardware/Firmware Requirements (continued)

3.7. IOP Hardware/Firmware Requirements

**IOP - Hardware/
firmware revision
list, continued**

Board Name	Part Number	Minimum Board Revision	Minimum Firmware Revision (*)
IOP, STI/MV, CC	51304516-250	A (20)	D (3.8) (Note l)
STI IOP R230	51304281-200	A (00)	C (3.6) (Note l)
IOP, STI/MV, CC	51304516-250	A (20) (Note m)	B (5.1)

Note l Rev. D or greater is required for IOP, STI/MV, CC Redundancy. Rev. C is allowed only in R4xx or R5xx systems with nonredundant STI IOPs.

Note m IOP, STI/MV, CC can be used as a replacement for nonredundant STI IOPs, but cannot be used in redundant configurations with STI IOPs.

Table 3-8 IOP Hardware/Firmware Requirements (continued)**Recommended
actions**

Before loading R6xx, check the hardware and firmware revisions of all your High Density IOPs on site; including your spares stock. The best way to check IOP hardware revisions online is to use the “IOPMDATA” menu from the “PERFMENU” screen or individual IOP “DETAIL STATUS” screens.

If any of your High Density IOPs do not meet the minimum revisions they should be retrofitted per Retrofit Announcement Letter (RAL) #96-04 distributed in June 1996.

If you have any questions concerning this recommendation or need replacement boards under retrofit 96-04, please contact your local Honeywell representative, Technical Assistance Center (TAC), or Customer Response Center (CRC).

3.8 FTA Hardware/Firmware Requirements

FTA - Hardware/ firmware revision list

Board Name	Part Number	Min. Board Revision	Min. Firmware Revision
FTA, A/O, Comp Term, CC	51304476-175	A	
FTA, Redundant A/O, Comp Term, CC	51304335-175	A	
AO FTA R230	51303953-100	A	
FTA, HD A/O W/STANDBY COMP TERM CE CC	51204172-175	A	
FTA, A/O, Screw Term	80364007-200	A	
HD FTA, A/O Redn, Comp Term, CC	80366177-250	A	
AO FTA Red (16) Screw without Stby/Man	80364007-200	A	
FTA, 24VDC D/I, Comp Term, CC	51304441-175	A	
DI FTA, 24 Vdc SS Iso R210	51401201-100	A	
FTA, 120VAC D/I, Comp Term, CC	51304439-175	A	
DI FTA, 120 Vac SS Iso R210	51401186-100	A	
DI FTA, 240 Vac SS Iso	51304219-100	A	
Relay, Dual Input, EC	51190523-125	A	
FTA, D/I 24VDC, Comp Term, CC, CE	51204160-175	A	
FTA, D/I 24VDC SCREW TERM CE CC	51204154-175	A	
FTA, 240VAC D/I, Comp Term, CC	51304431-175	A	
FTA, 3/30VDC D/O, Comp Term, CC	51304650-150	A	
DO FTA 3-30 Vdc SS R210	51401220-100	A	
FTA, 3/30VDC D/O, Comp Term, CC	51304650-150	A	
FTA, 31/200VDC D/O, Comp Term, CC, CE	51309154-175	A	
DO FTA 31-200 Vdc SS R210	51401223-100	A	
FTA, 24VDC Non-Isol D/O, Comp Term, CC	51304446-150	A	
DO FTA 24 Vdc 100MA non-Iso, R210	51303959-100	A	
FTA, 120/240VAC SS D/O, Comp Term, CC, CE	51304648-175	A	

Table 3-9 FTA Hardware/Firmware Requirements

3.8. FTA Hardware/Firmware Requirements**FTA - Hardware/ firmware
revision list, continued**

Board Name	Part Number	Min. Board Revision	Min. Firmware Revision
FTA, 120/240VAC SS D/O, Comp Term, CC, CE	51304648-175	A	
DO FTA 120-240 Vac SS R210	51401157-100	A	
FTA, DO RELAY 120V COMP TERM CEE	51309148-125	A	
DO FTA 120 Vac/125 VdcR210	51401191-100	A	
FTA, D/O 24VDC COMP TERM CC CE	51204162-175	A	
FTA, D/O 24VDC, Screw Term, CE, CC	51204156-175	A	
FTA, HD D/O RELAY COMP TERM CE CC	51204166-175	A	
HD FTA, D/O Relay, Screw Term, CE, CC	51204164-175	A	
FTA, 240V D/O Relay, Comp Term, CC, CE	51309150-175	A	
DO FTA 240 Vac/125 VdcR210	51401194-100	A	
I/O Link Module, Triple, CC	51201557-350	A	
PWA, IOP FIBER OPTIC CC	51304419-150	A	
HLAI FTA R230	51303956-100	A	
FTA, HLA/STI, Comp Term, CC	51304453-150	A	
HLAI/STI FTA Red w/compression	51304337-150	A	
FTA, Redundant HLA/STI, Comp Term, CC	51304337-250	A	
HD FTA, HLA/STI, Comp Term	51204170-200	C	
HD FTA, HLA/STI, Screw Term	51204168-100	E	
FTA, LLAI, Comp Term, CC	51304437-150	A	
LLAI FTA R230	51401107-100	A (Note o)	
LLAI FTA R230	51401183-100	A (Note o)	

Revision level designations in hardware and firmware correspond as follows:

A = 0, B = 1, C = 2, etc.

Note n The associated LLAI IOP must have a minimum of rev. H firmware.

Note o 51401183 LLAI FTA cable is limited to 20 meters from the IOP. Use 51401107 for distances up to 50 meters.

Table 3-9 FTA Hardware/Firmware Requirements (continued)

**FTA - Hardware/ firmware
revision list, continued**

Board Name	Part Number	Min. Board Revision	Min. Firmware Revision
FTA, LLA1 MUX/SDI/SI Pwr Adapter	51309204-125	A	
FTA, LLA1 MUX TC REM CJR COMP TERM CE	51309213-125	B	
TC FTA w/Remote CJR (daughter bd.)	51401564-100	A	B (Note p)
LLMUX Remote Sensor Assy, CC	51304589-150	A	
FTA, LLMUX TC, Comp Term, EC	51309223-125	C	
TC FTA w/Local CJR (daughter bd.)	51401564-100	A	B (Note p)
FTA, LLMUX2 RTD CE CC	51305907-175	A	
FTA, LLMUX2 RTD CE CC	51401376-100	A	
FTA, LLMUX2 RTD CE CC	51305907-175	A	A
FTA, Pulse input, Comp Term, CC	51304084-175	A	
FTA, Serial Device I/F Toledo Weigh Cell	51303932-226	B	A (Note q)
FTA, Serial Device I/F Manual Auto Station	51303932-227	B	A
FTA, SERIAL IF ALLEN BRADLEY	51303932-428	B	B (Note r)
FTA, SERIAL IF MODBUS RS-232	51303932-426	B	D (Note s)
FTA, Serial Device I/F UDC 6000	51303932-228	F	B

Revision level designations in hardware and firmware correspond as follows:
A = 0, B = 1, C = 2, etc.

- Note p The next valid revision is D. Do not use Rev. C.
Note q SDI T8142 Rev. A does not support negative setpoint operation.
Note r Rev. C firmware is recommended.
Note s Rev. E firmware is recommended.

Table 3-9 FTA Hardware/Firmware Requirements (continued)

3.9 Logic Manager Hardware/Firmware Requirements

Valid Logic Manager Module revisions The following tables show the components of each module in the Logic Manager, and their respective allowable board and firmware revisions.

Component/Module Name (Processor Rack Modules)	Model Number	Valid Board Revisions	Valid Firmware Revisions
Logic Mgr. Module	620-0071	HR05, 06	2.5
Redundancy Ctl. Module	620-0059	K, L, M, N, P	2.2
24 k Memory Module	620-0024	F1, H1, J1, L1	
4 k x 4 k Register Module	620-0056	A, B, C, D, E, F	
System Ctl. Module	620-0054	A2, Bn-Gn	1.0
Processor Module	620-0080	M, N, S, T, U	2.6, 3.1 (See Note 1)
I/O Control Module	620-0085	A1, B, C, D, E, F	
115/230 Vac Pwr. Supply	620-0036	K2, R2, X1, X5, Y	
24 Vdc Pwr. Supply Module	620-0047	G	
Serial Link Module	621-9939	H, J, K, M, N	2.2, 2.3, 2.4 (See Note 1)
Parallel Link Driver Module	620-0088	A1, B1, C, D	

Note 1 Latest firmware revision (Rev. 3.1 for the Processor Module or Rev. 2.4 for the Serial Link Module) incorporates Expanded Diagnostics.

Table 3-10 Logic Manager: Board and Firmware Revisions

3.9. Logic Manager Hardware/Firmware Requirements**Valid Logic Manager
Module revisions,**
continued

Component/Module Name (I/O Rack Modules)	Model Number	Valid Board Revisions	Valid Firmware Revisions
Enhanced Diagnostic Module	621-0021	A1, D, E	
Dual-Port Serial I/O Module	621-9938	L1, M, N1, P, R (See Note 2)	3.2
Dual-Port Serial I/O Module	621-9938R	A1, B1, C	3.2
Single-Port Serial I/O Module	621-9940	H1, G1, J1, K, L (See Note 3)	3.2
Serial Link Selector Module	621-9928R	A	
Serial Link Selector Module	621-9928	A, B	
Parallel Bus Selector Module	621-9929	C1, E1, F1, G	
24 Vdc I/O Power Supply	621-9932	E2, G2, H1, L	
24 Vdc I/O Power Supply	620-0046	G3, L, M, N, P, R, S, T	
115/230 Vac I/O Pwr 15 A	621-9933	N1, N2, R, S, T, U, V, W	
115/230 Vac I/O Pwr 8 A	620-0034	M1, N1, P1	
115/230 Vac I/O Pwr 8 A	620-0041	S, T, U, V, W, X, BB, CC	
Parallel I/O Module	621-9937	C2, D, E, F, G, J	
Control Network Module	620-0038	B9, C7, E9, F6, L3, R, S, T, U	1.0
Comm Interface Module	620-0043	G, H, J, K, L	1.1
Data Collection Module	620-0048	R, T, S, U, V	3.1
DMCS Comm. I/F Module	620-0044	F4, J1, M	2.0
Dual-Port Data Coll. Module	620-0052	E	2.0

Note 2 Recommended: L1, P

Note 3 Recommended: G1, K

Table 3-10 Logic Manager: Board and Firmware Revisions (continued)

3.10 Safety Manager-Tricon Hdwr/Fmwr Requirements

Lowest acceptable level for Tricon V8.2 modules

A Version 8.2 system is required to have revision levels equal to or later than the lowest acceptable levels shown in the tables below.

If your system has equipment that is at a lower revision level than the lowest acceptable level for that item, please contact Triconex Customer Satisfaction to arrange for an appropriate upgrade.

Module Type	Model Number	Hardware Revision	Firmware Label	Firmware Meta Number
MP	3004	B8 or C1	TSX	4505C
			IOC	4224
			COM	4556
MP	3005	C1	TSX	4505C
			IOC	4224
			COM	4556
EICM	4108	B7	ICX	4528
			ICM	4289
			VRTX	C3.06
EICM	4118	B1	IICX	4530
			ICM	4289
DCM	4318	A1	DCMX	4500
			ICM	4289
NCM	4328	A1	NCMX	4501
			ICM	4289
HIM	4508	B8	HCX	4716
			ICM	4289
SMM	4400	B04	SMMX	4508
			SICM	4684

Table 3-11 Version 8.2–Lowest Acceptable Level for Tricon Modules

3.10. Safety Manager-Tricon Hdwr/Fmwr Requirements**Tricon V8.2, lowest acceptable level equipment**

The following table describes the lowest acceptable level of equipment for Version 8.2 of Tricon.

Module Type	Model Number	Hardware Revision	Firmware Label	Firmware Meta Number
AI	3700	A6 or B	AI/NITC	3630
AI	3701	A6 or B	AI/NITC	3630
AIO	3800	C10	AIO	3803
AIO	3801	C8	AIO	3803
DI	3501	B2	DI	3287
DI	3502	B2	DI	3287
DI	3503	B2	DI	3287
EDI	3501E	C3	EDI	4299
EDI	3502E	C3	EDI	4299
EDI	3503E	C3	EDI	4299
EDI	3505E	C3	EDI	4299
EDI	3531E	C3	EDI	4299
EDI	3532E	C3	EDI	4299
EDI	3533E	C3	EDI	4299
EIAI	3703E	D4	EIAI/ITC	3429
EITC	3708E	D4	EIAI/ITC	3429
HDAI	3704E	A9 or B3	HDI	4843
HDDI	3504E	A9 or B3	HDI	4843
NITC	3706	A3 or B	AI/NITC	3630
PI	3510	B1	PI	4559

Table 3-12 Version 8.2 Tricon–Lowest Acceptable Level Equipment

**Tricon V8.2, lowest
acceptable level
equipment, continued**

Module Type	Model Number	Hardware Revision	Firmware Label	Firmware Meta Number
DO	3601	A10 or B	DO	3570
DO	3603	A6	DO	3570
DO	3603A	A9	DO	3570
DO	3603B	A12	DO	3570
DO	3604	A10 or C	DO	3570
DO	3604A	A13 or C	DO	3570
DO	3607	A10 or C	DO	3570
DO	3607A	A13	DO	3570
DO	3608	A10 or B	DO	3570

Table 3-12 Version 8.2 Tricon—Lowest Acceptable Level Equipment
(continued)

3.10. Safety Manager-Tricon Hdwr/Fmwr Requirements

**Tricon V8.2, lowest
acceptable level
equipment, continued**

Module Type	Model Number	Hardware Revision	Firmware Label	Firmware Meta Number
EAO	3805E	B6 or C1	EAO	4532
EDO	3601E	B1	EDO	4621
EDO	3604E	B1	EDO	4621
EDO	3607E	B1	EDO	4621
EDO	3608E	B1	EDO	4621
ERO	3635E	A	ERO	3568
ERO	3635R	A	ERO	3568
ERO	3636E	A or B1	ERO	3568
ERO	3636R	A or B1	ERO	3568
SDO	3611E	C5	SDO	3566
ESDO	3613E	B4 or C	ESDO	3565
ESDO	3614E	B4 or C	ESDO	3565
ESDO	3615E	B4 or C	ESDO	3565
ESDO	3617E	B4 or C	ESDO	3565
RXM	4200	A11 or B2	RXM	3310
RXM	4201	A11 or B2	RXM	3310

Table 3-12 Version 8.2 Tricon–Lowest Acceptable Level Equipment
(continued)

3.10. Safety Manager-Tricon Hdw/Fmwr Requirements

Version 8.2 Release Application software list Software products related to Version 8.2 release of Tricon can be found in the following table.

Version Number	Software Package	Meta Number
MSW 2.0.1	TRISTATION Multi-System Workstation	4502
	Moddef (included with MSW 2.0.1)	4518
	TSTools (included with MSW 2.0.1)	4404
	TSAccess (included with MSW 2.0.1)	4403
2.0	Sequence of Events Retrieval Application (SOE)	4442
1.1	Tricon Dynamic Data Exchange Server (DDE)	n/a
2.0	DOS 802.3 Driver Interface for TCP/UDP/IP	n/a
2.0.2	Sequence of Events Recorder (SER)	3449
2.0	Remote Diagnostic Application (RDS)	4399

Table 3-13 Software Products Related to Tricon Version 8.2

3.11 FSC Safety Manager Hrdwr/Fmwr Requirements

Overview This section identifies all the FSC hardware, software, and firmware requirements for R510.

FSC System Hardware Module List The following table shows the components of each Hardware Module and their minimum hardware and firmware revisions.

MODULE NAME	Model No.	Min. Hardware Revision	Min. Firmware Revision	Notes
Vertical Bus Driver (VBD) to control I/O in I/O rack(s)	10001/R/1	00400	na	
Processor module (CPU)	10002/1/2	08400	4.20	
Memory module (MEM)	10003/1/1	00600	na	3
Communication module (COM)	10004/1/1	00700	4.20	
Watchdog (WD) + WD-HBus print	10005/1/1	01300	na	
Diagnostic and Battery Module (DBM)	10006/2/1	08800	na	
Single Bus Driver (SBD) to control I/O modules in Central Part rack	10007/1/1	01500	na	
FSC Safety Manager Module	10008/2/U	14400	2.00	
Horizontal Bus Driver for FSC-1O1 and FSC-1O2	10100/A1	01600	na	
Horizontal Bus Driver for FSC-1O1R (controlling 1 redundant I/O rack)	10100/A21	04700	na	
Horizontal Bus Driver for FSC-1O1R (controlling 2 redundant I/O racks)	10100/A22	04800	na	

Note 3 Hardware module only available for spare parts and upgrades of installed FSC systems.

Table 3-14 FSC System Hardware Modules

3.11. FSC Safety Manager Hrdwr/Fmwr Requirements
**FSC System
Hardware Module
List, continued**

MODULE NAME	Model No.	Min. Hardware Revision	Min. Firmware Revision	Notes
Horizontal bus driver	10100/2/1	10900	na	2
16-channel Digital Input module Fail-Safe 24 Vdc - 7 mA	10101/1/1	01700	na	3
16-channel Digital Input module Fail-Safe 60 Vdc - 3.3 mA	10101/1/2	07200	na	3
16-channel Digital Input module Fail-Safe 48 Vdc - 4 mA	10101/1/3	07300	na	3
16-channel Digital Input module Fail-Safe 24 Vdc - 7 mA	10101/2/1	11000	na	2
16-channel Digital Input module Fail-Safe 60 Vdc - 3.3 mA	10101/2/2	11100	na	2
16-channel Digital Input module Fail-Safe 48 Vdc - 4 mA	10101/2/3	11200	na	2
4-channel Analog Input module Fail-Safe 0(4)-20 mA / 0(2)-10 V	10102/1/1	01800	na	3
4-channel Analog Input module Fail-Safe 0(1)-5 V / 0(2)-10 V	10102/1/2	01900	na	3
4-channel Analog Input module Fail-Safe 0(1)-5 V / 0(2)-10 V	10102/2/1	11300	na	2
16-channel Digital Input module Not Fail-Safe 24 Vdc - 7 mA	10104/1/1	02100	na	3
16-channel Digital Input module Not Fail-Safe 24 Vdc - 7 mA	10104/2/1	12700	na	2
8-channel Digital Output module Fail-Safe 24 Vdc - 400 mA	10201/1/1	02200	na	3

Note 2 Hardware module supports the FSC Plug and Play integration methodology.

Note 3 Hardware module only available for spare parts and upgrades of installed FSC systems.

Table 3-14 FSC System Hardware Modules (continued)

**FSC System
Hardware Module
List, continued**

MODULE NAME	Model No.	Min. Hardware Revision	Min. Firmware Revision	Notes
8-channel Digital Output module Fail-Safe 24 Vdc - 400 mA	10201/2/1	11500	na	2
4-channel Output module Fail-Safe with double switch off 24 Vdc - 1A	10203/1/2	04500	na	
2-channel Analog Output module Fail-Safe 0(4) - 20 mA	10205/1/1	02300	na	3
2-channel Analog Output module Fail-Safe 0(4) - 20 mA	10205/2/1	11600	na	2
12-channel Digital Output module Not Fail-Safe 24 Vdc - 400 mA	10206/1/1	02400	na	3
12-channel Digital Output module Not Fail-Safe 24 Vdc - 400 mA	10206/2/1	11700	na	2
12-channel Relay module Not Fail-Safe 36 Vdc - 2 A	10208/1/1	02600	na	3
10-channel Relay module Not Fail-Safe 36 Vdc - 2 A	10208/2/1	11800	na	2
16-channel Digital Output module Not Fail-Safe 24 Vdc - 100 mA	10209/1/1	02700	na	3
16-channel Digital Output module Not Fail-Safe 24 Vdc - 100 mA	10209/2/1	11900	na	2
4-channel FS/4 channel NFS output module 24 Vdc	10212/1/1	02800	na	3
4-channel Digital Output module Fail-Safe 110 Vdc - 25 mA	10213/1/1	02900	na	
4-channel Digital Output module Fail-Safe 60 Vdc - 675 mA	10213/1/2	07000	na	3
4-channel Digital Output module Fail-Safe 48 Vdc - 750 mA	10213/1/3	07100	na	3
4-channel Digital Output module Fail-Safe 60 Vdc - mA	10213/2/2	12100	na	2
4-channel Digital Output module Fail-Safe 48 Vdc - 50 mA	10213/2/3	02200	na	2
3-channel Digital Output module Fail-Safe 220 Vdc/35 W with Line monitoring	10214/1/2	03200	na	

Note 2 Hardware module supports the FSC Plug and Play integration methodology.

Note 3 Hardware module only available for spare parts and upgrades of installed FSC systems.

Table 3-14 FSC System Hardware Modules (continued)

FSC System
Hardware Module
List, continued

MODULE NAME	Model No.	Mini. Hardware Revision	Min. Firmware Revision	Notes
4-channel Digital Output module Fail-Safe 24 Vdc - 2 A, Max, 6 A per module	10215/1/1	05700	na	3
4-channel Digital Output module Fail-Safe 24 Vdc - 2 A, Max, 6 A per module	10215/2/1	02300	na	2
4-channel Digital Output module Fail-Safe 24 Vdc Line monitored testable - 1 A. Max. 3 A per module	10216/1/1	09100	na	3
4-channel Digital Output module Fail-Safe 24 Vdc Line monitored testable - 1 A, Max, 3 A per module	10216/2/1	02400	na	2
4-channel Digital Output module Fail-Safe 48 Vdc Line monitored testable - 500 mA	10216/2/3	13400	na	2
Power Supply Unit (PSU) 24 Vdc/5 Vdc - 12 Amp	10300/1/1	04600	na	
Watchdog Repeater (WDR)	10302/1/1	03400	na	3
Watchdog Repeater (WDR)	10302/2/1	02600	na	2
Power Supply Distribution (PSD) for VBD-102 function only	10303/1/1	03500	na	
16-channel Analog Input Converter mA / V	10305/1/1	08300	na	3
Vertical Bus Terminator	10307/1/1	na	4.00	
Earth leakage detector rail mounted	10310/1/1	09700	na	
Earth leakage detector rack mounted	10310/2/1	10800	na	

Note 2 Hardware module supports the FSC Plug and Play integration methodology.

Note 3 Hardware module only available for spare parts and upgrades of installed FSC systems.

Table 3-14 FSC System Hardware Modules (continued)

FSC Field Termination Assembly Module List

The following table lists the Field Termination assembly modules for R510.

MODULE NAME	Model No.	Revision	Notes
FTA with ELCO connectors for 10101/2/1, 10101/2/2, 10101/2/3, and 10104/2/1 modules	FTA-E-01	na	4
FTA with ELCO connector for 10102/2/1, 10213/2/2, 10213/2/3, 10216/2/1, 10201/2/1, 10215/2/1, and 10205/2/1 modules	FTA-E-02	na	4
FTA with ELCO connector for 10206/2/1 and 10209/2/1 modules	FTA-E-03	na	4
FTA with ELCO connector for 10208/2/1 modules (potential free contacts)	FTA-E-04	na	4
FTA with ELCO connector for 10208/2/1 modules (powered output contacts)	FTA-E-05	na	4
FTA with terminals for 0101/2/1, 10101/2/2, 10101/2/3, and 10104/2/1 modules	FTA-T-01	na	4
FTA with terminals for 10102/2/1, 110213/2/2, 10213/2/3, 10216/2/1, 10201/2/1, 10215/2/1, and 10205/2/1 modules	FTA-T-02	na	4
FTA with terminals for 10206/2/1 and 10209/2/1 modules	FTA-T-03	na	4
FTA with terminals 10208/2/1 modules (potential free contacts)	FTA-T-04	na	4
FTA with terminals for 10215/2/1 modules	FTA-T-05	na	4
FTA with terminals 115 Vac (Active inputs)	FTA-T-06	na	4
FTA with terminals 115 Vac (Passive inputs)	FTA-T-07	na	4
FTA with Quad relay contact with single read-back for 10201/2/1 and 10216/2/1 modules	FTA-T-08	na	4
FTA with octal relay output (NO, NFS) for 10209/2/1 and 10206/2/1 modules	FTA-T-09	na	4

Note 4 Only available with the FSC Plug and Play integration methodology.

Table 3-15 FSC Field Termination Assembly Modules

3.11. FSC Safety Manager Hrdwr/Fmwr Requirements

**FSC Dev. System
Hardware Module
List**

The following table lists the Development System Hardware Modules for R510.

MODULE NAME	Model No.	Revision
EPROM programmer (including 07170/A/1)	07170/1/1	00200
PC-print with RS-232 / RS-485 serial interfaces, hardware key, and EPROM programmer interface	07177/1/1	03600
PC-print with hardware key and EPROM programmer interface	07177/1/3	03700

Table 3-16 FSC Development System Hardware Modules**FSC Dev System
Software List**

The minimum release for the FSC Development System to support the FSC Safety Manager is 4.00. The following table lists the FSC Development System Software for this release.

DESCRIPTION	Notes
Basic package	
Programming package (included in basic package)	5
Verify application and revision controller	
On-line package	
SER and report editor	
FSCSER: SER logging software for one stand-alone FSC system (FSCSER point-to-point connection)	
FSCSER-NET: SER logging software for multiple FSC systems (FSC network or FSCSER multidrop network)	

Note 5 Delivered with Basic package with FSC system.

Table 3-17 FSC Development System Software, Release 4.00

3.12 Hardware/Firmware Upgrade Kits

Overview

Upgrade kits are available to replace or upgrade LCN boards, to get them to a level required for R6xx.

Some kits are offered through an exchange program, while others simply add boards to the existing node.

K2LCN Upgrade kit numbers

Upgrade kits are available for the K2LCN boards, through an exchange program, where you will exchange old boards for new ones.

The kits are applicable to all K2LCN-based modules, and can be used to upgrade other 68000 processor-based (such as an EMPU), or 68020 processor-based (such as HPK2 and HMPU) modules. In certain cases, additional hardware may also be required when upgrading these boards. Please note the K2LCN board cannot be used with any external memory.

NOTE:

If the current AM is using an HMPU and has AMCL02 configured as an external load module, it cannot be upgraded to a K2LCN. It must be upgraded to a K4LCN.

Contact your local Honeywell service branch to obtain the upgrade kits.

Description	Model Number
K2LCN PWA, 2 Mw Upgrade Kit	MP-ZPROC2
K2LCN PWA, 4 Mw Upgrade Kit	MP-ZPROC4
K2LCN PWA, 8 Mw Upgrade Kit	MP-ZPROC8

Table 3-18 K2LCN Memory Upgrade Kits



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

K4LCN Upgrade kit numbers K4LCN boards, with memory on the board, can be purchased using the following model numbers.

Description	Model Number
K4LCN Processor/Memory Upgrade – 16MW	MP-ZPSC16-100

Table 3-19 K4LCN Board (with Memory) Upgrade Kit

LCNP to LCNP4 Upgrade For upgrading GUS nodes.

NOTE:

Currently LCNP, LCNP4 and LCNP4e boards can be differentiated using only the display on the board. The SMCC/LVRLOG/AUDIT tools cannot differentiate between a LCNP4-GPS, LCNP4-M, and a LCNP4-E board as they have the same LCN FW version.

If you are using LCNP4-M card, you have to open the chassis to ascertain whether it is a LCNP4 or LCNP4-GPS or LCNP4-E.

Description	Model Number
P4 Upgrade Kit – 16MW (Required R521.4 or R530.4 plus R110 S/W. (S/W must be ordered separately).	MP-ZLCNP4-100

Table 3-20 LCNP to LCNP4 Upgrade

Zip drive upgrade kits The following Zip drive upgrade kits can be purchased using the following model numbers.

Description	Model Number
Single Bernoulli drive to single Zip drive upgrade instructions	MP-ZIP100
Single Zip drive to dual Zip drives upgrade instructions	MP-ZIP101
Dual Bernoulli drives to dual Zip drives upgrade instructions	MP-ZIP102
Floppy drive(s) to single Zip drive upgrade instructions	MP-ZFZIP1
Floppy drive(s) to dual Zip drives upgrade instructions	MP-ZFZIP2
UWS floppy drive(s) to dual Zip drives upgrade instructions	MP-ZFZIP3

NOTE:

Zip drive support is not available on SPC

Table 3-21 Zip Drive Upgrade Kits

68040 Upgrade kits for redundant AMs It takes two separate upgrade kits to upgrade a 68020 processor to a 68040 processor in redundant AMs.

Description	Model Number
K4LCN Processor/Memory Upgrade – 16MW	MP-ZPSC16-100
EAMR board (two per upgrade kit)(See Notes 1 & 2)	MP-ZEAMR1
Note 1: This assumes you have Rev. T (or later) LCN I/O boards, which were required to support 68020 processors. The Rev. T (or later) LCN I/O board is required for both the K2LCN and K4LCN processor boards.	
Note 2: The redundant K4LCN AM upgrade requires the EAMR board be installed before the AM will load. (See AM special considerations, Section 5.17)	

Table 3-22 68040 Upgrade Kits for Redundant AMs

**Firmware upgrade
for variable
deadband on HG**

The following firmware is required to use the variable deadband option. The option can be applied on a “per box” basis for LLPIU and DHP boxes on the Data Hiway.

Firmware Description	Part Number
LLPIU variable deadband firmware	38120017-100
DHP variable deadband firmware	38120019-100

Table 3-23 Firmware Upgrade for Variable Deadband on HG

**Optional firmware -
HG Alarm Deadband**

A special variable deadband firmware can be purchased at any time, and is not specifically required for Release 6xx.

If the special variable deadband firmware is already installed before Release 6xx is installed, steps must be taken to turn on the variable deadband option and to prevent initialization of PV alarm deadband values (PVALDB).

**EPNI and Color
 Palette upgrade kits**

The following kits are available to upgrade EPNI boards and the Color Palette.

Model Number	Description
MP-ZNMS03	EPNI Single NIM upgrade for Time Sync
MP-ZNMR03	EPNI Redundant NIM upgrade for Time Sync
MP-ZUSCP1	Color Palette Upgrade

Table 3-24 EPNI and Color Palette Upgrade Kits

**PM/APM/HPM
 upgrade kits**

The following kits are available to upgrade PMs to APMs, PMs to HPMs, and APMs to HPMs, in either redundant or nonredundant situations.

If Upgrading from...	To a...	Refer to Upgrade Kit...
Nonredundant PM or APM	Nonredundant HPM, or Nonredundant HPM (conformally coated)	MU-ZHPS01 MC-ZHPS01
Redundant PM or APM (File-to-file redundancy)	Redundant HPM in 15-slot card files, or Redundant HPM in 15-slot card files (conformally coated)	MU-ZHPR01 MC-ZHPR01
Redundant PM or APM (side-by-side redundancy)	Redundant HPM in 7-slot card files, or Redundant HPM in 7-slot card files (conformally coated)	MU-ZHPR02 MC-ZHPR02
Nonredundant PM or APM (in 15-slot card file only)	Redundant HPM in 7-slot card files, or Redundant HPM in 7-slot card files (conformally coated)	MU-ZHPR03 MC-ZHPR03

Table 3-25 PM/APM/HPM Upgrade Kits

3.12. Hardware/Firmware Upgrade Kits

If Upgrading from...	To a...	Refer to Upgrade Kit...
Nonredundant HPM in 15-slot card files	Redundant HPM in 15-slot card files, or Redundant HPM in 15-slot card files (conformally coated)	MU-ZHMR01 MC-ZHMR01
Nonredundant HPM in 7-slot card files	Redundant HPM in 7-slot card files, or Redundant HPM in 7-slot card files (conformally coated)	MU-ZHMR02 MC-ZHMR02

Table 3-25 PM/APM/HPM Upgrade Kits (continued)

Upgrade paths that are not supported The following possible upgrade paths from PMs or APMs to HPMs are not supplied as kits, nor are they supported.

Upgrading From a ...	To a ...
Redundant PM or APM in 15-slot card files (file-to-file redundant configuration)	Side by side Redundant HPM (in 7-slot card files)
Redundant PM or APM in side-by-side configuration	File to file HPM (in 15-slot card files)

Table 3-26 Possible Upgrade Paths Not Supported

ENIM upgrade kits The following kits are available to upgrade NIMs to ENIMs, in either redundant or non-redundant situations.

If Upgrading from...	To a...	Refer to Upgrade Kit...
Non-redundant NIM – CE compliant	Non-redundant ENIM – CE compliant	MP-ZENMSC
Non-redundant NIM – Non-CE compliant	Non-redundant ENIM – Non-CE compliant	MP-ZENIMS
Redundant NIM – CE compliant	Redundant ENIM – CE compliant	MP-ZENMRC
Redundant NIM – Non-CE compliant	Redundant ENIM – Non-CE compliant	MP-ZENIMR

Table 3 27 NIM Upgrade Kits

EHPM upgrade kits The following kits are available to upgrade PMs / APMs / HPMs to EHPMs, in either redundant or non-redundant situations.

If Upgrading from...	To a...	Refer to Upgrade Kit...
Non-redundant PM or APM	Non-redundant 15-slot EHPM, CE Compliant	MC-ZEHPS2
Redundant PM or APM	Redundant 15-slot EHPM, CE Compliant	MC-ZEHPR2
Non-redundant PM or APM	Non-redundant 7-slot EHPM, CE Compliant	MC-ZEHPS7
Redundant PM or APM	Redundant 7-slot EHPM, CE Compliant	MC-ZEHPR7
Non-redundant PM/APM in 15-slot card files	Non-redundant EHPM in 15-slot card files (Non-CE compliant)	MC-ZEPS2N
Redundant PM/APM in 15-slot card files	Redundant EHPM in 15-slot card files (Non-CE compliant)	MC-ZEPR2N
Non-redundant PM/APM in 7-slot card files	Non-redundant EHPM in 7-slot card files (Non-CE Compliant)	MC-ZEPS7N
Redundant PM/APM in 7-slot card files	Redundant EHPM in 7-slot card files (Non-CE Compliant)	MC-ZEPR7N
Non-redundant HPM CE Compliant or Non-CE Compliant	Non-redundant EHPM, CE Compliant	MC-ZH2EHS
Redundant HPM CE Compliant or Non-CE Compliant	Redundant EHPM, CE Compliant	MC-ZH2EHR

Table 3-28 HPM Upgrade Kits

CF9 switch upgrade kit The following kit is available to install a redundant pair of CF9 Ethernet switches for use with EHPM nodes in an HPM cabinet.

Model Number	Description
CC-ZCF9PM	Control Firewall Ethernet Switch kit with 8 ports and 1 uplink port

Table 3-29 CF9 Upgrade Kit

3.12. Hardware/Firmware Upgrade Kits

FTE switch used with LCN Cabinet upgrade kit The following kit is available to install the FTE switch that is used with the ENIM.

Model Number	Description
NE-ZFTEB2	FTE network media kit containing two switches each configured with 8 Level 1 copper ports, 8 Level 2 copper ports, and two dual-purpose ports.
NE-ZFTEB3	FTE network media kit containing two switches each configured with 8 Level 1 copper ports, 8 Level 2 fiber ports, and two dual-purpose ports.
NE-ZFTEB4	FTE network media kit containing two switches each configured with 8 Level 1 copper ports, 8 Level 2 copper ports, 8 Level 2 fiber ports, and two dual-purpose ports.

Table 3-30 FTE Switch Upgrade Kit

LM upgrade kits The following kit is available to upgrade LM to MLPLC.

Upgrade Kit	Part Number
Logic Manager to Master Logic Upgrade Kit	51195766-033

Table 3-31 LM Upgrade Kit

4. HM Volume Sizing

4.1 HM Volume Sizing

Volume sizing

Determine the volume configuration requirements for your HM(s). Keep the notes and comments below in mind.

- Wren I and Wren II are not supported in R6xx releases.
- The first USER volume can be the FAST USER VOLUME, with the maximum number of files being 9995, and the volume size of 0 to MAX DISK SIZE.
- If file descriptors are used on this user volume, the user volume will NOT be a FAST USER VOLUME.

ATTENTION

HVTS version 10.5 (or greater) must be used when formatting new disks.

Other sizing considerations to go from R43x to R6xx

The following table provides other sizing considerations that go from R43x to R6xx.

Volume	Consideration
&3np	The area database is larger
&2np	Journals are larger
&4np	Maintenance – support software is larger
&8np	NIM/UCN checkpoints are larger

Table 4-1 Other Sizing Considerations

Disk sizing terminology

The following is a description of the various terms relating to HM, Bernoulli, and Zip disk space.

Size/Term	Conversion Factor
1024 bytes	1 KB (kilobyte)
1024 * 1024	1,048,576 bytes or 1 MB (megabyte)
1 word	2 bytes
1 sector (block)	256 bytes
1 Track	32 sectors
20 MB Bernoulli Cartridge	21,417,984 bytes
35 MB Bernoulli Cartridge	35,662,848 bytes
150 MB Bernoulli Cartridge	150,998,016 bytes
100 MB Zip	100,649,000 bytes

Table 4-2 Disk Sizing Terminology

Winchester disk drives and capacities

The Winchester disk drives and capacities in the History Module are shown in the table below, in megabytes.

When configuring the HM disk drives, at the **VOLUME CONFIGURATION** screen, you have the choice of entering a disk type number (3, 4, 5, or 6) or the size of a single disk (in KB), in the **DISK TYPE/SIZE** field.

The HM disks, the disk type numbers, and their capacities are as follows

Disk Name	Model	Disk Type Entry	Capacity (in kilobytes)
Wren III	Wren 3 51195155-100	3	Single - 135,900 KB Dual - 271,800 KB
WDA 210	ST-3283 51196220-100	4	Single - 215,040 KB Dual - 430,080 KB
WDA 445	ST-3600 51196243-100	5	Single - 454,130 KB Dual - 908,260 KB
WDA 875	ST-31230 51196680-100 Replacement ST-32272 51196680-200	6	Single - 896,032 KB Dual - 1,792,064 KB Single - 896,032 KB Dual - 1,792,064 KB
WDA 1.8 (1.8 Gig)	ST-32272 51196895-100	7	Single - 1,836,736 KB Dual - 3,673,472 KB
Dual Logical 875MB	ST-39175LW 51191560-200 or ST-39236LW 51191579-200	6	Dual- 1,792,064 KB
Dual Logical 1.8GB	ST-39175LW 51191560-100 or ST-39236LW 51191579-100	7	Dual- 3,673,472KB

Table 4-3 Winchester Disk Drive Entries and Capacities

NOTE:

In R6xx, all history modules must be configured for the correct drive size. If the HM fails to initialize, check the Model number and Part number for correct size.

Disk Configuration

Disks larger than 529,208 KBs, configured for a smaller size in R4xx, now must be configured for the correct drive type.

4.2 Fast Search User Volume

Fast search user volume improves system performance

In R500 and later, an HM fast search user volume is available to provide directory searches 20 times faster than the standard linear volume/directory search method, by using a FAST (binary search) tree algorithm. This means quicker access to schematics and other files in the user volume. Also, the fast search volume usually improves performance of all system functions that rely heavily on directory searches, including copy file, delete file, create file, read file, and schematic access. The value of this function becomes even more significant as you move to larger History Modules, which will likely have more files to search.

Binary search vs. linear search methodology

Files not on a fast search user volume are located using a linear volume/directory search. This is very time consuming for user volumes configured for many files because all configured volume directory entries are searched to the requested file, even if these entries are empty. This is often a significant factor in slow schematic call-ups.

Honeywell's FAST (binary search) tree algorithm maintains a single fast search user volume with directories of frequently used files and does a binary search of the sorted files in that volume, significantly speeding the search process.

How fast search is configured

By setting up a fast search directory in the fast search volume, the HM can locate files very quickly. The fast search volume is always the first user volume defined in the user volume configuration page in the HM.

Fast search implementation

For maximum performance, Honeywell recommends the following:

- 1) Specify fast search directories first in the search path sequence.
- 2) Configure the fast search volume with as much space as possible.
- 3) Configure the fast search volume for the maximum number of files (9995).
- 4) Place performance-critical schematics, logs, CLs and other critical files in the fast search volume.

NOTE: The fast search volume is limited to 63 directories and 9995 files. File descriptors are not available if the fast access option is chosen.

CAUTION

FST_VOLZ Caution—The Area Data Base directories can be cleaned up and many operational schematics moved to this FAST volume. There is a negative to this proposal:

Remember the FST_VOLZ.EC will not automatically move files from this FAST volume to the Fast Load cartridge. You would need to modify the FST_VOLZ.EC to ensure that the required schematics were available if the HMs were not available and the Fast Load cartridge was to be the only source in a system emergency reload situation.

Recommendation

It is recommended that for new systems, set up the following directories to reside in the FAST user volume:

&CLX, &CUS, DIA1, &DOC, TFIL, and TLK1

On systems where the area volumes are not used for .DO files, set up the following directories to reside in the FAST user volume:

- all picture .DO files (.DS optional),
- all Free Format Logs (.FO files),
- all CL source and object files, and
- all .KO, .KS, and Button files.

NOTE:

If your schematic (.DO) and Free Format Log (.FO) files are already in the &Dan directory (an = area number), you do not need to move them into the FAST user volume.

4.3 HM Volume Size Estimator Tables and Figures

HM volume size estimating help

For help in estimating the amount of space required by each volume on your HMs, refer to section 7 – HM Configuration and Use in the *Engineers Reference Manual*.

Burst Buffer size variations

Actual events recorded in the Burst Buffer are different sizes. Buffer sizing uses the Process Alarms entry as the average size. Therefore, a buffer size of 7500 would contain 7500 Process Alarms. CL messages are about three times the size of the Process Alarms, so you get about one-third as many in the buffer.

In configuring the Burst Buffer size, a good rule of thumb is to allow for three times the number of events that will be received by the HM during any 1 minute period. Another is to set the value to the maximum number of events expected during a “worst case” 2 minute period.

AM changes due to new parameter – ASSOCDSP

The AM checkpoint volume changes are as follows:

Volume ID	Name/Content	R43x Size	R510 and Newer Size
&5np	AM Ck Pt		
	unit_ohd	28.7 Kb*units	28.7 Kb*units
	Unckpted_buf_sz (default)	68 KB	68 KB
	multi_CDS_sz (default)	82.6 KB	104.8 KB
	user mem 1.5 Mw	1018 KB	N/A
	2.0 Mw	1716 KB	N/A
	3.0 Mw	3398 KB	1986 KB
	4.0 Mw	5078 KB	3948 KB
	5.0 Mw	5138 KB	5908 KB
	6.0 Mw	8440 KB	7870 KB
	7.0 Mw	10122 KB	9832 KB
	8.0 Mw	11802 KB	11792 KB
	16.0 Mw	n/a	27482 KB

Table 4-4 AM Checkpoint Volume Changes

**HG Checkpoint
Volume Change**

The HG checkpoint volume change is as follows:

Volume ID	Name/Content	R4xx Size	R521 and Newer Size
&7np	HG Ck Pt / cluster KB for each Data Hiway	1075 KB	2146 KB

Table 4-5 HG Checkpoint Volume Changes

The number of file handles in the HG checkpoint volume increases by 20.

The new checkpoint file used to store the ASSOCDSP parameter is created automatically on the first save. The default parameter value is “blanks.” There is no “null” checkpoint file.

After migration, absence of the ASSOCDSP checkpoint file will prohibit the HG from loading. This is consistent with the current action taken when the checkpoint file is missing.

**HM quality
improvements**

The following HM quality improvements have been added for the R5xx/R6xx series releases:

- **The HM Load Time Has Been Significantly Improved**—In previous releases, HMs with large continuous history sometimes have taken a long time to load (up to 30 minutes). For R520 and later, the HM boot up time has been improved and will be approximately 3-5 minutes for a K4LCN, and 4-6 minutes for a 68020 HM.
- **NOTE:** The HM must run in the HMON (on-line) personality for 2 hours on R520, or later, for the boot speedup to take effect.
- **Maximum History Volume Size Increased**—In R510 and previous releases, the HM history capacity was clamped at approximately 1 GB. This restriction no longer applies, and the HM history can be up to (approximately) 1.6 GB with a 1.8 GB drive.
- **When using a K4LCN and WDA 445 or newer drives**—Prior to R520, the maximum recommended size was 2400 points (40 pps) and 168 hours for snapshots. For R520 and later, the HM can collect 3000 points (50 pps) and all limits and suggested restrictions on snapshot hours are removed.

ATTENTION

When migrating from R4xx/R5xx to R61x or newer, NIM Checkpoint Volume Size must be increased.

**NIM Volume
Checkpoint Size**

In R5XX, the recommended checkpoint size allocation for each HPM checkpoint is 14012 sectors.

In R61X (and newer), the recommended checkpoint size allocation for each HPM checkpoint is 16012 sectors.

If using Fieldbus, an additional 2000 sectors per NIM should be allocated in the checkpoint volume to support the Metadata file. Also, the recommended checkpoint size allocation for each Fieldbus IOP checkpoint is 1410 sectors. Users should use these new values when performing HM Volume Configuration to support future expansion.

There is a possibility that your current HM checkpoint volume is not large enough for your UCN(s). If this is the case, you will need to re-configure the HM that which contains the checkpoint volume. Please refer to the Engineer's Reference Manual, and System Startup Guide for detailed instructions on performing the steps necessary to backup, re-configure, and restore the affected HM.

Example:

The Engineer's Reference Manual, Section 7, contains information on estimating NIM checkpoint size. It is always good to allocate spare disk space for future growth.

5. Special Considerations

5.1 Overview

Overview

TotalPlant Solution (TPS) system network Release 6xx software contains certain items requiring special considerations, of which the customer should be aware when migrating from an earlier release of software or when loading R6xx software on a new system.

It is important that you read the applicable special considerations in this section. Where possible, they are grouped by product (HM, Universal Station, etc.), so you can skip those that do not apply to your system.

5.2 General System and Configuration

Maximum Concurrent Loads using HM

ATTENTION

Loading more than six nodes concurrently from an HM can lead to load errors. If more than six concurrent node loads are attempted using .EC files, LD command or the System and/or Node Status displays, there is a risk that some of the loads may be unsuccessful.

Additionally, this may adversely affect node load performance and exhaust the resources of the History Module. A node is in the process of loading until it reaches the state where the loading of all components as shown on the Detail Status Display is complete.

UCN and NIM Migration

CAUTION

UCN migration should not be attempted if any cable errors are present on the UCN.

If making system changes, new installations, or hardware upgrades, it is recommended that NIMs be assigned the lowest UCN address.

NIMs must be loaded with R6xx software before any other UCN nodes on the network are loaded. The NIMs have UCN cable-handling code that must be active before the other UCN nodes (PMS, APMs, etc.) are loaded. Also, UCN cable swapping must be disabled.

If you fail to follow this loading order, there is the potential for loss of view and/or loss of non-local, advanced and peer-to-peer control.

IMPORTANT! Should there be any need to regress NIM or UCN nodes back to R4xx, each node must be SHUTDOWN and reloaded, OFF-PROCESS.

**Clock source and
HMPU/HPK2**

If any nodes have HMPU or HPK2 boards, they must be used as clock sources. HMPU and HPK2 boards cannot receive clock messages from K2LCN and K4LCN boards. For more information, see binder TPS 3025, LCN Guidelines Implementation, Troubleshooting, and Service, section 2.5.

**User input of two-
Digit Years**

TPN year data entry is in the format of "YY", and four- digit entry is not used. Date values for the year range from 1979 to 2047; hence, the "YY" accepts values of 79 through 47.

TPS Applications will accept year data entry in the format "YYYY" and a shorthand notation of "YY" is acceptable input. When "YY" is used, it will behave consistent with the Microsoft standard, which assumes two-digit dates 00-29 belong to the year 2000. Therefore:

- A two-digit TPS date of 26 would represent the year 2026.
- A two-digit TPS date of 31 would represent the year 1931.

The TPN and TPS policies deviate in behavior in the years ranging 2030 - 2047.

Loading redundant LCN nodes during isolation

When node isolation occurs on redundant LCN nodes (HG, AM, NIM), care must be taken with the procedure used to resolve the problem. If both nodes are isolated from the US, both must be SHUT DOWN or use the LCN Reconnect procedure found in the *LCN Guidelines* manual before reloading. Otherwise, a reload to one node results in both nodes fighting to be the primary. This results in either a crash or having 2 primaries, which produces unpredictable results.

In R6xx, loading redundant nodes from the System Status display automatically includes both nodes in the SHUTDOWN list, and, if confirmed by the operator, reloading will occur.

NOTE: You cannot load nodes if an LCN Reconnect has been initiated and you are using mixed releases (such as R43x and R6xx).

Multiple Loads

When preparing to reload multiple nodes on the LCN, Honeywell recommends the user shutdown or reset all the nodes to be reloaded before starting the individual loads. If nodes are reset while loads are in progress, there is a possibility that the loading nodes will go into an isolated state and have to be reset and loaded again.

Load from removable media

If a target US is being loaded from removable media through a source US (which is running the Universal personality), and the &DSY volume is not mounted, a mount message will be issued to the source US.

If the source US has been changed to the Engineering Main Menu, the Command Processor, or any of the Main Menu primary functions, the mount message will not be visible until it is changed back to one of the operating displays. It is recommended the source US remain on the load display until the load is completed.

Block sizes greater
than 6xx

CAUTION

If File Descriptors are in use, do not format cartridges for block sizes greater than 600. Doing so may result in corruption of data on the cartridge. Conversely, do not use File Descriptors if block sizes are in excess of 600.

HG Checkpoint to
LCN ZIP Disk may
experience errors

ATTENTION

There have been reported random problems when using Zip Disks with volumes formatted for a “block Size” of 1700. The problem is more likely when two zip disks are used in the same Universal Station. Our recommendation is to use a “block Size” of 600, as shown in the following example:

CR \$fn>vol -BS 600 -MF 600

Copying the HM
Boot file

When copying the HM personalities, be aware the boot file must undergo a name change. It is on the cartridge as \$HB00T.MS and must be placed on the HM as \$B00T.MS.

When copying, perform a regular copy as shown below.

```
cp $Fs>&HMI>$HB00T.MS    PN:nn>!9np>$B00T.MS -D
```

— or —

```
cp $Fs>&HMO>$HB00T.MS    PN:nn>!9np>$B00T.MS -D
```

where

s = source drive

nn = node number

np = node pair

To avoid the problem, use command file LOC_VOLZ.EC. It is supplied with the Honeywell software in the directory &EC.

Reinitialize HM if backing out

In migrating to R6xx keep in mind that once the HM has been initialized on R6xx, should there be a need to “back out” to your previous release, the HM must be reinitialized on the previous release.

Then, all previous non-translated files must be copied back to the HM(s) and the executable files must be rerun, using the previous software release cartridges.

Change in error message when restoring APL files

In R4xx and previous releases, an attempt to copy APL00001.MM and APL00002.MM files (which contain history group point definitions) to the HM after it was newly initialized and before the HM was booted in the HMON personality, an error message would be received. This indicated the APL00001.MM and APL00002.MM files did not copy due to insufficient space.

This has changed in R5xx and R6xx, and the error message will no longer be received; therefore, the absence of this error message when running R6xx DOES NOT indicate your history group definitions have been restored.

The correct procedure for restoring history group definitions is described in section 10.1 of this manual, or in Section 7 of the R6xx *Engineer’s Reference Manual*.

Configuring History Groups in HM

Do not configure the HM for only a single 60 second History Group. If you initialize your HM to this configuration, an attempt to boot the On-line HM Personality (HMON) will result in the HM staying in the READY state for about ten minutes, and then crashing. All attempts to boot your HM will fail in this manner.

To correct this condition, configure the HM for two or more History Groups, rerun the Configurator, install the NCF, and then reinitialize the HM.

Using SHUTDOWN on HMs with K4LCN

ATTENTION

If your HM has a K4LCN processor board, the method of “failing” the HM to get an AUTOBOOT where you first shutdown the HM to the QUALIF state, then issuing a SHUTDOWN again to get it to the failed state, does not always work. Make sure that the Autoload target is Enabled. Follow the procedure in section 10.1.

Use of FST_VOLZ.EC on 20 MB cartridges

The FST_VOLZ.EC may fail to copy all personalities and checkpoints to a 20 MB cartridge, since the available file capacity is exceeded. Should this occur, there are two possible workarounds:

- If possible, use a 35 MB cartridge or Zip cartridge.
- Evaluate what is really needed on the FST_VOLZ cartridge. There may be UCN personalities and/or checkpoints that are not needed. By removing unessential files, additional space on the 20 MB cartridge will be made available.

5.3 LCN Faults

LCN reconnect

Once LCN reconnect has been initiated, new LCN faults cannot be reported. So, it is important to ensure that the LCN RECONNECTION terminates once normal LCN communications are restored. If the LCN RECONNECT action was initiated via the LOAD ISOL target on the System Status Display, termination should be automatic. Refer to the *LCN Guidelines-Implementation, Troubleshooting, and Service* manual in the *LCN Installation* binder.

CAUTION

The LOAD ISOL action may cause loss of view and/or loss of control in the attempt to restore normal LCN communication between all nodes.

5.4 Maintenance

Tool Kit problem

The cable status parameter shown in the CLOKSTAT display (refer to the *Customer Resource Manual* Section 15 for detailed instructions) correctly indicates the proper operation of the clock subsystem when all nodes consistently show ALTCABLA or ALTCABLB over several display updates (about 1 minute).

If problems in the clock subsystem exist, a node may show any of the possible status values. Do not depend on the cable status to indicate which cable is receiving time frames when the clock subsystem is having problems.

5.5 Hardware Considerations



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

RULA product on US The RULA product is no longer supported starting with TPN R630 and newer.

K2LCN/K4LCN Address pinning conflict If you are upgrading your processor board(s) to a K2LCN or K4LCN in a five or ten-slot module, **you must remove the addressing pins** on that board, so they do not conflict with the address pinning on the associated I/O board. See the upgrade kit instructions for details.

On dual node modules, address pinning is still done on the processor board, and not on the associated I/O board.

K4LCN processor board Placing a K4LCN or LCNP4 processor in your operator station(s) (US, UxS, UWS, GUS), can potentially overload data owner nodes (NIM, HG, PLCG/EPLCG, AM, A^xM) not having a K4LCN processor.

If any data users on the LCN (US, UxS, AM, A^xM, CG or GUS) are upgraded to the K4LCN processor, any other NIMs, HGs, AMs, and CGs on that LCN must also be upgraded to K4LCNs.

The K4LCN NIM is required if (a) one or more HPMs are installed on the UCN or (b) one or more K4LCN USs (or UxSs or GUSs) are installed that access the NIM.

The K4LCN provides system performance improvements, and you can now have up to 16MW of memory. With R500, AM Redundancy (AMR) is supported with the HMPU and AMR boards or with the K4LCN and EAMR board combination.

If GUSs use the LCNP4 then the Data Owners (NIMs, HGs, AMs, and CGs) must also be K4LCN boards.

CAUTION

For redundant node pairs, both nodes must have the same processor board and the same size memory. Redundant pairs will not operate with mismatched processors and/or memory, other than during a short migration period. The K4SDR and K4LCN-16 are functionally equivalent and can be mixed with matching memory size.

EXCEPTIONS:

1. **Redundant AM:** Both primary and backup AMs must be shut down at the same time before changing processor types or memory size.
2. **Redundant NIM and HG:** Mixed memory sizes are allowed for K4SDR and K4LCN-SS PWA boards only.

**K2LCN/K4LCN
firmware Revision
when using SMCC**

ATTENTION

For K2LCN and K4LCN boards only—SMCC REV/CONFIG STATUS may not display the correct revisions. You must visually inspect each board.

**Halt with Rev L
HPK2s**

When an EPNI is in the same node as a Rev L HPK2, do not run EPNI TEST2 of the Hardware Verification Test System in the subsystem or module modes. If TEST2 is conducted under these circumstances, the HPK2 issues a HALT command, causing HVTS in the NIM to crash. This is not a serious problem, as the NIM is, by definition, off-control. To fix the problem, replace the HPK2 with Rev M or later. (See Table 3-1)

Changing board types—effect on OPA

OPA (On-process Analysis) is a software function that tracks long term error aggregate rates for the boards in the LCN. However, once a board is replaced by a different type of board in an LCN node, the OPA testing and tracking stops for that board.

The only way to restart the OPA testing and tracking is to delete the current OPA error tracking files. When this is done, OPA will start over.

To delete the current OPA tracking files:

- MANUAL CLEAR all outstanding Active Maintenance Recommendations
- Set system HM to INIT mode (HMOFF OK).
- PROTECT the .CM file in !ESY [PROT PN:nn>!ESY>* .CM]
- DELETE the rest of the files in !ESY [DL PN:nn>!ESY>]
- UNPROTECT the .CM file in !ESY [UNPT PN:nn>!ESY>* .CM] (**CAUTION:** The HM will crash if you skip this step.)
- REBOOT HM to ONLINE mode HMON
- If drives are redundant, synch them at this time.

**A^xM layer Modules
Prevention of
Driver/Comm Errors
on K2/K4**

ATTENTION

Users of A^xM nodes with K2LCN/K4LCN processors must ensure the WSI-2 circuit board on the WSI-2 assembly is re-pinned to do a single bus transfer for DMA read instead to two. This will prevent LCN driver and communications errors that can occur under heavy bus loading conditions.

5.6 Universal Station

Checkpoint restore When using removable media to load an HG, the checkpoint media and checkpoint files must NOT be protected.

Equipment List Load module Do not configure the Equipment List loadable module MSF for the Operator Personality. If you do, the configurator does not detect the error and the Universal Station in Operator mode will produce a Status Detail error message on startup.

Schematic Access to the Documentation Tool The Schematic Access to the Documentation Tool package uses an existing “display file” routine for displaying query results on the screen. If a line in a file is greater than 240 characters, Display_File displays the following error message, instead of the data:

This line caused a buffer overflow

Queries to be displayed using the new QDisplay actor should be built with a width of 240 characters or less.

Initially load only one US to completion At the start of migration, be sure to load only one Universal Station or GUS with R6xx to completion, including the creation of a duplicate name file. A modification was placed in R500 that delays the creation of the duplicate name file even after the US goes to OK.

Use the **US Status Detail Display** to verify that the following message is displayed before continuing:

CUSTOM/STANDARD PARAMETER DUPLICATE Xref created

Modifying US Default Load Personality

ATTENTION

If the **DEFAULT LOAD PERSONALITY** has been modified (NCF change) for an existing US node using the MODIFY target, the change does not become effective until after the US has been reloaded with the new NCF. To get the desired personality the **first** time you reload the US after the NCF change, reload the US using the MANUAL load target and specify the personality.

5.7 US Alarm Displays

Circumstances that cause unsynchronized US alarm displays

US alarm displays and certain other event-driven displays and indicators (Examples: node/hiway status displays, keyboard LEDs, annunciator horns, contact outputs) can become unsynchronized (that is, will not match) across USs in the same console and area under the following circumstances:

- **Alarm/Message List Overflow** – Buffer space for US alarm/message lists is exhausted (600 alarms/100 messages, old alarms/messages are replaced by new). The US cannot recover when the number of outstanding alarms/messages exceeds the list size or if list overflow occurs during lost event recovery.
- **US Throughput Rate Exceeded** – Nonrecoverable events (for example, alarm ACK and unit alarm enable/disable) are lost due to US event manager job queue overflow.

SOLUTION:

- Perform event recovery if “lost events” are indicated.
- To resynchronize displays, perform an area change to a different area on each station, then reload each station in the console (one at a time).

Area Change

ATTENTION

Only one area change should be in progress on a console at any time.

Invalid alarm

When an HG point has an “INV” alarm, the HG will not re-arm the “INV” alarm for an \$AUXUNIT change or any other alarming parameter change.

Changes to ALPRIOR From R500 on, the alarm priority status (ALPRIOR) is no longer displayed on standard displays. It can still be fetched through schematics, and using CL.

CAUTION

Storing to ALPRIOR causes the value to be propagated down to all alarm priorities on the point. A CL compiler warning message describes the consequences of storing to ALPRIOR.

**Possible lost
alarms with RTJ
Only package**

It is possible that alarms with the priority of JOURNAL could be lost, if the printer is not operational at the time the alarm occurs.

**Area Data Base Unit
Assignment**

ATTENTION

Each unit selected for the Area Database should be unique. Do not assign the same unit multiple times to the same Area Database or the US may fail when acknowledging alarms for this unit.

R610 (and newer) prevents you from building an Area Database which contains the same unit assigned multiple times. However, R610 does not correct an existing Area Database with this errant condition.

To check your Area Data Base configuration, do one of the following:

- Call the Alarm Summary Display. Inspect the unit boxes at the bottom of the screen to ensure that no unit is repeated. Repeat for each configured Area Database.
- Call the Console Status Display, select Unit Assignment. Inspect the units listed to ensure that no unit is repeated. Repeat for each configured Area Database.
- Use the Area Data Base builder tool from the Engineer Main Menu to select and reconstitute the Area Database Unit Assignment. Inspect the units listed to ensure that no unit is repeated. Repeat for each configured Area Database.

If you find that an Area Database is improperly configured, use the Area Data Base builder tool to correct the unit assignment. Then, perform an Area Change operation for each station using this Area Database.

5.8 Universal Station — Engineering Functions

Using Command
Processor functions
while
HM is off-line

CAUTION

If one or more HMs on an LCN is running its off-line personality (&HMI), take care when using the Command Processor Create Directory command, because it is possible to create a duplicate directory on an HM that can make an HM inaccessible.

For example, if a duplicate &ASY directory is inadvertently created on a non-system HM and an NCF file was copied to that duplicate directory, an off-line HM could not be reloaded with its on-line personality (&HMO) because there is a duplicate NCF.

If a duplicate directory is created, you can delete it by using the Delete Directory command, using the PN:nn form to specify the physical node (HM).

An HM in the initialization personality (&HMI) checks all of its volumes for duplicate directory names. It DOES NOT check the LCN (NET) for duplicate names.

- If a duplicate name that matches another History Module's directory (NET) is created, the last HM to attempt to boot the on-line personality (&HMO) will go to SEVERE when the condition is detected by on-line diagnostics. This can be fixed by putting the HM with the duplicate directory in the off-line personality (&HMI) and using the Delete Directory command.
- If duplicate name is created in another root volume of the same HM, it is necessary to initialize the HM. No data from either directory can be saved.

While the duplicate directory condition exists, you can reliably access data from either directory by using the physical node syntax (PN:nn).

**Using Utility
functions while
HM is on-line**

Do not attempt to copy HM history files while the HM is running in the On-Process Personality (HMO).

If you do, the files copied are invalid, and you lose any history you have collected in those files. Collection of History may be suspended until you reboot the HM.

Such copy activities must be done when the HM is running the Off-Line Personality (HMI).

**Area Pathname
must be set**

The Area Database Pathname Catalog must be modified to access the volume(s) where the Free Format log files exist. Failure to do so causes an “Abstract Missing” message when a Free Format log is printed or displayed.

**Remote file
access**

While in the Command Processor, commands that access remote console devices across a Network Gateway (using the form nn\\$. . .) are not supported.

**Modifying
automatic
checkpoint
period**

If you wish to modify the default automatic checkpoint period, keep the value within the allowable range, which is 00:01 hours to 12:00 hours in the PERFMENU schematic CHKPTIME.

DO NOT use 00:00 hours in an effort to force continuous checkpointing. If automatic checkpointing is disabled on any node being checkpointed by that HM, having the period set to 00:00 hours will cause retrieval of history to be blocked.

**Disable TPN
Automatic Restart
during certain NCF
Changes**

It is recommended that the TPN Auto Restart function be DISABLED during NCF changes that add, modify or delete TPN nodes.

After the NCF changes are complete, the System HM should be rebooted to become synchronized with the new NCF. The Auto Restart function can then be ENABLED if desired.

**Picture Editor DDB
used for CROSSKBD
Parameters**

If the Picture Editor DDB enumeration variables are used to supply the value for the CROSSKBD parameters, the DDB variables' default values may produce unexpected results. The picture Editors default values for integers is zero and for enumerations is "NULL SET" which has a value of zero.

ATTENTION

These DDB variables can be used to supply the values for parameters of the CROSSKBD actor as shown in the following example.

```
CROSSKBD(G_INT(INT01),G_ENM(ENM01),G_INT(INT02),OPR,FALSE);
```

In this case, ENM01 and INT02 hold the value of the disconnect key and the idle minutes before disconnect, respectively. If the actor is invoked before values have been stored to ENM01 and INT02, then the default will be used.

According to the default values for integers (zero) and enumerations (Null Set = 0), the disconnect key will be the "zero" key (CNF1) and the "idle minutes" will be zero, meaning the connection will never timeout.

**Enhanced group edit
function**

Customers that have built pictures or buttons to perform group editing according to the instructions given in the Actors Manual (SW09-555), Table A-4, should review the target actions caused by modified Group Edit functionality in R6xx.

Before R530, the group edit action required actors to store to two parameters per point position on the group:

- \$GRPBASE(nnn).EXTIDLST(n)
- \$GRPBASE(nnn).INTIDLST(n)

Starting with R530, the group edit action requires actors to store to three parameters per point position on the group:

- \$GRPBASE(nnn).EXTIDLST(n)
- \$GRPBASE(nnn).INTIDLST(n)
- \$GRPBASE(nnn).KEYWORD(n)

Variant Corruption Problem

ATTENTION

Background: The Variant corruption problem which was reported under the R510 software is not unique to that particular release. It has been encountered on rare occasions as far back as R200. We do not know what conditions cause the problem nor have we ever been able to reproduce the failure, but we will continue to pursue a solution, as more data becomes available.

Scope: This instruction is to provide information for users to be able to recover from the variant corruption problem if it occurs.

Symptoms: Variants do not appear to work correctly on custom schematics. When this problem exists, typically what appears on the screen is the value of the last case limb plus possible remnants from previous case limbs. This occurs because all case limbs of corrupted variants are usually executed in order regardless of the actual value of the variant's conditional. The case limb values are overlaid on top of each other, the last case value being prevalent. For example:

```
IF DIGITAG.PV = ON THEN SUB "READY"  
ELSE SUB "NOTREADY"
```

The result is that Subpicture NOTREADY will always be displayed.

Note that any resultant targets or conditions associated with Subpicture NOTREADY will also be active.

Verification that the Problem Exists: Read the schematic source file into the Picture Editor. Select the suspect variants and print them. Corrupted variants will show each case limb twice instead of once. (E.g., Normal is bad value limb, case 1 limb, case 2 limb. Corrupted is bad value limb, bad value limb, case 1 limb, case 1 limb, case 2 limb, and case 2 limb.).

(E.g., Normal is bad value limb, case 1 limb, case 2 limb. Corrupted is bad value limb, bad value limb, case 1 limb, case 1 limb, case 2 limb, and case 2 limb.)

Corrective Action: It is unknown at this time what the precise condition is that causes the corruption, however reloading the Universal Station that was used to compile the corrupted schematic usually corrects the condition and should avoid future schematics from being corrupted.

Fixing a corrupted schematic requires reading the schematic source into the Picture Editor and doing a select and modify of all variants (simply re-enter each variant, you do not have to change anything). It is suggested that you re-enter all variants to ensure that an undetected corrupted variant was not overlooked. The variants can then be checked using the PRINT function. If the variants are OK, the schematic must be re-compiled. If corruption still exists after re-loading the Universal Station and modifying the variants, please take a dump of the node and perform the clean-up effort on a different Universal Station.

**“Hidden” Picture
Editor Compiler
Errors**

ATTENTION

An error occurring on any page of a multiple-page target, condition, variant, or “define actions” will be indicated on the **first page only**. If the picture editor compiler marks valid entries in red (error) on page 1, be sure to check subsequent pages, at the same coordinates, for an error.

Picture editor

Before release 530, under certain conditions picture “condition” logic that required multiple pages may have compiled with no error, but would not execute the logic on the second page of the condition when the picture object was on display. If the picture conditions work properly now, there is nothing further to do.

If it does not work properly, the picture source must be recompiled using the R6xx Picture Editor software to correct the error. No picture source file changes are necessary. Be sure to take whatever steps are necessary to make the US use the newly compiled picture object.

**Region Number in
CHG_ZONE Actor
must be correct
when calling
Enhanced Change
Zone**

If the incorrect “region” is used when calling the enhanced change zone, the change zone will not operate correctly. In some cases, this has caused the US to fail. If you experience problems with the Enhanced Change Zone, ensure that the schematic is using the correct “region” in the CHG_ZONE actor.

**Executable
command files and
wildcards**

You should review the Executable Command files (.EC) for proper use of wildcards based upon the new rules defined in R6xx.

R43x enhancements to the Command Processor “LS” command had the side-effect of causing “LS” to handle wildcards (* and ?) in the file name differently than all other Command Processor commands. This led to undesirable situations where a command action was taken on more files than anticipated per use of the LS command with the same wildcard pattern. For example,

<u>Command</u>	<u>Description</u>
LS *LOG?.XX	Find all files that match the pattern ? means any nonblank character
DL *LOG?.XX	Deletes all files that match the pattern ? means any nonblank character or no character

The result was that more files could be deleted by the **DL** command than were listed by **LS** command.

In R530 the meaning of the ? wildcard in non-LS commands has changed to match the definition used by the LS command. Additionally, non-LS Command Processor commands will no longer accept *? nor ?* as a valid use of wildcards in the file name field. A filename field can still have both * and ?, but they cannot be adjacent to each other.

Building databases while loading node

Do Not Build Database While Loading Nodes. Loading a secondary node (HG/NIM/AM) while database building can cause the secondary node to crash **or the database to be invalid.**

Simultaneous Point Loads from Multiple USs to the Same NIM/HG May Corrupt the NIM/HG Database. – Alert BW-2001-06

CAUTION

This problem is very rare but may cause database corruption, which prevents you from adding points to the NIM/HG. To avoid this problem do not run large exception build files that add/modify points on the same NIM/HG from multiple stations.

Database migration R43x to R6xx only

DB files (databases) cannot be directly migrated from R43x to R6xx, with the exception of History Group DB files and Area Database files.

If you need to migrate any other R43x DB files, they will have to be converted to an EB file, the EB file will have to be translated to R6xx, then it can be converted back to a DB file.

If the .EB files are used for backup, once you have migrated from R43x to R530 or later, reprint the entities into the EB files on R6xx.

No file response in remote node.

The maximum length of any file on an LCN System is 65,535 records. When using the Command Processor to copy very large files between HM and removable media you may receive the error message NO FILE RESPONSE IN REMOTE NODE error message. This message indicates that the Command Processor has timed-out while waiting for the copy command to complete. However, the copy action does continue and will complete, although it may take several more minutes. Use the LS command to check for completion.

5.9 System Operation

Scheduling	Hourly logs and reports should be scheduled to run 5 minutes after the hour in order to avoid the time during which the continuous history file updates are occurring.
System time	Exercise care when setting the system time backward. Normal time changes are expected to be between 1 and 6 minutes (for minor time corrections), and between 54 and 66 minutes for seasonal time changes. Violation of these guidelines may render some history data inaccessible.
Changing system time	If the system time is set back (e.g., 1 hour) the Real Time Journal entry of a time change is not output to the printer until the elapsed time expires or until the printer output buffer fills up.
HM disk redundancy	If the on-line personality for an HM is loaded from a backup drive with the primary drive(s) in the formatted state, a control track error can be reported in the Real Time Journal. <u>In this situation, the error is normal and can be ignored.</u>, if there are no redundant drives, this error is serious. Hardware maintenance is required. If an HM that is running the on-line personality is "SHUTDOWN, " then "RESET" while part of a set of redundant drives has power off, the STA command does not detect the drives with power off and the HM status display shows OK for this HM.
History Module fragmentation	It is possible to get a "history module full" message while it actually has adequate storage space. This may occur after some time of usage, where the HM storage becomes fragmented, and the disk appears full (files that have been transferred are fragmented as the HM finds available space to store them). This does not occur very often. When it does occur, it is usually on the user-configured volume. Refer to the Engineer Reference Manual Section 7.9 for more information.

**Multiple copies to
History Module**

Copying from more than one media at the same time, to a single volume on the History Module, can result in fragmentation of the files on the HM.

CAUTION

If this should happen on a volume that starts with a “I,” treat it as a system problem. **DO NOT DELETE ANY FILES!** Gather the needed data, following the procedures in section 14.2, then call the Technical Assistance Center (TAC) in Phoenix at 1 (800) 822-7673

**Hairline Trend Values
may not Match
Historical Data**

ATTENTION

The implementation of the hairline trend enhancement to the trend functionality has uncovered instances where the value versus time relationship of the trace data does not represent the historical data collected from the HM correctly. This error in data representation is a result of either multiple values or the lack of any values existing because of system time changes (backward and forward time changes, respectively). Therefore, the display of trace data which has been collected shortly after a system time change may not accurately correspond to the actual data archived by the History Module.

**Different Behavior
of HM starting
with TPN R630.3**

Periodic checking of disk status used to occur every minute (nominal). Starting with TPN R630.3, the disk status check has been slowed to once every two minutes (nominal). This means that it will now take up to two minutes to see a status change in an HM disk. (example: turning a disk on or off)

HM PSDP parameter \$PRSTSxx.HMDRVSTS(20) is no longer spare. It will now provide the “number of disks” for this HM as defined in the NCF.CF.

The HM will check disks periodically as follows:

- HM with one disk and no redundancy: Check disk 5
- HM with one disk and redundancy: Check disks 5 and 4
- HM with two disks and no redundancy: Check disks 5 and 3
- HM with two disks and redundancy: Check disks 5, 4, 3 and 2

Please note that disk redundancy is a system-wide option. If your NCF is configured for HM disk redundancy, individual HMs that do not have redundant disks will suffer a slight performance degradation as the SPC controller times-out looking for non-existent disks during the periodic check of disk status.

**Extra History Data
Returned to CM50 or
AM/CL program**

A backwards retrieval of snapshot data that is over 8 hours old causes the HM to return duplicate data records for the requested snapshot values. For example, if 10 snapshot values are requested, the HM will return 10 data records with a status of OK, plus 10 more data records with a status of “not in history”. The user program must account for this when sizing data arrays, and in the logic used to test for valid data values.

**MODATTR
parameter**

Before R530, the MODATTR parameter of an AM point could be modified from a user’s picture regardless of the value of MODEPERM for the selected point (i.e., even if MODEPERM = NotPerm, the operator could change an AM point’s MODATTR from a picture change zone). Starting with R530, the MODEPERM function for AM points is enforced from user pictures. This is consistent with the MODEPERM function on HG and NIM points.

Pictures used to modify AM point’s MODATTR parameter will return a “not permitted” error if the point’s MODEPERM = NotPerm and the station is in OPERATOR key level.

5.10 Network Interface Module (NIM)

PNI/EPNI boards - release compatibility

The Release 6xx NIM software will support the earlier released NIM hardware with the PNI board. The PNI board may be used on R6xx (and older systems), only if the SOE Time Sync is disabled.

You must replace PNI boards with EPNI boards if SOE Time Sync is to be used.

5.11 UCN Operation

UCN Checkpointing Checkpointing to floppy disks is no longer supported.

New volume for UCN personalities UCN device personalities have been moved to a common directory, &UCN, starting with R500.

LCN Clock Source Node actions with UCN S.O.E LCN systems that include the UCN Sequence Of Events function must perform the following steps.

Step	Action
1	Ensure the LCN Clock Source Node be provided with both the Precision Clock Source/Repeater (PCS/R) and the Precision Clock Cable. These functions are included by default in systems that are 100% K2LCN, 100% K4LCN, or a mix of K2LCN and K4LCN.
2	Ensure the LCN Clock Source Node power supplies and the NIM power supplies are pinned for External Clock. This requirement is for all SOE systems, with or without the K2LCN/ K4LCN board.

Table 5-1 LCN Clock Source Node Setup

METADATA.FF In order to provide UCN support for fieldbus in the R6xx Platform; information about custom parameters on fieldbus devices is presented to the NIM in the form of a new METADATA file. The TPS-Builder updates this file whenever a new fieldbus device type/version is utilized

5.12 UCN Configuration Cabling and Cable Faults

Proceed with
Caution, when
Configuring and
installing the UCN
network

CAUTION

Proceed with **CAUTION** when configuring and installing the UCN network. Refer to the *PM Site Planning* manual. This section should be read and understood thoroughly prior to network installation. Also reference the *UCN Guidelines manual*.

Double cable
faults

R43x, and newer UCN Nodes (NIMs & PMs) behave differently under double cable fault conditions than do UCN Nodes of earlier releases. The major differences are:

Variant	Description
Display	When a NIM cannot communicate on the UCN (double cable fault), its state changes to COMMFAIL. When both NIMs of a redundant pair or a nonredundant NIM experiences a loss of communication (COMMFAIL), then all other nodes are reported as UNKNOWN . UNKNOWN in this case means the NIM can make no judgment about the communication integrity of other nodes in the network when the NIM itself cannot participate in the network due to cable faults. NOTE Recovery from dual cable faults on the UCN may result in an intermediate state, which exhibits a total UCN loss-of-view, or node specifics loss-of-view until both faults are repaired. For additional information, refer to the <i>UCN Guidelines</i> manual.

Table 5-2 Double Cable Fault Variances

Double cable faults, continued

Variant	Description
NIM	In releases prior to R420/R430, a double cable fault condition as detected by the NIM would invariably lead to the failure of one of the two nodes in a redundant pair. For a nonredundant NIM, node failure can only occur under rare circumstances. Starting with R420, the following was implemented: a) If a nonredundant NIM detects a dual-cable fault, it periodically retries to establish communication until the fault is repaired. b) If the primary node of a redundant pair cannot communicate on the UCN but the secondary can, then a failover occurs and the new primary acts as above. The original primary (which has failed) must be reloaded. This operation is the same as in previous releases. c) If the secondary node of a redundant pair cannot communicate on the UCN, but the primary can, then the secondary enters the COMMFAIL state and attempts to re-establish communication. If the primary subsequently loses communication and the secondary regains communication capability prior to the primary, then actions take place as in b) above. If the primary regains communication first, the secondary can also re-establish communication without the necessity of a reload. To summarize, the only time a NIM can fail (failover) due to UCN communication problems is when a secondary NIM can communicate on the UCN and the primary cannot.
PM/APM/HPM	The action here is similar to the NIM. The only time a failure (failover) occurs due to cable faults is when the primary cannot communicate on the UCN, but the secondary can. A secondary node's detection of double cable faults does not lead to node failure. NOTE, there is no COMMFAIL state. UCN nodes that cannot communicate on the UCN are shown as OFFNET.

Table 5-2 Double Cable Fault Variances (continued)

Loss of UCN TimeSynch

Double UCN cable faults may cause loss of UCN TimeSynch. The symptoms are as follows:

- APMs report TimeSynch states of DEGRADED and eventually of FAILED
- More than one NIM reports a TimeSynch state of LCN_OK

The only remedy is to reload both NIMs. This can be accomplished through reloading the secondary, shutting down the primary, and reloading again.

Failure indication variances

The Communication Processor error block from a PM, APM, or HPM Maintenance Diagnostic Display will have either

AA AA 0A 20 or AA AA 0A 21

as the first characters in the display, when a failure is caused by lost UCN communications. In R6xx, the AA AA 0A 21 error will appear only if the secondary node experienced a hardware fault. In previous releases (prior to R41x) this error was announced whenever the secondary went into “alone-in-ring” caused by network communication faults.

The following recovery procedure can be used in this case.

Step	Action
1	Select the FAILED node.
2	Select RECOVER (once). NOTE: SHUTDOWN is a part of the RECOVER command, and when SHUTDOWN is used, the node either comes back to a running state, or goes ALIVE.

Table 5-3 Recovering from Lost UCN Communications

Lost communication failure

If the failure was due to lost communication, then the node will recover within 2 minutes. If the other node was in a state other than OFFNET, the recently recovered PM, APM, or HPM becomes BACKUP, otherwise a failover occurs and the failed node (whose state was OFFNET) is recovered via the SHUTDOWN command.

CAUTION

An OFFNET UCN node does not necessarily indicate that it has ceased local control.

If a UCN node has a status of OFFNET, do not cycle power unless either the COMM STATUS LED or CONTROL STATUS LED are off or flashing twice a second. If the status is FAILED, the PM, APM, or HPM should be SHUTDOWN.

Obtaining UCN Network statistics

A display is available that summarizes UCN Network statistics for PMs, APMs, or HPMs. The display can be used to troubleshoot media faults in the network, and can be accessed by performing the following steps.

Step	Action
1	From the UCN status Display Select UCN CABLE STATUS target.
2	Select UCN COMM STATUS target. The statistics displayed are a subset of those available from each nodes diagnostic display. Cable noise and silence are indicated as well as statistics like number of noise bits, frame fragments and checksum errors. The network statistics can be reset on a network- wide basis.

Table 5-4 Obtaining UCN Network Statistics - PMs/APMs/HPMs

5.13 Process Manager (PM) and Advanced Process Manager (APM)

PM/APM ESD/Surge Test/Calibration The PM and APM have an ElectroStatic Discharge (ESD) rating of 15 kV, and the electrostatic discharge is made to the PM cabinet with the door open or closed.

The ESD rating for PM hardware other than the cabinet is 5 kV. To prevent ESD damage when performing servicing inside the PM cabinet with the dc power on, the user should wear an ESD wrist strap connected to the master-reference-ground bus bar inside the cabinet.

When working on field devices connected to PM/APMs, caution must be taken not to allow ESD directly into field connector terminals. Any unconnected leads should be taped or insulated to prevent contact with objects at other potentials.

PM/APM Sequence Programs - Sequence Overrides **Sequence Overrides**—The Process Module Point displays on the US allow the user to perform phase, step, and statement overrides when the sequence execution mode is Fail, Error, or Pause.

However, when the state is **Off/Loaded/Single-Step**, the step and statement numbers are not visible on the Process Module Display. Therefore, the Forward/Backward commands do not work.

To prevent an error from occurring, the PM points no longer allow forward/backward commands when the state is **Off/Loaded/Single-Step**.

Solution:

When in the **Off/Loaded/Single-Step** state, performing a Start command immediately stops the sequence with the status changed to Normal-Pause. The Override functions may then be executed.

5.13. Process Manager (PM) and Advanced Process Manager (APM)

PM/APM Sequence Programs - CL/PM Compiler switch

To aid in the migration of PMs to APMs, a switch has been added to the CL Compiler. This allows the recompilation of sequence programs before the actual upgrade of the module to an APM.

The switch, `—UPGAPM`, relaxes the type matching between the sequence header identifier and the bound data point, allowing an APM header with a PM bound data point. The sequence object is created in the APM object structure and has the suffix `.NO`.

To make use of the compiler switch, be sure of the following:

- The LCN MUST be converted to R6xx before recompiling the sequence programs.
- Between the time the sequence programs are compiled, and the time they are loaded on the APM, there must be no box or point configuration changes.

Upgrading a PM to an APM?

All CL Programs for PM nodes, which are being upgraded to APMs, must be recompiled after changing the point descriptor in the Sequence Header from PM to APM. This recompilation may be done in one of two ways:

- Either upgrade the PM to an APM, then recompile all sequence programs for that node, or
- Recompile the sequence program before upgrading to APM, using the `—UPGAPM` compiler switch on the invocation line.

Programs running on other nodes that access points in this APM should not have to be recompiled, as long as the slot numbers of the points in the upgraded nodes are not changed.

NOTE: APM/CL code also supports R520 and later features, such as access to time and string parameters, Device Control points, and Array points.

Check your application(s) for special requirements

Since applications running on the system may have special requirements, check your application(s) for specific System Stability requirements. Some control packages require the final controllers be placed in a specific mode prior to performing a software migration.

5.13. Process Manager (PM) and Advanced Process Manager (APM)

CL/PM problems

Certain problems can be encountered when using the PVNORMAL, OP, and OPFINAL parameters of 2 and 3 state Digital Composite and Device Control points in APMs.

If the OP or PVNORMAL parameter is passed as a parameter of a CL subroutine, the state NONE must be added to the end of the parameter state list for 3 state devices. The states \$NULL/NONE must be added to the end of the parameter state list for 2 state devices. This is true for the OP parameter even though the OP may never take on this value.

ATTENTION

When Digital Composite PVs or OPs are passed to a subroutine, an error may occur. For example, if you build a Digital Composite with single OP with STATE0=off and STATE1=on, the PM code would look like this:

```
CALL SUB1 (DIGCOMP.OP)
.
.
.

SUBROUTINE SUB1 (A; IN OFF/ON)
```

If you attempt to compile this code in the APM, it returns the DATA TYPE MISMATCH message under the CALL statement. To make it work in the APM, you must add \$NULL and NONE to the State list as follows.

```
SUBROUTINE SUB1 (A; IN OFF/ON/$NULL/NONE)
```

The PM to APM translator does not pick this up so it is up to the programmer to modify the state list by hand.

**Use of "GOTO"
Statement for
Backward Braching
with xPM/CL
Programs may cause
Program Preemption**

CAUTION

To avoid this problem, the following steps need be followed:

1. Set the CLBACK parameter (number of loops without preemption) to the value desired.
2. The GOTO destination must be a label.
3. The label must not immediately follow a STEP statement, but have some code between the STEP and the label.

5.13. Process Manager (PM) and Advanced Process Manager (APM)

**APM/HPM
TimeSync
Latch Failure**

With a TimeSync Latch Failure on an APM or an HPM, the DISOE events may have bad time stamps. The latch failure must be corrected (that is, the soft failure must be cleared).

This is accomplished by manually swapping the redundant APM or HPM, or by replacing the APM COMM board, or the HPM COMM/CTRL board.

**PM and APM
Analog Output
supports 0
Milliamp (0mA)**

A new feature has been added to the Analog Output IOP that now supports 0 Milliamp operation. To provide this function without requiring system software support, the Analog Output will treat all output requests of -6.9% as a request for 0mA for direct operation, +106.9% will cause physical output of 0mA for reverse. For other output values, the AO output characteristics remain unchanged.

To use this feature, the Analog Output board prom must minimally be at Revision E (3.9).

DI IOPs/LLAI IOPs

DI IOPs can now support Change of State (COS) Alarming in APMs and HPMs. To use the DI COS alarming, you must get the new DI IOP firmware, as shown in the notes in Table 3-8. DISOE IOPs already support COS alarming; therefore, no firmware update is required.

Also, LLA IOPs can now support Open Sensor Detect for RTDs. When enabled, Open Sensor Detect detects open TCs and/or RTDs. It is an extension of the already existing "Open TC" alarm in the LLA I-8 IOP. To use this feature, you must get the new LLA IOP firmware as shown in the notes in Table 3-8.

Both the LLA I and DI IOP firmware are backward compatible. This means they will not cause problems if placed in R3xx or R4xx systems.

5.13. Process Manager (PM) and Advanced Process Manager (APM)

PID Reg point output connection to an APM Array point

The PID Reg point, as long as it uses the external reset feedback algorithm, can write to a numeric in an APM or HPM Array point.

Ideal form PID equation has changed for PM/APM

Adding this function to an existing PM or APM will impact migration of PMs and APMs from R43x versions of software to R520 and later releases. Because the ideal form PID equation is being changed, any existing PID loops that use the ideal form PID and that have derivative time (T2) configured for nonzero values, will notice a change in the PID action and may require retuning.

To ensure against any possible upsets to the process while migrating PMs or APMs to R6xx (from R43x versions of software), any PID RegCtl points which use the ideal form PID (and T2 configured for non-zero values) will have to be put in MANUAL mode before migration.

PM to APM or HPM upgrades

REMINDER: Upgrading from a PM to an APM or HPM requires a hardware upgrade kit and must be performed OFF-LINE. See section 8.21 for more information.

PM / APM / HPM upgrade to EHPM

REMINDER: Upgrading from a PM / APM or HPM to EHPM requires hardware upgrade kit and must be performed OFF-LINE. See section 10.21 for more information.

Array Points and CL**ATTENTION**

Be careful when using Array points in Peer-to-Peer communication. If you do use them, be sure to test the status of the Array point data first. Refer to the following table.

IF...	THEN...
You are doing peer reads ,	first check the Array point parameter BADPVFL.
BADPVFL is Off,	the data is good and can be read.
BADPVFL is On,	reading the data will result in a communication error, which will fail CL programs.
IF...	THEN...

5.13. Process Manager (PM) and Advanced Process Manager (APM)

IF...	THEN...
You are doing peer writes ,	first check the Array point parameter INITREQ.
INITREQ is Off,	the Array point can be written to.
INITREQ is On,	an error will be returned on attempted writes.

APM SOE Journal problems

An Area SOE Journal will historize/report SOE data only for the units that are assigned to that History Module. If units assigned to an Area are assigned to different HMs for journalizing purposes, additional Area SOE journals can be configured. Up to 20 Area SOE journals can be configured, (i.e., you could have one for each configured HM).

APM Uncommanded Change Alarming

Swedish Electrical Regulations requires that cause of a motor failure to be displayed upon detection. Changes have been implemented in the APM Digital Composite and Device Control points to skip PV alarm processing for one scan, in favor of interlock processing whenever the Interlock Alarm Option is configured other than None and an interlock is pending.

This results in Alarm Interlocks always having precedence over Process Feedback alarms. This new behavior is in effect starting from R510.3.2, with APM control software version 51.02 or greater.

5.14 High-Performance Process Manager (HPM)

**Voltage Margin
plug positioning**

This caution concerns the yellow-colored "Voltage Margin" shorting plug located in the "service area" behind the front bezel door of the HPM I/O Link Card.

This plug must ALWAYS be in the "Normal (+5 Vdc)" position during normal HPM operation.

CAUTION

DO NOT MOVE the yellow-colored Voltage Margin shorting plug from the **NORMAL** position unless under TAC supervision, and **NEVER** while "on control."

**Nonredundant
PKGOPT
eliminated**

The Nonredundant PKGOPT has been eliminated for the HPM, since the HPM hardware is always "redundant-capable" (even when running as a nonredundant node). Therefore, all HPMs, even if physically/logically nonredundant, should be configured as follows:

If the HPMM is in a . . .	Choose . . .
7-slot backplane	PKGOPT=REDUN
15-slot backplane	PKGOPT=REDUN_2F

**HPM Migration from
R5xx to R6xx**

CAUTION

Ensure that the HPM is running communication personality 50.03 or greater to go to R6xx. If the HPM communication personality is less than 50.03, the HPM cannot be migrated on-line. Both Primary/Backup HPMs must be shut down before the HPM(s) can be upgraded.

**HPM
Uncommanded
Change Alarming**

Swedish Electrical Regulations requires that cause of a motor failure to be displayed upon detection. Changes have been implemented in the HPM Digital Composite and Device Control points to skip PV alarm processing for one scan, in favor of interlock processing whenever the Interlock Alarm Option is configured other than None and an interlock is pending.

This results in Alarm Interlocks always having precedence over Process Feedback alarms. This new behavior is in effect starting from R510.3.2, with APM control software version 51.02 or greater.

HPM Totalizer

Starting from HPM control software version 53.03 (in R530.3) and greater, there are changes to the Regulatory PV Totalizer algorithm to support true double precision floating point calculation. Due to those changes, it is recommended that the migration to that release software and beyond should only be done when all totalizers, if any, are stopped and reset although other remaining control functions in the node could be done on process.

Migrating any totalizer on process or not being reset would incur any error accumulating up to that point or over. On the other hand, failing back from this software release to any pre-totalizer fix software would put back the totalizer accumulation inaccuracy problem.

**Migrating PM/APM
Checkpoint to HPM**

A checkpoint of PM and APM to be used in a 7 slot HPM should be translated using the split-rack option, i.e., the switch -S. .

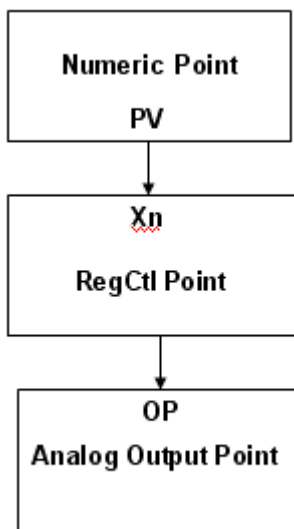
**HPM 15-slot
checkpoint on 7-slot
backplane**

A 15-slot HPM checkpoint cannot be re-used on a 7-slot backplane. The workaround for hosting the 15-slot HPM database to 7-slot HPM is to save the 15-slot HPM box configuration in an EB file. Modify the package option in the EB file from REDUN_2F to REDUN. If there is an IOP present in slot 8 of the 15-slot HPM configuration, physically move the IOP and the FTA connection to a vacant slot in the 7-slot HPM configuration and appropriately modify the file and slot numbers in the EB file corresponding to this IOPs Module number. Load the modified box configuration to the 7-slot HPM. Reload all points from IDF with overwrite option

OP of a REG CTL Point bumps with AUTOMAN and OPSELECT algorithm whenever the point's input recovers from the NaN value

For the AUTOMAN and OPSELECT algorithms, the regulatory control point's output bumps whenever the regulatory point's input recovers from the NaN value.

As an example, consider the following scenario that uses the control loop shown below.



□

The regulatory control point has a control-input connection (Numeric.PV) to a Numeric point, and a control-output connection (AO.OP) to an Analog Output point

Because the control input connection is accomplished in the RegCtl point there will not be any Anti-Reset Windup or Back initialization available.

As shown below, the Numeric.PV value is initially 25 and then is changed to an NaN value while the mode of the regulatory control point is CAS. When the Numeric.PV value is changed back to 25, the output bumps to 100 and then ramps down to 50.

OP of a REG CTL Point bumps with AUTOMAN and OPSELECT algorithm whenever the point's input recovers from the NaN value
(continued)

Action

Numeric.PV = 25
 XEULO/HI = 0.0 - 50.0
 OP = 50.0
 BADCTLOP = No_Shed
 Numeric.PV is changed to NaN.
 Numeric.PV is changed back to 25.

Response

Mode remains in CAS
 OP Bumps to 100 and ramps back to 50

A regulatory control point with the OPSELECT algorithm responds in the same way as described above for the AUTOMAN algorithm.

Workaround: Set the BADCTLOP parameter to a value other than “No Shed.”

xPM Point Detail Connection page status might be misleading

The connection status of GOOD or BAD on an xPM Detail Connection page is based on the value of parameters BADOC1FL, BADOC2FL, BADOC3FL and BADOC4FL. These parameters reside in the xPM and are updated by xPM only when the particular connection is accessed. If configuration changes are made to the destination point, you must retest the connection.

Changes to the HPM's SI ScanRate (SCANPER) via the DEB are blocked by the HPM if any SI Array Points exist in the HPM

ATTENTION

If the HPM contains only Non-SI Array points (i.e., Array points with "External Data Option" set to "NONE"), ignore this notice. If the HPM contains SI Array points, please read the following information:

If modifying the HPM's SI scan rate (SCANPER), remember the following:

Changing the SCANPER will affect the number of SI Array points that can be scheduled, per the established configuration rules.

1. Create EB files for all SI Array points in the affected HPM. Manually record local array data that cannot be reloaded from the DEB. Note all user applications (schematics, CLs, groups, etc.) that refer to these points.
2. Delete all SI Array points. Note that if HPM is not in the RUN state, the actual delete actions (invisible to the user) will be delayed until HPM is put back to the RUN state. This may have some unexpected effects regarding the SI Array point building. (Also see item 5 below.)
3. Command the HPM to the IDLE state and load the SCANPER change for the HPM.
4. Reload the SI Array points and restore the local array data. While loading the SI array points, if HPM returns an error message regarding the available SI array slots and the user is certain that the number of SI array points has not exceeded the limit per configuration rules, it is recommended that the HPM be put to RUN state for a few seconds to allow the previous delete actions to complete.
5. Check all user applications that refer to the reloaded SI Array points and restore as required. (The internal "local routing code" for the points may now be different.)

**Schedule Info
Display of HPM Not
Updated with
Proper Flag Slot
Number after
Migrating to R650
(or newer)**

With TPN R650, the number of Alarmable flag points available in HPM has been increased from the first 128 to the first 1024. However, upon migration of HPM from a prior release to this one by online migration using the hot failover method, the schedule info display in the detail display of HPM continues to show the number as 128 instead of 1024.

Note: This problem does not occur in other ways of HPM migration (For example, Checkpoint Migration and Offline Migration.) This does not impact the processing of any of the 1024 flag points.

As a workaround, after migrating the HPM by the online migration using the hot failover method bring the HPM to the IDLE state. Now, perform a Point mix change on the HPM box and load the changes. This will update the schedule display with the correct number of HPM flag points available (1024). Note that a Point mix change can be done by either incrementing or decrementing any of the non-process affecting parameters, namely NFASTCTL, NFASTDC, NFASTLOG, NFASTPV, and NFASTDEV. These changes can be reverted back, if required.

5.15 Fieldbus IOP

Checkpoint Save can fail if Fieldbus IOP errors encountered Under some circumstances, parameter errors in Fieldbus IOPs can cause the failure of an entire HPM checkpoint. Users are cautioned to ensure that Fieldbus IOPs that have experienced loading errors be excluded from checkpointing. Usually this can be done by not forcing the DBVALID parameter to its TRUE condition.

Problem retrieving new Fieldbus Device META data The Metadata for a Fieldbus point remains in memory until a point with a different block id is invoked. For example, if a SECM point is on the display, calling up a PECM point will refresh the Metadata memory. This is only a problem if a Fieldbus point is being displayed at the time a new Metadata file is put onto the system.

**Fieldbus MetaData
File Revision
Handling**

MetaData file revision checking has been added to the NIM and the Universal Station/Native Window. When a MetaData file revision number is verified as correct by the NIM, the MetaData file revision information is now reported on the NIM Detail Status page.

MetaData file revision mismatches are reported on the Status Detail page of the NIM. The NIM will transition to a WARNING state and display the following error message:

METADATA.FF file revision error - NIM read MMM, expecting NN

where MM = the major and minor revision numbers in the METADATA.FF file read, and NN = the revision numbers expected by the NIM.

File revision mismatches found in the Universal Station/Native Window are reported on the point detail view page, which is attempting to access the MetaData. The following error message is displayed:

METADATA NOT AVAILABLE

File ver MM, expecting NN

where MM = the major and minor revision numbers in the METADATA.FF file that was read, and NN = the revision numbers expected by the Universal Station/Native Window displays.

Note: The US/GUS/EST may display the message, "METADATA NOT AVAILABLE," without the line showing the file version information. If this occurs, the MetaData file revision number is correct. The error relates to the VIEW page's attempt to access unavailable MetaData information for the point being processed. The problem involves using a MetaData file that is incompatible with the point database that is loaded.

Segment Status Display does not present the complete status of a given FBus Device.

For a given device on the Fieldbus Segment Status Display, the indication “DEV OK,” which is derived from the parameter PDSTAT, is an indication that the device resources are available. It does not give a full indication of the state of the device. Pending changes to the display, users are advised to check the parameter DEVSTAT on the PECM Detail Display.

First Call-Up of SECM Detail Display produces a Display Data Base Error indication

If, after the initial load of a US, the detail display for a Fieldbus SECM point is invoked from the point detail display (i.e., by using the sequence “DETAIL”-key, enter SECM ID, select “FBUS SEGMENT STATUS” target), the display will be invoked. However, the status message “Updating display..” will remain in the upper left hand corner of the screen, and the error “Display Data Base Error” will be displayed. This appears to be a result of incomplete initialization of the station’s global display database at startup.

The error is non-fatal, only occurs on the first invocation after the load of the US, and will not reappear on subsequent repetitions of the invocation sequence.

The error does not appear if the Link detail is invoked by invoking the IOP Status Display (via the UCN STATUS, HPM STATUS path) and then selecting the Link of interest.

**CL Compiler
problems with
Fieldbus data**

FBus Device Specific Data Retrieval Response adversely affects AM/CL programs – AM/CL programs that reference Fieldbus Device data, need to be run in the background mode with background delays, otherwise the AM/CL execution would abort.

CL Compiler failure on FBus enumerations – Due to the inability of the CL compiler to handle enumerations having more than 255 members, the following Fieldbus parameters (which use a 500-member enumeration, \$FFEUIDX) will generate compiler errors if they are included in CL programs.

The parameters: FFFFUIDX, FFOPUIDX, FFPVUIDX, FFDXUIDX

AM/CL is unable to read/write LX-1400 Device Mapped Parameters – Accessing Generic Parameters from an AM/CL program must be run in the background mode. If they are run in the foreground run time errors will be generated.

AM/CL program unable to write to Fieldbus CM Point Mapped Parameters – Users should be aware that there are a number of restrictions when accessing Fieldbus CM point mapped parameters FLOAT[n], US16[n], US8[n], and VS[n] via AM/CL. Specific restrictions noted to date:

1. CALL MOVE_PARAMETER will return an error for US8 or US16 arrays if the array is not completely filled with the proper element type.
2. An attempt to SET LOCAL_VARIABLE = <FB>,US8(n) where the nth element is not the proper type will produce a runtime CNFERR.
3. Attempts to write to mapped parameters with the builder set to ENGINEER generated KEYLEVEL errors; the DATACHNG schematic was able to perform the same writes.

Other errors may exist which have not been specifically reported. Users should exercise caution when incorporating CM point mapped parameters in AM/CL.

REDTAG REDTAG is not supported for TPN Fieldbus CM points.

Special Characters If the Fieldbus special characters are encountered on a US without the FieldbusFont set, then blanks are displayed.

PVSOURCE and MODATTR Parameters Although PVSOURCE, PVSRCOPT and MODATTR/NMODATTR parameters exist for FB CM points, they are not operational. Changing their value will not impact display, or control operations.

The RESTORE ALL Function does not Check errors on Individual Fieldbus IOP Restores

CAUTION

If the user wishes to use the RESTORE ALL function from the HPM Status Display to restore checkpoint files, the user has to make sure that all the underlying FBIOPs truly need to be restored. If so, the user then also has to make sure that all the Links on every FBIOP that would be restored are SHUTDOWN. Otherwise, such an FBIOP database might potentially be corrupted.

Potential Fieldbus IOP failure on Writing Generic Publisher Parameters

CAUTION

The User should not attempt to write to any of the Generic Publisher parameters IN[] or IND[] via TPS components such as AM/CL, without configuring these in a control strategy using TPS Builder. Writing to these without a valid configuration may result in a failure of Fieldbus IOP.

Several Devices Cycle between VERIFYING AND BAD CONN during Download

CAUTION

DO NOT change the First Unpolled Node and Number of Unpolled Nodes Parameters in the TPS Builder Forms. Use the ones defaulted by Builder. If the devices are out of this address range, please use NI configurator to bench configure the addresses.

Extraneous Characters are Added to the display of Fieldbus Generic Strings

CAUTION

When Fieldbus generic parameters of type “String” are written from the DATACHNG display, they are displayed with extraneous trailing characters.

Corruption of Display buffers by METADTAx parameters

CAUTION

Users should avoid including Fieldbus parameters METADTAF, METADTAR and METADTAT in schematic display requests, such as DATACHNG. These parameters are blind records, and corrupt the Data Access buffers supplied to the displays resulting in data display errors. There is currently no automatic mechanism in the system to exclude these parameters.

Enabling FF Devices as Link Master Capable Devices on Installation

CAUTION

In order to use Link-Master capable FR3051, FR3244 devices or the Yokogawa EJA100 device as a Link Master, use the NI configurator to bench-configure the BootOperaFunctionalClass parameter of these devices with a value of 2 (LinkMaster).

Note, this step is necessary only if the manufacturer does not default the BootOperaFunctionalClass parameter of these Link-Master capable devices to Link Master. Please check with the manufacturer for default settings.

No Devices on a Fieldbus Link should have a Device Address of 20 (14Hex)

CAUTION

This address is exclusively reserved by the Fieldbus IOP interface and the FBIOP will fail to start the link.

Cold Restart of HPM does not put Reg Control MODE to MAN on FIELDBUS Points

CAUTION

For a HPM cold-start, if a RegCtl point has an output connection to a Fieldbus device, its Mode remains unchanged and will not be set to MAN. This behavior is different from that of a RegCtl point, which has output connection(s) to conventional AO/DO IOP(s). This is by design to take full advantage of Foundation Fieldbus device functionality.

**Default Fieldbus
Device Addressing
Concern****CAUTION**

If devices of the same addresses are connected to a Fieldbus Link, such device(s) with duplicate addresses assume default addresses. These default addresses are limited to four which are 0xF8, 0xF9, 0xFA, 0xFB. Once these four default addresses are occupied any more devices with duplicate addresses would not be visible. This is a limitation of Fieldbus as it has specified only four default addresses. As and when devices at these default addresses get assigned to a unique address the other “invisible” devices start occupying the default addresses and can then be seen. This will continue until all the devices have a unique address. Thus, a user is advised to have a maximum of 4 duplicate addressed devices. Any more than four will not be visible on the link. The user can, however, rely on device matching function for any combination of duplicate addressed devices. Device matching will continue uninterrupted.

**Trip Limits shown in
the Alarm Summary
displays are not
always correct. They
could get out of Sync
Under Certain
Specific Conditions****CAUTION**

If a User changes alarm limits on a Function Block in FF device via detail displays or via other schematics the FBIOP does not cache the new alarm limit until it receives a data change report from the FF device. However, if an alarm has been generated before the time the FF device sends the alarm limit change report; the FBIOP supplies the previous limit value.

**FBIOP Does Not
Recover transducer
Block Alarms****CAUTION**

The FB IOP does not recover Transducer Block alarms. This could happen if the device and the IOP for some reason do not communicate while the device is reporting Transducer Block Alarms. The miscommunication could happen by either the device or FBIOP physically getting disconnected or PTEXECST of the corresponding PECM and/or FBCM set to INACTIVE. After establishing communication, the IOP recovers all the Process and Block Alarms, but not the Transducer Alarms.

Simultaneous Loading of the same Link on the same IOP from Two Different TPS Builders

CAUTION

Simultaneous downloading of the same Link on the same FBIOP from two different TPS Builders will result in an erroneous download and the Link and FF Devices may not be functional. A re-download may be required from a single TPS Builder.

Poor Throughput on Mapped Parameters

ATTENTION

Generic mapping parameters will experience a performance problem when displaying on any schematics or accessing them via other TPN entities. It is recommended to not have more than 5 Class C parameters being accessed at any given time by a given TPN entity such as schematics on a particular device.

Softfailures generated During Download

ATTENTION

It is possible to see Softfailures being generated during a download to a Fieldbus link. These could be transitory Softfailures as the FF devices come up on the link as they are being configured.

First Detail Display Page for the Fieldbus AO FBCM Point Incorrectly Displays the Value of the PVAUTO Parameter

ATTENTION

The first detail display page for the Fieldbus AO FBCM point shows the value of the PV AUTO parameter. The PVAUTO parameter value does not apply to the AO FBCM point and should be ignored.

ATTENTION

PVAUTO Does Not Show on the Detail Display for a Fieldbus CM Point Using the PID Function Block

**Solution for Clearing
Out Soft Failures
Produced By
Downstream
Function Blocks in
Reaction to failed
Upstream Devices**

ATTENTION

If an OUT from a function block (in the failed device) was providing an input (IN, FF) to a downstream function block (example PID located in a FBus device that continues to operate), then that downstream function block will generate a IN FAIL block error. This block error, in turn, generates a soft failure against the FBCM of that downstream function block. Subsequently, the SYST_STAT LED again comes On. Setting the PECM of the failed device to inactive does not get rid of this soft failure, since the soft failure is originating from an input failure condition being detected by a working FBus device. The only means by which an operator can extinguish the SYST_STAT LED in this condition is to perform the following steps:

Inactive the PECM of the failed device - This causes the Device Missing soft failure (on the failed device) to be removed.

Disable Block Alarms on the downstream function block - This cause the IN FAIL soft failure to be disabled. At that time the SYST_STAT LED would be extinguished.

The procedure for disabling block alarms on downstream function block is:

Call up the FBCM's Detail Display for that block.

Proceed to FB View page, subpage 2

- Select the Alarm_Summary target on the display, this causes the Alarm enable/disable selections to appear to the left.
- Select the last entry in this list (Block_Alarm box), and press ENTER.
- Wait a few seconds - the display updates. One will see the "Block_Alarm" text appear in blue in the disable field of the displayed Alarm_Summary record.

This procedure disables all block errors in a function block. (Fieldbus Foundation does not support individual disabling of each block error.) Therefore, once the failed upstream device is corrected, and again functioning, the operator must go back and enable the block alarms on this downstream function block. This needs to be done so that proper soft failure reporting can then again occur (by this downstream device) for input failure conditions. The operator will not be prompted about re-enabling the block error detection in the downstream device's function block.

**Fieldbus IOP –
ST3000 PID Block
Consideration**

ATTENTION

The first detail display page for the Fieldbus AO FBCM point shows the value of the PVAUTO parameter. The PVAUTO parameter value does not apply to the AO FBCM point and should be ignored.

**Fieldbus Link
Loading
Considerations**

ATTENTION

AM/CL ACCESS RESTRICTIONS – THE HPM PARAMETERS THAT ALLOW AM/CL TO ACCESS TO THE FIELDBUS PARAMETER VALUES HAVE BEEN CLASSIFIED AS CLASS A, CLASS B, AND CLASS C PARAMETERS AS DESCRIBED BELOW.

Class A parameter	IOP parameter that is contained in the IOP. Both read and write requests are directed to the IOP database.
Class B parameter	A device parameter that is cached (and possibly mapped) in the IOP. Read requests are directed to the IOP database. Write requests are directed to the device.
Class C parameter	A device parameter that is not cached in the IOP. Both read and write requests are directed to the device.

Please note that Write Access to Class B parameters and Read/Write access to Class C parameters must be factored into FF Link loading and bandwidth usage.

For more information on AM/CL access restrictions, please refer to the CL/AM Reference Manual (AM27-610), and to Section *Fieldbus Mapped, Projected, and Converted Parameters* of the PM Family Parameter Reference Dictionary (XPM09-620A). Both documents are located on the TPS documentation CD media.

HPM/CL Access Restrictions - The HPM parameters that allow HPM/CL to access to the Fieldbus parameter values have been classified as Read-Only and Read/Write. For more information on HPM/CL access restrictions, please refer to the CL/HPM Reference Manual (HP27-610), and to Section *Fieldbus Mapped, Projected, and Converted Parameters* of the PM Family Parameter Reference Dictionary (XPM09-620A). Both documents are located on the Knowledge Builder CD.

HPM Corrupted Checkpoint File

ATTENTION

Under certain conditions of HPM IOP configuration and Fieldbus IOP failure, it is possible to create a corrupted HPM checkpoint file which will result in a NIM range error failure during a restore operation. The scenario which results in this corrupted checkpoint:

1. There must be at least 25 IOPs of all types configured on the HPM.
2. The scenario assumes one successful save of all IOPs has occurred.
3. If the next save occurs when a Fieldbus IOP has failed, the resulting checkpoint (which is now the most recent) will be corrupted.

Recovery:

1. Ensure that the Fieldbus IOP is operational and perform one more checkpoint operation.
2. If the failed Fieldbus-IOP was recovered by resetting it, then its data will not be there. In that case, after the successful save checkpoint operation, you can copy the older good "Fieldbus-IOP" checkpoint file (not the HPM checkpoint file whose master record file is corrupted) on top of (delete then copy) the just saved but blank Fieldbus-IOP checkpoint file (blank because the Fieldbus-IOP was recovered).

AI Alarm Limit Settings from View pages are Not Checked

ATTENTION

The AI alarm limits on the Function Block view pages are sent to the Foundation Fieldbus device, which may, or may not perform any data validation. If it is necessary to change Function Block Alarm limits, and Traditional TPN data validation is desirable, then Operators should use the 1st page of the Detail display.

Note:

Any data owned by the Fieldbus device and changed through the View displays is not validated by TPS. It is the responsibility of the specific Fieldbus device to validate per FF specifications the data that is written to it. However, data owned by the Fieldbus device that is modified through TPS mapped parameters would undergo the standard TPS validity checks.

**Fieldbus IOP Not
same as Existing
Other IOPs**

ATTENTION

This release of TPN does not allow SECM, or PECM Points to be configured in Group

ATTENTION

This release of TPN requires the following interaction to perform Foundation Fieldbus IOP restore Operations (i.e., shutdown and reload both primary and secondary HPMs with out errors):

IDLE HPM and All IOPs < Traditional TPN >

For Each Fieldbus IOP and for each active Link on the IOP

Set SECM.Ptexecsts = Inactive

Set SECM.COMMAND = ShutDown

For Last SECM on IOP Wait for SECM.Status to go to STK LDED

Note:

If any Link is still running when you reload the HPM, you will get errors indicating the IOP database was not restored.

**Fieldbus Operator
Display Problems**

ATTENTION

Standard Trend display of Fieldbus points is not operational in this Product Release. Upon callup, the error message "Invalid Point Type To Trend" will be displayed.

Changes of SHED_RCAS, SHED_ROUT from the View displays is not operational. On configuration, these default to a value of 32000 (1 sec) in the device. As a Workaround the user would need to change them from a running remote cascade control loops from a HPM of 1 sec macrocycle to the desired value.

Resource Block parameters can not be changed from the Operator view display. The NI configurator tool would be required to make changes to these parameters.

**PECM Detail Display
Problem**

ATTENTION

If the PECM point is accessed directly (Detail key, tag name) the displayed information is incorrect. The PD TAG is not representative of the tag detailed. The PECM point must be detailed by using the IOP Slot Summary display or selecting the device through the segment display to provide correct data.

5.16 Logic Manager (LM)

Using external timers on LM

If any external shutdown timers are in use in conjunction with the LM, be sure to check the timer function and make appropriate adjustments. The time will vary as a function of the amount of ladder logic in the LM. As you increase your ladder logic, you should adjust the timer function as necessary.

Performing a swap of LM primaries requires the external timers to be set at 6.5 seconds. Set external timers to a time of 6.5 seconds or greater to eliminate problems in this area.

LM I/O diagnostic

As of R400, the Logic Controller (LC) program alters the original design of the LMM point algorithm scan. It no longer stops its scan when a segment of the serial I/O system fails. It continues to scan using the I/O, which is still active. This continual scanning operation renders the previous “scan loss detect mechanism” ineffective.

To adequately diagnose serial I/O faults in the Logic Manager (LM), use the LC control program found in the R500 revision of the Ladder Logic Program shown in the *Logic Manager Implementation Guidelines* manual, *Appendix A*.

For nonredundant LMs, refer to the *Logic Manager Implementation Guidelines*.

For redundant LMs, refer to the *Logic Manager Implementation Guidelines*.

Choosing PM CL Sequence Program instead of Ladder Logic File

If an operator mistakenly selects a PM CL sequence file while performing a Ladder Logic load into the LM, the message “File FMT” will appear at the US and the load will not be performed.

**Logic Manager
Digital Input Points
can Default to OFF
State During
Swap/Failover
Operations**

CAUTION

During Logic Manager failure or swap operations, there is a brief time window, prior to the first scan of the PLC data base, during which the PVFL of Digital Input points can be defaulted to its OFF state regardless of the actual state of the input. This can result in inappropriate actions by any points, CL programs, or schematics, which use the affected points as inputs. If the point is configured for change of state alarming, alarms will be generated.

Workaround:

Use the BADPVFL to determine the validity of the PVFL value. Note that the BADPVFL is correctly set to ON during the swap/failover operation.

5.17 Fail Safe Controller – Safety Manager Module (FSC-SMM)

FSC-SM and time synchronization	You may <u>not</u> use a 10006/2/2 (DCF) in an FSC network that is connected to a TPS UCN. When a 10006/2/2 is configured in one of the FSC systems in the network, the TPS time synchronization will be overridden.
Switching off a central part	The only way to shut down a Central Part is described in the FSC Software Manual - Appendix D - Step 8 - Switching off a Central Part: Trip the Central Part by pulling the WD flat cable, then turn the CPU keyswitch to STOP.
FSC-SM Control Functions Manual, Section 4.7 Logic point	Input Connections: When the input to the logic slot is obtained from a flag point in another node then the PVFL parameter must be used instead of the PV parameter.
DC point parameters that are not supported	The DC point processing does not include the following parameters: • CMDFALFL • CMDFALTM • OVRDALOP • OVRDCONF • OVRDDESC • OVRDIOFL • OVRDI1FL • OVRDI2FL
Force count on detail display	The value of Force Count on the FSC-SMM detail display may be higher than the actual number of forced variables. This is because the Force Count is also incremented for each watchdog group on an output module in which a forced variable is located.
Parameter write requests during OLM	During on-line modification (OLM) of the Control Processor application of an FSC-SM, it is not allowed to execute parameter write requests (or restore a checkpoint). This may result in a crash of the secondary SMM.
Control program save/restore	The CTL PRG LOAD and CTL PRG SAVE options on the US, that are meant to save and reload different revisions of the FSC application program to and from the HM, are not supported.

**FSC-Safety Manager
(FSC-SM)****ATTENTION****Changing PVSOURCE for DI, AI, and DC points**

When changing PVSOURCE from AUTO to MAN in a redundant node, the point PV in the primary node will reflect the last value obtained from FSC while PVSOURCE was still set to AUTO (as expected). However, the PV in the secondary node will not be updated. This will cause the PV to drop to zero (digital points) or NaN (analog points) when a primary node swap occurs. The workaround is to manually write the PV directly after changing PVSOURCE from AUTO to MAN.

Dual Alarms in Journal

When a primary node swap occurs, the new primary node will report all active alarms as it starts processing points. In addition, a node failover causes an event recovery process to be started. This has no effect on any alarms in the alarm summary that had already been reported before. However, the alarm journals will show one or two alarms per point with timestamps close to the time of the node failover. These are re-reported alarms (with no return from the alarm state since it was first reported) and can safely be ignored.

5.18 Triconex Safety Manager

Triconex –SMM Digital Sequence of Events (DISOE)

ATTENTION

- **Triconex Time Must Be Set to UCN Time Before Starting the SM-TRICON):**

In order to insure proper operation of SOE points associated with SM-TRICON modules, it is important that the Trichina time be synchronized with the UCN time prior to loading/starting the SM-TRICON. Synchronization of the Triconex to the UCN time is automatic, PROVIDED the Triconex time is within 5 - 10 minutes of the current UCN time.

The Triconex time can be changed using the appropriate Triconex display. The Triconex me need not be set exactly. It is sufficient to set the Triconex time to within 5 -10 minutes of the current UCN time before loading/starting the SM-TRICON.

- **DISOE following customer ladder logic changes:**

Changes to large ladder logic sequences in the Triconex PLC may cause the PLC to halt the reporting of Digital Sequence of Events (DISOE) changes to the Triconex-SMM. Field experience has shown that the "download changes" option, when applied to large ladder logic sequences, interrupts the processing of the Triconex DISOE buffers due to a timing problem in the PLC. A workaround is available: Rebuilding of any DISOE point will restart the reporting of DISOE changes.

Triconex V9 CPU Low-density Chassis

ATTENTION

When a TRICONEX V9 CPU is configured with a low-density chassis, it will display on the SMM Detail Display as "TRICON8".

The erroneous platform indication is a result of the SMM misinterpreting the low-density chassis as an indicator of a TRICON8 CPU. There are no indications that other areas of the display are affected. The anomaly is limited to the display of the platform type.

Triconex V10

ATTENTION

The Detail Display of SMM shows the prior version of Triconex in the "PLATFORM" field even if the Triconex is updated to Triconex version 10

Workaround: Perform the following steps to verify the Triconex Version 10.

1. Run Triconex Diagnostic Monitor, and connect to Triconex Node.
2. From the **Commands** drop-down menu, select **Display Firmware Version**.
3. The version for Triconex Enhanced Main Processor displayed should be 6056 in the first column of "Version(s)" display."

5.19 Hiway Gateway (HG)

Point Building limits

There should not be more than 512 point aliases for each hiway box; there is a potential for HG failures if this limit is exceeded. The limit is not enforced because aliasing is sometimes a useful way to temporarily simulate a process when limited hiway equipment is available.

Undocumented errors

ATTENTION

If regulatory input connection parameter names (PVSLT SRC, SPSLT SRC, etc.) are not correctly configured, an error will not be reported until the point is loaded to the box. The user should verify input connections are properly built when point loading is completed.

For example, if the user enters an invalid tag, the slot source is not updated in the process box.

When reloading an analog input Multifunction Controller entity that is simultaneously on display on another node, the Real Time Journal intermittently includes a series of software errors (D\$ _DH_RETRV4) along with the ENTITY RELOADED message. These errors can be ignored.

HG points must be in manual mode

HG points must be in manual mode, and they cannot have any linkage to Contact Cutout, before they can be deleted.

Pre-R500 Exception Build files won't load

Exception Build (.EB) files created before R500, containing HG alarming point definitions, will not load in R520 and later releases, because of alarm (ALPRIOR) changes.

Translation of these EB files is covered in section 7.6, under the label: EBX Translator.

5.20 Application Module (AM)

Standard enumeration set state name changes

Some standard enumeration set names change between releases. As a result, if an attempt is made to link a CL/AM block that was compiled previous to the new release, and the enumeration set has changed, the link will fail.

The link error produced is:

```
<point.parameter> Compile Type Does Not Match Actual Type.
```

The parameters which are affected by these changes and their corresponding enumeration type are listed for the various migrations:

MIGRATING FROM R2XX, R3XX TO R400 OR HIGHER	
Parameter	Enumeration Type
HIGHAL	ALMTYPE
DISPTYPE	DISPTYPE
NODESTS	DSPSTAT
NODETYP	HPNNDTY
PNTNODTY	HPNNDTY
IOMLHMST	IOMHF
PNTTYPE	PNTTYPE
PMMCMD	\$UMCMD
PNTMODTY	\$UPCMDTY
MIGRATING FROM R4XX TO R5XX	
Parameter	Enumeration Type
PNTTYPE	PNTTYPE
MIGRATING FROM R5XX TO R60X OR R61X	
ENT_TYPE	ENTYTYPE
DISPTYPE	DISPTYPE
PNTTYPE	PNTTYPE

Table 5-5 Changed Standard Enumeration Set State Names

To correct the situation, perform the following steps.

Step	Action
1	Recompile the CL/AM block.
2	Retry the link.

Table 5-6 To Eliminate Errors from Enumeration Set Changes

Do not use the –OCD compiler switch unless you want to change your Custom Data Segment definition.

If a Custom Data Segment parameter is any of the above Enumeration Types, a reference to the parameter in a CL/AM block also causes the block not to link. Use the same solution shown above.

Memory Sizes

The total amount of memory in an AM can range from 3 Mw to 16 Mw (Mw = one million words).

The amount of **user** memory available in an AM (PSDP parameter AMMEMAOP) is a function of the total AM memory size.

Available User Memory

The following table shows how the AM memory available to the user is related to the total AM memory size.

NOTES:

- Options such as AM redundancy and K4LCN processor reduce available AM user memory (see Figures 2-3 and 2-4 in the *Implementation Guidelines Manual, AM12-510* for additional information).
- Memory mapping is different for an A^xM.
- AM's maximum memory used is dependent on the hardware used. For example, the AM memory with:
 the HPK2/HMPU processors uses a maximum of 7 Mw,
 the K2LCN processor uses a maximum of 7 3/4 Mw, and
 the K4LCN processor exists in 4 Mw, 8 Mw, and 16 Mw size only.

The available AM user memory with 7 3/4 Mw is 5,786,245 words.

- The A^xM available user memory for the: 6 Mw A^xM corresponds to the 3 Mw AM; 7 Mw A^xM corresponds to the 4 Mw AM; 8 Mw A^xM corresponds to the 5 Mw AM; and 16 Mw A^xM is 11,058,224 words.

Total AM Memory Size	Available AM User Memory (R6xx)
3.0 Mw	1,016,358 words
4.0 Mw	2,020,545 words (2,009,435 for K4LCN)
5.0 Mw	3,024,731 words
6.0 Mw	4,028,918 words
7.0 Mw (HPK2/HMPU maximum)	5,033,105 words
7.75 Mw (K2LCN maximum)	5,786,245 words
8.0 Mw	6,037,291 words (6,020,211 for K4LCN)
16.0 Mw	14,070,784 words (14,045,430 for K4LCN)

Table 5-7 Available AM User Memory

The User Memory for the “minimum” sized AM (3 Mw) is:

1,016,358 words = 993 Kwords (nmbr. of words ÷ 1024) =

1986 Kbytes (nmbr. of Kwords x 2)

To calculate memory in...	Use this formula...
Words	$1,016,358 + [(Nmbr\ of\ Megs\ over\ 3) \times (1048576 \times .957667)]$
Kwords	$1,016,358 + [(Nmbr\ of\ Megs\ over\ 3) \times (1048576 \times .957667)] \div 1024$
Kbytes	$1,016,358 + [(Nmbr\ of\ Megs\ over\ 3) \times (1048576 \times .957667)] \div 1024 \times 2$
Where: 1048576 = words per Mword, 1024 = words per Kword, 2 = Bytes per word	

Table 5-8 Calculations for User Memory in Larger AMs

Exceeding available user memory

The total AM memory used increases with each point loaded into the AM and with each CL block linked to a point. If one of these operations results in the total memory used exceeding the total user memory available, the operation is aborted and an error message is generated.

**R6xx AM
CALCULTR.PV
algorithm and AM
Rolling Average
Checkpoint
incompatibility**

In accordance with the Load And Go philosophy, after certain functions are used they modify the checkpointed database in such a way that it is no longer compatible with the old software. The AM PV Calculator Algorithm and AM Rolling Average Algorithm are such functions.

If you load an AM software release before R530 with an R6xx checkpoint that contains an AM Regulatory point build with the CALCULTR.PV algorithm, the AM will fail when the Calculator algorithm point is accessed.

If you load an AM software release before R640 with an R640 (or newer) checkpoint that contains an AM point using the AM Rolling Average Algorithm option, the AM will fail when this point is accessed.

**Avoid potential AM
failure**

Honeywell strongly recommends you archive your old AM checkpoints before upgrading to R6xx. Then, if for any reason you need to revert to a prior release, these archived checkpoints will not contain the CALCULTR.PV or AM Rolling Average and therefore avoid the problem.

If you do not have an archive of old AM checkpoints and still must revert back to an earlier release, do the following:

- Delete any Regulatory point built for the R6xx AM that contains the CALCULTR.PV algorithm.
- Build a replacement Regulatory point using the prior release
- Delete the AM Rolling Average option from any point that uses this option.

**Sizing of Units in
16-megaword AMs**

The AM 16-megaword K4LCN has a special user memory requirement regarding the checkpoint buffer size for the largest unit (MEMCKPT).

If the AM point database is very large in only one unit, point building may get out-of-memory errors even when a large amount of user memory is available (1-2 megawords of MEMFREE). This is due to a hardware constraint that restricts the checkpoint buffer for the largest unit to a portion of the lower 8-megawords of memory only.

This out-of-memory condition will not occur on other memory sizes of AM K4LCN, or if the point database is distributed across multiple units. When this condition occurs the free memory remains available for other AM optional uses other than point building.

Redundant AMs

Important note for redundant AM (CLM) nodes using the “FURTHER EXTERNAL DIRECTIVE” option (third page of the NCF node configuration): Each node of the redundant pair must have its own APCFGnn.X. file. (where nn = node number). Prior to R611, only the primary AM (CLM) of a redundant pair needed this file.

5.21 Network Gateway (NG)

Network Gateway Interface

Network Gateway Interface (NGI) “A” must be installed in the lowest available slot. NGI “B” must be in a higher slot than “A.” When “A” is removed or is not present, its slot must be left empty. Any variance from these rules results in driver and communication errors and maintenance recommendations being reported against the wrong cable.

If an error occurs during a transaction, e.g., File Copy, across the Network Gateway, an error will be returned to the initiator of the request.

Note, that by design, there is no attempt to inform the destination node.

ATTENTION

R6xx Network Gateway can only communicate with R5xx and R6xx systems, not R4xx systems.

CAUTION

Global addressing of Network Gateways is not supported in this release. The user should select "LOCAL" PIN addressing from the SYSTEM-WIDE VALUES and NG LOCAL SYSTEM ID configurator display.

Access of Remote Network History Disallowed

With R400 through R430, certain unit configurations of LCNs would allow a local LCN to access history data directly on a remote LCN HM via PV History Retrieval. This was a misapplication of the Network Gateway’s capability and did not always work satisfactorily. Starting with R431 and R500, the LCN’s data access mechanism has been changed to block such requests.

System Title

The “System Title” for both local and remote LCN has to consist of at least one Alpha character (non-numeric, non-blank).

Network Gateway users must observe the following TPN Interoperability Restriction PAR #1-V01M

CAUTION

A TPN system running R60x software (R600.1 through R601.5) must not attempt to access Fieldbus point/parameter data on a remote R61X or newer system via Network Gateway. Doing so could cause the requesting node (US, AM, CG) on the R60x system to fail.

5.22 Universal Station^X & Application Module^X

ATTENTION

Please refer to the *TPN software Change Notice* for current information on U^XS software. The U^XS was withdrawn from sale November, 1997.

ATTENTION

Please refer to the *TPN software Change Notice* for current information on A^XM software. The A^XM was withdrawn from sale December, 1999.

5.23 GUS Special Considerations

GUS picture restart	All GUS pictures should be terminated while a GUS node is being updated. After installing the LCN Personality it is recommended that before attempting to run GUS pictures, restart the entire GUS node.
GUS picture validation	GUS pictures do NOT require revalidation for an LCN upgrade from R5xx to R6xx.
GUS restart for LCN operations	GUS use for standard LCN operations that do not involve GUS pictures, only require that they be reloaded with the newly installed pers_US.
GUS minimum configuration	Refer to the TPN and TPS Software Change Notices for minimum software requirements of non-TPN software on TPS nodes.

Be sure to use the correct Version of GUS Back-Plane Modules

ATTENTION

Reference the current TPN and TPS/GUS Software Change Notices for **minimum** revision levels of GUS back-plane modules. Be sure to update your system HM with the correct versions for proper operation of your GUS node.

MSCHEM.LO, CSCHEM.LO (&Z14)

GUSALA.LO, GUSALAG.LO, YGOCX.LO (&Z12)

These back-plane modules often increase in size with each new version. Ensure that your NCF configuration has allowed sufficient space for back-plane software. If not, make the appropriate NCF changes for your custom configuration.

6. Distribution Media

6.1 R6xx Software Distributed on Zip Disks

Overview

On an all US system, R6xx software is provided on Zip disks only. Software will no longer be distributed on Bernoulli.

On an all GUS system, R6xx software is provided on CD-ROM only. Software will not be distributed on Zip or Bernoulli.

6.2 R6xx Zip Disk Contents

R6xx Zip disk contents

The following table lists the contents of the Zip disks. Your planned R6xx system configuration determines which Zip disks you will receive. Possible Zip disks are &Z1, &Z3, &Z5, &Z6, &Z13, &Z14, &Z16, &Z17.

Zip	Description	Control No.	Directories	
&Z1	Personality and Support Files	MP-M01000-n* System 68020 and 68040	&AMG	AM GDFs
			&ARG	Area GDFs, initial area 00 files
			&CIO	Computing Module Personality
			&CUS	Backplane Modules
			&DS2	HM Resident Displays backup
			DIA1	LCN Cable Statistics Displays
			&DSY	Standard displays & display abstracts
			&EC	Migration and other command files
			&HGG	HG GDFs
			&HGO	Hiway Gateway Personality
			&HMO	History Module Personality
			&HMI	History Module Initialization Personality
			&LDR	Boot loaders and QLT personalities
			&UCN	UCN Device Personality
			&NMG, &NM2	NIM GDFs
			&NGO	Network Gateway Personality
			&NMO	NIN On-Process Personality
			&OP1	UNP Overlays
			&OPR	US Operator Personality
			&OVG	Equipment List Builder overlay
			TLK1	Toolkit
			&UNP	Universal Personality
			&UXS	U ^X S/GUS/ES-T Personality

* Where (n) is specific to the series release (i.e., R6xx MP6ZIP1, etc.)

Table 6-1 Zip Disks Contents and Control Numbers

**R6xx Zip disk
 contents, continued**

Zip	Description	Control No.	Directories
&Z3	Application Module	MP-M03000-n*	&AMO Application Module Personality &AXM A ^X M/APP/ESVT Personality &LDR Boot loaders and QLT personalities &CUS Backplane Modules &EC Migration and other command files TLK2 CL file read/write option package &CLX AM Sets Syntax files
&Z5	HVTS	n/a	Test Diagnostic Software
&Z6	SYS Startup &ASY	MP-M06000-n*	&ASY and &D01 System files, 64-node NCF null Area 01 and Button files &I01 - &I20 Null Hiway/process network checkpoint files &KJF - Kanji font files &AMC Application Module null checkpoint &CIC Computing Module null checkpoints
&Z12	MS	MP-M12000-n*	GUS / Integration Node &CUS, &EC
&Z13	HPM/APM I/O Simulators	MP-M13000-n*	Simulators &UCN, &EC
&Z14	MS	MP-M14000-n*	Multiple Schematics &CUS, &EC
&Z16	SKFO Software Option Key File	MP-M16000-n*	&KFO Software Options Key-file &EC
&Z17	Triconex Safety Manager	MP-M17000-n*	&UCN Triconex Safety Manager Personality &EC

* Where (n) is specific to the series release (i.e., 650, 660, etc.)

Table 6-1 Zip Disks Contents and Control Numbers (continued)

6.3 R6xx Emulated Disk Files/CD-ROM Contents

GUS-TPS network software on CD-ROM The GUS-TPS network software is provided to you on CD-ROM. The following emulated disk files, and personality images can be found on CD-ROMs for this release.

Emulated Disk File Name	Description	Control No.	Directories	
disk_&Z1.LCN	GUS – TPN Software	MP-TPN000-n* System 68020 and 68040	&AMG	AM GDFs
			&ARG	Area GDFs, initial area 00 files
			&CIO	Computing Module Personality
			&CUS	Backplane Modules
			&DS2	HM Resident Displays backup
			DIA1	LCN Cable Statistics Displays
			&DSY	Standard displays & abstracts
			&EC	Migration and other command files
			&HGG	HG GDFs
			&HGO	Hiway Gateway Personality
			&HMO	History Module Personality
			&HMI	History Module Init Personality
			&KJF	Kanji Font files
			&LDR	Boot loaders and QLT personalities
			&UCN	UCN Device Personality
			&NMG, &NM2	NIM GDFs
			&NGO	Network Gateway Personality
			&NMO	NIN On-Process Personality
			&OP1	UNP Overlays
			&OPR	US Operator Personality
			&OVG	Equipment List Builder overlay
			TLK1	Toolkit
			&UNP	Universal Personality
			&UXS	U ^X S/GUS/ES-T Personality

* Where (n) is specific to the series release (i.e., 650, 660, etc.)

Table 6-2 CD-ROM Contents for TPS Network Software

Emulated Disk File Name	Description	Control No.	Directories
disk_&Z3.LCN	TPN Application Module (68020 & 68040)	MP-AM3000-n*	&AMO Application Module Personality &AXM A ^X M/APP/ESVT Personality &LDR Boot loaders and QLT personalities &CUS Backplane Modules &EC Migration and other command files TLK2 CL file read/write option package &CLX AM Sets Syntax files
disk_&Z5.LCN	Hardware Verification Test Suite (HVTS)	n/a	Off-Line Verification & Test Systems
disk_&Z6.LCN	Sys Start-up &ASY (on the GUS-TPN Software CD)		&ASY 64-node start-up NCF &D01 null Area 01 and Button files &I01 - &I20 Null Hiway/process network checkpoint files &KJF Kanji font files &AMC Application Module null checkpoint &CIC Computing Module null checkpoints
disk_&Z12.LCN	GUS/ESVT Load Modules Button File/ADB		Sample Button file and area database &CUS – GUS/ESVT load modules on the Utilities and Load Modules CD
disk_&Z13.LCN	GUS-UCN Simulators	MP-SWS000-n*	USIM TPS Simulators for the APM and HPM.
disk_&Z14.LCN	GUS load modules		&CUS – GUS load modules on the Utilities and Load Modules CD
disk_&Z16.LCN	&KFO Software Options Key File		&KFO Software Options Key-file &EC files (this is a PC diskette)
disk_&Z17.LCN	Triconex Safety Manager	MP-TSM000-n*	&UCN Triconex Safety Manager Personality

* Where (n) is specific to the series release (i.e., 650, 660, etc.)

Table 6-2 CD-ROM Contents for TPS Network Software (continued)

NOTE

disk_&Z12.LCN contains a blank area database that references a sample button file configuration. The button file was created for use with a PC-101 keyboard and can be used as a template for creating your own custom button file.

GUS/ES-T Workstation Personality installation files are found under directory GUS\PERS_US on the GUS-TPN CD.

APP/ESVT/IntegrationNode personality installation files are found under the directory APP\PERS_AM on the Application Module CD. AMW personality installation files are found under the directory AMWPERS_AM on the Application Module CD.

6.4 DAT Tape Contents

DAT Tape/CD-ROM contents The following DAT Tapes and CD-ROMs are supplied with the related optional hardware.

Product Name	Control No.	Related Product	Contents
VAX (TK50 media)	MP-CM50S9	Computing Module	VAX platform
VAX (CD-ROM)	MP-C50S10	Computing Module	Alpha platform
A ^X M Personality (DAT tape)	MP-SWAXM1	A ^X M	A ^X M personality files
Patches tape	51153380	HP-UX	HP-UX 10.20 patches and Y2K bundle
HP-UX Install (DAT tape)	51153333	A ^X M and U ^X S	
OpenDDA developer's kit (DAT tape)	MP-SWDDA1	A ^X M	
Multiple Schematics (DAT tape and Bernoulli)	MP-ZUXSTM	Multiple Schematics	All required Multiple Schematics files
FORTTRAN compiler	MP-SWDDA2	A ^X M	
ANSI C Compiler	MP-SWDDA3	A ^X M	
C++ Compiler	MP-SWDDA4	A ^X M	
U ^X S Windows U ^X S Personality	MP-ZUXSTM	U ^X S	DAT tapes (delivered together)
DMCIX interface	MP-DMCIX	DMC	
PCDE NetDDE run-time for HP-UX	MP-SWXDR1	PC Data Exchange	
PCDE Data Exchange	MP-SWXDS1	PC Data Exchange	PCDE server for X layer, NetDDE run-time for HP-UX
PCDE for X layer servers	MP-SWXDS2	PC Data Exchange	NetDDE not run-time included
NetDDE Runtime for Windows	MP-SWXDR2	NetDDE	
OpenDNA DECnet protocol	MP-SWDNP1	DECnet	
OpenDNA VT340 emulator	MP-SWTRM1	VT340 Terminal	

Table 6-3 DAT Tape/CD-ROM Contents

7. Setup for Migration from R43x

7.1 Overview

Migration from R43x

ATTENTION

Section 7 is for migrations from R43x. If you are migrating from R5xx, please skip this section and proceed to Section 8.

To migrate from R43x to R6xx, you must first put a station (US, UxS, GUS, EST) in R53x to do translations and other preparations. The rest of the LCN/UCN nodes can stay in R43x until you reach instructions to upgrade in Section 9. Follow the steps in Section 7.

Prerequisite steps done?

Before continuing, ensure that the items below have been checked and completed as necessary.

(√)	Prerequisite Step
	Have read the <i>Read Me First!</i> section in this manual
	Have read all applicable Special Considerations in Section 5
	Have read Section 2 - <i>What To Do Before Migrating Your Software</i>
	Have parts available to perform all required Hardware upgrades/additions
	Have parts available to perform all required Circuit board revision updates
	Have parts available to perform all required Firmware revision level updates
	Have parts available to perform all required Memory size adjustments (or memory available for use during migration), including possible additional memory required for External Load Modules
	Have checked to ensure all required R6xx software is available , according to Section 6.
	Have obtained R6xx versions of all currently configured External Load Modules certified to run on R6xx
	Have obtained necessary R53x software for R43x to R53x database translation.

Table 7-1 Steps Prerequisite to Continuing with This Section

On-process migration

Migrations can be performed without removing the system from on-line control. The process of migrating UCN devices can be lengthy and exacting, and errors could have process impact. For this reason, Honeywell strongly recommends against an on-line migration of a system controlling a volatile process.

NOTE: The MINIMUM CONFIGURATION THAT CAN BE MIGRATED ON-PROCESS AND STILL RETAIN a “VIEW OF THE PROCESS” IS:

3 – US with Engineering Keyboards (two must have dual cartridge or zip drives)

1 - HM

1 – redundant module (HG or NIM)

If your subsystem doesn't have at least this configuration, on-process migration can not be done.

Migration - how does it apply to your system?

It is Honeywell's recommendation that each migration (including all steps in this guide) be examined and reviewed for its specific requirements as they apply to your system.

Command Processor commands

The following is a quick reference to the Command Processor commands that you may use when migrating your system. If you are unfamiliar with using any of these commands, please refer to the *Command Processor Operation* manual, found in the TPS 3032-1 Implementation Engineering Operations - 1 binder.

If you need to...	Use	Valid Device ID Form(s)	Valid Options (see descriptions on next page)	Reference Section (SW11-507)
Copy a file	CP	NET, \$Fn, PN:nn	-D	5.3
Copy a volume	CPV	NET, \$Fn, PN:nn	-A, -D, -I	5.4
Create a directory	CD	NET, \$Fn, PN:nn		5.6
Create a new volume	CR	\$Fn	-BS, -MF, -F, -MD	5.5
Delete files	DL	NET, \$Fn, PN:nn	-D	5.9
End a Command Processor session	END			5.11
List file attributes	LS	NET, \$Fn, PN:nn	-A, -D, -FD, -M, -N, -P, -REC, -S	5.17
List volumes and directories in a selected HM	LSV	NET, PN:nn		5.18
Protect a file	PROT	NET, \$Fn, PN:nn		5.24
Protect a Zip disk	PROTV	\$Fn	-NQ	
Select a printer to receive output	P0	NET, \$Pn, \$Fn, PN:nn		5.23
Select device(s) to receive data	D0	NET, \$Pn, \$Fn, PN:nn		5.7
Send messages to a US <u>and</u> a selected printer	PD		ON, OFF	5.22
Translate	TR			
Unprotect a file	UNPT	NET, \$Fn, PN:nn		5.24
Unprotect a Zip disk	UNPTV	\$Fn	-NQ	

Table 7-2 Command Processor Commands - Quick Reference

7.1. Overview**Command Processor command options**

The following is a list of the valid options that can be used in conjunction with the previously listed Command Processor commands.

NOTE: The option may have different meanings for different commands, so follow the descriptions as they apply to each command.

Option	As Used by this Command	Means (for this command)...
- D	CP, CPV, DL, LS	Detail option (provides for command details)
- A	CPV	Include all directories in the volume if the destination is a cartridge
- I	CPV	Enable media initialization
- BS	CR	Block size [26 to 1700 if not using -FD; 26 to 300 (migrating from R4xx) or 26 to 6xx (migrating from R500 or R510) if -FD is also being used]
- F	CR	Removable media format option
- MD	CR	Memory-resident Directory Option (NOTE: Do not use at same time as -BS)
- MF	CR	Maximum files option (1 to 32767)(Default for cartridge = 3020)
- X	CR	Extended Directory Support. Allows for creation and tracking of up to 2046 directories/virtual volumes on a cartridge. Do not use on disks that are used for - Translations, - Find Names, - AM/CL file I/O, or CDS Read/Write.
- A	LS	List all files and all directories in this volume
- BF	LS	Display limited attribute information
- F	LS	(-F DATE, TIME) List only those files with date/time after those on command line
- FD	LS	List all files along with their file descriptors
- M"xx"	LS	Only display files whose descriptors match "xx".
- N	LS	List only file info., no header, footer information
- P	LS	(-P DATE, TIME) List only those files with date/time prior to those on command line
- REC	LS	List all files and their record information
- S	LS	Sort file attributes by DIR, NAME, EXT, DATE, or DESC
ON	PD	Prompts sent to printer
OFF	PD	Prompts not sent to printer
- NQ	PROTV UNPTV	Suppresses list of directories on device and need for operator to respond

Table 7-3 Command Processor command Options - Quick Reference

Enough time allotted?

Be sure you have allotted enough time to complete the migration process. It is difficult to say how long it should take, but it often requires in excess of ten hours to complete the on-line portion of the migration, depending on the number of nodes you have, and how experienced you are with software migrations.

Allow 4 hours per HM and 4 hours to load the LCN nodes; and add an additional 5 minutes per PMM/APM/LMM/HPM/SM/FSC on each UCN.

Step layout

This migration procedure is organized in several major steps, which are divided up into individual steps, to allow break points where possible. A column is provided to the right of each step to check (✓) off each step upon its completion.

Please read every step and associated CAUTIONS, ATTENTIONS, and NOTES, to help ensure a successful and time efficient migration. It is very likely that not doing so will cause a failed migration attempt.

Locate extra cartridge labels

Some of the procedures in this migration require that you create additional cartridges that will require labels to be created and physically attached to the cartridge.

Be sure you have blank cartridge labels available to you before starting the migration.

ATTENTION

The NCF for AM, US/GUS/EST, and HG nodes must be configured as NO for "further external directives," unless otherwise directed by Honeywell. Failure to do so will cause the loader to look for nonexistent files, and the node will crash.

7.2 Database Preparation for 43x to R53x Migrations

Media used in database preparation

ATTENTION

Honeywell recommends that you use Zip media for all actions in Section 7.

Identify NET volumes Perform the following steps to capture the HM Volume configuration for each HM, and print the HM Volume usage.

Step	Action	Done (✓)
1	Perform a list volume to the printer, as follows: TYPE DO \$Pn where n = printer number TYPE PD ON	
2	TYPE LSV NET	
3	TYPE LS PN:nn -d where nn = HM node number This will print the current volume usage for that HM.	
4	Repeat Step 3 for each remaining HM on your system. Use this information to find the space available to reconfigure/ re-allocate space for R530.	

Table 7-4 Identify NET volumes

Stabilize process

NOTE: Boxes with communications errors, and boxes in a state where they cannot be checkpointed, will not take checkpoints and **cannot be migrated to R6xx.**

Verify there are no boxes in the wrong state, or with communication errors on HIWAY(s) or UCN(s).

Take demand checkpoints

Follow steps 1 through 4 to take demand checkpoints to NET for all data owners on the LCN.

Step	Action	Done (√)
1	Take demand checkpoints to NET for all HG devices.	
2	Take demand checkpoints to NET for all UCN devices.	
3	Take demand checkpoints to NET for all AMs.	
4	Take demand checkpoints to NET for all CGs.	

Table 7-5 Take Demand Checkpoints

7.2. Database Preparation for 43x to R53x Migrations**Build R43x FAST load disks**

Use the following steps to build custom fast load media. This media can be used to quickly reestablish R43x paths to critical values should a system re-establishment be disrupted.

Step	Action	Done (✓)
1	MOUNT R43x source disk &Z1 in \$Fs (source drive).	
2	MOUNT destination disk in \$Fd (destination drive)	
3	DISABLE auto checkpointing for the target process network (if enabled).	
4	Printer output should already be directed to the printer. If not, do the following: TYPE: DO \$Pn (where n= printer number) TYPE: PD ON	
5	RUN the FAST_VOLZ.EC (example shown below). EC \$Fs>&ec>FST_VOLZ.EC \$Fd an hn nnnn Where s = Source drive number d = Destination drive number an = Area number (01 to)10 hn = Hiway or UCN number nnnn = Volume name on cartridge (if it is to be initialized)	
6	If desired, ENABLE auto checkpointing.	
7	Label disk as: "Fast Load for Console ## Rxxx Hiway ##" or "Fast Load for Console ## Rxxx UCN ##"	
8	REPEAT Steps 2 through 7 for each remaining Area, Hiway and UCN.	
9	To ensure there are no write errors on your fast load cartridges, COPY each newly created disk to one scratch disk (use the same one over and over), using the FCOPY command.	

Table 7-6 Build R43x FAST Load Disks

**Save UNP Load
Support files**

Follow steps 1 through 6 in the following table to save the area database files for future translations to Zip disks.

Step	Action	Done (√)
1	Initialize a Zip disk to be used for system startup. Create a directory for each area. Example syntax: CR \$Fd>CNCF> -MF 4500 Label Zip disk "Temporary Files".	
2	If any area has memory resident schematics, BUILD directory(s), and COPY to disk. NOTE: In the process of accessing the memory resident schematics, you may automatically create an uninstalled Area Database work file (.WA). If so, you should DELETE it at this time.	
3	Printer output should already be directed to the printer. If not, do the following: TYPE DO \$Pn (where n = printer number) TYPE PD ON	
4	Copy the area files from the NET onto the disk under the same directory names. EXAMPLES: CD \$Fd>CNCF>&Dan UNPT NET>&Dan>AREA* .DA CP NET>&Dan>*. * \$Fd>&Dan>= -D where an = Area Database Number (01 to 10) d = Destination drive number	
5	REPEAT step 4 for each additional area.	
6	If the button file is not in &Dan, then BUILD the correct directory on disk, COPY button.ko file to the correct directory.	

Table 7-7 Save UNP Load Support Files

7.2. Database Preparation for 43x to R53x Migrations

Create "Temporary Files" disk

Use the following steps to copy all the .PL files and R43x user files of .ER, .SE, .SP, and .XX onto a Customer NCF disk.

These saved files will be used to do the initial loading of an R530 UNP with the translated NCF for preparation to go to R6xx

Using the "Temporary Files" disk created in step 1 on Table 7-7, Save UNP Load Support Files do the following:

Step	Action	Done (✓)
1	On the disk, CREATE a directory called &ASY. SYNTAX EXAMPLE: CD \$Fd>CNCf> &ASY	
2	COPY all &ASY files (* . *) to the disk. SYNTAX EXAMPLE: UNPT NET>&ASY>NCF.* CP NET>&ASY>*. * \$Fd>&ASY>= -D	

Table 7-8 Create "Temporary Files" Disk

Create "Temporary Files" disk,
continued

Use the following steps to copy the \$*.PL files to the cartridge, using the Honeywell-provided **R530** disk &Z6 SYSTEM STARTUP &ASY.

Step 6 in the following table is required because the LCN file manager cannot detect write errors. So, it is necessary to verify that the data on a disk is good by copying it to another disk.

Step	Action	Done (√)
3	UNPROTECT and DELETE \$* .PL files from "Temporary Files" disk. SYNTAX: UNPT \$Fd>&ASY>\$* .PL DL \$Fd>&ASY>\$* .PL where d = "Temporary Files" disk	
4	MOUNT Honeywell-provided R530 disk &Z6 (64-node NCF) in another external drive.	
5	COPY \$* .PL files from &Z6 to "Temporary Files". Disk SYNTAX: CP \$Fs>&ASY>\$* .PL \$Fd>&ASY>= -D where s = Source drive &Z6 d = "Temporary Files" disk	
6	To ensure there are no write errors on your backup disks, COPY the newly created disk to one scratch disk (use the same one over and over), using the FCOPY command.	

Table 7-8 Create "Temporary Files" Disk (continued)

Save IDFs as .EBs (optional)

If you need to maintain a current loadable image of your database, it is recommended that you save Intermediate Data Files (IDFs) as Exception Build (EB) files, to be rebuilt later. Follow steps 1 through 7 to accomplish those saves. Make sure your IDFs are current.

The .EB files require that the EB translator be used to translate the ALPRIOR parameter into the appropriate separate alarm priorities (SAP) for each point type. AM, HG, and NIM point types are affected. EB translation must be performed after R530 US is loaded and before R6xx is installed.

If you are rebuilding the database from the .EB files, the .EB files must be translated before using.

ATTENTION Quotation marks (") in .EB files - User-entered quotation marks (") in .EB files must be removed in the PTDESC, EUDESC, or KEYWORD of any point types, before the .EB files can be translated.		
Step	Action	Done (✓)
1	INITIALIZE a zip for the database files. CREATE directories as appropriate. LABEL disk "EB Files."	

Table 7-9 Save IDFs as .Ebs

Save IDFs as .EBs (optional), continued

Step	Action	Done (√)
2	On a US with R43x Engineering or Universal Personality, use the Text Editor to CREATE a DEB command file (name.EC). There should be one PI command line per IDfname.DB file. name.EC file EXAMPLE: <pre>V1 = NET>IDF V2 = \$F1>EB PI V1>IDfname1.DB &B V2>EBsourc1.EB PI V1>IDfname2.DB &B V2>EBsourc2.EB PI V1>IDfname3.DB &B V2>EBsourc3.EB</pre> ATTENTION The .EC command file should not have the same filename as the IDF or .EB files because it interferes with the automatic naming of the error file.	
3	From the Engineering Main Menu, SELECT BUILDER COMMANDS	
4	From the Builder Commands Menu, SELECT EXECUTE COMMAND FILE	
5	TYPE in the path and name for your .EC file on the pathname for COMMAND FILE line.	

Table 7-9 Save IDFs as .EBs (continued)

**Save IDFs as .EBs
(optional), continued**

Step	Action	Done (✓)
6	PRESS <ENTER> to execute. The output is your set of text (ASCII) Exception Build Files. Exception Build File output format: <pre> &t point type &n entity name Parameter name = value Parameter name = value . . Parameter name = value &t point type &n entity name Parameter name = value Parameter name = value . . Parameter name = value </pre>	
7	Repeat steps 1-6 until all points are saved in EB files on removable media.	

Table 7-9 Save IDFs as .EBs (continued)

Build and save all History Group Point definitions list

If the History Group Point definitions do not already exist in the EB files, perform the steps below to build and save the History Group Point definitions to EBs.

Step	Zip Disk Action	Done (√)
1	Using the Text Editor and following the syntax example, BUILD History Group Point definitions list. EXAMPLE SYNTAX: <pre>\$CHui(gn) where ui = Unit Index number (from UNIT NAMES under the Engineering Main Menu, Unit Names, under the column labeled "NO") gn = History Group number in unit</pre> EXAMPLE: <pre>\$CH01(01) \$CH01(02) \$CH01(03) \$CH01(04) \$CH01(05) \$CH02(01) \$CH02(02) \$CH02(03) \$CH02(04)</pre> NOTE: Place four spaces between each definition	
2	PERFORM DEB PRINT SYSTEM ENTITIES using the saved selection list. This saved file should have a .EL extension. OUTPUT to a file in the net with .EB extension.	

Table 7-10 Build and Save All History Group Point Definitions List

7.3 System Backout Backup for R43x

Create HM backups

Use the following steps to copy all the directories and files on each HM to a cartridge. This system backout backup should be performed to allow you to back out to R43x from R6xx.

CAUTION This backup does not save any history or journals.		
Step	Action	Done (√)
1	If this HM has dump volumes, it is recommended they be deleted before you perform the backup. DL PN:nn>&OPD>*.* (&NMD, &HGD, &CGD, &AMD, &HMD)	
2	DISABLE automatic checkpointing from all checkpointed nodes (HG, NIM, CG, and AM) and do not allow demand checkpoints.	
3	CREATE volumes on all of the backup cartridges, either with or without FD (File Descriptors). Follow one of the examples below, depending on your situation. EXAMPLE WITH FD: CR \$Fd>HMnn> -FD -MF 3000 where d = destination drive number nn = HM node number EXAMPLE WITHOUT FD: CR \$Fd>HMnn> -MF 3000 -BS 1500 Where d = destination drive number nn = HM node number	
4	Through the Command Processor, EXECUTE the BACKUP command to create backup. EXAMPLE SYNTAX: BACKUP PN:nn \$Fd where nn = HM node number d = destination drive number	
5	To ensure there are no write errors on your backup cartridge, COPY each newly created cartridge to one scratch cartridge (use the same one over and over), using the FCOPY command, and the same scratch cartridge.	
6	Honeywell recommends you have two backup copies. Use FCOPY to copy each cartridge to another cartridge.	
7	Repeat steps 1 through 6 for each HM.	

Table 7-11 Create HM Backups

Save maintenance recommendation journal (optional)

All History Modules must be initialized for R6xx. So, if maintenance history is required, you must perform the following steps. This saves the Maintenance Recommendation Journal to the printer.

Step	Action	Done (✓)
1	From the Engineering Personality Main Menu, SELECT SMCC MAINTENANCE	
2	From the SMCC Main Menu, SELECT SYSTEM MAINT JOURNAL	
3	From the System Maintenance Journal screen, SELECT PRINT ENTIRE SMJ	
4	SELECT Ctrl/Page Back	
5	From the SMCC Main Menu, SELECT ACTIVE MAINT JOURNAL	
6	From the System Maintenance Journal screen, SELECT PRINT ACTIVE MAINTENANCE RECOMMENDATIONS	
7	From the System Maintenance Journal screen, SELECT PRINT ALL ERROR AGGREGATES	
8	LABEL printer output appropriately and preserve as a record of your maintenance history.	

Table 7-12 Save Maintenance Recommendation Journal

7.3. System Backout Backup for R43x

**Save Process/
System Journal
(optional)**

Process/System Journal files are deleted as a part of the migration. If you need to save your Process/System Journal events, perform the following steps to save them to hard copy.

Step	Action	Done (√)
1	From the Operator Keyboard or Engineering Main Menu, SELECT SYSTEM MENU	
2	From the System Menu, SELECT EVENT HISTORY MENU	
3	From the Event History Display, SELECT each of the seven types of history and print them.	
4	From the System Menu, SELECT REPORT/LOG/TREND/JOURNAL MENU	
5	From the menu displayed, SELECT all the items there you wish to print, and SELECT PRINT	
6	LABEL printout appropriately and preserve as record of your unit journal history.	

Table 7-13 Save Process/System Journal

Save Continuous History (History Data and Group Point definitions)

Perform this procedure if you want both History Data and Group Point definitions saved. Skip to the next procedure if you only want Group Point definitions saved.

The Backout Backup process does not save Continuous History. So, if

- you wish to restore it to R43x, or
- you wish to restore it forward to R6xx,

Perform the following steps to save Continuous History.

Step	Action	Done (✓)
1	LOAD the HM with the R43x OFF-LINE personality.	
2	FORMAT sufficient cartridges to save history. To determine number of cartridges you will require, divide the number of sectors by the number given for your cartridge size (below): 20 MB cartridge: Number of Sectors ÷ 83,000 (rounded up) 35 MB cartridge: Number of Sectors ÷ 122,500 (rounded up) 150 MB cartridge: Number of Sectors ÷ 525,000 (rounded up) Zip 100 MB disk: Number of Sectors ÷ 394000 (rounded up)	
THIS IS <u>NOT</u> A STEP -- This is an example of the output when calculating number of cartridges needed. <pre> LS PN:nn>!0np> -D where nn = HM node number np = HM node pair number Note number of sectors is displayed EXAMPLE: VOLUME: !001 VOLUME DESC: Continuous History MAX FILES: 416 TOTAL SECTORS: 195552 SECTORS IN USE: 194330 <-Sectors used VOLUME SUBSTATE: INITIALIZED DIRECTORY: MEMORY RESIDENT AVL SUPPORTED ON VOLUME: ENABLED ACCESS LEVEL: NETWORK_READ_LOCAL_WRITE FILE DESCRIPTORS: DISABLED CURRENT DIRECTORIES ASSIGNMENT: !A00 Catalog Complete So, 194330 ÷ 83,000 = 2.34 (or 3 "20 MB" cartridges, rounded up) </pre>		

Table 7-14 Save Continuous History

7.3. System Backout Backup for R43x**Save Continuous History (History Data and Group Point definitions),**
continued

Step	Action	Done (√)
3	ENSURE the HM is in the Off-line personality.	
4	SAVE history for the volume. EXAMPLE SYNTAX: CPV PN:nn>!0np> \$Fd>!0np> -A -D where nn = HM node number np = HM node pair number d = destination drive number Continue to insert cartridges until backup is complete.	
5	Honeywell recommends you have two copies of your backup. Use FCOPY to COPY each cartridge to another cartridge.	
6	LABEL each cartridge, as follows: !0np #1 of ____ !0np #2 of ____ !0np #3 of ____ , etc.	
7	REPEAT Steps 2 through 6 for !1np, if that volume exists on the specified HM.	
8	ADD final "total" number of cartridges in the "____" (blank part of each label).	
9	REPEAT Steps 1 through 8 for each HM that has Continuous History.	

Table 7-14 Save Continuous History (continued)

Save Continuous History (Group Point definitions only)

Only perform this procedure if you did not perform the previous procedure [Save Continuous History (History Data and Group Point definitions)], but you wish to save only the History Group Point definitions.

Perform the following steps to save the History Group Point definitions. These steps are performed for !0np only.

Step	Action	Done (✓)
1	LOAD HM into the R43x OFF-LINE personality.	
2	Format one blank cartridge, using the syntax below. SYNTAX: CR \$Fd>!0np where d = destination drive number np = HM node pair number	
3	COPY the Group Point definition files from NET to cartridge. SYNTAX: CP PN:nn>!0np>APL *.mm \$Fd>!0np>= -D where nn = HM node number d = Destination drive np = HM node pair number	
4	LABEL the cartridge !0np (History Group Point definition files)	
5	To ensure there are no write errors on your !0np cartridge, COPY the newly created cartridge to a scratch cartridge, using the FCOPY command.	

Table 7-15 Save Continuous History (Group Point definitions only)

**Backout Backup
(optional)**

If you need to assure that all of the copies above worked, perform the following step to copy each cartridge to another scratch cartridge.

By doing so, any copy errors that have occurred will become evident.

Step	Action	Done (√)
1	To ensure there are no write errors on your backup cartridges, COPY each newly created cartridge to a scratch cartridge, using the FCOPY command, and using the same scratch cartridge (use the same scratch cartridge over and over).	

Table 7-16 Backout Backup

7.4 Save R43x System Files for R6xx

Backup HM

In the following steps, you will use the Command Processor to backup all non-history files with the exception of personality images, from each HM. It is not required for the HM to be in the OFF-LINE personality to perform the following backup.

If desired, the HM can at this point be rebooted in the R43x ON-LINE personality. If this is done, it is normal that HMs with redundant drives shows a SEVERE status. This is not a problem.

ATTENTION

The backup command does not backup the personality images or the continuous history files. It saves the Continuous History Group definitions.

Step	Action	Done (√)
1	RECORD all activity that follows on the printer. TYPE D0 \$Pn TYPE PD ON	
2	If this HM has dump volumes, to reduce the time it takes and the amount of disk space it occupies, it is recommended that all dump files in each dump subdirectory be deleted before you perform the backup. DL PN:nn>&OPD>*. * (&NMD, &HGD, &CGD, &AMD, &HMD)	
3	Using the Command Processor BACKUP -UPGRADE400 command, COPY all non-history files from the HM to cartridge. EXAMPLE SYNTAX: BACKUP PN:nn \$Fd -UPGRADE400 where nn = HM node number d = Destination drive number NOTE: The BACKUP command formats each cartridge, if needed, according to your responses to prompts. The BACKUP command can be used in parallel for all HMs.	

Table 7-17 Backup HM

Step	Action	Done (√)
4	LABEL the cartridges "BACKUP PN:nn #y". where nn = HM node number y = Sequential number for multiple media	
5	LOAD one of the BACKUP cartridges in \$Fd, and LOAD R530 &Z1 in \$Fs where d = Destination drive number s = Source drive number	
6	DELETE the files that are not migrated to R530. EXAMPLE SYNTAX: EC \$Fs>&EC>CLEANUPZ.EC d where s = Source file d = Destination cartridge you are cleaning up NOTE: Upon completion, the system will display Delete Complete, followed by Please Wait. Do not continue until the waiting period has completed.	
7	REPEAT steps 4 through 6 for each BACKUP cartridge.	
8	To ensure there are no write errors on your backup cartridges, COPY the newly created cartridge to a scratch cartridge, using the FCOPY command.	
9	REPEAT steps 1 through 8 for each remaining HM. REMINDER: The BACKUP command can be used in parallel for all HMs.	

Table 7-17 Backup HM (continued)

Preparation for translation

Prior to translating, ensure your process is as stable as possible.

Step	Action	Done (✓)
1	NOTE: Boxes with communications errors, and boxes in a state where they cannot be checkpointed, <u>will not take checkpoints</u>. Correct any box or I/O errors on HIWAY(s) or UCN(s).	

Verify the LCN RECONNECT is terminated, as follows:

Step	Action	Done (✓)
1	From the Console Status Display, SELECT LCN OVERVIEW	
2	If the SYSTEM VERSION port (upper-right corner) contains INIT LCN RECONNECT, (meaning it is initiated), use the LCN Overview Display to ensure all nodes have the same network view. SELECT TERM LCN RECNCT, and PRESS <ENTER>	

Table 7-18 Preparation for Translation

7.5 R43x to R53x Translation

Translators used The following is a list of translators, along with a brief description and the syntax to be used to run each translator. (s=source drive; d=destination drive; an= area number; uu=UCN node number; hh=Hiway node number).

Do not use the “-X” option when formatting disks that are used for translation.

Name	Description	Syntax
NCF	Network configuration	TR NCF \$Fs \$Fd
ADB	Area database	TR ADB an \$Fs \$Fd
NIM	NIM checkpoints	TR NM \$Fs \$Fd uu
HG	HG checkpoints	TR HG \$Fs \$Fd hh
EB	EB files	TR EB \$Fs>SRCS>* .EB \$Fd>SRCS>= -A

Table 7-19 Translators Used

Save checkpoints to cartridge Perform the following steps to save the NIM and HG checkpoints to cartridge for translation. Using the “Temporary Files” disk, checkpoint all data owners.

Step	Action	Done (✓)
1	PERFORM CHECKPOINTS for HGs and all UCN-controlled devices to cartridge.	
2	PERFORM CHECKPOINTS for all AMs and CGs to cartridge.	
3	CREATE a list of the ACP points, to be used in a later procedure for migrating the CGs.	

Table 7-20 Save Checkpoints to Cartridge

Load US with R530 Migrate the US. Note that once it is loaded with R530 software, it cannot communicate with any nodes running the older R43x software.
NOTE: When the first node is loaded with R530, the System Status Display may show CABLE SUSPECT. This can be ignored.

ATTENTION Be sure to load the US with R530 <u>to completion</u>, including the creation of a duplicate name file. A modification was placed in R500 that delays the creation of the duplicate name file until after the US goes to OK. Use the US Status Display to verify the message "CUSTOM/STANDARD PARAMETER DUPLICATE Xref created" is displayed before continuing.		
Step	Action	Done (✓)
1	CHOOSE a Universal Station that has at least one (preferably two) zip drives, an Engineering Keyboard, a working printer, and a keyswitch for placing the US in engineer keylevel access,. If your US has no keyswitch please see section 14.7 for a work around.	
2	SHUTDOWN US <u>twice</u> to make it go to FAIL.	
3	This step must be performed because: - the QLTs have changed, and the node hardware must be re-qualified with the new QLTs; and - now is the time to update the hardware, as required. Once the US goes to FAIL state, go to the physical US node, POWER OFF the node and WAIT 5 seconds. Then, if not upgrading US hardware, POWER ON the node.	
4	Otherwise, UPGRADE US hardware to R53x UNP minimums as needed, then POWER ON the node.	
5	From Honeywell-provided disk &Z1 and &Z6 (for the Button files-and Area Database).LOAD the R53x Universal Personality (UNP), and the 64-node NCF.	
6	VERIFY the date and time in the upper right corner of the Engineering Main Menu is correct. CORRECT date and time, if necessary.	
7	From SUPPORT UTILITIES, SET device path to removable media.	

Table 7-21 Load US with R530

7.5. R43x to R53x Translation

Perform the following steps to translate the NCF file.

Translate NCF

Step	Action	Done (√)
1	MOUNT the R53x translator disk (&Z1) in the lowest numbered removable drive.	
2	From the Command Processor, TYPE TR, and PRESS <ENTER>. NOTE: The translator loads, and a list of possible entries, is displayed with a message "ILLEGAL / NOT IMPLEMENTED". THIS IS OK.	
3	Make a copy of the "Temporary Files" disk. EXAMPLE SYNTAX: FCOPY \$Fs \$Fd where s = source drive number d = destination drive number Label the disk "Customer Support Files - R43x to R53x"	
4	TRANSLATE the NCF file. EXAMPLE SYNTAX: TR NCF \$Fs \$Fd where s = source drive number (temporary files) d = destination drive number (customer support files)	

Table 7-22 Translate NCF

Translate Area Databases

Perform the following steps to translate the Area Databases.

Step	Action	Done (✓)
1	TRANSLATE <u>each area</u> . EXAMPLE SYNTAX: TR ADB an \$Fs \$Fd where an = Area Database number (01-10) S = source drive number d = destination drive number	
2	If any nodes on your system have nonstandard custom External Load Modules, CREATE directories &CUS and &CLX, and COPY all (both standard and nonstandard) R530 External Load Modules to destination disk.	
3	Make sure the button file directory is created on the customer support files and the button file is copied to that directory.	

Table 7-23 Translate Area Databases

Translate NIMs

Perform the following steps to translate the NIM checkpoints.

Step	Action	Done (✓)
1	TRANSLATE <u>each NIM checkpoint</u> . NOTE: Only the NIM checkpoint will be translated. The PM/APM/LMs do not need to be translated. EXAMPLE SYNTAX: TR NM \$Fs \$Fd uu where S = source drive number d = destination drive number uu = UCN network number (01-20)	
2	REPEAT step 1 for each additional UCN.	

Table 7-24 Translate NIMs

**Translate
HG/PLCGs**

Perform the following steps to translate the HG/PLCG checkpoints.

Step	Action	Done (√)
1	TRANSLATE <u>each</u> HG/PLCG checkpoint. EXAMPLE SYNTAX: TR HG \$Fs \$Fd hh where S = source drive number d = destination drive number hh = Hiway number (01-20)	
2	REPEAT step 1 for each additional HG/PLCG.	

Table 7-25 Translate HGs/PLCGs

7.6 EBX Translator

Quotation marks (“)
in .EB files

ATTENTION

Quotation marks (“) in .EB files must be removed in the PTDESC, EUDESC, or KEYWORD of any point types, before the .EB files can be translated.

**EBX Translator for
separate alarm
priorities**

Separate alarm priorities have been implemented on R53x and R6xx which expand the ALPRIOR parameter into additional priority parameters. These added parameters depend upon node type, point type, and visibility picks in the DEB (Data Entity Builder).

During a migration to R530, the older release exception build files (.EB) must be parsed to find any lines containing the ALPRIOR parameter. Each ALPRIOR parameter must be replaced with the correct separate alarm priority parameters for the point currently being defined in the EB file.

The exception build file translator (EBX), is used to translate these EB files.

EBX Translator
Command line

You can call up the EBX translator using the following syntax.

Syntax For EBX Translator
TR EB <input pathname><output pathname> <switch> Where: input pathname = pathname to use for input file(s) output pathname = pathname to use for the (translated) output file(s). switch = an option for specifying all sub \\ directories on a volume (-A)
EXAMPLES: TR EB \$F1>SRCS>TEST1 \$F2>SRCS>TEST1T TR EB \$F1>SRCS>* .EB \$F2>SRCS>= TR EB \$F1>SRCS>TEST? .EB \$F2>SRCS>= TR EB \$F1>SRCS>* .EB \$F2>SRCS>= -A
NOTES: If the output pathname is not specified, then the input file(s) will be replaced with the translated data from the translator output. If the output pathname is specified, then it must include the filename or an equal sign (=). If the input pathname includes a wildcard, then the output pathname must include an equal sign (=). If the -A switch is used, then the destination volume must contain the same directories as the input volume. Any directory that only exists on one volume will not be translated.

Table 7-26 Syntax for EBX Translator

7.7 Load Universal Station with New Files

First R53x Universal Station load On the first R53x Universal Station load, the NCF cartridge must be left in the drive for approximately five minutes after the station goes to OK. This allows the `duplcat.SP` file to be created. Use the `US Status Display` to verify the message “CUSTOM/STANDARD PARAMETER DUPLICATE XRef created” is displayed before continuing.

Reboot US Perform the following steps to reboot the US with R53x UNP, Customer NCF, and translated area databases.

Step	Action	Done (✓)
1	SHUTDOWN and RESET the Universal Station that you have been using for the translation.	
2	When loading this node, you will be asked by the system to mount a sequence of R53x cartridges, all in the same cartridge drive, during the load. The R53x Zip disk to be used: Customer Support Files R43x to R53x IMPORTANT! The "CUSTOMER SUPPORT FILES - R43x to R53x " must remain on the same drive until the US is fully loaded. See the NOTE above.	
3	VERIFY the date and time in the upper right corner of the Engineering Main Menu is correct. CORRECT date and time if necessary.	
4	SET device paths to removable media.	

Table 7-27 Reboot US

You have completed the translations and preparations for R53x.

7.8 Prepare for Migration to R6xx

Using FCOPY, make a copy of disk “Customer Support Files R43x to R53x”.

Label this new disk “Customer Support Files R53x to R6xx”

Use the following steps to copy the \$.PL files to the cartridge, using the Honeywell-provided **R6xx** disk &Z6 SYSTEM STARTUP &ASY or R6xx TPN software CD DISK_&Z6.LCN.

Step 5 in the following table is required because the LCN file manager cannot detect write errors. So, it is necessary to verify that the data on a disk is good by copying it to another disk.

Step	Action
1	UNPROTECT and DELETE \$.PL files from “Customer Support Files” disk. SYNTAX: UNPT \$Fd>&ASY>\$.PL DL \$Fd>&ASY>\$.PL where d = “Customer Support Files R53X to R6xx” cartridge
2	MOUNT Honeywell-provided R6xx disk &Z6 (64-node NCF), or the CD DISK_6.LCN in another external drive.
3	COPY \$.PL files from &Z6 to “Customer Support Files” disk SYNTAX: CP \$Fs>&ASY>\$.PL \$Fd>&ASY>= -D where s = Source drive &Z6 d = “Customer Support Files” disk
4	Save External Load Modules to Customer Support Files Create directories &CUS and &CLX and copy all (both standard and nonstandard) R6xx External Load Modules to the destination disk.
5	To ensure there no write errors on your backup disks, COPY the newly created disk to one scatch disk (use the same one over and over), using the FCOPY command.

Table 7-28 Create “CUSTOMER NCF” DISK

Continue with the migration to R6xx

Please continue with the migration to R6xx by going to Section 9.

8. Setup for Migration from R5xx

ATTENTION

Section 8 is for migrations from R5xx to R6xx. If you are migrating from R43x, please begin with Section 7. If migrating from R43x and have already done Section 7, continue with Section 9.

8.1 Overview

Prerequisite steps done?

Before continuing, ensure that the items below have been checked and completed as necessary.

Done (√)	Prerequisite Step
	Have read the <i>Read Me First!</i> section in this manual.
	Have read all applicable Special Considerations in Section 5.
	Have read Section 2 - <i>What To Do Before Migrating Your Software</i>
	Have parts available to perform all required Hardware upgrades/additions .
	Have parts available to perform all required Circuit board revision updates .
	Have parts available to perform all required Firmware revision level updates .
	Have parts available to perform all required Memory size adjustments (or memory available for use during migration), including possible additional memory required for External Load Modules.
	Have checked to ensure all required R6xx software is available , according to Section 6.
	Have obtained R6xx versions of all currently configured External Load Modules certified to run on R6xx.

Table 8-1 Steps Prerequisite to Continuing with this Section

8.1. Overview

Migrations can be performed without removing the system from on-line control. The process of migrating UCN devices can be lengthy and exacting, and errors could have process impact. For this reason, Honeywell strongly recommends against an on-line migration of a system controlling a volatile process.

It is Honeywell's recommendation that each migration (including all steps in this guide) be examined and reviewed for its specific requirements, as they apply to your system.

Command Processor commands

The following is a quick reference to the Command Processor commands that you may use when migrating your system. If you are unfamiliar with using any of these commands, please refer to the *Command Processor Operation manual*, SW11-507, found in the TPS 3032-1 *Implementation Engineering Operations - 1* binder.

If you need to...	Use	Valid Device ID Form(s)	Valid Options (see descriptions on next page)	Reference Section (SW11-507)
Copy a file	CP	NET, \$Fn, PN:nn	-D	5.3
Copy a volume	CPV	NET, \$Fn, PN:nn	-A, -D, -I	5.4
Create a directory	CD	NET, \$Fn, PN:nn		5.6
Create a new volume	CR	\$Fn	-BS, -MF, -F, -MD	5.5
Delete files	DL	NET, \$Fn, PN:nn	-D	5.9
End a Command Processor session	END			5.11
List file attributes	LS	NET, \$Fn, PN:nn	-A, -D, -FD, -M, -N, -P, -REC, -S	5.17
List volumes and directories in a selected HM	LSV	NET, PN:nn		5.18
Protect a file	PROT	NET, \$Fn, PN:nn		5.24
Protect a Zip disk	PROTV	\$Fn	-NQ	
Select a printer to receive output	PO	NET, \$Pn, \$Fn, PN:nn		5.23
Select device(s) to receive data	DO	NET, \$Pn, \$Fn, PN:nn		5.7
Send messages to a US <u>and</u> a selected printer	PD		ON, OFF	5.22
Translate	TR			
Unprotect a file	UNPT	NET, \$Fn, PN:nn		5.24
Unprotect a Zip disk	UNPTV	\$Fn	-NQ	

Table 8-2 Command Processor Commands - Quick Reference

8.1. Overview

Command Processor command options The following is a list of the valid options that can be used in conjunction with the previously listed Command Processor commands.

NOTE: The option may have different meanings for different commands, so follow the descriptions as they apply to each command.

Option	As Used by this Command	Means (for this command)...
-D	CP, CPVDL, LS	Detail option (provides for command details)
-A	CPV	Include all directories in the volume if the destination is a cartridge
-I	CPV	Enable media initialization
-BS	CR	Block size [26 to 1700 if not using -FD; 26 to 300 (migrating from R4xx) or 26 to 6xx (migrating from R500 or R510) if -FD is also being used]
-F	CR	Removable media format option
-MD	CR	Memory-resident Directory Option (NOTE: Do not use at same time as -BS)
-MF	CR	Maximum files option (1 to 32767)(Default for cartridge = 3020)
-X	CR	Extended Directory Support. Allows for creation and tracking of up to 2046 directories/virtual volumes on a cartridge. Do not use on disks that are used for - translations, - Find names, - AM/CL file I/O, or CDS Read/Write.
-A	LS	List all files and all directories in this volume
-BF	LS	Display limited attribute information
-F	LS	(-F DATE, TIME) List only those files with date/time after those on command line
-FD	LS	List all files along with their file descriptors
-M "XX"	LS	Only display files whose descriptors match "XX".
-N	LS	List only file info., no header, footer information
-P	LS	(-P DATE, TIME) List only those files with date/time prior to those on command line
-REC	LS	List all files and their record information
-S	LS	Sort file attributes by DIR, NAME, EXT, DATE, or DESC
ON	PD	Prompts sent to printer
OFF	PD	Prompts not sent to printer
-NQ	PROTV UNPTV	Suppresses list of directories on device and need for operator to respond

Table 8-3 Command Processor command Options - Quick Reference

Enough time allotted?

Be sure you have allotted enough time to complete the migration process. It is difficult to say how long it should take, but it often requires in excess of ten hours to complete the on-line portion of the migration, depending on the number of nodes you have, and how experienced you are with software migrations.

Allow 4 hours per HM and 4 hours to load the LCN nodes; and add an additional 5 minutes per PMM/APM/LMM/HPM/SM/FSC on each UCN.

Step layout

This migration procedure is organized in several major steps, which are divided up into individual steps, to allow break points where possible. A column is provided to the right of each step to check (✓) off each step upon its completion.

Please read every step and associated CAUTIONS, ATTENTIONS and NOTES to help ensure a successful and time efficient migration. It is very likely that not doing so will cause a failed migration attempt.

Locate extra cartridge labels

Some of the procedures in this migration require that you create additional cartridges that will require labels to be created and physically attached to the cartridge.

Be sure you have blank cartridge labels available to you before starting the migration.

For all GUS systems, you may want to backup the system to its local hard drive. Ensure enough space is available.

ATTENTION

NCFs for AM, US, and HG nodes must be configured as NO for "further external directives," unless otherwise directed by Honeywell. Failure to do so will cause the loader to look for nonexistent files, and the node will crash.

8.2 Database Preparation for R5xx to R6xx Migrations

Identify NET volumes Perform the following steps to capture the HM Volume configuration for each HM, and print the HM Volume usage.

Step	Action	Done (√)
1	Perform a list volume to the printer, as follows: TYPE DO \$Pn where n = printer number TYPE PD ON	
2	TYPE LSV NET	
3	TYPE LS PN:nn -d where nn = HM node number This will print the current volume usage for that HM.	
4	Repeat Step 3 for each remaining HM on your system. Use this information to find the space available to reconfigure/ re-allocate space for R6xx.	

Table 8-4 Identify NET Volumes

Stabilize process

NOTE: Boxes with communications errors, and boxes in a state where they cannot be checkpointed, will not take checkpoints and therefore **cannot be migrated to R6xx**.

Verify there are no boxes in the wrong state, or with communication errors on HIWAY(s) or UCN(s).

Take demand checkpoints

Perform steps 1 through 4 in the following table to take demand checkpoints to NET for all data owners on the LCN.

Step	Action	Done (√)
1	Take demand checkpoints to NET for all HG devices.	
2	Take demand checkpoints to NET for all UCN devices.	
3	Take demand checkpoints to NET for all AMs.	
4	Take demand checkpoints to NET for all CGs.	

Table 8-5 Take Demand Checkpoints

**Build R5xx FAST
load disks**

Use the following steps to build custom fast load media. This media can be used to quickly reestablish R5xx, paths to critical values, should a system reestablishment be disrupted.

Step	Action	Done (√)
1	DISABLE auto checkpointing for the target process network (if enabled).	
2	MOUNT R5xx, source cartridge &Z1 or R5xx CD emulated disk in \$Fs (source drive).	
3	MOUNT destination cartridge in \$Fd (destination drive).	
4	Printer output should already be directed to the printer. If not, do the following: TYPE DO \$Pn (where n = printer number) TYPE PD ON	
5	RUN the FST_VOLZ.EC (example shown below). EC \$Fs>&EC>FST_VOLZ.EC \$Fd an hn NET nnnn where s = Source drive number d = Destination drive number an = Area number (01 to 10) hn = Hiway or UCN number nnnn = Volume name on cartridge (if it is to be initialized)	
6	If desired, ENABLE auto checkpointing.	
7	LABEL cartridge: "Fast Load for Console ## Rxxx Hiway ## " -or- "Fast Load for Console ## Rxxx UCN ## "	
8	REPEAT Steps 2 through 7 for each remaining AREA, Hiway and UCN.	
9	To ensure there are no write errors on your fast load cartridges, COPY each newly created cartridge to one scratch cartridge (use the same one over and over), using the FCOPY command.	

Table 8-6 Build R5xx Fast Load Disks

8.2. Database Preparation for R5xx to R6xx Migrations**Save UNP Load Support files**

Follow steps 1 through 6 in the following table to save the area database files to Zip disks.

Note: If using a GUS for the upgrade the “Customer Support Files – R53x – R6xx” zip disk created in Tables 8-7 and 8-8 must be an emulated device file (.lcn). The .lcn file will be used in Section 9.3.

Step	Action	Done (✓)
1	Initialize a Zip disk or emulated disk to be used for system startup. Create a directory for each area. Example syntax: CR \$Fd>CNCf> -MF 4500 Label Zip disk "Customer Support Files – R53x to R6xx".	
2	If any area has memory resident schematics, CREATE directory(s), and COPY to disk. NOTE: In the process of accessing the memory resident schematics, you may automatically create an uninstalled Area Database work file (.WA). If so, you should DELETE it at this time.	
3	Printer output should already be directed to the printer. If not, do the following: TYPE DO \$Pn (where n = printer number) TYPE PD ON	
4	Copy the area files from the NET onto the disk under the same directory names. EXAMPLE: CPV NET>&Dan> \$Fd>&Dan> -D where an = Area Database Number (01 to 10) d = Destination drive number	
5	REPEAT step 4 for each additional area.	
6	If the button file is not in &Dan, then CREATE the correct directory on disk, COPY button.ko file to the correct directory.	

Table 8-7 Save UNP Load Support Files

Create “CUSTOMER NCF” disk

Use the following steps to copy all the .PL files and R5xx user files of .ER, .SE, .SP, and .XX onto a Customer NCF cartridge. These saved files will be used to do the initial loading of an R6xx UNP with the translated NCF, and it will be copied to the R6xx system HM after initialization.

Using the "Customer Support Files" disk created in step 1 of Table 8-7, “Save UNP Load Support Files,” execute the following steps:

Step	Zip Disk Action	Done (✓)
1	On the disk, CREATE a directory called &ASY. EXAMPLE SYNTAX: CD \$Fd>CNCf> &ASY	
2	COPY all &ASY files (* . *) to the disk. EXAMPLE SYNTAX: CPV NET>&ASY> \$Fd>&ASY> -D	
3	Save External Load Modules to Custom Support Files. If any nodes on your system have nonstandard custom External Load Modules, CREATE directories &CUS and &CLX and COPY all (both standard and nonstandard) R6xx External Load Modules to the destination disk.	

Table 8-8 Create “CUSTOMER NCF” Disk

8.2. Database Preparation for R5xx to R6xx Migrations

Create “CUSTOMER NCF” disk, continued

Use the following steps to copy the \$*.PL files to the cartridge, using the Honeywell-provided **R6xx** disk &Z6 SYSTEM STARTUP &ASY or R6xx TPN software CD DISK_&Z6.LCN.

Step 7 in the following table is required because the LCN file manager cannot detect write errors. So, it is necessary to verify that the data on a disk is good by copying it to another disk.

Step	Action	Done (√)
4	UNPROTECT and DELETE \$* .PL files from “Customer Support Files” disk. SYNTAX: UNPT \$Fd>&ASY>\$* .PL DL \$Fd>&ASY>\$* .PL where d = “Customer Support Files R53X to R6xx” cartridge	
5	MOUNT Honeywell-provided R6xx disk &Z6 (64-node NCF), or the CD DISK_6.LCN in another external drive.	
6	COPY \$* .PL files from &Z6 to “Customer Support Files”. disk SYNTAX: CP \$Fs>&ASY>\$* .PL \$Fd>&ASY>= -D where s = Source drive &Z6 d = “Customer Support Files” disk	
7	To ensure there are no write errors on your backup disks, COPY the newly created disk to one scratch disk (use the same one over and over), using the FCOPY command.	

Table 8-8 Create “CUSTOMER NCF” Disk (continued)

Save IDFs as .EBs (optional)

If you need to maintain a current loadable image of your database, it is recommended that you save Intermediate Data Files (IDFs) as Exception Build (EB) files, to be rebuilt later. Follow steps 1 through 7 in the following table to accomplish those saves.

Step	Action	Done (✓)
1	INITIALIZE a cartridge for the database files. CREATE directories as appropriate. LABEL disk "EB Files."	
2	On a US with R5xx Engineering or Universal Personality, use the Text Editor to CREATE a DEB command file (name.EC). There should be one PI command line per IDfname.DB file. name.EC file. EXAMPLE: <pre>V1 = NET>IDF V2 = \$F1>EB PI V1>IDfname1.DB &B V2>EBsourc1.EB PI V1>IDfname2.DB &B V2>EBsourc2.EB PI V1>IDfname3.DB &B V2>EBsourc3.EB</pre> CAUTION The .EC command file should not have the same filename as the IDF or .EB files because it interferes with the automatic naming of the error file.	
3	From the Engineering Main Menu, SELECT BUILDER COMMANDS	
4	From the Builder Commands Menu, SELECT EXECUTE COMMAND FILE	
5	TYPE in the path and name for your .EC file on the pathname for COMMAND FILE line.	

Table 8-9 Save IDFs as .EBs (optional)

**Save IDFs as .EBs
(optional), continued**

Step	Action	Done (✓)
6	PRESS <ENTER> to execute. The output is your set of text (ASCII) Exception Build Files. Exception Build File output format: &t point type &n entity name Parameter name = value Parameter name = value . . Parameter name = value &t point type &n entity name Parameter name = value Parameter name = value . . Parameter name = value	
7	Repeat steps 1-6 until all points are saved in EB files on removable media.	

Table 8-9 Save IDFs as .EBs (optional) (continued)

Build and save all History Group Point definitions list

If the History Group Point definitions do not already exist in the EB files, perform the following steps to build and save the History Group Point definitions to EBs.

Step	Zip Disk Action	Done (✓)
1	Using the Text Editor and following the syntax example, BUILD History Group Point definitions list. EXAMPLE SYNTAX: <pre>\$CHui(gn) where ui = Unit Index number (from UNIT NAMES under the Engineering Main Menu, Unit Names, under the column labeled "NO") gn = History Group number in unit</pre> EXAMPLE: <pre>\$CH01(01) \$CH01(02) \$CH01(03) \$CH01(04) \$CH01(05) \$CH02(01) \$CH02(02) \$CH02(03) \$CH02(04)</pre> NOTE: Place four spaces between each definition	
2	PERFORM DEB PRINT SYSTEM ENTITIES using the saved selection list. This saved file should have a .EL extension. OUTPUT to a file in the net with .EB extension.	

Table 8-10 Build and Save all History Group Point Definitions List

8.3 System Backout Backup for R5xx

Create HM backups Use the following steps to copy all the directories and files on each HM to a cartridge. This system backout backup should be performed to allow you to back out to R5xx, should that need arise.

Honeywell provides a procedure to back out of an R6xx migration, returning to R5xx. If you should need to back out of the R6xx migration, you will need the cartridges created in the steps below.

CAUTION

This backup does not save any history or journals.

Step	Action	Done (✓)
1	If this HM has dump volumes, it is recommended they be deleted before you perform the backup. DL PN:nn>&OPD>*.* (&NMD, &HGD, &CGD, &AMD, &HMD)	
2	DISABLE automatic checkpointing from all checkpointed nodes (HG, NIM, CG, and AM) and do not allow demand checkpoints.	
3	CREATE volumes on all of the backup cartridges or emulated disks, either with or without FD (File Descriptors). Follow one of the examples below, depending on your situation. EXAMPLE WITH FD: CR \$Fd>HMnn> -FD -MF 3400 where d = destination drive number nn = HM node number EXAMPLE WITHOUT FD: CR \$Fd>HMnn> -MF 3400 -BS 1700 where d = destination drive number nn = HM node number	

Table 8-11 Create HM Backups

Create HM backups,
continued

Step	Action	Done (√)
4	Through the Command Processor, EXECUTE the BACKUP command to create backup. EXAMPLE SYNTAX: BACKUP PN:nn \$Fd where nn = HM node number d = destination drive number or emulated disk	
5	To ensure there are no write errors on your backup cartridge, COPY each newly created cartridge to one scratch cartridge (use the same one over and over), using the FCOPY command, and using the same scratch cartridge.	
6	Honeywell recommends you have two copies of your backup. Use FCOPY to copy each cartridge to another cartridge.	
7	Repeat steps 1 through 6 for each HM.	

Table 8-11 Create HM Backups (continued)

8.3. System Backout Backup for R5xx

Save maintenance recommendation journal (optional)

All History Modules must be initialized for R6xx. So, if maintenance history is required, you should perform the following steps. This saves the Maintenance Recommendation Journal to the printer.

Step	Action	Done (√)
1	From the Engineering Personality Main Menu, SELECT SMCC MAINTENANCE	
2	From the SMCC Main Menu, SELECT SYSTEM MAINT JOURNAL	
3	From the System Maintenance Journal screen, SELECT PRINT ENTIRE SMJ	
4	SELECT Ctrl/Page Back	
5	From the SMCC Main Menu, SELECT ACTIVE MAINT JOURNAL	
6	From the System Maintenance Journal screen, SELECT PRINT ACTIVE MAINTENANCE RECOMMENDATIONS	
7	From the System Maintenance Journal screen, SELECT PRINT ALL ERROR AGGREGATES	
8	LABEL printer output appropriately and preserve as a record of your maintenance history.	

Table 8-12 Save Maintenance Recommendation Journal (optional)

**Save Process/
System Journal
(optional)**

Process/System Journal files are deleted as a part of the R6xx migration. If you need to save your Process/System Journal events, perform the following steps to save them to hard copy.

Step	Action	Done (√)
1	From the Operator Keyboard or Engineering Main Menu, SELECT SYSTEM MENU	
2	From the System Menu, SELECT EVENT HISTORY MENU	
3	From the Event History Display, SELECT each of the seven types of history and print them.	
4	From the System Menu, SELECT REPORT/LOG/TREND/JOURNAL MENU	
5	From the menu displayed, SELECT all the items there you wish to print, and SELECT PRINT	
6	LABEL printout appropriately and preserve as record of your unit journal history.	

Table 8-13 Save Process/System Journal (optional)

Save Continuous History (History Data and Group Point definitions)

Perform this procedure if you want both History Data and Group Point definitions saved. Skip to the next procedure if you only want Group Point definitions saved.

The Backout Backup process does not save Continuous History. So, if

- you wish to restore it to R5xx, or
- you wish to restore it forward to R6xx,

Perform the following steps to save Continuous History.

Step	Action	Done (✓)
1	LOAD HM with the R5xx OFF-LINE personality.	
2	FORMAT sufficient cartridges to save history. To determine number of cartridges you will require, divide the number of sectors by the number given for your cartridge size (below): For 20 MB cartridge: Number of Sectors ÷ 83,000 (rounded up) For 35 MB cartridge: Number of Sectors ÷ 122,500 (rounded up) For 150 MB cartridge: Number of Sectors ÷ 525,000 (rounded up) For Zip 100 MB disk: Number of Sectors ÷ 394000 (rounded up) For 20 MB emulated disk: Number of Sectors ÷ 83,000 (rounded up)	
THIS IS <u>NOT</u> A STEP—This is an example of the output when calculating number of cartridges needed. LS PN:nn>!0np> -D where nn = HM node number np = HM node pair number Note number of sectors is displayed EXAMPLE: VOLUME: !001 VOLUME DESC: Continuous History MAX FILES: 416 TOTAL SECTORS: 195552 SECTORS IN USE: 194330 <-Sectors used VOLUME SUBSTATE: INITIALIZED DIRECTORY: MEMORY RESIDENT AVL SUPPORTED ON VOLUME: ENABLED ACCESS LEVEL: NETWORK_READ_LOCAL_WRITE FILE DESCRIPTORS: DISABLED CURRENT DIRECTORIES ASSIGNMENT: !A00 Catalog Complete So, 194330 ÷ 83,000 = 2.34 (or 3 “20 MB” cartridges, rounded up)		

Table 8-14 Save Continuous History

Save Continuous History (History Data and Group Point definitions),
continued

Step	Action	Done (✓)
3	ENSURE the HM is in the Off-line personality.	
4	SAVE history for the volume. EXAMPLE SYNTAX: CPV PN:nn>!0np> \$Fd>!0np> -A -D where nn = HM node number np = HM node pair number d = destination drive number Continue to insert cartridges until backup is complete.	
5	Honeywell recommends you have two copies of your backup. Use FCOPY to COPY each cartridge to another cartridge.	
6	LABEL each cartridge, as follows: !0np #1 of ____ !0np #2 of ____ !0np #3 of ____ , etc.	
7	REPEAT Steps 2 through 6 for !1np, if that volume exists on the specified HM.	
8	ADD final "total" number of cartridges in the "____" (blank part of each label).	
9	REPEAT Steps 1 through 8 for each HM that has Continuous History.	

Table 8-14 Save Continuous History (continued)

8.3. System Backout Backup for R5xx

Save Continuous History (Group Point definitions only)

Only perform this procedure if you wish to save only the History Group Point definition and if you did not perform the previous procedure *Save Continuous History (History Data and Group Point definitions)*.

Perform the following steps to save the History Group Point definitions. These steps are performed for !0np only.

Step	Action	Done (✓)
1	LOAD HM into the R5xx OFF-LINE personality.	
2	Format one blank cartridge or emulated disk, using the syntax below. SYNTAX: CR \$Fd>!0np where d = destination drive number np = HM node pair number	
3	COPY the Group Point definition files from NET to cartridge. SYNTAX: CP PN:nn>!0np>APL*.mm \$Fd>!0np>= -D where nn = HM node number d = Destination drive np = HM node pair number	
4	LABEL the cartridge !0np (History Group Point definition files)	
5	To ensure there are no write errors on your !0np cartridge, COPY the newly created cartridge to a scratch cartridge, using the FCOPY command.	

Table 8-15 Save Continuous History (Group Point definitions only)

**Backout Backup
(optional)**

If you need to assure that all of the copies above worked, perform the following step to copy each cartridge to another scratch cartridge.

By doing so, any copy errors that have occurred will become evident.

Step	Action	Done (√)
1	To ensure there are no write errors on your backup cartridges, COPY each newly created cartridge to a scratch cartridge, using the FCOPY command, and using the same scratch cartridge (use the same scratch cartridge over and over).	

Table 8-16 Backout Backup

8.4 Save R5xx System Files for R6xx

Backup HM

In the following steps, you will use the Command Processor to backup all non-history files with the exception of personality images, from each HM.

It is not required for the HM to be in the OFF-LINE personality to perform the following backup. If desired, the HM can at this point be rebooted in the R5xx ON-LINE personality. If this is done, it is normal that HMs with redundant drives shows a SEVERE status. This is not a problem.

ATTENTION The backup command <u>does not</u> backup the personality images or the continuous history files. It saves the Continuous History Group definitions.		
Step	Action	Done (✓)
1	RECORD all activity that follows on the printer. TYPE DO \$Pn TYPE PD ON	
2	If this HM has dump volumes, to reduce the time it takes and the amount of disk space it occupies, it is recommended that all dump files in each dump subdirectory be deleted before you perform the backup. DL PN:nn>&OPD>*. * (&NMD, &HGD, &CGD, &AMD, &HMD)	
3	Using the Command Processor BACKUP command, COPY all non-history files from the HM to cartridge. EXAMPLE SYNTAX: BACKUP PN:nn \$Fd where nn = HM node number d = Destination drive number NOTE: The BACKUP command formats each cartridge, if needed, according to your responses to prompts. The BACKUP command can be used in parallel for all HMs.	

Table 8-17 Backup HM

**Backup HM,
continued**

Step	Action	Done (√)
4	LABEL the cartridges "BACKUP PN:nn #y". where nn = HM node number y = Sequential number for multiple media	
5	LOAD one of the BACKUP cartridges in \$Fd, and LOAD R6xx &Z1 or DISK_&Z1.LCN in \$Fs where d = Destination drive number s = Source drive number	
6	To ensure there are no write errors on your backup cartridges, COPY the newly created cartridge to a scratch cartridge, using the FCOPY command.	
7	NOTE: <u>Do Not</u> delete the .DA and .WA files. Answer " <u>No</u> " to the following question: "All area definition files 01-10 (.DA and .WA) in &D directories are about to be deleted. Continue (Yes/No)" DELETE the files that are not migrated to R6xx. EXAMPLE SYNTAX: EC \$Fs>&EC>CLEANUPZ.EC d Where s = Source file d = Destination cartridge number you are cleaning up NOTE: Upon completion, the system will display Delete Complete, followed by Please Wait. Do not continue until the waiting period has completed.	
8	REPEAT steps 4 through 7 for each BACKUP cartridge or emulated disk.	
9	REPEAT steps 1 through 8 for each remaining HM. REMINDER: The BACKUP command can be used in parallel for all HMs.	

Table 8-17 Backup HM (continued)

Preparation for translation

Prior to translating, ensure your process is as stable as possible.

Step	Action	Done (✓)
1	NOTE: Boxes with communications errors, and boxes in a state where they cannot be checkpointed, <u>will not take checkpoints</u> . Correct any box or I/O errors on HIWAY(s) or UCN(s).	

Verify the LCN RECONNECT is terminated, as follows:

2	From the SYSTEM STATUS Display, SELECT A Node Then SELECT LCN OVERVIEW	
3	If the SYSTEM VERSION (upper-right corner) contains INIT LCN RECONNECT, (meaning it is initiated), use the LCN Overview Display to ensure all nodes have the same network view. SELECT TERM LCN RECNECT, and PRESS <ENTER>	

Table 8-18 Preparation for Translation

Save checkpoints to cartridge

Perform the following steps to save the NIM and HG checkpoints to cartridge for translation using the “Customer Support Files” R53x to R6xx disk.

Note:

If using a GUS for the upgrade, the “Customer Support Files” disk referenced below must be an emulated device file (.lcn). The .lcn file will be used in Section 9.

Step	Action	Done (✓)
1	PERFORM CHECKPOINTS for HGs and all UCN-controlled devices to cartridge.	
2	PERFORM CHECKPOINTS for all AMs and CGs to cartridge or emulated disk.	
3	CREATE a list of the ACP points, to be used in a later procedure for migrating the CGs.	

Table 8-19 Save Checkpoints to Cartridge

9. Loading R6xx Universal Station

9.1 R6xx Translation

Translators used

Ensure that you have completed Section 7 actions if migrating from R43x. Ensure that you have completed Section 8 actions if migrating from R5xx.

Below is a list of translators along with a brief description and the syntax to be used to run each translator. (s=source drive; d=destination drive; an= area number; uu=UCN node number; hh=Hiway node number).

Do not use the “-X” option when formatting disks that are used for translations.

Name	Description	Syntax
NCF	Network configuration	TR NCF \$Fs \$Fd
NIM	NIM checkpoints	TR NM \$Fs \$Fd uu
HG	HG checkpoints	TR HG \$Fs \$Fd hh

Off-Line translators for xPM:

NOTE: Only R6xx checkpoints can be used as source when running these off-line translators.

PM to HPM	UCN, converting from a PM to an HPM	TR PM HPM \$Fs \$Fd uu nn [-S]
APM to HPM	UCN, converting from an APM to an HPM	TR APM HPM \$Fs \$Fd uu nn [-S]
PM to APM	UCN, converting from a PM to an APM	TR PM APM \$Fs \$Fd uu nn

Table 9-1 Translators Used for R6xx

NOTE:

If using a GUS to perform the migration to R6xx, skip to section 9.3.

Load US with R6xx

Migrate the first Universal Station. Note that once it is loaded with R6xx software, it cannot communicate with any nodes running the older R5xx software.

NOTE: When the first node is loaded with R6xx, the System Status Display may show **CABLE SUSPECT**. This can be ignored because the system now has an LCN cable master node on the previous release and an LCN cable master node on R6xx.

ATTENTION

Be sure to load only one US with R6xx to completion. Expect the US to go to a WARNING STATUS. Use the Console Status Display to verify the message "Custom/Standard Parameter duplicate XREF Creation Failed." This happens because the US cannot write to the &ASY directory at this time since the &Z6 zip is write-protected. This error will not adversely affect translations.

Step	Action	Done (√)
1	CHOOSE a Universal Station that has two zip drives, an Engineering Keyboard, a working printer, and a keyswitch for placing the US in engineer keylevel access. If your US has no keyswitch please see section 14.7 for a workaround.	
2	SHUTDOWN US.	
3	This step must be performed because: - the QLTs have changed, and the node hardware must be requalified with the new QLTs, and - now is the time to update the hardware, as required. Once the US goes to QUALIF state, go to the physical US node, POWER OFF the node, WAIT 5 seconds , then, if not upgrading US hardware, POWER ON the node. Otherwise:	
4	UPGRADE US hardware to R6xx UNP minimums as needed, then POWER ON the node.	
5	From Honeywell-provided disk &Z1 and &Z6 (for the Button files-and Area Database). LOAD the R6xx Universal Personality (UNP), and the 64-node NCF.	
6	VERIFY the date and time in the upper right corner of the Engineering Main Menu is correct. CORRECT date and time if necessary.	
7	From SUPPORT UTILITIES , SET device path to removable media.	

Table 9-2 Load US with R6xx

Translate NCF

Perform the following steps to translate the NCF file.

Step	Action	Done (✓)
1	MOUNT the R6xx translator disk (&Z1) in the lowest numbered removable drive.	
2	From the Command Processor, type TR, and press <ENTER>. NOTE: The translator loads, and a list of possible entries, is displayed with a message "ILLEGAL / NOT IMPLEMENTED". THIS IS OK.	
3	MOUNT "Customer Support Files – R53x to R6xx" in \$Fs.	
4	Label a disk "Customer Support Files – R6xx" and mount it in \$Fd.	
5	Make a copy of the "Customer Support files" FCOPY \$Fs \$Fd	
6	Translate the NCF file. EXAMPLE SYNTAX: TR NCF \$Fs \$Fd where S = source drive number d = destination drive number	
7	If any nodes on your system have nonstandard custom External Load Modules, CREATE directories &CUS and &CLX, and copy all (both standard and nonstandard) R6xx External Load Modules to the destination disk.	

Table 9-3 Translate NCF

9.1. R6xx Translation**Translate NIMs**

Perform the following steps to translate the NIM checkpoints.

Step	Action	Done (√)
1	INITIALIZE a cartridge for the translated NIM checkpoints in \$Fd (destination drive) and create a directory for each UCN network number. EXAMPLE SYNTAX: CD \$Fd>CNCf> &Iuu where d = destination drive number uu = UCN network number (01-20)	
2	MOUNT the "Customer Support Files – R6xx" in \$Fs (source drive).	
3	TRANSLATE <u>each NIM checkpoint</u> . EXAMPLE SYNTAX: TR NM \$Fs \$Fd uu where s = source drive number d = destination drive number uu = UCN network number (01-20)	
4	REPEAT step 3 for each additional UCN.	
5	LABEL media "TRANSLATED NIM CHECKPOINT R6XX".	
6	PT &Fd>&Iuu>*. * MOUNT the R5xx UCN checkpoint. CP \$Fs>&Iuu>*. * \$Fd>&Iuu>= UNPT \$Fd>&Iuu>*. * where s = source drive number d = destination drive number uu = UCN network number (01-20)	

Table 9-4 Translate NIMs

**Translate
HG/PLCGs**

Perform the following steps to translate the HG/PLCG checkpoints.

Step	Action	Done (√)
1	INITIALIZE a cartridge for the translated HG/PLCG checkpoints in \$Fd (destination drive) and create a directory for each HG/PLCG network number. EXAMPLE SYNTAX: CD \$Fd>CNCF> &Ihh where d = destination drive number hh = Hiway number (01-20)	
2	MOUNT the "Customer Support Files" in \$Fs (source drive).	
3	TRANSLATE <u>each HG/PLCG checkpoint</u> . EXAMPLE SYNTAX: TR HG \$Fs \$Fd hh where s = source drive number d = destination drive number hh = Hiway number (01-20)	
4	REPEAT step 3 for each additional HG/PLCG.	
5	LABEL media "Translated HG checkpoints R6xx" for HGs, and "Translated PLCG checkpoints R6xx".	
See NOTE on the next page for How to recover from R6xx Off-line Hiway Gateway Translator errors .		

Table 9-5 Translate HGs/PLCGs

NOTE**How to recover from the following R6xx Off-line Hiway Gateway Translator errors:**

FATAL ERROR : Could Not Copy ASSOCDSP Data File

FATAL ERROR : Could Not Copy Aux Data File

FATAL ERROR : Could Not Copy OP Data File

These errors occur when the HG master checkpoint file references an "ASSOCDSP Data" file, an "Aux Data" file, or an "Output Alarm Data" file, but the HG Off-line translator cannot find these files on the source medium.

1. The most likely source of this error is the failure to copy all checkpoint files to the source medium. Ensure that all checkpoint files have been copied and try the translation again.

The "ASSOCDSP Data" files are named as follows: HG0nnEXy.CP

The "Aux Data" files are named as follows: HG0nnAXy.CP

The "Output Alarm Data" files are named as follows: HG0nnOPy.CP

where: nn = hiway number 01 through 20

y = file version 1 or 2

2. Make certain that the original HG checkpoint did not contain any HG0nnEXy.CP, HG0nnAZy.CP or HG0nnOPy.CP files, then copy "default versions" of these checkpoint files to the source medium from the R6xx &Z6 zip disk and try the translation again.

If the "ASSOCDSP Data" files are missing, use the following Command Processor command:

```
CP $Fxx>&Inn>HG0nnEX?.CP $Fzz>&Inn>= -D
```

If the "Aux Data" files are missing, use the following Command Processor command:

```
CP $Fxx>&Inn>HG0nnAX?.CP $Fzz>&Inn>= -D
```

If the "Output Alarm Data" files are missing, use the following Command Processor command:

```
CP $Fxx>&Inn>HG0nnOP?.CP $Fzz>&Inn>= -D
```

Where: nn = hiway number 01 through 20
 xx = drive number for &Z6 zip disk
 zz = drive number for your checkpoint source medium

The need to perform step #2 should be very rare! Do this ONLY if it is certain that the original HG checkpoint did not contain these files. Also, copy ONLY the file types that caused the translator error.

Load the NCF Cartridge**ATTENTION**

On the first R6xx Universal Station load, the NCF cartridge must be left in the drive for approximately five minutes after the station goes to OK. This allows the duplicat.SP file to be created.

Use the US Status Display to verify the message "CUSTOM/STANDARD PARAMETER DUPLICATE XRef created" is displayed before continuing.

Reboot US

Perform the following steps to reboot the US with R6xx UNP, Customer NCF, and translated area databases.

Step	Action	Done (√)
1	SHUTDOWN and RESET the Universal Station that you have been using for the translation.	
2	When loading this node, you will be asked by the system to mount a sequence of R6xx cartridges, all in the same cartridge drive, during the load. The R6xx Zip disk to be used: Customer Support Files R6xx IMPORTANT! The "CUSTOMER SUPPORT FILES - R6xx" must remain on the same drive until the US is fully loaded. See the NOTE above.	
3	VERIFY the date and time in the upper right corner of the Engineering Main Menu is correct. CORRECT date and time if necessary.	
4	SET device paths to removable media.	

Table 9-6 Reboot US

9.2 NCF Configuration Changes

Upgrade NCF Action

NOTE:

Ensure NCF.CF is not write protected when doing the “Upgrade NCF” action and/or other “install” action.

Update R6xx NCF

ATTENTION

You may get an error after pressing <ENTER>, and making a change to the NCF. If so, check to ensure each HM volume size is correct for the following entries:

- Burst Buffer size must be less than 7501.
- Journals sizes for:
 - PROCESS ALARMS must be less than 15001
 - PROCESS CHANGES must be less than 10001
 - OPERATOR MESSAGES must be less than 5001
 - SYSTEM STATUS must be less than 12501
 - SYSTEM ERRORS must be less than 15001
 - SYSTEM MAINTENANCE must be less than 15001
 - AREA SOE must be less than 15001

The order of changes to modify an NCF configuration is the same as for building the NCF. Some of the more important relationships to keep in mind while changing the NCF.CF are:

- New Units Names should be added first. They are used in the configuration of AM & CG nodes and the HM Volumes.
- New Area Names may be used in a Universal Station configuration and in HM Volumes. Honeywell recommends you configure one more area than the number required.
- Add GUS Node(s) according to the TPS System Implementation Guide.
- Add APP Node(s) according to the TPS System Implementation Guide.
- NCF for the AM must be configured as NO for "further external directives," unless otherwise directed by Honeywell. Failure to do so will cause the loader to look for nonexistent files, and the AM will crash.
- System Wide Value configuration can affect HM Volume configuration.

New checkpointed nodes can be used in HM Volume configuration.

Update R6xx NCF,
continued

NOTE:

The UCN Checkpoint Volume size (&801) has increased. Make sure you allow sufficient space to accommodate automatic checkpointing. To calculate the size of the UCN checkpoint volume &8np for each primary and redundant NIM on the LCN, refer to the HM Volume size Estimator Table in Section 7 of the engineer's Reference Manual. Use the equations provided in the HM Volume Size Estimator table for volume &8np **to calculate the UCN checkpoint size of each primary and redundant NIM**. Then, enter the new UCN checkpoint size for each primary and redundant NIM node into the Physical Node for Checkpoint Configuration portion of the NCF.

There are two changes to US node configuration in R6xx. The default settings are:

Function	Default Setting
STATION DEFAULT LOAD PERSONALITY	OPR
EXTERNAL CUSTOM SCHEMATIC MEMORY (Kilowords)	0 Kilowords

Table 9-7 Default Setting NCF.CF

NOTE: Do not use the **ALL** option if you have memory configured for Custom Backplane Modules.

The LCN Node Version/Revision Logging Function can also be added to Universal Station Operator and Universal Personalities.

There are two limit changes on page two of the AM configuration.

Function	New Limit
CVB SIZE FOR FAST & SLOW POINT PROCESSORS	32700
REDUNDANCY BUFFER INCREASE (# OF 32KW BLOCKS)	20 blocks

Table 9-8 New Limits

Update R6xx NCF,
continued

There is a new installation unit in System Wide Values.

SYSTEM WIDE VALUES MENU	
TARGET	ACTION
AM MULTIPLE PRIMMOD ALARMING OPTIONS	Calls up the AM Multiple Primmod Alarming Options Display.

Table 9-9 System Wide Values

AM MULTIPLE PRIMMOD ALARMING OPTIONS DISPLAY	Default
AM MULTIPLE PRIMMOD ALARMING OPTIONS: • Selecting DISABLED causes alarms to be reflected only in PRIMMOD parameter alarm groups. • Selecting ENABLED EXCLUSIVE MODE causes alarms to be reflected in alarm groups for those \$MPROD parameters which have values. If none of the \$MPROD parameters have values, the alarms will be reflected in the PRIMMOD alarm group. • Selecting ENABLED INCLUSIVE MODE causes alarms to be reflected in the alarm groups for all the point's \$MPROD parameters which have values and also in the alarm group for the PRIMMOD parameter.	DISABLED

Table 9-10 AM Multiple Primmod Alarming Options

Update R6xx NCF,
continued

Page 1 NCF CONSOLE DATA	Default
REAL TIME JOURNAL SIZE The US RTJ buffer can be sized up to 400 events to minimize event loss caused by US buffer overflow during alarm floods. You must eventually reload all USs in order to implement new RTJ buffer sizing.	100 events
REAL TIME JOURNAL HEADER/TRAILER MESSAGES Selecting BEG/END causes the RTJ header and trailer messages to behave exactly as they currently do with a header and trailer being output every time the RTJ buffer is emptied on 5 second intervals. Selecting BEG ONLY causes only one RTJ header message to be output per page, allowing more efficient use of the printer and making the RTJ page easier to read.	BEG/END
CONTACT CUTOFF ALARMS ON ALARM SUMMARY This NCF option specifies how alarms should be handled when a point goes into the Contact Cutoff state.	CLEAR WHEN ACK'D
Page 2 HORN ANNUNCIATION POLICY	Default
CONTACT x OUTPUT STATE (x = 1,2,3) Selecting MOMENTARY configures the US alarm contacts as momentary contacts. The horn will sound for two seconds, then automatically silence. The duration is not configurable.	STEADY
PRINTER ALARMS ANNUNCIATED Selecting NO prevents printer alarms from being sent to the operator (off-line, paper out, power off, jam).	YES

Table 9-11 NCF CONSOLE DATA Changes

Update R6xx NCF, continued

Page 3 ALARM MANAGEMENT CONFIGURATION	Default
RED COLOR ALARM PRIORITY This entry has not changed for R530.	EMERGENCY
PRIORITY INDICATORS ON ALARM DISPLAYS Selecting SYMBOLS replaces the letters for indicating priority (E, H, L) with symbols (■ ▲ ▼) on the Alarm Summary and Alarm Annunciator displays.	CHARACTERS
ALARM PRIORITY COLORS Selecting 3 COLORS USER SELECTABLE displays additional targets, allowing the user to select from any of the seven Picture Editor colors for each alarm priority, including low priority.	RED, YELLOW
LOW PRIORITY ALARMS IN AREA ALARM SUMMARY Selecting YES causes low priority alarms to be included on the Area Alarm Summary.	NO
INITIAL ALARM SUMMARY SORT STATE Specifies how alarms will initially be sorted on the Area and Unit Alarm Summary displays (chronological or by priority).	CHRONOLOGICAL
AUDIBLE ALARM ANNUNCIATION SUPPRESSION TIME-OUT (SECONDS) A new target, AUDIBLE, on the Alarm Summary and Alarm Annunciator displays, allows operators to disable the alarm horn on a priority basis (configurable keylock access) for all stations in the console. If you leave the timeout value at the default of zero seconds, the target will not appear on the Alarm Summary and Alarm Annunciator display.	0 seconds
ALARM SUMMARY FREEZE TIME-OUT (SECONDS) A new target, DISPLAY FREEZE, on the Alarm Summary display, allows operators to stop incoming alarms from scrolling into the display (configurable keylock access). If you leave the timeout value at the default of zero seconds, the target will not appear on the Alarm Summary display and the ramp keys will not cause the display to freeze.	0 seconds

Table 9-11 NCF CONSOLE DATA Changes (continued)

Update R6xx NCF,
continued

Page 3 ALARM MANAGEMENT CONFIGURATION,	Default
CONSOLE-WIDE SILENCE BUTTON This NCF option specifies which horns on the console will be silenced when the SILENCE button is pressed. Selecting NO will enable silencing of horns on the SILENCE button's console in the SILENCE button's Area. Selecting YES will enable silencing of horns anywhere on the SILENCE button's console, regardless of the areas loaded on stations in the console.	NO
Page 4 CONSOLE DATA - RTJ ALARM COLOR DISPLAY	Default
Select from any of the 8 printer colors for each RTJ alarm type.	
EMERGENCY PRIORITY	BLACK
HIGH PRIORITY	BLACK
LOW PRIORITY	BLACK
PRINT ONLY	BLACK
JOURNAL PRINT	BLACK
RETURN TO NORMAL	BLACK
LCN CABLE SWAP INHIBIT TIME IN MINUTES Specifies the length of time, in minutes, the system is to remain on the same cable. The default value is 0 and indicates the function is disabled.	0
ACCESS LEVEL TO INHIBIT LCN CABLE SWAPPING Specifies the access level required to execute this function.	SUPER

Table 9-11 NCF CONSOLE DATA Changes (continued)

Update R6xx NCF,

Page 5 ALARM COMMAND ACCESS TABLE	Default
CHANGE ALARM SUMMARY SORT STATE Specifies the lowest keylock access level required to change the sort state between chronological sorting and sorting by priority using the SORT target on the Area and Unit Alarm Summary displays.	OPER
CHG ALARM PRIORITY FILTER ON ALARM SUMMARIES Specifies the lowest keylock access level required to filter displayed alarms using the DISPLAY target on the Area and Unit Alarm Summary displays. This target allows the operator to specify the priority of alarms that are to be displayed.	OPER
CHG AUDIBLE ALARM ANNUNCIATION SUPPRESSION Specifies the lowest keylock access level required to suppress audible alarms using the AUDIBLE target on the Alarm Summary and Alarm Annunciator displays.	OPER
ALARM SUMMARY DISPLAY FREEZE Specifies the keylock access level required to freeze the Alarm Summary display using the DISPLAY FREEZE target or the ramp keys.	OPER
AUXILIARY UNIT CHANGE Specifies the lowest keylock access level required to change the \$AUXUNIT parameter on a point	ENGR
GROUP DETAIL DISPLAY PARAMETER CHANGES/CONTROL ACTIONS Specifies the lowest keylock access level required to make parameter modifications/control actions for all point types on group and point detail displays invoked from Operator Keyboard buttons "GROUP" and "DETAIL."	OPER
Alarm Limits Specifies the lowest keylock access level required to make changes to Alarm Limit Parameters. Default means to use the access level as defined in the Parameter Reference Dictionary.	DEFAULT
Control Limits Specifies the lowest keylock access level required to make changes to Control Limit Parameters. Default means to use the access level as defined in the Parameter Reference Dictionary.	DEFAULT
Range Limits Specifies the lowest keylock access level required to make changes to Range Limit Parameters. Default means to use the access level as defined in the Parameter Reference Dictionary.	DEFAULT
Tuning Parameters Specifies the lowest keylock access level required to make changes to Tuning Parameters. Default means to use the access level as defined in the Parameter Reference Dictionary.	DEFAULT

Table 9-11 NCF CONSOLE DATA Changes (continued)

Verify HM configuration for R6xx

Because of the size increase in the boot volume **when migrating from R43x or R5xx, all History Modules must be initialized. This is not an option.** Failure to initialize History Modules will cause unpredictable results. In preparing to migrate the software, any WREN-I or WREN-II drives must have been upgraded, and the drive definitions for any HM configured for WREN-I or WREN-II drives, must have been changed. To do so, follow the steps under “Migrate HMs.”

HM Fast User Volume

Please read and follow the guidelines for HM Configuration presented in Section 4 of this (CRG) Customer Release Guide.

Update R6xx NCF. configuration

Perform the steps in the following table to update the system NCF.CF file with the following:

- Changes to the System Wide Values/Console Data information described above.
- Update the HM configuration

NOTE: Even if no configuration changes are made, the HM checker must be run to verify the HM will hold everything specified for R6xx.

Update the system NCF.CF file.

Step	Action	Done (✓)
1	Create a Backup &ASY directory on another cartridge. CR \$Fb>&ASY> (Where: b = backup cartridge) Remove the cartridge from the drive. LABEL the cartridge “BACKUP &ASY R6xx”	
2	LOAD the “Customer Support Files R6xx” disk in \$Fs	
3	LOAD &Z1 in \$Fd.	
4	SET paths for NETWORK CONFIG to \$Fs>&ASY> (where “Customer Support Files R6xx” resides)	
5	SET paths for CONFIGURE OVRLY \$Fd>&OP1 (where: d = &Z1 will reside)	
6	Set the path for the EXT LOAD MODULE as follows: \$Fs>&CUS> (Where: s = Customer Support Files R6xx)	
7	Set the path for the NCF backup as follows: \$Fd>&ASY> (Where: d = backup cartridge)	

Table 9-12 Update R6xx NCF.CF Configuration

9.2. NCF Configuration Changes**Update R6xx NCF
configuration.,
continued**

Step	Action	Done (√)
8	From the Engineering Main Menu, SELECT SYSTEM WIDE VALUES, then SELECT CONSOLE DATA. VERIFY ON- LINE appears in the upper-right corner	
9	REMOVE &Z1 , and REPLACE it with "BACKUP &ASY R6xx"	
10	If you are not adding GUS nodes to your system, skip to step 12, otherwise SET the following to NUMERIC for Printed Trend Format	
11	SELECT BEG/END for the following: Real Time Journal Header/Trailer	
12	MAKE any desired changes to the default Console Data Settings described at the beginning of section 9.2.	
13	Run the Checker and Perform Install	
14	Return to the Engineering Main menu. If you are not adding GUS nodes to your system, skip steps 14 through 20, otherwise SELECT LCN Nodes. VERIFY ON- LINE appears in the upper-right corner	
15	ADD the GUS to the NCF by selecting a node number, then SELECT UNIVERSAL STATION . CONFIGURE Cartridge Drive numbers for emulated disk access. CONFIGURE a printer number (if applicable). SELECT the following options: UP PERSONALITY, OPERATOR and ENGINEERING KEYBOARD SELECT TOUCHSCREEN (If applicable)	
16	From the LCN Node configuration (for a newly added GUS node), SELECT Modify Node	
17	ENTER the following information for the External Load Module Names: <u>Name</u> UPBASE MSCHEM CSCHEM <u>Pers</u> UP UP UP <u>Name</u> YGOCX GUSALA GUSALAG <u>Pers</u> UP UP UP	

Table 9-12 Update R6xx NCF.CF (continued)

**Update R6xx NCF
configuration.,**
continued

Step	Action	Done (√)
18	Press <ENTER>	
19	To the total number displayed above the Maximum External Module Memory port (modules plus additional memory), ADD 10,000. ENTER the result in the Maximum External Module Memory port. Press <ENTER>	
20	CHECK and INSTALL the changes. REPEAT steps 14 through 19 for each GUS node to be added.	
The following procedure (steps 21 through 24) is optional and is used if the new LCN Version/Revision Logging Function will be run on a previously configured US station.		
21	From the UP Main Menu, Select the LCN NODES target, the desired US Node, and the MODIFY NODE target.	
22	Perform this step to run the function in the Operator personality: In one of the MODULE ports, KEY in the name OPLVR In the corresponding PERS port, KEY in the name OPR.	

Table 9-12 Update R6xx NCF.CF (continued)

9.2. NCF Configuration Changes

**Update R6xx NCF
configuration.,**
continued

Step	Action	Done (√)
23	Perform this step to run the function in the Universal Personality: In one of the MODULE ports, KEY in the name UPLVR In the corresponding PERS port, KEY in the name UP.	
24	PRESS ENTER to add the module. If the total exceeds the maximum memory size, Make the maximum memory size at least 2000 greater to allow for maintenance (10,000 greater if this is a GUS). CHECK and INSTALL the changes.	
25	From the Engineering Main Menu, SELECT VOLUME CONFIGURATION, then SELECT this HM node pair. VERIFY ON - LINE appears in the upper-right corner.	
26	BECAUSE THERE ARE CHANGES IN ALL HMs, INPUT configuration information. Reference Section 4.	
27	If you will be using 68040 personalities, you will have to add them under the PROGRAM IMAGE target on the VOLUME CONFIGURATION.	
28	RUN HM checker to verify that this HM will hold everything specified for R6xx (even if no configuration changes were made). Press F1 <CTRL + F1> To see the volume mapping and loading information for the HM, contained in the current engineering reference file, Press F8 <CTRL + F8> Make sure there are no warnings or errors listed in this file before proceeding.	

Table 9-12 Update R6xx NCF.CF (continued)

**Update R6xx NCF
configuration.,
continued**

Step	Action	Done (√)
29	If there are errors in the engineering reference file (&ASY>NCFnp.ER), or if the checker fails, ADJUST this HM configuration until there are no errors.	
30	When checker reports no errors for this HM, PERFORM the on-line INSTALL of the NCF. PRESS F2 <CTL + F2>	
31	REPEAT Steps 25 through 30 for each HM on your system.	
32	SHUTDOWN and RELOAD this R6xx US with the UNP personality and this migrated Customer NCF (Customer Support Files. This will ensure the UNP is on the correct version of the NCF.	

Table 9-12 Update R6xx NCF.CF (continued)

It is not necessary, but if possible, load a second Universal Station with R6xx. This station would be used for loading and viewing R6xx nodes. The original R6xx station can stay in the Engineering function.

Proceed to Section 10.

9.3 Migrate R5xx Systems Using GUS

Load the GUS Station from R6xx TPS Network Software CD

These procedures are for migrating R5xx systems using GUS to R6xx software using the GUS-TPN Network Software and TPS System Software CD's and emulated disks.

The following procedure will reference emulated disks &Z1, &Z6 (found on the CD labeled "R6xx GUS TPN SOFTWARE"); &Z12, &Z14 (found on the CD labeled "UTILITIES & LOAD MODULES") and supporting files. Once these procedures have been followed, the steps will allow you to migrate the rest of the TPS Network to R6xx.

NOTE: The following procedures identify the CD drive as drive D:. This is an example only. Use whatever drive letter is appropriate for your system in the steps where the D: drive is mentioned.

Initially all the necessary work can be done in Universal Personality. Use the 64-node NCF, default Area01 and button files (&D01) found on DISK_&Z6.LCN and the DISK_&Z1.LCN for the Universal Personality.

Shutdown and reload the Native Window

The following procedures will shutdown the Native Window and load the Native Window using the R6xx UNP Personality and the 64 Node Network Configuration to allow translations and creation of the "R6xx PREPARATION.LCN" emulated disk.

Note: Starting in Step 21, you will need to use the emulated device file (.lcn) "Customer Support Files R53x – R6xx" created in Tables 8-7, 8-8 and 8-19.

Shutdown and reload the Native Window

Perform the following steps to shutdown and reload the Native Window.

Step	Action	Done (√)
1	Log on using "LOCAL" administrator privileges.	
2	Dismount ALL mounted EMULATED DISK(s).	
3	Shutdown the Native Window Workstation and RESET the LCNP. NOTE: If Hardware or Firmware upgrades are required, Close all applications, Shutdown and power down and make the required changes using ESD protection.	
4	Mount the R6xx GUS-TPN Software CD into the CD-ROM drive.	
5	From the active Native Window: a) select Access b) select Mount/Dismount Emulated Disks... c) select Create.... d) select the CD-ROM drive e) select DISK_&Z1.LCN f) select Open g) select Create.... h) select DISK_&Z6.LCN i) select Open j) select D:\DISK_&Z1.LCN and mount it in the LEFT drive k) select D:\DISK_&Z6.LCN and mount it in the RIGHT drive. l) select OK	

Table 9-13 Shutdown and Reload the Native Window

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
6	Using the Native Window, select or press LOAD. Type 1 and press <ENTER> Prompt OPR,UNP ? Type U and press <ENTER> Wait for NCF? N, 1, 2, 3, 4, X ? Type 2 and press <ENTER> Wait for ABST N, 1, 2, 3, 4, X, ? Type 1 and press <ENTER> Wait for BUTT N, 1, 2, 3, 4, x ? Type 2 and press <ENTER>)	
7	IF the System Status display appears with a MOUNT REQUEST at the top of the screen MOUNT STD ABST.... a) Select the MOUNT STD ABST... target b) Select DEFAULT SOURCE c) Select EXECUTE COMMAND	
8	Wait for the load to complete. a) Select Displays b) Select System c) Select Console Status ATTENTION Be sure to load only one GUS with R6xx to completion. Expect the GUS to go to a WARNING STATUS. Use the Console Status Display to verify the message: "Custom/Standard Parameter duplicate XREF Creation Failed." This happens because the GUS cannot write to the &ASY directory at this time. This error will not adversely affect translations.	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
9	Verify drive numbers for this station to allow engineering functions to emulated disks. a) Select Access... b) Select Mount/Dismount Emulated Disks... c) Select Dismount on RIGHT drive. d) Select Dismount on LEFT drive. e) Select Create... f) Select a device/location (such as C:\TEMP)	
10	Change the File name: From: newfile.lcn To: R6xx PREPARATION.LCN a) Select Open b) Change the Size to 95 MB c) Select Dynamic d) Select OK	
11	Select the Emulated disk "R6xx PREPARATION.LCN" file and place it in the Right Drive and Select OK	
12	Check the Console Status display and verify the drive addresses.	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
	NOTE: The following procedures will create the emulated disk with the R6xx supporting files. It will contain the emulated disk name R6xx PREPARATION.LCN disk with the Volume name of R6xx. The files and directories required are: &ASY (NCF.CF, *.SP, *.SE, and other custom files i.e.; VPCONFcc.XX, PRFAILcc.XX, etc. files) &DSY (R6xx supporting files) &Daa (AREAAa.DA and support files i.e.; Memory Resident Schematics, button.KO, etc.), &CUS, &CLX (all R6xx required External Load Modules i.e.; MSCHEM.LO, CSCHEM.LO, XYPLOT.LO, etc.) NOTE: If the button.KO file resides in another area pathname search, create the directory and copy required supporting files to the "R6xx PREPARATION.LCN" emulated disk.	
13	On the Native Window, select Engineering	
14	Select MENU	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
15	Select Command Processor a) Type CR \$Fn>R6xx> -F -MF 3000 -BS 1700 (where n represents the RIGHT drive #) b) Press <ENTER> c) Type CD \$Fn> R6xx> &ASY d) Press <ENTER> e) Type CD \$Fn> R6xx> &Inn (where "nn" represents the hiway/network # required) f) Press <ENTER> NOTE: Create each additional &Inn directories required for your system. g) Type CD \$Fn> R6xx> &Daa (where "aa" represents the GUS LCN area data base number) NOTE: Create each additional &Daa directories required for your system. h) Type CD \$Fn> R6xx> &CUS i) Type CD \$Fn> R6xx> &DSY (which will contain the R6xx &DSY files to be copied later)	
16	Type SP and press <ENTER>	
17	Select SET DEVICE PATH TO REM. MEDIA	
18	Select the NCF BACKUP PATH and type \$Fn>&ASY> (where n = the RIGHT drive #)	
19	Press <ENTER>	
20	Select EXIT	
21	Mount the "Customer Support Files R5xx to R6xx" emulated disk that contains the files to be translated into the LEFT Drive. Translate the files from LEFT to RIGHT. Only the Network Configuration File (NCF.CF in directory &ASY), Hiway Gateway Checkpoint and Network Interface Module Checkpoint files (&lxx directories) need translated from R5xx to R6xx.	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
22	From the Native Window Menu Bar: • Select Access... • Select Mount/Dismount Emulated Disks... Select the "Customer Support Files" emulated disk and mount it on the LEFT drive.	
23	Type CP \$Fs>&ASY>*.* \$Fd>&ASY>= -D (where "s" is the Customer Support Files emulated disk and "d" represents the " R6xx PREPARATION.LCN" disk).	
24	Type DL \$Fd>&ASY> NCF.CF type DL \$Fd>&ASY>NCF.WF (where "d" is the "R6xx PREPARATION" disk (right drive))	
25	Copy any &CUS R6xx "compatible" External Load modules to the " R6xx PREPARATION.LCN" disk.	
26	Type CP \$Fs>&Daa>*.* \$Fd>&Daa>= -D (where "s" is the R5xx area data base disk and "d" is the "R6xx PREPARATION.LCN" disk. NOTE: If the area data base BUTTON file is on a different disk or directory, mount and create the directory(s) and copy the .KO file and possibly other supporting files to the R6xx PREPARATION disk.	
27	Dismount "Customer Support Files" from the LEFT drive and Mount Disk_&Z1.LCN on the LEFT drive	
28	From the Command Processor Type: TR press <ENTER>	
29	Dismount Disk_&Z1.LCN from the LEFT drive and Mount "Customer Support Files" on the LEFT drive	
30	From the Command Processor: Type TR NCF \$Fs \$Fd (where "s" is the "Customer Support Files" emulated disk and the "d" is the "R6xx PREPARATION.LCN" emulated disk.)	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
31	From the Command Processor: a) Type TR HG \$Fs \$Fd hh (where "hh" is Highway number (01-20) and "s" is the Customer Support Files emulated disk and the "d" is the "R6xx PREPARATION.LCN" emulated disk for the Hiway checkpoint(s)). b) Repeat for each additional Hiway. (HG or (E)PLCG. Note: If errors occur on this translation, see Table 9-5 notes for procedure to recover. c) Type TR NM \$Fs \$Fd uu (where "uu" is UCN Network Number (01-20) and "s" is the Customer Support Files emulated disk and the "d" is the "R6xx PREPARATION.LCN" emulated disk for the UCN checkpoint(s)). PT \$Fd>&Iuu>*. * MOUNT the R5xx UCN checkpoint CP \$Fs>&Iuu>*. * \$Fd>&Iuu>= -D UNPT \$Fd>Iuu>*. * where s = source drive number d = destination drive number uu = UCN network number (01-20) d) Repeat for each additional UCN.	
NOTE: At this point, the "R6xx PREPARATION.LCN" disk located in the RIGHT emulated disk drive should contain the translated Checkpoint and NCF.CF files.		
32	Select Access	
33	Select Mount/Dismount Emulated Disks....	
34	Select Dismount on LEFT Drive.	
35	Select Create	
36	Select the CD-ROM drive	
37	Select DISK_&Z1.LCN	
38	Select Open	
39	Select D:\DISK_&Z1.LCN and mount it in the LEFT drive	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
40	Type CP \$Fs>&DSY>*. * \$Fd>&DSY>= -D (where "s" is the GUS-TPN Software CD and "d" is the R6xx PREPARATION disk.)	
41	a) Type UNPT \$Fd>&CUS>*. * (where "d" is the R6xx PREPARATION disk.) b) Type CP \$Fs>&CUS>*. * \$Fd>&CUS>= -D (where "s" is the GUS-TPN Software CD and "d" is the R6xx PREPARATION disk.)	
42	Select Access	
43	Select Mount/Dismount Emulated Disks	
44	Select Dismount on LEFT Drive	
45	Remove the CD ROM from the drive	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
46	Insert into the CD ROM drive the CD ROM for the Utilities and Load Modules Rxxx, (where xxx is the latest release.)	
47	Select Access	
48	Select Mount/Dismount Emulated Disks.	
49	Select Create	
50	In the Look In, select the down arrow.	
51	Select the TPN_R6xx folder where R6xx is the latest one available.	
52	Select DISK_&Z12.LCN	
53	Select Open. Select OK.	
54	Mount DISK_&Z12.LCN in the LEFT drive.	
55	From the Native Window, select the Engineering (at the top of the screen), select Menu.	
56	Select the Command Processor Target.	
57	Execute the following: CP \$Fs>&CUS>*. *\$Fd>&CUS>=-D (where "s" is the CD-ROM emulated disk and "d" is the emulated R6xx PREPARATION disk.)	
58	Dismount the DISK_&Z12.LCN Emulated Disk.	
59	Select Create	
60	In the Look In, select the down arrow.	
61	Select the TPN_R6xx folder where R6xx is the latest one available.	
62	Select DISK_&Z14.LCN	

Table 9-13 Shutdown and Reload the Native Window (continued)

**Shutdown and
reload the Native
Window, continued**

Step	Action	Done (√)
63	Select Open. Select OK	
64	Mount DISK_&Z14.LCN in the LEFT drive.	
65	From the Native Window, select the Engineering (at the top of the screen), select Menu.	
66	Select the Command Processor Target.	
67	Execute the following: CP \$Fs>&CUS>*. *\$Fd>&CUS>=-D (where "s" is the CD-ROM emulated disk and "d" is the emulated R6xx PREPARATION disk.)	
68	Dismount the DISK_&Z14.LCN Emulated Disk and remove the Utilities and Load Modules CD.	
69	Close the Mount/Dismount Emulated Disks screen by selecting OK.	
70	Where we stand: The "R6xx PREPARATION.LCN" is located in the RIGHT drive. The "R6xx PREPARATION.LCN" disk should contain the following required files: &ASY R6xx support files (*.SP, *.SE, customers translated NCF.CF file, and custom files) &CUS (GUS supporting external load modules) &Daa AREAAa.DA, any memory resident schematics, button.KO (if area database is to contain it in the area directory, if not, it should reside on the directory) &DSY (R6xx supporting files) &lxx (all translated Hiway/UCN checkpoints)	

Table 9-13 Shutdown and Reload the Native Window (continued)

Shutdown and reload the GUS station using translated files

The following procedure will load the GUS with Universal personality from the CD ROM (located in the Left Drive) and the translated Network Configuration File (NCF) and required files (located in the Right Drive).

NOTE: If hardware or firmware changes are required, close all applications, log off and shutdown. Power down and use ESD procedures when installing hardware or firmware. Power up when completed and log on with administrator privileges.

Step	Action	Done (√)
1	Shutdown the GUS Native Window and RESET the LCNP.	
2	Mount GUS-TPN Software CD into the CD-ROM drive (if it isn't already.)	
3	From the active Native Window: a) Select Access b) Select Mount/Dismount Emulated Disks.... c) Select D:\DISK_&Z1.LCN and mount it in the LEFT drive. d) Select OK	
4	Using the Native Window, select or press LOAD. a) Type 1 and press <ENTER> Prompt OPR,UNP ? b) Type U and press <ENTER> Wait for NCF? N, 1, 2, 3, 4, X ? c) Type 2 and press <ENTER> Wait for ABST N, 1, 2, 3, 4, X, ? d) Type 2 and press <ENTER> e) Wait for BUTT N, 1, 2, 3, 4, X ? f) Type 2 and press <ENTER>	
5	If the System Status display appears with a MOUNT REQUEST at the top of the screen MOUNT STD ABST.... a) Select the MOUNT STD ABST... target b) Select ALTERNALT SOURCE c) Select the LEFT drive number. d) Select EXECUTE COMMAND	
6	Wait for the load to complete. a) Select Displays b) Select System c) Select Console Status	
7	Verify drive numbers for this station. NOTE: A warning status for this GUS may be ignored at this time.	

Table 9-14 Shutdown and Reload the GUS Station

**Make required
Network
Configuration
modifications**

The following procedure will allow the configuration of the new features to take place. The procedures will reference sections in the US migration path. The procedures will also allow the GUS TPS Network personality to be upgraded to the R6xx version.

NOTE:

Use the "R6xx PREPARATION.LCN" located in the RIGHT drive as the Network Configuration File path and use another emulated disk located in the LEFT drive as the BACKUP path. Mount/Dismount emulated disks as required.

Step	Action	Done (√)
1	Refer to section 9.2 for required changes, i.e.;	
	a) HM Volume Configuration	
	b) System Wide Values (Console Data)	
	c) LCN Node "MAXIMUM MODULE MEMORY (WORDS)" may require modifications. Run the CHECKER and perform an INSTALL for each GUS node to ensure the value is 10000 (WORDS) greater than the TOTAL MEMORY)	
	d) New functionality requirements	
2	After all required modifications have been implemented, the TPS Network Software can be installed.	
	Shutdown the GUS Native Window and RESET the LCNP.	

Table 9-15 Make Required Network Configuration Modification

Install TPS Network Software to the PC Local Hard Drive

The following procedure uses the R6xx GUS-TPN Software CD ROM. The software from the CD is copied to the LOCAL HARD drive. These installation instructions can be used for ALL TPS Network PC-Based Nodes, such as GUS, PHD, APP nodes, etc

Step	Action	Done (✓)
1	Log on to Windows as administrator (or as another user with administrator privileges).	
2	Load the distribution media for the package to be installed into the CD-ROM drive (GUS-TPN Software CD-ROM).	
3	Use the Windows Explorer, select the drive which contains the CD-ROM	
4	If the program in Explorer window does not show extensions (such as .exe) make them visible by executing the following steps: a) Select Tools b) Select Folder Options c) Select the View tab d) Deselect the Hide extensions for known file types check box.	
5	Select the directory containing the installation by performing the following steps:	
6	Change to folder \GUS\PERS_US.	
7	Change the folder to disk1.	
8	Double Click file SETUP.EXE to start the installation.	
9	Once the Setup program completes, The Welcome window is displayed.	
10	Read and follow the Welcome information before continuing	
11	Select Next to continue the setup program.	
12	The software License Agreement page is displayed.	
13	Select Yes to continue.	

Table 9-16 Install TPS Network Software to Local Hard Drive

**Install TPS Network
Software to the PC
Local Hard Drive,**
continued

Step	Action	Done (√)
14	This window entails the Name, Company and Serial information. If the information is correct for Name and Company, type 1 for the Serial and select Next. If it is correct, select Yes and skip to step 16.	
15	If Name and Company fields are incorrect, select No and enter the correct information. a) Once the information has been entered, select Next . a) Verify that the registration is now correct. b) Enter 1 for the serial number on the user information dialogue. c) Select Yes .	
16	The Target Directory Information window appears to identify where the software will be installed NOTE: The location of the software should NOT be located in any other location other than C:	
17	Click Next .	
18	For Startup Type, ensure Typical is selected and select Next .	
19	The Start Copying Files dialog box is displayed.	
20	Select Next .	
21	The Copying Progress window is displayed.	
22	When the copy is complete, the message Setup Complete is displayed.	
23	Select Finish to complete the installation.	
24	Remove the CD-ROM from the drive.	

Table 9-16 Install TPS Network Software to Local Hard Drive (continued)

Load the GUS TPS Network node

Load the GUS TPS Network node with the R6xx GUS-TPN Software and the "R6xx PREPARATION.LCN" emulated disk (which contains the &ASY, &Daa, &CUS, etc., customer support files).

Step	Action	Done (✓)
1	Shutdown the GUS Native Window and RESET the LCNP.	
2	Wait for the status to go to PWR_ON.	
3	Type W and press <ENTER> Wait for NCF? N, 1, 2, 3, 4, X ?	
4	Type 2 and press <ENTER> Wait for &CUS N, 1, 2, 3, 4, X, ?	
5	Type 2 and press <ENTER> NOTE: You will be prompted multiple times for CUS. This does not mean that the system cannot find the CUS directory or the external load modules. Due to the internal CUS loading code, each external load module requires it's own prompt. So, you will be prompted once for each external load module configured. Press <ENTER> each time until prompts cease.	
6	Wait for the load to complete. a) Select Displays b) Select System c) Select Console Status NOTE: If WARNING or SEVERE is displayed in the CONSOLE STATUS display, resolve the situation before continuing.	

Table 9-17 Load the GUS TPS Network Node

10. Migration from R43x, 53x to R6xx

10.1 Migrate HMs

Start the migration This section is the beginning of the actual R6xx software migration. A checklist has been provided on this page to help keep track of HMs migrated on the system. If migrating from R43x, be sure have completed Sections 7 and 9. If migrating from R5xx, be sure you have completed Sections 8 and 9.

R6xx HM migration checklist If your system has multiple HMs, use this checklist to help you keep track of what procedures have been completed, as you migrate each HM. Place a check (3) next to each procedure, after having performed that procedure for the given HM. If you have more than six HMs, make the required number of copies of this page before starting.

HM Procedure	HM LCN Addr./Node pair nmbr. _____	HM LCN Addr./Node pair nmbr. _____	HM LCN Addr./Node pair nmbr. _____	HM LCN Addr./Node pair nmbr. _____	HM LCN Addr./Node pair nmbr. _____	HM LCN Addr./Node pair nmbr. _____
Prepare HM						
Initialize HM						
Load HM with personalities						
Restore History Data and Definitions.						
Re-establish user directories						
Move &EC files to NET						
Restore Data						
Copy Custom External Load Modules						
Copy Personality Image files						
Copy Tool Kit Displays						
Copy translated HG/NIM checkpoints						
Copy translated Area Database files						
Synchronize redundant HM drives						

Table 10-1 HM Migration Checklist

10.1. Migrate HMs

Warnings prior to migrating HMs

CAUTION

With the exception below, when replacing failed drives from your History Module with replacement Winchester disk drives from another HM, they must be formatted using the SCMD Test from HVTs, version 10.5 or greater without rewriting the Bad Sector Block, before being used as redundant drives on this HM.

If this is not done, the existing file system on your HM may become corrupted or crash when synchronized. This is not a problem with drives received from the factory, because they have been formatted prior to shipment.

When initializing the History Module drives, both the primary and secondary drives must be powered on. Because the HM Winchester disk drives have been redundant on a volume-by-volume basis since Release 400, it is important to realize that the HM may be fully functional while there are corrupted volumes on any or all of the disks.

Therefore, prior to setting any disks off-line for replacement purposes, the user should reallocate bad sectors on the good drive using SMCC, and synchronize the disk (if possible) to ensure the best possible condition of data integrity.

The recommended migration procedure causes loss of HM access to the previous release side. Set up media paths to allow operator access to operational schematics during the migration period. History, NET checkpoints, journals, etc., will not become available until your system is up on R6xx software. Also the standard displays will not be available, so have a &DSY set up in a removable media.

Your system will be without a System HM until your system is up on Release 6xx software.

Prepare HM for software migration

For each HM, perform the steps described in Tables 10-2, 10-3, and 10-4.

Start with the System HM. Note that some steps apply only to the System HM.

Step	Action	Done (√)
1	From a US on previous release, CONFIRM that automatic checkpoints are disabled for any data owners checkpointing to this HM.	
2	IDENTIFY the System HM. (It is the HM with the &0np volume where np = node pair number)	
3	From the previous release US, ENB AUTO LOAD/DMP and set to DISABLE AUTOLOAD SHUTDOWN this HM, and WAIT for it to go to "QUALIF".	
4	Go to the physical HM node, POWER OFF the node, (leave off at least 5 seconds), upgrade hardware if required. (Refer to section 3.4 for minimum memory requirements.) Then, POWER ON the node. When the node address is displayed in the LEDs (approx. 30 seconds), press the RESET button. The status then goes to PWR_ON	
5	If the HM has disk redundancy, ENSURE all drives are powered up. THIS IS REQUIRED IN ORDER TO HAVE DRIVE SYNCHRONIZATION.	

Table 10-2 Prepare HM for Software Migration

10.1. Migrate HMs

**Prepare HM for
software migration,
continued**

Step	Action	Done (√)
6	Use "Custom Support Files R6xx" for US or, use "R6xx PREPARATION.LCN" emulated disk for GUS. From the R6xx US/GUS in UNP, and using the &Z1 disk provided, perform a MANUAL LOAD of this HM using the INIT program, with the R6xx HM Off-line personality (HMI), as follows: <pre> SELECT SYST STATUS SELECT HM (so it is backlit) SELECT NODE STATUS SELECT LOAD DUMP SELECT MANUAL LOAD SELECT INIT PROGRAM SELECT ALTERNATE SOURCE use &Z1 disk for the personality, and "Customer Support Files R6xx" for the data source, <u>updated by the On-line HM NCF migration steps</u> (see note below). </pre> NOTE: For a nonsystem HM, the data can come from NET rather than from the "Customer Support Files R6xx" cartridge.	
7	WAIT for the HM status to go to "HMOF OK" .	

Table 10-2 Prepare HM for Software Migration (continued)

Initialize HM

Perform Steps 1 through 4 in the following table to initialize the selected HM.

Step	Action	Done (√)
1	For the system HM, set the paths to removable media to use &Z1 for the overlay, and the "Customer Support Files R6xx" for the &ASY. For non-system HMs, set the paths to NET. INVOKE the Volume Configuration from the R6xx US in UNP.	
2	SELECT this HM node Pair.	
3	CAUTION Pressing <CTRL + F6> will INITIALIZE the HM, and destroy all data! To INITIALIZE this HM, PRESS <CTRL + F6>.	
4	When HM INITIALIZATION COMPLETE message appears, ABORT NCF modification, as follows: PRESS <CTRL + F5>. PRESS <ENTER>.	

Table 10-3 Initialize HM

10.1. Migrate HMs**Load HM with R6xx
HM personalities**

Using the &Z1 disk, load the selected HM disk with the R6xx personality(s), as shown in the following steps.

Step	Action	Done (√)
1	RECORD all command processor operations that follow. TYPE DO \$Pn TYPE PD ON (Hard copy records are required to evaluate any problems with this migration)	
2	MOUNT the &Z1 disk in \$Fs (source drive).	
3	RUN the LOC_VOLZ.EC. EXAMPLE SYNTAX: EC \$Fs>&EC>LOC_VOLZ.EC \$Fs nn np where S = source drive number nn = HM node number np = HM node pair number	
4	IF this is the System HM, LOAD the system configuration files by running SYS_VOLZ.EC EXAMPLE SYNTAX: EC \$Fs>&EC>SYS_VOLZ.EC \$Fs nn where S = source drive number nn = HM node number Respond NO to the question "Copy volume &ASY?" Respond NO to "Do you wish to skip the &DSY or any of the optional directories?" (The EC file will copy all files from the &DSY and the four GDF volumes.)	

Table 10-4 Load HM with R6xx HM Personalities

**Load HM with R6xx
HM personalities.**
continued

Step	Action	Done (✓)
5	IF this is the System HM, MOUNT the “Customer Support Files R6xx” cartridge, and copy the &ASY directory from this cartridge to the HM. For GUS, MOUNT the “R6xx PREPARATION.LCN” emulated disk, and copy the &ASY directory from this cartridge (US) or emulated disk (GUS/ EST) to the HM. EXAMPLE SYNTAX: COPY \$Fs>&ASY>*.* PN:nn>&ASY>= -d where S = source drive number nn = HM node number	
6	AUTOBOOT this HM as follows: SELECT SHUTDOWN from the System Status Display. WAIT for QUALIF to be displayed by the HM. Go to the HM, TURN OFF POWER to HM and WAIT 5 seconds, then POWER ON HM. If continuous history is to be restored, allow the HM to run at least 5 minutes.	
ATTENTION HMs with redundant drives will go to SEVERE status because they are out of synchronization. Do not synchronize redundant disks at this time. HMs with nonredundant drives will go to OK status. NOTE: &DS2 is used to provide alternative access to system displays. If you only have one HM, skip the following steps.		
7	For only one HM, which cannot be the system HM CREATE directory &DS2.	

10.1. Migrate HMs

Step	Action	Done (✓)
8	IF this HM contains the &DS2 directory, INVOKE the DS2_BKPZ.EC from &Z1 EXAMPLE: EC \$Fs>&EC>DS2_BKPZ.EC \$Fs where S = source drive number	

Table 10-4 Load HM with R6xx HM Personalities (continued)

Load HM with R6xx HM personalities, continued

CAUTION

You cannot restore Continuous History or Continuous History Group Definitions if the Continuous History has been modified on this HM (i.e. adding Groups, modifying History Collection, etc.). EB files must be used to restore History Group Definitions.

Restoring History Data and Definition Files – Choices

If you are restoring Continuous History, you have four choices on how you do the restoration. **Use only one of these procedures:**

IF you want to ...	THEN go to ...
use an EC to restore all history data and the history definition files (Requires two Universal Stations) (Honeywell recommends using the Honeywell-provided EC for this step.)	Restore History Data and Definitions using EC (alternative 1)
use an EC to restore just the APL files (History Group Definitions) (Requires two Universal Stations) (Honeywell recommends using the Honeywell-provided EC for this step.)	Restore APL Files using EC (alternative 2)
use manual process to restore all history data and the history definition files	Restore History Data and Definitions using manual process (alternative 3)
use the manual process to restore just the APL files (History Group Definitions)	Restore APL Files using manual process (alternative 4)
wait, and rebuild user files from an EB on R6xx	Re-establish user-created directories and you will later perform the history restores from R6xx.

Table 10-5 Restore History Data and Definition Files Choices

NOTE: Go to the procedure that applies to your situation. Remember, you should perform only one of the procedures listed above for a given HM. Repeat the correct procedure for each HM that contains History.

10.1. Migrate HMs

Restore History Data and Definitions using EC (alternative 1) Perform the following steps if you are using an EC to restore History Data and Group Definitions.

Step	Action	Done (√)
1	LOAD a second US/GUS with the R6xx personality.	
2	From the R6xx US in UNP, and using the &Z1 disk provided, LOAD this HM with the R6xx HM Off-line personality (HMI) For all HMs, the data will come from default NET.	
3	INVOKE the RSCNTHSZ.EC EXAMPLE: SP \$Fs>&EC EC RSCNTHSZ . EC where S = source drive number (&Z1)	
4	SELECT SHUTDOWN HM	
5	WAIT for QUALIF to be displayed by the HM.	
6	Go to the HM, TURN OFF POWER to HM and WAIT five seconds, then POWER ON HM. If the HM does not autoboot to OK within 1 hour, call TAC. DO NOT PROCEED!	
7	When HM autoboots and goes to HMON OK, continue with Re-establish user-created directories.	

Table 10-6 Restore History Data and Definitions Using EC

**Restore APL
files using EC
(alternative 2)**

Perform the following seven steps if using an EC to restore the APL files (History Group definitions).

CAUTION

You cannot restore Continuous History or Continuous History Group Definitions if the Continuous History has been modified on this HM (i.e. adding Groups, modifying History Collection, and so on). EB files must be used to restore History Group Definitions.

Step	Action	Done (√)
1	From an R6xx US, on the System Status Display, SHUTDOWN the HM, and wait for QUALIF status.	
2	From the R6xx US in UNP, and using the &Z1 disk provided, LOAD this HM with the R6xx HM Off-line personality (HMI) For all HMs, the data will come from default NET.	
3	INVOKE the RSHSDFZ.EC EXAMPLE: SP \$Fs>&EC EC RSHSDFZ.EC where S = source drive number	
4	WAIT for the EC to finish, then SELECT SHUTDOWN HM	
5	WAIT for QUALIF to be displayed by the HM.	
6	Go to the HM, TURN OFF POWER to HM and WAIT five seconds, then POWER ON HM. If the HM does not autoboot to OK within 1 hour, call TAC. <u>DO NOT PROCEED!</u>	
7	When the HM autoboots and goes to HMON OK, continue with the procedure under "Re-establish user-created directories."	

Table 10-7 Restore APL Files Using EC

10.1. Migrate HMs**Restore History Data and Definitions using manual process.
(alternative 3)**

Perform the following steps if you are using a manual process to restore History Data and Group Definitions.

CAUTION

You cannot restore Continuous History or Continuous History Group Definitions if the Continuous History has been modified on this HM (i.e. adding Groups, modifying History Collection, etc.). EB files must be used to restore History Group Definitions.

Step	Action	Done (√)
1	From an R6xx US, on the System Status Display, SHUTDOWN the HM, and wait for QUALIF status.	
2	From the R6xx US in UNP, and using the &Z1 disk provided, LOAD this HM with the R6xx HM Off-line personality (HMI) The data will come from the default NET, updated by the On-line HM NCF migration steps above.	
3	DELETE all files in !0np. EXAMPLE SYNTAX: DL PN:nn>!0np>*. * where nn = HM node number np = node pair number	
4	DELETE all files from each directory under!0np and !1np. EXAMPLE SYNTAX: DL PN:nn>!Ahg>*. * where nn = HM node number hg = history group number	

Table 10-8 Restore History Data and Definitions Using Manual Process

**Restore History Data
and Definitions using
manual process.
(alternative 3),
continued**

Step	Action	Done (√)
5	NOTE: This is a mandatory step if performing this manual process. This step prevents fragmentation. From the System Status Display on an R6xx US in UNP, and using the &Z1 disk provided, SHUTDOWN the HM, then RELOAD the HM with the R6xx HM Off-line personality (HMI). The data will come from the default NET.	
6	For each saved continuous history data cartridge "!0np #n of _____", MOUNT it in \$Fs, and COPY all of the continuous history and APL files to the HM. EXAMPLE SYNTAX: CPV \$Fs>!0np PN:nn -A -D CPV \$Fs>!1np PN:nn -A -D (if needed) where S = source drive number np = node pair number nn = HM node number NOTE: Make sure the APL files, which were previously saved, are copied back to this HM. ATTENTION The APL*. * files are protected, when File Transfer versions are used for transferring the files back to the HM. These files are to be unprotected; otherwise the modifications to the history groups cannot be performed. To unprotect the APL*. * files, perform the following steps: 1. Load the HM into HMOF personality. 2. Enter the command: UNPT PN:hm>!0np>APL*. * (Where hm is the HM LCN node # and np represents the HM node pair#). 3. When the files are unprotected, reboot the HM.	

10.1. Migrate HMs

Step	Action	Done (√)
AUTOBOOT the HM as follows:		
7	SELECT SHUTDOWN HM	
8	WAIT for QUALIF to be displayed by the HM.	
9	Go to the HM, TURN OFF POWER to HM and WAIT five seconds, then POWER ON HM. If the HM does not autoboot to OK within 1 hour, call TAC. <u>DO NOT PROCEED!</u>	
10	When the HM autoboots and goes to HMON OK, continue with the procedure under "Re-establish user-created directories.	

Table 10-8 Restore History Data and Definitions Using Manual Process (continued)**Restore APL files using manual process (alternative 4)**

Perform the following steps if using a manual process to restore the APL files (History Group definitions).

CAUTION

You cannot restore Continuous History or Continuous History Group Definitions if the Continuous History has been modified on this HM (i.e., adding Groups, modifying History Collection, etc.). EB files must be used to restore History Group Definitions.

Step	Action	Done (√)
1	From an R6xx US, on the System Status Display, SHUTDOWN the HM, and wait for QUALIF status.	
2	From the R6xx UNP, and using the &Z1 disk provided, LOAD this HM with the R6xx HM Off-line personality (HMI) The data will come from the default NET.	

Step	Action	Done (√)
3	As a safety measure, CREATE a temporary cartridge with an “APL” directory and COPY the R6xx APL files from the HM to the cartridge. EXAMPLE SYNTAX: CP PN:nn>!0np>APL0*.MM \$Fd>APL>= -D where nn = HM node number np = node pair number d = destination drive number ATTENTION The APL*. files are protected, when File Transfer versions are used for transferring the files back to the HM. These files are to be unprotected; otherwise the modifications to the history groups cannot be performed. To unprotect the APL*. files, perform the following steps: 1. Load the HM into HMOF personality. 2. Enter the command: UNPT PN:hm>!0np>APL*. (Where hm is the HM LCN node # and np represents the HM node pair#). 3. When the files are unprotected, reboot the HM.	
4	DELETE the default history group, point definition files. EXAMPLE SYNTAX: DL PN:nn>!0np>APL0*.MM -D where nn = HM node number np = node pair number	
5	NOTE: This is a mandatory step if performing this manual process. This step prevents fragmentation. From the System Status Display on an R6xx US in UNP, and using the &Z1 disk provided, SHUTDOWN the HM, then RELOAD the HM with the R6xx HM Off-line personality (HMI). The data will come from the default NET.	

Table 10-9 Restore APL Files Using Manual Process

10.1. Migrate HMs

**Restore APL
files using
manual
process
(alternative 4),
continued**

Step	Action	Done (√)
6	COPY the history group, point definition files from the "!0np #n of ___" cartridges. EXAMPLE SYNTAX: CP \$Fs>!0np>APL0*.MM PN:nn>!0np>= -D where s = source drive number np = node pair number nn = HM node number	
7	SELECT SHUTDOWN HM	
8	WAIT for QUALIF to be displayed by the HM	
9	Go to the HM, TURN OFF POWER to HM and WAIT five seconds, then POWER ON HM. If the HM does not autoboot to HMON OK within 1 hour, call TAC. <u>DO NOT PROCEED!</u>	

Table 10-9 Restore APL Files Using Manual Process (continued)

Re-establish user-created directories

Using steps 1 and 2 in the following table, restore the user-created directories on each HM.

Step	Action	Done (√)																		
1	Perform a list volume to the printer, as follows: TYPE DO \$Pn (where \$Pn = printer number) TYPE PD ON TYPE LSV PN:nn (where nn = HM node number)																			
2	COMPARE this Volume List with that taken in Section 7.2 or 8.2, Database Preparation, under Identify NET Volumes. <u>With the exception of directories listed below</u>, RECREATE directories, as required. CD NET>USER> DIR where USER = User volume DIR = Directory The following directories <u>do not</u> need to be recreated: <table> <tr> <td>&ENG</td><td>&OP2</td><td>&OV2</td></tr> <tr> <td>&OP4</td><td>&OV1</td><td>&OV4</td></tr> </table> NOTE: Some user-created directories may be needed by the system and may be recreated at this time. They are: <table> <tr> <td>CDSG</td><td>&CUS</td><td>&CLX</td></tr> <tr> <td>&OVG</td><td>&DOC</td><td>TLK1</td></tr> <tr> <td>DIA1</td><td>TFIL</td><td>&DS2 (for non-system HM only)</td></tr> <tr> <td>TLK2</td><td></td><td></td></tr> </table> For faster call-ups, place these directories under the FAST User volume.	&ENG	&OP2	&OV2	&OP4	&OV1	&OV4	CDSG	&CUS	&CLX	&OVG	&DOC	TLK1	DIA1	TFIL	&DS2 (for non-system HM only)	TLK2			
&ENG	&OP2	&OV2																		
&OP4	&OV1	&OV4																		
CDSG	&CUS	&CLX																		
&OVG	&DOC	TLK1																		
DIA1	TFIL	&DS2 (for non-system HM only)																		
TLK2																				

Table 10-10 Re-establish User-Created Directories

10.1. Migrate HMs

Move &EC files to NET

NOTE: The following steps should be performed to only one HM.
Perform the following steps to move the &EC files to the NET.

Step	Action	Done (√)
1	CREATE an &EC directory. EXAMPLE SYNTAX: CD NET>USER> &EC where USER = Volume name of your choice	
2	MOUNT &Z1 disk in \$Fs, and COPY the &EC files to the &EC directory on the HM. EXAMPLE SYNTAX: COPY \$Fs>&EC>*.* NET>&EC>= -D where S = Source drive number	

Table 10-11 Move &EC Files to NET

Restore data

When the directories all exist, use the RESTORE command in the steps below to copy the saved HM data to the respective directories.

CAUTION

Do not synchronize the drives during a RESTORE. This can cause drive problems.

Step	Action	Done (√)
1	This function is for the SYSTEM HM ONLY . Before performing the RESTORE function, you should ensure the NCF.CF (Latest 6xx file) is protected to ensure it is not overwritten. To do that, perform the following: PT PN:nn>&ASY>NCF.CF (where nn represents the HM node #)	
2	Mount in succession, the saved cartridges "BACKUP PN:nn#y." (created in section 7.4 or 8.4 Backup HM, Step 3.)	
3	COPY each to the HM, using the RESTORE command. EXAMPLE SYNTAX: RESTORE \$Fs PN:nn Where s = source drive number nn = HM node number There will be error messages that occur during this step. If error messages appear that you do not understand, call Honeywell TAC. NOTE: Copying from more than one media at the same time, to the History Module, can result in fragmentation on the HM.	
4	This function is for the SYSTEM HM ONLY . Once the RESTORE function has completed, you should ensure the NCF.CF file is unprotected for future modifications. To do that, perform the following: UNPT PN:nn>&ASY>NCF>CF (where nn represents the HM node#)	

Table 10-12 Restore Data

10.1. Migrate HMs

**Copy Custom
External Load
Modules (optional)**

There are a wide variety of applications and repeatable specials that contain External Load Modules, so you must follow the instructions that come with each respective user manual. If you have site-specific (custom) External Load Modules, perform the following steps.

Step	Action	Done (✓)
1	Locate all of your Application/Repeatable Special User Manuals (such as Loop Tune or XYPLOT).	
2	<u>If it does not already exist</u> , CREATE the directory &CUS.	
3	After having located the appropriate directory name in the Application/Repeatable Special User Manual, CREATE the directory (giving it the name stated in the user manual) that will contain files associated with the application or repeatable special.	
4	Follow the procedures in the Application/Repeatable Special User Manual to copy those modules and associated files from the compatible media to the appropriate directories.	
5	REPEAT steps 3 and 4 until all External Load Modules have been copied into each new directory.	

Table 10-13 Copy Custom External Load Modules (optional)

**Copy Personality
Image files to the
NET**

The following steps copy the R6xx Personality Image files to the NET. This step also allows you to load the DIA1, TLK1 and TLK2 Tool Kit Displays, if desired.

Step	Action	Done (√)
1	Copy the personality files configured on the NET HM using the PRS2VOLZ.EC command file for 68020 personalities, and/or using the PRS4VOLZ.EC command file for 68040 personalities. EXAMPLE SYNTAX EC NET>&EC>PRS2VOLZ.EC \$Fs EC NET>&EC>PRS4VOLZ.EC \$Fs where S = source drive number (&Z1 etc.)	

Table 10-14 Copy Personality Image Files

10.1. Migrate HMs

**Copy translated
HG/NIM checkpoints**

Perform the following steps to copy the HG/NIM translated checkpoint files to NET.

Step	Action	Done (√)
1	MOUNT the checkpoint data disk in \$Fs. The cartridge needed is: "Customer Support Files – R6xx" This disk contains translated NIM and HG checkpoints. For GUS, MOUNT the "R6xx PREPARATION.LCN" emulated disk.	
2	RUN EC file for each HG/NIM on your system. EC file command EXAMPLE SYNTAX: EC NET>&EC>GWY_VOLZ.EC \$Fs un where S = source drive number un = the 2 digit network number (01-20) If the &EC is not on the NET, it can be found on &Z1. ANSWER "Y" when the following question comes up: COPY THIS UCN'S UCN NODE CHECKPOINT FILES (YES/NO)?	

Table 10-15 Copy Translated HG/NIM Checkpoints

**Copy translated
Area Database files**

Perform the following steps to copy the translated Area Database files to NET.

Step	Action	Done (√)
1	MOUNT the Area Database disk in \$Fs. (where s = source drive number) The disk needed is "Customer Support Files R6xx"	
2	Copy translated areas. EXAMPLE SYNTAX: CP \$Fs>&Dan>*.* NET>&Dan>= -D where S = \$Fs source drive number an = area number (01-10)	

Table 10-16 Copy Translated Area Database Files**Copy &EC Directory
Files to HM
(OPTIONAL)**

Some customers choose to copy the &EC files found on the &Z1 zip disk to an HM-resident directory for convenience. If you wish to do this, now is a good time! Here are typical commands to do this:

```
UNPT NET>&EC>*.*
CP $Fn>&EC>*.* NET>&EC>= -D
```

10.1. Migrate HMs

Synchronize redundant HM drives If you have redundant disk drives on an HM, perform the following steps to synchronize the drives.

CAUTION

Do not synchronize the drives during a RESTORE. This can cause drive problems.

Step	Action	Done (√)
1	CHECK to ensure the secondary drive(s) is powered up at this time.	
2	From the Command Processor, ENTER SYN PN:nn where nn = HM node number NOTE: It may take several hours for the HMs to synchronize and the status to change from SEVERE to OK.	
3	Repeat steps 1 and 2 for each HM with redundant drives.	

Table 10-17 Synchronize Redundant HM Drives

10.2 Migrate Selected Universal Stations or GUS

Enable Alarming for Units

Before continuing with the migration, we recommend that alarming is enabled for all units, both console-wide and system-wide. Enable the alarming from the R4xx/R5xx side of the system and from the US/GUS running R6xx. The Unit Display for the R6xx US/GUS will display??? for the System Alarm State until a Data Owner is loaded.

Step	Action
1	Callup the CONSOLE STATUS display.
2	Select UNIT DISPLAY.
3	Ensure each unit's CONSOLE and SYSTEM alarm state are set to ENABLE.
4	If the unit is NOT ENABLE for CONSOLE ALARM STATE, select the UNIT, select CONSOLE ALARM STATE, and select ENABLE.
5	If the unit is NOT ENABLE for SYSTEM ALARM STATE, select the UNIT, select SYSTEM ALARM STATE, and select ENABLE.
6	Repeat for all consoles.

When the migration is complete, ensure that alarms are functioning normally. If desired, you may then INHIBIT or DISABLE alarms for units on a Console-wide or System-wide basis.

Table 10-18 Enable Alarming for Units

NOTE: If migrating to R640 (or newer), do not enable the Automatic TPN Restart function until ALL nodes on the LCN have been upgraded to R640 (or newer).

Obtain R6xx operational view of process

Using the following steps, you will load selected Universal Stations with the R6xx Universal Personality or Operator Personality.

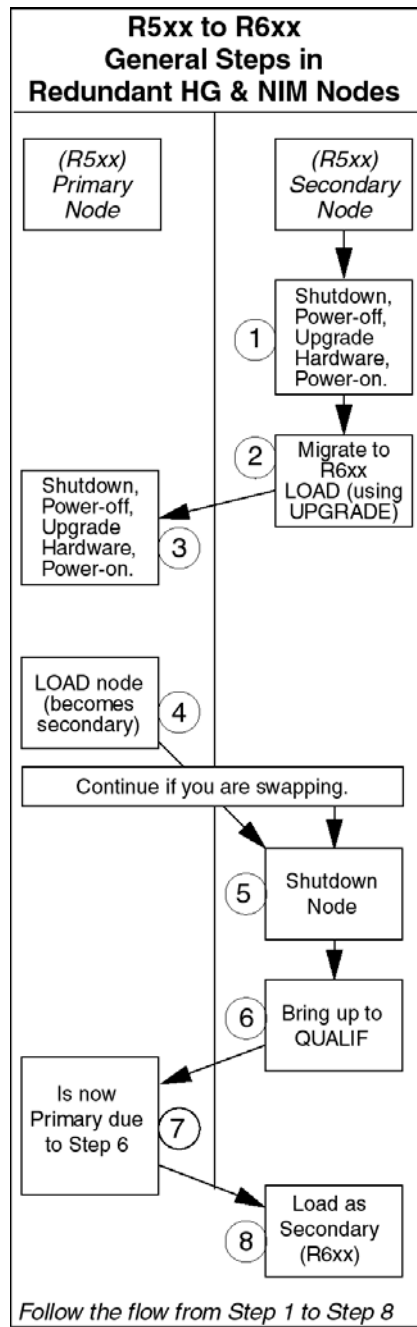
Ensure there are enough Universal Stations running on R43x or R5xx and R6xx to maintain view and control of your process.

10.2. Migrate Selected Universal Stations or GUS

Step	Action	Done (√)
1	SHUTDOWN an R43x or R5xx US or GUS two (2) times, then POWER IT DOWN for at least five (5) seconds.	
2	UPGRADE hardware to R6xx requirements, if necessary. (Refer to section 3.4 for minimum memory requirements.)	
3	POWER ON the node.	
4	For GUS nodes: Use the instructions from Section 9.3 titled "Install PC Network Software to PC Local Hard Drive".	
5	LOAD that US from the NET or GUS from "W", getting the NCF from the NET with the R6xx program image for the Universal Personality or Operator Personality. NOTE: Shutting down and reloading a US that has a printer attached may cause an originally ONLINE printer to return a status of OFFLINE after the load. Toggling the printer's ONLINE switch or button should cause it to return to ONLINE status.	
6	REPEAT steps 1 through 5 for each selected R43x or R5xx US you intend to load to an R6xx personality at this time. <u>Be sure to leave at least one US or GUS, with dual removable media or emulated disks on R43x or R5xx, until migration is completed. This station would be used to transition back to R43x or R5xx if necessary.</u>	

Table 10-19 Obtain R6xx Operational View of Process**Migrating redundant HGs and NIMs on-process - concepts**

The following drawing conceptually shows the general flow of migrating redundant HG or NIM nodes on a system while still on-process. The most important point is that you retain view of the process through either the primary or secondary node throughout the migration.

10.2. Migrate Selected Universal Stations or GUS**Figure 10-1 Overview of Migration from R5xx to R6xx**

10.3 Migrate Redundant HGs and EPLCGs

Prepare to Migrate Redundant HGs and EPLCGs

The following steps prepare the HGs and EPLCGs for the migrations.

If there are no **Redundant** HGs or EPLCGs, skip to section 10.4.

CAUTION

If USs or GUSs are K4 type processors, the HGs **MUST** be K4 processors also.

Step	Action	Done (✓)
1	ENSURE you have a list of all HG and EPLCG points being controlled from the AM and CG, so they can later be restored to the proper mode.	
2	If there are AM/CL points/programs that set the HIWAY to FULL control, then those AM/CL points/programs should be set to INACTIVE.	
3	From the previous release US or GUS, SET the R43x or R5xx Hiway in BASIC CTRL. SUGGESTION: The LCN Overview display could be used for tracking the movement of nodes from the previous release to R6xx.	

Table 10-20 Redundant HGs and EPLCGs

CAUTION

When the OUTIND is set to Reverse, there is a difference between the actual values used in the boxes and the displayed values of the affected parameters on the LCN nodes. Hence, Honeywell recommends that you use care in designing the logic of the affected parameters as described in the following:

- For LCN nodes, before setting the OUTIND to Reverse, modify the AM/CL programs, schematics, and TPS applications to accommodate the modified behavior.
- For boxes on the same Hiway, no modification is required.

Load HGs and/or EPLCGs

Skip this procedure for nonredundant HGs/EPLCGs.

Load Migration Software—Perform steps 1 through 8 in the following table to load the migration software into an HG pair or a EPLCG pair

Step	Action	Done (√)
1	From the HG Status Display: SHUTDOWN the BACKUP HG from an R43x or R5xx US. (NOTE: At this point, the R4xx HG will go to WARNING, due to loss of redundancy.)	
2	VERIFY that all R43x or R5xx HG checkpoint files from the &luu directories have been translated and are on R6xx NET.	
3	Once the HG goes to QUALIF state, go to the physical HG node, POWER OFF the node (leave off for at least 5 seconds).	
4	UPGRADE hardware to R6xx requirements, as necessary. (Refer to section 3.4 for minimum memory requirements.)	
5	POWER ON the node. The node status will go to PWR_ON.	
6	From the R6xx System Status Display, SELECT the HG, then SELECT the NODE STATUS target. This will invoke the PROCESS NETWORKS NODE STATUS display.	
7	CAUTION You cannot perform a NET LOAD in this step, you must select the MANUAL LOAD target. To START the load of the HG with the new software, SELECT the LOAD SFW UPGRADE target for the selected HG, Use NET for the PGM SOURCE and DATA SOURCE. The status will go to UPGRADE.	

Table 10-21 Load HGs and/or EPLCGs

Load HGs and/or EPLCGs, continued

Verify Operation of R6xx HG—When the load is complete, both HGs of each HG pair act as primaries, one on the previous release LCN, the other on the R6xx LCN.

At the R6xx US, the Gateway Status display shows the just-loaded HGs status as UPGRADE. At both the previous release US and the R6xx US, the status of the other HG in the pair is shown as OFF, even though both HGs are acting as primaries for their respective LCN.

Before migrating the process networks, ensure there is an adequate number of USs on the previous release to allow view and control of the process.

Step	Action	Done (✓)
8	At an R6xx US or GUS, VERIFY the R6xx HG is operational, by performing benign process actions, such as: • calling up Group Displays • calling up schematics • checking hiway status, box status, etc.	

Load HGs and/or EPLCGs, continued

Move the Alarming Functions—Perform steps 9 through 11 in the following table, to move the alarming functions from the previous release HG to the R6xx HG, and prepare the R6xx HG to take over the process.

The ability to monitor the previous release alarming will now disappear. The operator should move to a US running on R6xx to monitor alarms.

Step	Action	Done (✓)
9	From the previous release US, on the GATEWAYS AND INTERFACES DISPLAY, DISABLE alarming for the previous release HG.	
10	From an R6xx US or GUS, ENABLE alarming for the migrated (R6xx) HG.	
11	VERIFY alarming on this HG is working on R6xx. Continue monitoring the alarms until you are comfortable with your view of the process.	

Table 10-21 Load HGs and/or EPLCGs (continued)

10.3. Migrate Redundant HGs and EPLCGs**Load HGs and/or EPLCGs, continued**

Load the New Software to the Primary HG—Perform Steps 12 through 17 in the following table, to load the new software to the original primary HG, which then becomes the backup HG. This is being done to establish redundancy on the R6xx side, and to drop the previous release side.

Step	Action	Done (✓)
12	Operators will be LOSING VIEW OF THE PROCESS. They should move to a US or GUS running on R6xx software. From a previous release US, on the HG STATUS Display, SHUTDOWN the previous release primary HG.	
13	This step must be performed because: - the QLTs have changed, and the node hardware must be re-qualified with the new QLTs, and - now is the time to update the hardware, as required. Once the HG goes to QUALIF state, go to the physical HG node, POWER OFF the node. (If not changing out hardware, leave node powered off for at least 5 seconds)	
14	Change hardware, as required.	
15	POWER ON the node.	
16	From an R6xx US or GUS, once status goes to BACKUP, SELECT the HG node status showing UPGRADE. SELECT the ENABLE UPGRADE target, and PRESS <ENTER> . Status goes from UPGRADE to OK to WARNING due to loss of redundancy.	
17	From an R6xx US or GUS, LOAD the R6xx software to the PWR_ON HG using the AUTOLoad NET target on the PROCESS NETWORKS NODE STATUS DISPLAY. It now comes up as the backup HG and redundancy is restored. This HG pair now has full functionality.	

Table 10-21 Load HGs and/or EPLCGs (continued)

Load HGs and/or EPLCGs, continued

Enable Automatic Checkpointing (optional)—Perform Steps 18 through 20 in the following table to enable Automatic Checkpointing for the hiway. Repeat this process for any remaining HG pair.

Step	Action	Done (√)
18	REQUEST a checkpoint of the HG (and ALL BOXES) data to the NET, <u>TWICE</u> . SELECT DEFAULT SOURCE to direct the checkpoint to NET.	
19	If desired, automatic checkpointing for this hiway can now be enabled.	
20	REPEAT Steps 1-20 for each remaining previous release <u>redundant</u> HG.	

Correct the Trip Point Values—Perform Step 21 in the following table to correct the Trip Point values in the EXCEEDED LIMIT ALARM DISPLAY.

21	In using the Alarm Summary Display, the Trip Point value may be bad. To fix the problem: A. Acknowledge all unit alarms in the Alarm Summary B. SHUTDOWN the primary (R6xx) HG C. RELOAD the HG (from Step B) D. In the UNIT ALARM display, check each entity for Invalid Trip point value. (SELECT the entity, then PRESS the <DETAIL> key.) E. SELECT the ALENBST parameter (unique for each point), and CHANGE it from its current state to INHIBIT, then back to its configured state. F. Repeat steps D and E for each entity that has the wrong trip point value	
-----------	--	--

Table 10-21 Load HGs and/or EPLCGs (continued)

As HGs are migrated, you may want to migrate related USs to R6xx.

10.4 Migrate Nonredundant HGs and EPLCGs

Prepare to migrate HGs and EPLCGs The following steps prepare the HGs and EPLCGs for the migrations. If there are no Nonredundant HGs or PLCGs, skip to the next section, “HG Alarm Deadband Option.”

CAUTION

If USs or GUSs are K4 type processors, the HG/EPLCGs **MUST** be K4 processors also.

Step	Action	Done (✓)
1	ENSURE you have a list of all HG and/or PLCG points being controlled from the AM and CG, so they can later be restored to the proper mode.	
2	From the previous release US or GUS, SET the Hiway in BASIC CTRL. SUGGESTION: The LCN Overview display could be used for tracking the movement of nodes from the previous release to R6xx.	

Table 10-22 Prepare to migrate HGs and EPLCGs

**Load nonredundant
HG and/or EPLCGs**

Load migration software into a HG and/or EPLCG.

CAUTION

In performing the following steps, you will lose all view and control of the affected HG.

In Steps 1 through 6 you will prepare the HG to be powered down, powered off, the hardware changed, then powered back up.

Step	Action	Done (√)
1	DISABLE automatic checkpointing for this hiway.	
2	From the previous release US, SHUTDOWN the HG.	
3	INFORM the operators they must now move to the R6xx US(s) or GUS.	
4	Once the HG goes to QUALIF state, go to the physical HG node, POWER OFF the node. Leave off for at least 5 seconds.	
5	Change hardware, as required. (See section 3.4 for memory size requirements.)	
6	POWER ON the node.	

Table 10-23 Load Nonredundant HGs and/or EPLCGs

10.4. Migrate Nonredundant HGs and EPLCGs

Load nonredundant HGs and/or EPLCGs, In the following steps you will reload the HG, set up the hiway and box control states, request checkpoint of the HG data, and if desired, enable automatic checkpointing.

continued

Step	Action	Done (✓)
7	From an R6xx US, LOAD the HG software from the NET with the translated checkpoint. Status goes to OK.	
8	For EPLCGs only: Perform hiway command from Hiway Status Displays. NOTE: All DHP boxes will be in ADDR ERR state. Perform INIT address command to the hiway. The status of the boxes will go to RESET. Load data to all boxes from NET. Status goes to OK.	
9	SET UP the hiway and box control states as you desire.	
10	REQUEST checkpoint of the HG (and all boxes) data <u>TWO TIMES</u> to NET.	
11	If desired, automatic checkpointing for this hiway can now be enabled.	
12	REPEAT steps 1-11 for each remaining <u>nonredundant</u> HG. If there are no NIMs to migrate you can now move the previous release AMs and CGs to R6xx, so you can do Advanced Control. See sections 10.14 and 10.15 for instructions on how to do this. NOTE: Do not restore the mode of the HG points controlled by an AM or CG (identified in Step 2) until after the AM and CG are migrated.	

Table 10-23 Load Nonredundant HGs and/or EPLCGs (continued)

10.5 HG Alarm Deadband Option

For optional HG Alarm Deadband package only

If you do not have the HG Alarm Deadband option, skip this procedure, and continue with the next section.

The special variable deadband firmware may be purchased at any time, and is not specifically required for Release 6xx.

If the special variable deadband firmware existed prior to Release 6xx being installed, the following steps must be taken to turn on the variable deadband option, and to prevent initialization of PV alarm deadband values (PVALDB).

Step	Action	Done (✓)
1	Be sure the HG has already been loaded with Release 6xx software.	
2	Prior to loading the Hiway boxes, RECONSTITUTE each LLPIU and/or DHP box configuration for each box that contains the special variable deadband firmware, and TURN ON the BOXVDB option, and load each entity.	
3	Perform a SAVE DATA <u>two times</u> to NET.	

Table 10-24 Optional HG Alarm Deadband Package

At this time if there are no NIMs to migrate, you can, at your option, migrate the previous release AMs and CGs to R6xx, so you can do Advanced Control. See sections 10.14 and 10.15 for instructions.

10.6 Migrate Redundant NIMs

Migration Procedures

CAUTION

The UCN servicing these NIMs may include multiple LCNs. If your system has multiple LCNs serviced by this UCN, you must contact the Honeywell Technical Assistance Center for special procedures. You cannot follow the procedures in this section.

A NIM in the R6xx UPGRADE mode will lose its UPGRADE status at the System Status display, if a node with the next sequential LCN address is SHUTDOWN. Its displayed status goes to WARNING, but this is a false indication, and the NIM is actually still in UPGRADE.

If USs are K4-type processors, the NIMs **MUST** also be K4 processors.

If GUS, APP, EAPP, EST or ESVT is present on the network, then NIMs **MUST** be K4 processor based LCN cards.

ATTENTION

Throughout the sections 10.6 ~ 10.9, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms **EHPM**, **ENIM**, and **EUCN** respectively.

Where we stand

At this point in the R6xx migration, two logically independent LCNs have been formed on one physical LCN with at least one US and one HM running on R6xx software. The R6xx personalities and translated NIM checkpoints have been loaded on the R6xx HM (NET). Automatic checkpointing for all NIMs has been disabled.

**Migration
Procedures,**
continued

The following procedures describe the steps required to migrate both the previous release NIM hardware and software to R6xx. If there are no redundant NIMs, skip to section 10.7, Migrate Nonredundant NIMs.

Step	Action	Done (√)
1	On the previous release US or GUS, CHECK to be sure all UCN nodes are operating normally before proceeding. NOTE: It is recommended the installation of new or upgraded IOPs be put off until after the migration.	
2	MAKE A LIST of all UCN points being controlled from an AM or CG, so they can later be restored to their proper mode of advanced control.	
3	VERIFY automatic checkpointing is disabled on this UCN.	
4	At the previous release US or GUS, SWITCH the UCN to BASIC control. If there are AM/CL points/programs that set the UCN to FULL control, then those AM/CL points/programs should be set to INACTIVE.	
5	At the previous release US, DISABLE cable swapping for all UCN nodes on this UCN.	

Table 10-25 Migration Procedures**CAUTION**

If any external shutdown timers are in use in conjunction with the NIM, be sure to check the timer function to ensure the timers are set at a minimum of 3 seconds. Performing a swap of NIMs requires that these timers be set to 3 seconds or longer.

10.6. Migrate Redundant NIMs

Migration Procedures, continued

A change was made to the regulatory control point, using the PIDFORM in IDEAL and nonzero T2s, that will require retuning as part of the migration. Refer to the PM/APM Special Considerations (section 5.12) regarding PIDFORM in IDEAL and nonzero T2s. If External Shutdown timers are in use in conjunction with the NIM, be sure to check the timer function to ensure that the timers are set at a minimum of 3 seconds. Performing a swap of NIMs requires that these timers be set to 3 seconds or longer.

CAUTION

When the OUTIND is set to Reverse, there is a difference between the actual values used in the xPM nodes and the displayed values of the affected parameters on the LCN nodes. Hence, Honeywell recommends that you use care in designing the logic of the affected parameters as described in the following:

- For LCN nodes, before setting the OUTIND to Reverse, modify the AM/CL programs, schematics, and TPS applications to accommodate the modified behavior.
- For xPM nodes on the same UCN, no modification is required.

Step	Action	Done (√)
6	From the previous release US or GUS, SHUTDOWN the backup NIM. NIM status now goes to QUALIF . NOTE: At this point, the previous release NIM will go to WARNING, because of loss of redundancy.	
7	NOTE: The QLTs have changed, and the node hardware must be requalified with the new QLTs. Go to the physical NIM node, and POWER OFF the node, (leave off at least 5 seconds), UPGRADE hardware if required. (Refer to section 3.4 for minimum memory requirements.) Then, POWER ON the node. (Status goes to PWR_ON)	
8	CAUTION You cannot perform a NET LOAD in this step, you must select the MANUAL LOAD target. On an R6xx US, use the LOAD SFW UPGRADE target of the PROCESS NETWORKS NODE STATUS to: LOAD the NIM with the new software. Use NET for the PGM SOURCE and DATA SOURCE. The status should go to UPGRADE.	

Table 10-25 Migration Procedures (continued)

10.6. Migrate Redundant NIMs

**Migration
procedures,
continued**

When the load is complete, both NIMs of each NIM pair act as primaries, one on the previous release LCN portion, the other on the R6xx LCN portion.

At the R6xx US or GUS, the Process Networks Node Status display shows the just-loaded NIM status as UPGRADE. At both the previous release US or GUS and the R6xx US, the UCN Status Display shows both the previous release NIM and the R6xx NIM as OK/BACKUP, even though both NIMs are acting as primaries for their respective LCN portions.

Step	Action	Done (√)
9	At an R6xx US or GUS, VERIFY the R6xx NIM is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc. CHECK to ensure the IDEAL types are operating correctly.	
10	VERIFY the alarming is working correctly through the R6xx NIM.	
NOTE: Since the UCN process nodes are still on the previous release, the following checkpoints are the previously released versions.		
11	From an R6xx US or GUS, CHECKPOINT each PM, APM, and LM for this UCN, once to the NET, then to cartridge.	
12	REPEAT steps 3 through 11 for all remaining REDUNDANT backup NIMs.	
13	INFORM the operators they must now move to the R6xx US(s).	

Table 10-25 Migration Procedures (continued)

Migration Procedures,
continued

The following steps complete the migration for the other half of the redundant NIM pairs. The steps involve freeing up the odd UCN address from the previous release NIM, upgrading the hardware, and finally loading the R6xx software.

Step	Action	Done (√)
14	From the previous release US or GUS SHUTDOWN the NIM. The NIM status now goes to QUALIF.	
15	To force the R6xx NIM to take the odd UCN address, from an R6xx US or GUS, Select the R6xx NIM on the PROCESS NETWORKS NODE STATUS display, then CAUTION Step 13 must be completed before selecting ENABLE UPGRADE for the R6xx NIM! Select ENABLE UPGRADE. Its status will go from UPGRADE to OK to WARNING, due to the loss of redundancy.	

Table 10-25 Migration Procedures (continued)

10.6. Migrate Redundant NIMs**Migration Procedures,**
continued

At an R6xx US, perform the following steps by using the Gateway Status display to load the R6xx software.

Step	Action	Done (√)
16	POWER DOWN the previous release NIM, (leave off for at least 5 seconds) Upgrade hardware as required. (Refer to section 3.4 for minimum memory requirements.) Then, POWER ON the NIM.	
17	At an R6xx US or GUS, ensure the NIM is in PWR_ON status, then, using AUTOLOAD NET, LOAD the NIM with R6xx software.	
18	When the load is complete, the NIM status is BACKUP, and it has the even (secondary) UCN address. The primary NIM status will go from WARNING to OK (redundancy is restored). NOTE: When both NIMs of the pair are migrated to R6xx, the primary/secondary position of the pair has been reversed from the original position. You may wish to failover via primary SHUTDOWN and AUTOLOAD to restore them back to their original positions. NOTE: Do not restore the mode of the UCN points controlled by an AM or CG (identified in step 2) until after the AM and CG are migrated. NOTE: From an R6xx US, CHECKPOINT all devices for this UCN excluding HPMS, once to the NET, then to cartridge.	
19	REPEAT steps 14 through 18 for the second half of any remaining <u>redundant</u> NIM pairs.	

Table 10-25 Migration Procedures (continued)**Where to go from here**

After having migrated all of your redundant NIMs, continue with one of the following UCN-related procedures (as appropriate):

- Section 10.7 (Migrate Nonredundant NIMs), or
- Section 10.8 (Migrate Redundant PMs/APMs/HPMs), or
- Section 10.9 (Migrate Nonredundant PMs/APMs/HPMs), or
- Section 10.11 (Migrate Redundant LMs), or
- Section 10.12 (Migrate Nonredundant LMs).
- Section 10.13 (Migrate Safety Manager)

10.7 Migrate Nonredundant NIMs

Important!

CAUTION

The UCN servicing these NIMs may include multiple LCNs. If your system has multiple LCNs serviced by this UCN, you must contact the Honeywell Technical Assistance Center for special procedures. You cannot follow the procedures in this section.

If any USs or GUSs have K4 processor type, the NIM MUST also be a K4 processor.

Where we stand

At this point in the R6xx migration, two logically independent LCNs have been formed on one physical LCN, with at least one US or GUS and one HM running on R6xx software.

The R6xx personalities and translated checkpoints have been loaded on the R6xx HM (NET). Automatic checkpointing for all NIMs has been disabled.

Migration Procedures

The following procedures describe the steps required to migrate both the previous release NIM hardware and software to R6xx. If there are no nonredundant NIMs, skip to section 10.8, Migrate Redundant PMs/APMs/HPMs.

CAUTION

In performing the following steps, you will lose all view and control of the affected NIMs.

CAUTION

If any external shutdown timers are in use in conjunction with the NIM be sure to check the time function to ensure the timers are set at a minimum of 3 seconds. Performing a swap of NIMs requires that these timers be set to 3 seconds or longer.

10.7. Migrate Nonredundant NIMs

R43x/R5xx to R6xx A change was made to the regulatory control point, using the PIDFORM in IDEAL and nonzero T2s, that will require retuning as part of the migration. Refer to the PM/APM Special Considerations (section 5.12) regarding PIDFORM in IDEAL and nonzero T2s.

Step	Action	Done (√)
1	At the previous release US, CHECK to be sure all UCN nodes are operating normally before proceeding. NOTE: It is recommended that the installation of new or upgraded IOPs be put off until after the migration.	
2	MAKE A LIST of all UCN points being controlled from an AM or CG, so they can later be restored to their proper mode.	
3	VERIFY automatic checkpointing is disabled on this UCN.	
4	At the previous release US, SWITCH the UCN to BASIC control.	
5	INFORM the operator(s) that they must now move to the R6xx US(s).	
6	At the previous release US or GUS, SHUTDOWN the previous release NIM. The NIM status now goes to QUALIF (or possibly FAIL).	
7	POWER DOWN previous release NIM, (leave off for at least 5 seconds) Upgrade hardware as required. (Refer to section 3.4 for minimum memory requirements.) Then, POWER ON NIM.	

Table 10-26 R43x and R5xx Upgrade to R6xx

R43x/R5xx to R6xx,
continued

Step	Action	Done (√)
8	At an R6xx US or GUS, ensure the NIM is in POWER_ON status, then, using AUToload NET, LOAD the NIM that was on the previous release with R6xx software.	
9	At an R6xx US or GUS, VERIFY the R6xx NIM is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc.	
10	VERIFY the alarming is working correctly through the R6xx NIM.	
11	NOTE: Do not restore the mode of the UCN points controlled by an AM or CG (identified in step 2) until after the AM and CG are migrated. From an R6xx US, CHECKPOINT each UCN device excluding HPMS, twice to NET, THEN TO CARTRIDGE.	
12	If other Nonredundant NIMs require upgrading, repeat Steps 1 through 12 for each remaining <u>nonredundant</u> NIM.	

Table 10-26 R43x and R5xx Upgrade to R6xx (continued)

10.8 Migrate Redundant PMs/APMs/HPMs

Migrate redundant PMs/APMs/HPMs to R6xx

If you do not have redundant PMs, APMs, or HPMs, skip this procedure and continue with section 10.9,

Migrate Nonredundant PMs/APMs/HPMs.

CAUTION

The following precaution must be followed when migrating redundant APMs and HPMs that use SI Array points which reference external data. Failure to observe this procedure could result in a loss of data from one or more SI Array points.

If you have any questions regarding this migration precaution, please contact your local Honeywell National Response Center (NRC), Technical Assistance Center (TAC), or Customer Response Center (CRC).

If your redundant APM / HPM are currently running below the minimum level shown, you are at risk and must follow this special precaution.

	Minimum R5xx Version
APM-cntrl (umf)	53.5
APM-comm (umc)	53.0
HPM-cntrl (hmf)	53.6
HPM-comm (hpmc)	53.1

Prepare a list of all APM/HPM SI Array points whose “External Data Option” is NOT equal to “NONE”. Use the Data Entity Builder “Print Entities” function to create “exception build” (.EB) files for these points.

After performing an on-line upgrade of the APM/HPM, we recommend that your SI Array points be reloaded from the .EB files you created in the previous step. Otherwise, it is possible that one or more SI Array points will be displaying invalid data (i.e., NaN). Reloading the SI Array points on the new software, and taking a checkpoint, will correct the problem for future migrations.

Your particular application may require additional precautions to prevent possible plant upset.

This problem will not occur if you perform an off-line upgrade of the APM/HPM. (i.e., shut down the redundant pair and reload with new software.) However, this causes a loss of view and control for the affected points.

Redundant previous release PMs, APMs, and HPMs can be migrated to R6xx Online. All PMs, APMs, and HPMs on the physical UCN must be migrated to R6xx software. No offline translators are required for this migration. Database translation will be performed online during the migration.

The Online migration requires that the secondary PM, APM, or HPM be migrated first while the primary PM, APM, or HPM continues to control the process.

NOTE: If you have External Shutdown Timer

If External Shutdown timers are in use in conjunction with the PM, APM, or HPM, be sure to check the timer function to ensure the timers are set at a minimum of 3 seconds. Performing a swap of these devices requires that the timers be set to 3 seconds or longer.

There is a concern that it may take longer than 3 seconds for the HPM on-line translation to complete due to major control database expansion and re-organization in R6xx.

CAUTION

Once all PM/APM/HPM nodes have been transitioned to R6xx, use the PERFMENU, UCNVERSN schematic to verify that those nodes are running R6xx software.

10.8. Migrate Redundant PMs/APMs/HPMs

Migrate redundant PMs/APMs/HPMs to R6xx, continued

CAUTION

A change was made to the regulatory control point, using the PIDFORM in IDEAL and nonzero T2s, that will require retuning as part of the migration. Refer to the PM/APM Special Considerations (section 5.12) regarding PIDFORM in IDEAL and nonzero T2s.

NOTE:

Steps 1 and 2 are for R43x to R6xx migrations ONLY.

Step	Action	Done (✓)
1	(NOTE: DOC TOOL from the Engineering Main Menu, provides an easy interface to this information.) Use DOC TOOL to capture all ideal form for the PM/APM and its tuning constants. Print this file for possible retuning. PLACE any REGCTL points that are using the IDEAL form of the PID and nonzero T2, in MAN mode.	
2	NOTE: Only perform this step If you had REGCTL points affected by Step 1 above. Otherwise, skip to Step 3. CHECKPOINT (if any) the R6xx PM (or APM) pair to NET, if you have REGCTL points in MANUAL.	
3	From an R6xx US or GUS, on the System Status Display: SELECT the BACKUP PM/APM/HPM, and perform a SHUTDOWN. The PM/APM/HPM status will go to ALIVE on the UCN Status Display.	

Table 10-27 Migrate redundant PMs/APMs/HPMs for R6xx

**Migrate redundant
PMs/APMs/HPMs to
R6xx, continued**

Step	Action	Done (√)
4	At an R6xx US or GUS, LOAD the secondary PM/APM/HPM manually, using NET as source for the PGM SOURCE and the DATA SOURCE. The UCN status indicates LOADING until the load is complete. The status for this PM/APM/HPM will then go to BACKUP. The color of the letters goes to cyan when load is completed. CAUTION WAIT 5 minutes <u>after</u> the color has gone to cyan. (This allows time for background diagnostics to run.) If the Node Type ID letters remain in yellow, <u>do not continue</u>. Call TAC at 1-800-822-7673 as soon as possible.	
5	At an R6xx US or GUS, SELECT the primary PM/APM/HPM, and perform the SWAP PRIMARY. The PM/APM/HPM on R6xx becomes the primary, and the status of the PM/APM/HPM on the previous release goes to ALIVE.	
6	At an R6xx US or GUS, LOAD the secondary PM/APM/HPM using NET as source for the PGM SOURCE and the DATA SOURCE. The UCN status indicates LOADING until the load is complete. The status for this PM/APM/HPM will then go to BACKUP.	
7	At an R6xx US or GUS, VERIFY the R6xx PM/APM/HPM is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc.	

Table 10-27 Migrate redundant PMs/APMs/HPMs for R6xx (continued)

10.8. Migrate Redundant PMs/APMs/HPMs

**Migrate redundant
PMs/APMs/HPMs to
R6xx, continued**

Step	Action	Done (√)
8	One by one, place the IDEAL form and non-zero T2 REGCTL point types from step 1 back into automatic mode. CONFIRM that the loop is operating correctly. RE-TUNE as needed.	
9	CHECKPOINT the R6xx PM/APM/HPM to NET <u>two times</u> .	
10	REPEAT steps 1-9 for all redundant PM/APM/HPM being migrated on each UCN.	
CAUTION Once all PM/APM/HPM nodes have been transitioned to R6xx, use the R6xx PERFMENU, UCNVERSN schematics to verify those nodes are running R6xx software. After migration to R6xx, examine all APM/HPM SI Array Points to verify that they show valid data. If not, reload the points in question.		

Table 10-27 Migrate redundant PMs/APMs/HPMs for R6xx (continued)

10.9 Migrate Nonredundant PMs/APMs/HPMs

Migrate Nonredundant PMs/APMs/HPMs to R6xx

If you do not have nonredundant PMs/APMs/HPMs, skip this section, and continue with section 10.11, Migrate Redundant LMs.

Nonredundant previous release PMs/APMs/HPMs can be migrated only offline. All PMs/APMs/HPMs on the physical UCN must be migrated to R6xx software.

Since this procedure is offline, as each PM/APM/HPM is shut down, you lose the view to points, all local control, and all peer-to-peer communication stops. Putting the PM/APM/HPM in IDLE and restoring a checkpoint places all control points in IDLE and causes outputs for component points to go to the FAILSAFE position.

CAUTION

A change was made to the regulatory control point, using the PIDFORM in IDEAL and nonzero T2s, that will require retuning as part of the migration. Refer to the PM/APM Special Considerations regarding PIDFORM in IDEAL and nonzero T2s.

In performing the following steps, you will lose all view and control of the affected PMs.

NOTE:

Steps 1 and 2 are for R43x to R6xx migrations ONLY.

Step	Action	Done (√)
1	PLACE any REGCTL points that are using the IDEAL form of the PID and nonzero T2, in MAN mode. (NOTE: DOC TOOL from the Engineering Main Menu, provides an easy interface to this information.) Use DOC TOOL to capture all ideal form for the PM/APM/HPM and its tuning constants. Print this file for possible retuning.	
2	NOTE: Only perform this step IF you had REGCTL points affected by Step 1 above. Otherwise, skip to Step 3. CHECKPOINT (if any) the previous release PM/APM/HPM to NET, if you have REGCTL points in MANUAL.	
3	From an R6xx US, on the System Status Display, SET the PM/APM/HPM, to IDLE state, and perform a SHUTDOWN. The PM/APM/HPM status will go to ALIVE on the UCN Status Display.	

Table 10-28 Migrate Nonredundant PMs/APMs/PMs to R6xx

10.9. Migrate Nonredundant PMs/APMs/HPMs

Step	Action	Done (√)
4	At an R6xx US, LOAD the PM/APM/HPM using NET as source for the PGM SOURCE and the DATA SOURCE. The UCN status indicates LOADING until the load is complete. The status for this PM/APM/HPM will then go to IDLE.	
5	At an R6xx US, STARTUP the PM or APM/HPM using WARM START or whatever means is best suited for your system. Its status will go to OK.	
6	At an R6xx US, VERIFY the R6xx PM (or APM/HPM) is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc.	
NOTE: Step 7 is for R43x to R6xx Upgrade Only.		
7	One by one, place the IDEAL form and non-zero T2 REGCTL point types from step 1 back into automatic mode. CONFIRM the loop is operating correctly. RE-TUNE as needed.	
8	CHECKPOINT the R6xx PM (or APM/HPM) to NET <u>two times</u> .	
9	REPEAT steps 1-8 for all nonredundant PMs/APMs/HPMs being migrated.	

Table 10-28 Migrate Nonredundant PMs/APMs/HPMs to R6xx (continued)**CAUTION**

Once all PM/APM/HPM nodes have been transitioned to R6xx, use the R6xx PERFMENU, UCNVERSN schematics to verify those nodes are running R6xx software.

After migration to R6xx, examine all APM/HPM SI Array Points to verify that they show valid data. If not, reload the points in question.

10.10 Loading Fieldbus IOP Software

Downloadable Fieldbus IOP Software Personality

If you do not have Fieldbus IOP on your system, skip this section and proceed to section 10.11.
 The fieldbus IOP Software Personality, unlike its predecessors, is downloadable. This software is also flashed into static memory. To check the currently loaded Software Personality version, and/or the next Software Personality to be loaded, follow the two steps below:

Step	Action	Done (√)
1	Select the fieldbus IOP Detail Display.	
2	Select the "PI Load Info".	

Load New Software Personality into Fieldbus IOP

To load the new Software Personality into the fieldbus IOP, continue with the following procedure:

Step	Action	Done (√)
3	Make sure that the desired fieldbus IOP Software personality file exists on the HM or removable media under the &UCN directory. Usually the name has the form FBIOxxxx (where xxxx is the version number).	
4	Select the HPM Detail Display.	
5	Select the fieldbus IOP to be loaded.	
6	Select the "LOAD/SAVE RESTORE" target, then the "PROGRAM LOAD" target (at the bottom left hand corner of the display).	
7	A Software Personality File will appear in front of the "PI FILE TO BE LOADED:" string. This may be the currently loaded version, a blank or the last user entered string.	
8	Select this target. A data entry target preceded by "ENTER FILE NAME" should appear on the top left-hand corner of the screen.	

Table 10-29 Load New Software Personality into Fieldbus IOP

10.10. Loading Fieldbus IOP Software

Step	Action	Done (√)
9	Type the file name for the Software Personality File desired, then press the ENTER Key on the Keyboard. (IMPORTANT: It must be the ENTER key on the keyboard and NOT the one shown on the screen. The latter will clear the entry).	
10	Select ENTER from the screen; then select the location where the file resides.	
11	The progress can then be monitored and the status of the load can be viewed from the "PI Load Info" display. • During the Software Personality Load, the IOP status will become invalid momentarily when the IOP begins flashing its new personality. • After the Status indicates SUCCESS, the IOP will restart with the new Personality at which time it will show NR (No Response) for a brief time.	

Table 10-29 Load New Software Personality into Fieldbus IOP (continued)

10.11 Migrate Redundant LMs

Migrate Redundant LM to R6xx

If you do not have redundant LMs on your system, skip this section and continue with section 10.12, Migrate Nonredundant LMs.

Redundant LMs can be migrated to R6xx On-line. All LMs on the physical UCN must be migrated to R6xx software. No translators are required for this migration. Therefore, the migration will be performed using R43x checkpoints.

The On-line migration requires that the secondary LM be migrated first while the primary LM continues to control the process. Ensure the primary LM is the preferred controller before proceeding.

NOTE: If upgrading from R5xx, use PERFMENU, UCNVERSN schematics to determine the software version of your running LM. Compare this version with the new LM version shown on your R684 Software Change Notice. Continue with the migration procedure in this section only if a newer version of LM software is now available. If a newer version is not available, continue with the next section.

CAUTION

If any external shut down timers are in use in conjunction with the LM, be sure to check the timer function and make appropriate adjustments. The time will vary as a function of the amount of ladder logic in the LM. As you increase your ladder logic, you should adjust the timer function as necessary.

Performing a swap of LM primaries requires the external timers to be set at 6.5 seconds. Set external timers to a time of 6.5 seconds or greater to eliminate problems in this area.

10.11. Migrate Redundant LMs**Migrate Redundant LM to R6xx, continued**

Step	Action	Done (√)
1	Determine if the LEAD (PRIMARY) LMM is the "PREF" by executing the following steps: At the UCN Status Display, SELECT the "ODD" (OK & PRIMARY) LMM node SELECT DETAIL STATUS target. VERIFY P/S is PRIMARY, and the FILE POS is PREF. IF it is PREF and "SYNCHST" is SYNCHED, continue with the remainder of this step. If you find that the "NON-PREF" is the LEAD (PRIMARY), go directly to Step 2. SELECT the LMM SELECT RUN STATES target SELECT SWAP PRIMARY NOTE: After the swap, the non-preferred LM should assume the lead, and will be performing the ladder logic control. WAIT seven minutes.	
2	Shutdown BACKUP (This is the preferred LM), as follows: SELECT the EVEN numbered LM (NOTE: you may have to select it twice) SELECT RUN STATES target SELECT SHUTDOWN target SELECT ENTER target. The status of the BACKUP LM goes to OFFNET then to the ALIVE state	
3	At an R6xx US, SELECT LOAD/SAVE/RESTORE SELECT PROGRAM LOAD SELECT ENTER target SELECT DEFAULT SOURCE for the LMM program source SELECT DEFAULT SOURCE for the data source When the load completes, the LMM goes to BACKUP. WAIT 5 minutes to allow synchronization to complete. VERIFY the detail status of the backup LMM shows "PGM MISMATCH." This status is okay.	
4	Swap primaries (Preferred LM now becomes the LEAD again). SELECT RUN STATES target SELECT SWAP PRIMARY SELECT ENTER target Once swap is complete, WAIT another seven minutes.	

Table 10-30 Migrate Redundant LM to R6xx

**Migrate
Redundant LM to
R6xx, continued**

Step	Action	Done (√)
5	Shutdown BACKUP (This is the non-preferred LM), as follows: SELECT the EVEN numbered LM (NOTE: you may have to select it twice) SELECT RUN STATES target SELECT SHUTDOWN target SELECT ENTER target. The status of the BACKUP LM goes to OFFNET then to the ALIVE state	
6	RELOAD with new personality (to the BACKUP LM) SELECT LOAD/SAVE/RESTORE SELECT PROGRAM LOAD SELECT ENTER target SELECT DEFAULT SOURCE for the LMM program source SELECT DEFAULT SOURCE for the data source When the load completes, the LMM goes to BACKUP. WAIT 7 minutes to allow synchronization to complete.	
7	At an R6xx US, VERIFY the R6xx LM is capable of full operation, by - calling up Group Displays, - calling up schematics, - changing modes, etc. If you should detect any errors at this time, call Honeywell TAC. Also, if the LM shows a status of OK/NOSYNC (which means there is no synchronization between the primary and secondary LMs) - call Honeywell TAC. The LMM is now on R6xx with its database synchronized to its R6xx partner.	
8	CHECKPOINT the R6xx LM to NET, <u>twice</u>	
9	SAVE the Ladder Logic to the HM and to removable media for EACH LM.	
10	REPEAT steps 1-9 for all redundant LMs being migrated.	

Table 10-30 Migrate Redundant LM to R6xx (continued)**CAUTION**

Once all LM nodes have been transitioned to R6xx, use the R6xx PERFMENU, UCNVERSN schematics to verify those nodes are running R6xx software.

10.12 Migrate Nonredundant LMs

Migrate Nonredundant LM to R6xx

If you do not have nonredundant LMs on your system, skip this section and continue with section 10.13, Migrate FSC Safety Manager.

Nonredundant LMs can only be migrated off-line.

All LMs on the physical UCN must be migrated to R6xx software.

NOTE: If upgrading from R5xx, use PERFMENU, UCNVERSN schematics to determine the software version of your running LM. Compare this version with the new LM version shown on your R684 Software Change Notice. Continue with the migration procedure in this section only if a newer version of LM software is now available. If a newer version is not available, continue with the next section.

Perform the following steps to migrate nonredundant LMs:

Step	Action	Done (√)
1	At an R6xx US, from the UCN Status Display, idle and shut down the LMM, as follows. At the UCN STATUS display SELECT the LM to be migrated SELECT RUN STATES SELECT IDLE SELECT ENTER (wait for IDLE to be displayed) SELECT SHUTDOWN SELECT ENTER (wait for the status to go to ALIVE)	

Table 10-31 Migrate Previous Release Nonredundant LM to R6xx

**Migrate
Nonredundant LM
to R6xx, continued**

Step	Action	Done (√)
2	At an R6xx US or GUS, Load the LM: At the UCN STATUS display, SELECT LOAD/SAVE RESTORE SELECT PROGRAM LOAD SELECT ENTER SELECT NET for the PGM source SELECT NET for the DATA source and when load is complete, continue with step 3.	
3	Change the IDLE status to STARTUP.	
4	At an R6xx US or GUS, once the LM status is OK, VERIFY the R6xx LM is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc. If you should detect any errors at this time, call Honeywell TAC.	
5	REPEAT steps 1-4 for all nonredundant LMs being migrated.	
6	CHECKPOINT the R6xx LMM to NET, <u>twice</u> .	
7	SAVE the Ladder Logic to the HM and to removable media for EACH LM.	

Table 10-31 Migrate Previous Release Nonredundant LM to R6xx (continued)**CAUTION**

Once all LM nodes have been transitioned to R6xx, use PERFMENU, UCNVERSN schematics to verify those nodes are running R6xx software.

10.13 Migrate FSC Safety Manager or Triconex SMM

**Honeywell HSMS
upgrade
document**

If you do not have FSC Safety Manager or Triconex SMM on your system, skip this section and continue with section 10.14, Auto Checkpointing.

CAUTION

Once you migrate the FSC you cannot backout and preserve View.

R6xx FSC (SMM)

The R6xx FSC SMM cannot be loaded by the R6xx NIM using an R5xx checkpoint. If an attempt is made to load an R6xx FSC - which does not have a redundant partner using an R6xx NIM and an R5xx checkpoint - a “FILE ACCESS ERROR” will be generated. Under these conditions the FSC database will have to be rebuilt using EB or DB files.

The preferred procedure is to load the R6xx NIM while the FSC is still at the R5xx level, create a checkpoint using the R6xx NIM then proceed with the migration of the FSC to R6xx..

R6xx Triconex (SMM)

The R6xx Triconex SMM can be loaded using the R6xx NIM and the R5xx checkpoint.

If you are currently using Triconex SMM software prior to version 65.0 please be aware of the following anomaly:

During failover or swap primary of a Triconex SMM, a peer data request could return incorrect digital state data. This could cause applications to take action on false data. **This has been corrected in Triconex Safety Manager version 65.0.** Before migrating from a pre-65.0 version of Triconex Safety Manager ensure that your applications cannot or will not take undesired action should an incorrect digital state value be read from the Triconex Safety Manager.

CAUTION

If any external shutdown timers are in use in conjunction with the FSC Safety Manager or Triconex SMM, be sure to check the timer function and make appropriate adjustments.

CAUTION

Once all SMM nodes have been transitioned to R6xx, use PERFMENU, UCNVERSN schematics to verify those nodes are running R6xx software.

Follow instructions for Logic Manager

Observing the cautions presented above, the instructions in Sections 10.11 and 10.12 can be followed to update the “TPN portion” of the FSC Safety Manager or Triconex-SMM (i.e., the part that connects directly to the UCN and runs TPN software.)

Additional SM upgrades if needed

If additional software/firmware/hardware upgrades are planned, please refer to appropriate instructions provided by the vendor. For FSC Safety Manager, refer to the Honeywell HSMS upgrade document and the FSC Software Manual.

10.14 Auto Checkpointing

Enable Auto Checkpointing for NIMs and UCN devices

OPTIONAL STEPS—Perform the following Auto Checkpointing steps **only if you want to enable Auto Checkpointing on the NIMs and UCN devices**. Otherwise, continue with the next section.

Using the following steps, enable Auto Checkpointing only after the entire logical UCN network has been migrated to R6xx. Auto checkpointing may be selected for the entire NIM or for individual devices. For each box that should be supported by auto checkpointing, enable checkpointing as follows.

Step	Action	Done (√)
1	FOR each NIM: At an R6xx US or GUS, from the PROCESS NETWORKS NODE STATUS menu, SELECT NIM SELECT AUTO SAVE SELECT ENABLE SAVE SELECT NTWK STATUS SELECT AUTO CHECKPT SELECT UCN CKPT ENABLE SELECT ENTER	
2	FOR UCN DEVICES: On the same screen, SELECT DEV CKPT ENABLE	
3	SELECT a device, and SELECT ENTER	
4	SELECT another device, and SELECT ENTER	
5	REPEAT Step 4 until auto checkpointing is enabled for all desired UCN devices.	

Table 10-32 Enable Auto Checkpointing for NIMs and UCN Devices

10.15 Migrate Application Modules (AM, AMW, APP)

Migrate Application Modules

Perform the following steps to upgrade hardware, and load the AM with R6xx software.

Step	Action	Done (√)
1	CAUTION Choosing cold, warm, or hot restart: If you use totalizers in your AM and you wish to retain those values, screen print each AM totalizer to record the values. If you restart with COLD or WARM, the totalizer values will go to zero. To retain the totalizer values in R6xx, load the AM using the HOT restart. NOTE: Your AM checkpoints should already be on the NET. See section 10.1	
2	From a previous release US or GUS, SHUTDOWN an AM (both AMS, if redundant). The AM status now goes to QUALIF.	
3	Once the AM goes to QUALIF state, POWER OFF the AM node(s). Leave off for at least 5 seconds.	
4	UPGRADE the hardware to R6xx requirements, as needed. (Refer to section 3.4 for minimum memory requirements.)	
5	POWER ON the AM node(s).	
6	LOAD the AM using NET for PGM SOURCE, and DATA SOURCE. You can use a cold, warm or hot restart. When loaded, the AM goes to the WARNING status until the CDS synchronization is complete. This can take as long as 10 minutes. Review the status using the Node Status Display for the AM. Also, a redundant AM will remain in the WARNING status until its BACKUP AM is loaded.	
7	If this AM is redundant, LOAD the BACKUP AM. The AM status goes to BACKUP and the primary goes to OK.	

Table 10-33 Migrate Application Modules

10.15. Migrate Application Modules (AM, AMW, APP)

**Migrate
Application
Modules,**
continued

Perform the following steps to enable AM control, start AM control, take checkpoints to the NET for each AM, and enable Auto Checkpointing (if Auto Checkpointing is desired).

Step	Action	Done (✓)
8	From an R6xx US or GUS, SET control status for the hiway and its boxes to FULL CONTROL, to enable AM control (no points in cascade at this time).	
9	From an R6xx US or GUS, SET control from the UCN and its devices to FULL CONTROL to enable AM control.	
10	From an R6xx US or GUS, START AM control by placing Hiway/UCN points in CASCADE mode. Nodes on the UCN and Extended Controllers on Data Hiways retain cascade requests previously issued. Basic Controllers, and Multifunction Controllers (MCs and AMCs) do not retain former cascade requests, and resume operation as with a cold restart.	
11	REPEAT steps 1 through 10 for each remaining AM.	
12	If desired, ENABLE auto checkpointing for each AM.	
13	CHECKPOINT the R6xx AM to the NET.	

Table 10-33 Migrate Application Modules (continued)

Load the R6xx AMW personality

With TPN 683 the AMW package installer has changed. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

The R6xx AMW personality is part of the R6xx TPN media and can be found on the GUS-TPN Application Module CD ROM. Perform the following steps to load the AMW personality.

Note: The AMW personality is not part of the TPN media from TPNR684.

Step	Action	Done (✓)
1	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs → Command Prompt , Type “ phdctl stop ” press Enter .	
2	Using the Windows Task Bar: START → HONEYWELL TPS → LCNP STATUS . Select Reset button, Select Yes , and wait until the Node goes to PWR_ON state.	
3	Using the Windows Task Bar: Select Settings → Control Panel → Add/Remove Programs .	
4	From Add/Remove Programs Window Select TPS AMW Personality , Select Add/Remove... button. Verify Uninstallation Window.	
5	Select AMW Personality , Select OK . When the uninstallation window returns, select OK . Select OK from Add/Remove Programs window.	
6	Insert the TPN – Application Module CD .	
7	Using Windows Explorer, go to the <CD drive>:\AMW and run the amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package.	
8	Using Windows Explorer, check under c:\HWIAC\TPS\PERSONALITIES\AM directory to make sure that the personality files have been loaded.	
9	Do an AUTOLOAD NET on the GANT node. It should come up as an AMW with an OK status.	
10	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs → Command Prompt , Type “ phdctl start ” then press ENTER .	

Table 10-34 Load the R6xx AMW Personality

10.15. Migrate Application Modules (AM, AMW, APP)

Load the R6xx APP personality

With TPN 684 the APP package installer has changed. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

The R6xx APP personality is part of the R6xx TPN media and can be found on the GUS-TPN Application Module CD ROM.

Perform the following steps to load the APP personality.

Step	Action	Done (✓)
1	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs →Command Prompt . Type " phdctl stop " then press Enter .	
2	Using the Windows Task Bar: START→ HONEYWELL TPS → LCNP STATUS . Select Reset button, Select Yes and wait until the Node goes to PWR_ON state. Note: Ensure that the LCNP autorestart status is disabled.	
3	Using the Windows Task Bar: Select Start→Control Panel→Add/Remove Programs .	
4	From Add/Remove Programs Window Select TPS AMW Personality (AM) , Select Add/Remove... button. Verify Confirm File Deletion Prompt. "Are you sure you want to completely remove "TPS – amw Personality (AM)" and all of its components?	
5	Select Yes . When complete, select OK then select OK from Add/Remove Programs window.	
6	Insert the TPN – Application Module CD .	
7	Using Windows Explorer, go to the <CD drive>:\APP and run the tps - amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package.	
8	Do an AUTOLOAD NET on the APP node. It should come up as an APP with an OK status.	
9	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs →Command Prompt , Type " phdctl start " then press ENTER .	

Table 10-35 Load the R6xx APP Personality

10.16 Migrate Computer Gateways/PLNMs

Migrate Computer Gateways/PLNMs

Perform the following steps to migrate the Computer Gateways/PLNMs from previous release to R6xx software.

Step	Action	Done (√)
1	From the previous release US or GUS, SHUTDOWN the CG, and wait until it goes to QUALIF state.	
2	Once the CG goes to QUALIF state, POWER OFF the CG node. Leave off for at least 5 seconds.	
3	UPGRADE the hardware to R6xx requirements, as needed. (Refer to section 3.4 for minimum memory requirements.)	
4	POWER ON the CG node.	
5	From an R6xx US or GUS, LOAD the CG, using the checkpoint saved on the previous release.	
6	CHECKPOINT the R6xx CG to the NET.	
7	Repeat steps 1 through 6 for remaining CGs/PLNMs.	

Table 10-36 Migrate Computer Gateways PLNMs

10.17 Migrate Network Gateways

Migrate NGs

Perform the following steps to migrate the Network Gateway(s) from the previous release to R6xx software.

CAUTION

The NG(s) on the other LCN(s) must be at a minimum software release of R5xx before migrating the NG(s) on this LCN to R6xx.

Step	Action	Done (√)
1	From the previous release US or GUS, SHUTDOWN the NG, and wait until it goes to QUALIF state.	
2	Once the NG goes to QUALIF state, POWER OFF the NG node. Leave off for at least 5 seconds.	
3	UPGRADE the hardware, as needed. (Refer to section 3.4 for minimum memory requirements.)	
4	POWER ON the NG node.	
5	From an R6xx US or GUS, LOAD the NG. Its status will go to OK.	
6	REPEAT steps 1 through 5 for each remaining NG.	

Table 10-37 Migrate NGs

10.18 Migrate Remaining USs

Migrate Remaining USs Use the following steps to upgrade the hardware, and migrate the remaining USs from the previous release to R6xx software.

Step	Action	Done (✓)
1	From the previous release US or GUS, SHUTDOWN all USs, one at a time, that have not yet been loaded with R6xx software. This step must be performed because the QLTs have changed, and the node hardware must be re-qualified with the new QLTs.	
2	Once the US goes to QUALIF state, go to the physical US node, POWER OFF the US (for at least 5 seconds), UPDATE hardware, as required, (see section 3.4 for minimum memory requirements), then POWER ON the node.	
3	For GUS: Use the instructions from Section 9.3 titled "Install TPS Network Software to PC Local Hard Drive".	
4	RELOAD each of these USs with R6xx software.	

Table 10-38 Migrate Remaining US

10.19 Load EBs and Enable History Collection

Restores using EBs If you have not restored Continuous History data, or if you have not restored History Group Point Definitions, and would like to restore from EBs (Exception Builds), do so now.

Enable History Collection

ATTENTION

Do not perform the following steps until the history groups have been restored.

Perform the following steps to enable History Collection

Step	Action	Done (✓)
1	SELECT HM node number from which History Collection will be enabled.	
2	ENABLE History Collection.	

Table 10-39 Enable History Collection

10.20 Migrating X-Layer Nodes

ATTENTION

Refer to the Software Change Notice for TPN and U^xS for migration instructions. The U^xS was withdrawn from sale November, 1997.

A^xM nodes cannot be migrated by following the procedures for migrating AM nodes. A^xM nodes have both LCN and UNIX code that must be upgraded as part of the migration.

10.20. Migrating X-Layer Nodes**Migrate A^XM**

You cannot directly migrate the previous release A^XM to R6xx software. Initially, the A^XM must be brought up to R6xx as an AM, then you upgrade it on the UNIX side.

After migrating the A^XM as an AM (see section 10.15), perform the 10 steps in the following table to complete the migration.

Step	Action	Done (√)
1	Following the <i>Application Module^X Customer Release Guide</i> and <i>Application Module^X System Administration</i> Install the R6xx A ^X M DAT tapes into the UNIX processor on this A ^X M.	
2	IF UPGRADING the A^XM processor to a K4LCN, LOG into the A ^X M through a terminal window as ROOT, then INPUT the following two UNIX commands: (If not upgrading processor board, skip to Step 3) rm /etc/opt/TDC_Open/common/personality.config Input the next command all in one string, with a SPACE after axmk: ln -s /opt/TDC_Open/common/newconfig/personality.config.axmk /etc/opt/TDC_Open/common/personality.config	
3	SHUTDOWN the previous release A ^X M node.	
4	QLTs have changed from the previous release to R6xx. To ensure the new QLTs run, POWER OFF the A ^X M.	
5	If you are upgrading from a K2LCN to a K4 LCN, REPLACE the K2LCN-8 with a K4LCN-8	
6	POWER UP the A ^X M.	
7	When the node goes to PWR ON state, Go to the AM NODE STATUS menu, SELECT this node, SELECT AUTONET LOAD target	
8	When the node status goes to OK, TAKE a demand checkpoint to NET.	
9	SHUTDOWN the AM personality running in this A ^X M.	
10	When the node goes to QUALIF, SELECT: Auto Load Net This will cause the A ^X M to boot from its local UNIX drive.	

Table 10-40 Migrate AxM

10.21 Upgrade PMs/APMs to APMs/HPMs

Upgrade the previous release PM or APM to R6xx APM or HPM

As a part of the R6xx migration, you may also be upgrading a PM to an APM or to an HPM, or you may be upgrading an APM to an HPM. You should follow upgrade kit installation instructions that come with each kit.

Important—Upgrading the hardware in a PM or APM to an HPM must be done AFTER the R6xx migration. You must migrate the PM or APM to R6xx, save the checkpoint, then translate to the HPM.

Upgrading PMs to APMs, PMs to HPMs, or APMs to HPMs must be done OFF-LINE. The upgrade involves

- saving the current PM configuration,
- shutting down and upgrading the hardware,
- reconfiguring the node, and
- reloading with the R6xx software and translated checkpoints.

The node hardware upgrade can be performed anytime after the R6xx system migration is complete. In that case, be sure the translations are performed using a current R6xx PM checkpoint (if converting from a PM), or using a current R6xx APM checkpoint (if converting from an APM).

CAUTION

Do not add any additional IOPs until after the R6xx migration is completed!

A table in section 3.12 references the applicable Upgrade Kit for each upgrade. Use the instructions provided with each Upgrade Kit to perform the upgrade.

10.22 Upgrade PMs/APMs/HPMs to EHPMs

Upgrade the PM or APM or HPM to EHPM

As a part of the R684 migration, you may also be upgrading a PM or an APM or an HPM to EHPM. You should follow the upgrade kit installation instructions that come with each kit.

Important—Upgrading the hardware in a PM or an APM or an HPM to EHPM must be performed AFTER the R684 migration. You must migrate the PM or APM to R6xx, save the checkpoint, and then translate to the HPM.

Upgrading PMs or APMs to EHPMs, or HPMs to EHPMs, must be performed OFF-LINE. The upgrade involves

- saving the current PM/APM/HPM configuration,
- shutting down and upgrading the hardware,
- reconfiguring the node,
- reloading the R684 software and the translated checkpoints,
- checkpoint reconfigured EHPM along with ENIM,
- reboot the ENIM to update the Authentication Group with the Device Index of the EHPMs.

CAUTION

If you do not reboot ENIMs, the EHPM status remains “OFFNET”.

The node hardware upgrade can be performed anytime after the R684 system migration is complete. In that case, be sure the translations are performed using a current R6xx PM checkpoint (if converting from a PM), or using a current R6xx APM checkpoint (if converting from an APM).

CAUTION

Do not add any additional IOPs until after the R684 migration is completed!

A table in section 3.12 references the applicable Upgrade Kit for each upgrade. Use the instructions provided with each Upgrade Kit to perform the upgrade.

11. Backing Out of R6xx

11.1 Regressing to R43x Software

Regressing back to R43x software

For migrations from R43x to R6xx, should you for any reason need to back out of an R6xx software migration, the following information must be understood.



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Number	To Back Out to an R43x Release, Follow This Guideline...
1	Using the media saved in section 7.3, System Backout Backup, you must re-initialize each HM on the system, and restore it with the data saved on the BACKOUT BACKUP cartridge(s).
2	Every node on the UCN and LCN will have to be SHUTDOWN and reloaded. THIS CANNOT BE PERFORMED ON-PROCESS.
3	All R6xx hardware upgrades that will not work with pre-R500 software must be removed, and the previously removed hardware reinstalled. (NOTE: Increased memory sizes can remain)

Table 11-1 Regressing Back to R43x Software

K4LCN to K2LCN regression

For migrations from R43x to R6xx, if you have installed a K4LCN processor board in any X-layer node (UXS or AXM), you must take specific steps to go back to the previous type processor board.

Steps to take an X-layer node (UXS or AXM) K4LCN back to a K2LCN processor board are detailed in section 10.4.

11.2 Regressing to R5xx Software

Regressing back
to R500 software

ATTENTION

Using GUS - you must uninstall before going back if regressing to R500 or R510, call TAC. **If you are regressing a GUS, call TAC. See Section 14 for the correct TAC telephone number for your area.**

For migrations from R5xx to R6xx, should you for any reason need to back out of an R6xx software migration, the following information must be understood.

Number	To Back Out to an R5xx Release, Follow This Guideline...
1	Using the media saved in section 8.3, System Backout Backup, you must re-initialize each HM on the system, and restore it with the data saved on the BACKOUT BACKUP cartridge(s).
2	Every node on the UCN and LCN will have to be SHUTDOWN and reloaded. This cannot be performed on-process
3	When reverting back to R5xx, you should use the SAVED R5xx checkpoint files, Area Database, Schematics, etc.

Table 11-2 Regressing Back to R5xx Software

11.3 Taking UXS K4LCN Back to K2LCN on X-Node

Taking UXS K4LCN back to a K2LCN board

If you are regressing your software from R6xx back to R43x, follow the procedure below to convert a UXS K4LCN back to a K2LCN processor board.

In this procedure, you will have reconfigured UNIX to use the K2LCN personality, in place of the K4LCN personality.



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Step	Action	Done (√)
1	From the UNIX prompt, INPUT the following two UNIX commands: <pre>rm /etc/opt/TDC_Open/common/personality.config</pre> Input the next command all in one string, with a <SPACE> after uxs : <pre>ln -s /opt/TDC_Open/common/newconfig/personality.config.uxs /etc/opt/TDC_Open/common/personality.config</pre>	
2	POWER OFF the UXS.	
3	If you are changing from a K4 LCN back to a K2LCN, REPLACE the K4LCN-8 with a K2LCN-8	
4	POWER UP the UXS.	
5	WAIT for UNIX to get up and running, then Perform a normal WSI load of this UXS Station.	

Table 11-3 Taking UXS K4LCN Back to a K2LCN Board

11.3. Taking UXS K4LCN Back to K2LCN on X-Node

Taking A^XM K4LCN back to a K2LCN board

If you are regressing your software from R6xx back to R43x, follow the procedure below to convert an A^XM K4LCN back to a K2LCN processor board.

In this procedure, you will have reconfigured UNIX to use the K2LCN personality, in place of the K4LCN personality.

Step	Action	Done (✓)
1	LOG into the A ^X M through a terminal window as ROOT.	
2	TYPE the following two commands: <pre>rm /etc/opt/TDC_Open/common/personality.config</pre> Input the next command all in one string, with a <SPACE> after axm: <pre>ln -s /opt/TDC_Open/common/newconfig/personality. config.axm /etc/opt/TDC_Open/common/personality. config</pre>	
3	LOG OUT of the A ^X M.	
4	POWER OFF the A ^X M node.	
5	REPLACE the K4LCN with a K2LCN.	
6	POWER UP the A ^X M. The system will now load the proper personality for the K2LCN into the A ^X M.	

Table 11-4 Taking A^XM K4LCN Back to a K2LCN Board

12. Migration from R6xx to R684

12.1 Migration Position Statement

Overview

Release 684 is a functional release that contains several significant function changes if you are migrating from an earlier R6xx release; however, its installation follows the simple "load-and-go" approach used for functional releases, with the exceptions noted under "Special Considerations." Its installation requires that the instructions in this section be followed. It can be installed with minimum disruption to on-line operations.

12.2 Migration Overview

Introduction

This section outlines the general procedures that must be used to install this software release in a running system. For assistance in interpreting these instructions or for help in the event of problems, contact the Honeywell Technical Assistance Center. Refer to "Support and other Contacts" listed in the beginning of this document. Note that the examples used may not apply exactly to your system, but can serve as a guide for planning.

General migration sequence

The following scenario is covered in the R6xx to R684 migration procedure:

These migration scenarios are described as follows.

Migration to R684 – The **TotalPlant®** Solution (TPS) system software is distributed on Zip disks if no GUS or APP nodes are used. Otherwise, the TPS system software is distributed on CD-ROM. No changes are required to the migration process.

12.2. Migration Overview

The general migration sequence is as follows:

- Save configuration and checkpoint data files
- Copy the new personality software for each module-type to the HM(s)
- Load other History Modules
- Load USs
- Load the Hiway Gateway and/or Network Interface Module and Process Manager/Advanced Process Manager/High Performance Process Manager/Logic Manager/Safety Manager
- Load any Application Modules and Computer Gateways
- Load any Network Gateways
- Load X-Layer nodes
- Update GUS personality on GUS/EST and reload
- Update APP personality on ESVT/ACE-T and reload

ATTENTION

R684 US/GUS/EST personalities must use the R684 version of &DSY, otherwise the US/GUS/EST may fail.

&DSY is copied to the HM in Section 12.4.

Expertise level to perform maintenance release migration

Minimum Knowledge—the person(s) performing this migration is expected to have the following skills.

- Proficient in manipulation through all standard **TotalPlant** Solution (TPS) System screens on a Universal Station, Universal Work Station, or Universal StationX, or GUS/EST
- Proficient in using Command Processor
- Proficient in using Data Entity Builder
- Know how to “back out” of a migration that cannot be completed

Knowledge For Updating GUS/EST and APP/ESVT/ACE-T Nodes—the user is expected to have the following background:

- Familiar with Windows 2000 or Windows 2003 Server or Windows XP.
- Basic GUS/EST navigation skills
- Proficient in Native Window operations
- Proficient in LCNP display operations
- Proficient in Emulated Disk operations

Migration steps in this manual are written to the level of a person with the above skills. Do not attempt to perform this migration if you do not have these skills, or have someone available who has these skills.

EC files used in this migration

The .EC command files used in this procedure are on &Z1 in directory &EC.

Redundant AM pairs

Note that for redundant AM pairs, both AMs must be shutdown and each reloaded if the release version of the AM is previous to 60.2.
Hint: Use SMCC to check the version of your AM.

12.3 Back Up Honeywell-Provided Media and System Data

Copy Honeywell-provided media (optional)

Zip Disk only—use the Command Processor Utilities to copy each of the Honeywell-provided media to a “just-initialized” removable medium.

Detailed instructions for this activity are provided in the *Command Processor Operation* Manual; section 5.5 under “Create Volume,” and section 5.4 under “Copy Volume.” (Be sure to use the “all directories” option, -A). NOTE: Do not use the FCOPY command. It does not validate the file directory and structure.

Save System Data

Before beginning the migration process, you must checkpoint all affected nodes to cartridges, so that if necessary, you can restore the system to the previous release.

Following the steps below, checkpoint your HG/PLCG/EPLCGs and all boxes, AMs (including APP, ESVT and ACE-T), AXMs, CGs, NIMs, and all PM/APM/LMM/SM/HPMs to removable media.

Alternately, if you are on R660 or later, you can also use the TPN Backup-Restore Utility to save the system data. Detailed instructions for this activity are provided in the *TPS/TPN Backup and Restore User's Guide*.

Step	Action	Done (✓)
1	Take demand checkpoints to \$Fd for all HG/PLCG/EPLCGs and boxes.	
2	Take demand checkpoints to \$Fd for all NIMs and other UCN devices.	
3	Take demand checkpoints to \$Fd for all AMs (including APP, ESVT and ACE-T).	
4	Take demand checkpoints to \$Fd for all CGs.	

Table 12-1 Checkpoint all Nodes

12.3. Back Up Honeywell-Provided Media and System Data

**Documentation Tool
Pre-Build Queries****CAUTION**

The Doc Tool pre-built queries found in the file “sysqry.dc” must be saved prior to migration to R684 in case it is necessary to back out to a previous release. If a backout is necessary then this file must be restored from the previously saved pre-migration version. Selecting a pre-built query, adding a new pre-built query, or deleting an existing pre-built query causes an automatic migration of the “sysqry.dc” file to the R630 or newer environment. Once migrated, this file becomes incompatible with the previous version of a Universal Personality.

**AM/CL and other
applications****CAUTION**

When the OUTIND parameter of an xPM REG or an AO point or an HG REG point is set to reverse, the output (OP) and output alarm parameters are reversed/swapped. Any AM/CL, custom graphics, Documentation tool queries, Find name queries or other TPN/TPS applications that use these parameters must be evaluated and possibly modified to accommodate the new behavior.

12.4 Copy R684: Software to History Module(s)

OPTIONAL - Set up system to enable other new options

NCF for the AM must be configured as NO for "further external directives," unless otherwise directed by Honeywell. Failure to do so will cause the loader to look for nonexistent files, and the AM will crash.

CAUTION

LOST FUNCTIONALITY—As each History Module is converted to R684, certain functionality is lost until the History Module transitions back to OK. In that period, you may be without functionality such as:

- **HM Resident Displays/Schematics**
- **Area changes**
- **Node loads**
- **History collection**
- **Automatic checkpointing**
- **Journals**

As part of your migration plan, you should either make a duplicate function for each of the above functions or migrate in a sequence that minimizes loss of these functions.

Prepare HM for software migration

Use the following steps to prepare the History Module for migration.
Be sure to print all of the activities on the printer

Step	Action	Done (✓)
1	If the HM has disk redundancy, ENSURE all drives are powered up. THIS IS REQUIRED IN ORDER TO HAVE DRIVE SYNCHRONIZATION.	
2	NOTE: The system HM is the HM with the &0np volume (np = node pair number). SELECT the system HM, and DISABLE History Collection	
3	From a US/GUS/EST, disable auto checkpointing on all nodes, as follows: From each HG, NIM, AM (including APP, ESVT and ACE -T), A ^X M, and CG node's status display, SELECT AUTO SAVE SELECT the DISABLE target	

Table 12-2 Prepare History Module for Software Migration R6xx to R684

12.4. Copy R684: Software to History Module(s)

CD-ROM users

Mount the appropriate CD in the CD-ROM drive to use the following emulated disks:

Use the GUS-TPN Software CD for

DISK_&Z1.lcn, DISK_&Z6.lcn

Use the GUS-TPS Application Module CD for

DISK_&Z3.lcn

Use the Utilities and Load Modules CD for

DISK_&Z14.lcn (MSCHEM.LO and CSCHEM.LO)

DISK_&Z12.lcn (YGOCX.LO, GUSALA.LO, GUSALAG.LO, EST.LO)

Perform the following steps when a disk change is required during the execution of the PERS?VOL .EC file:

Step	Action	Done (√)
1	DISMOUNT the currently mounted emulated disk and corresponding CD.	
2	MOUNT the required emulated disk and corresponding CD and continue as usual.	

Notes

The minimum configuration that can be migrated online and still retain a “view of the process” is:

- 3 – Universal Stations/GUSs with engineering keyboards (two must have dual cartridge drives/emulated disks)
- 1 – HM
- 1 – redundant module (HG or NIM)

In the case of redundant HM drives, both drives are shut down. Redundancy is not in effect again until the drives are synchronized.

At this point, alternate access to critical schematics on removable media is required.

Migrate System HM Perform the following steps to migrate the System HM.

Step	Action	Done (✓)
1	From a US/GUS/EST, CONFIRM that automatic checkpoints are disabled for data owners who are checkpointing to this HM and that no checkpoints are currently in progress.	
2	IDENTIFY the System HM. (It is the HM with the &Onp volume where np = node pair number)	
3	From the US/GUS/EST, SHUTDOWN the HM, and WAIT for it to go to QUALIF before proceeding.	
4	From the US/GUS/EST and using the &Z1 disk provided, Perform a MANUAL LOAD of this HM using the INIT PROGRAM, with the R684 HM Off-line personality (HMI), and DEFAULT for the data source.	
5	WAIT for the HM status to go to "HMOF OK . "	

Table 12-3 Migrate HM R6xx to R684

12.4. Copy R684: Software to History Module(s)**Load System HM
with R684 HM
personalities**

Using the &Z1 disk (DISK_&Z1.1cn), load the System HM with the R684 personality(s), as described in the following steps.

Step	Action	Done (✓)
1	ENSURE that the &Z1 disk is mounted in \$Fs (source drive).	
2	RUN the LOC_VOLZ.EC EXAMPLE SYNTAX: EC \$Fs>&EC>LOC_VOLZ.EC \$Fs nn np Where s = source drive number nn = HM node number np = HM node pair number	
3	MOUNT the &Z1 (disk in \$Fs (source drive)).	
4	LOAD the system configuration files by running SYS_VOLZ.EC EXAMPLE SYNTAX: EC \$Fs>&EC>SYS_VOLZ.EC \$Fs nn where s = source drive number nn = HM node number Respond N0 to the question "Copy volume &ASY?" Respond N0 to "Do you wish to skip the &DSY or any of the optional directories?" (The EC file will copy all files from the &DSY and the four GDF volumes.)	
5	AUTOBOOT the HM.	

Table 12-4 Load HMs with R684 HM Personalities**ATTENTION**

HMs with redundant drives go to SEVERE status because they are out of synchronization.

HMs with nonredundant drives goes to OK status.

**Migrate remaining
HMs**

Skip the following procedure if you have only one HM (the System HM), and continue with “Enable History Collection” on the next page.

Perform steps 1 through 11 in the following table to migrate the remaining HM(s) on your system.

Step	Action	Done (✓)
1	On the HM Node Status display, SELECT the node number of the HM you want to load.	
2	From any US/GUS/EST, SHUTDOWN this HM, and WAIT for it to go to QUALIF before proceeding.	
3	From the US/GUS/EST, and using the &Z1 disk provided, Perform a MANUAL LOAD of this HM using the INIT program, with the R684 HM Off-line personality (HMI), as follows: SELECT HM (so it is backlit) SELECT NODE STATUS SELECT LOAD DUMP SELECT MANUAL LOAD SELECT INIT PROGRAM SELECT ALTERNATE SOURCE using &Z1 for the personality, and the data source will be from the DEFAULT SOURCE (NET).	
4	WAIT for the HM status to go to "HMOF OK"	
5	RUN the LOC_VOLZ.EC EXAMPLE SYNTAX: EC \$Fs>&EC>LOC_VOLZ.EC \$Fs nn np where s = source drive number nn = HM node number np = HM node pair number	

Table 12-5 Migrate Remaining HMs R6xx to R684

12.4. Copy R684: Software to History Module(s)

**Migrate remaining
HMs, continued**

Load the HM with the R684 Personality, as follows

Step	Action	Done (✓)
6	From the History Modules Node Status Display, SELECT LOAD/DUMP	
7	SELECT MANUAL LOAD	
8	SELECT OPERATOR PROGRAM	
9	SELECT DEFAULT SOURCE for both the program source and data source.	
10	PRESS <ENTER>	
11	Wait for HM to go to HMON OK (or HMON SEVERE if redundant drives exist)	
12	IF this HM contains the &DS2 directory, INVOKE the DS2_BKPZ.EC from &Z1 EXAMPLE: EC \$Fs>&EC>DS2_BKPZ.EC \$Fs where S = source drive number	
13	REPEAT Steps 1 through 12 for each remaining HM.	

Table 12-5 Migrate Remaining HMs R6xx to R684 (continued)**Enable History
Collection**From a US/GUS/EST, perform the steps below to enable history
collection

Step	Action	Done (✓)
1	SELECT the History Collection HM's node number	
2	SELECT HIST COLLECT	
3	SELECT ENABLE COLLECT Current State becomes ENABLE.	

Table 12-6 Enable History Collection R684

**Copy Personality
Image files**

Using the steps below, copy the Personality Image files to one or more personality volumes. This step also allows you to load the DIA1, TLK1 and TLK2 Tool Kit Displays and their optional software, if desired.

Step	Action	Done (√)
1	If this HM has Personality Image File(s), Copy the personality files configured on the NET by using the PRS2VOLZ.EC command file for a 68020 personality, and/or using the PRS4VOLZ.EC command file for a 68040 personality. EXAMPLE SYNTAX: EC \$Fs>&EC>PRS2VOLZ.EC \$Fs EC \$Fs>&EC>PRS4VOLZ.EC \$Fs where S = source drive number	

Table 12-7 Copy Personality Image Files R684

12.4. Copy R684: Software to History Module(s)

CD ROM users:**Disk change may be required**

Perform the following steps when a disk change is required during the execution of the PRS?VOLZ .EC file:

Step	Action	Done (✓)
1	DISMOUNT the currently mounted emulated disk and corresponding CD.	
2	MOUNT the required emulated disk and corresponding CD and continue as usual.	

Copy Custom External Load Modules to HM (OPTIONAL - only for GUS/EST and APP/ESVT/ACE-T nodes)

Use this procedure only if you have GUS/EST or APP/ESVT/ACE-T nodes on your network. Otherwise, skip this procedure and proceed with the last procedure in section 12.4, “Copy R684 Software to History Module(s)” in this document.

Step	Action	Done (✓)
1	Install Utilites and Load Modules CD ROM. Select and mount disk_&Z14.lcn into an emulated drive.	
2	Enter the following command EC \$Fs>&EC>MSCHEMZ .EC \$Fs Note: Follow the procedure to copy the Load Modules required for TPS Nodes or EST/ESVT. This procedure allows you to copy the load modules from disk_&Z12 also.	

Table 12-8 Copy External Load Modules to HM (Optional for GUS/EST)

Copy &EC Directory Files to HM (OPTIONAL)

Some customers choose to copy the &EC files found on the &Z1 zip disk to an HM-resident directory for convenience. If you wish to do this, now is a good time! Here are typical commands to do this:

```
UNPT NET>&EC>*.*
CP $Fn>&EC>*.* NET>&EC>= -D
```

**Synchronize
redundant HM drives**

If you have redundant disk drives on the HM, perform the steps below to synchronize the drives

Step	Action	Done (√)
1	CHECK to ensure the secondary drive(s) are powered up at this time.	
2	From the Command Processor, ENTER SYN PN:nn where nn = HM node number NOTE: It may take up to several hours for the HMs to synchronize and the status to change from SEVERE to OK.	

Table 12-9 Synchronize Redundant HM Drives**Exit Mount/Dismount
Emulated Disks
Dialog Page**

Step	Action	Done (√)
1	On the GUS/EST From the Mount/Dismount Emulated Disk dialog page, REMOVE existing CD from drive.	
2	Dismount all emulated disks.	
3	Select OK	
4	Select OK when asked "Are you sure you want to apply changes and exit".	

Table 12-10 Dismount Emulated Disks

The only operations that remain to be done are to load all the rest of the TPS Network and enable, as desired, auto checkpoints, history collection, and journals.

12.5 Migrate USs/GUS/EST

Enable Alarming for Units

Before continuing with the migration, we recommend that alarming is enabled for all UNITS, both Console-wide and System-wide.

Step	Action
1	Callup the CONSOLE STATUS display.
2	Select UNIT DISPLAY.
3	Ensure each unit's CONSOLE and SYSTEM alarm state are set to ENABLE.
4	If the unit is NOT ENABLE for CONSOLE ALARM STATE, select the UNIT, select CONSOLE ALARM STATE, and select ENABLE.
5	If the unit is NOT ENABLE for SYSTEM ALARM STATE, select the UNIT, select SYSTEM ALARM STATE, and select ENABLE.
6	Repeat for all consoles.

When the migration is complete, ensure that alarms are functioning normally. If desired, you may then INHIBIT or DISABLE alarms for units on a Console-wide or System-wide basis.

Table 12-11 Enable Alarming for Units

Migrate USs

Use the following steps to migrate the USs from R6xx to R684 software.

Step	Action	Done (√)
1	From any US/GUS/EST, SHUTDOWN all USs/GUS/EST stations, one at a time, and load, using the R684 software.	
2	RELOAD each of the USs with R684 software. All nodes continue to be fully operational during this load-and-go procedure. There is no loss of view at any node or event except during the node load time.	
3	For each GUS/EST station, install the R684 GUS-TPN (GUS Personality) Software on the local hard drive. Note : Refer to Table 12-13 for installing GUS Personality on the local hard drive of a GUS Node and Table 12-14 for installing EST Personality on the local hard drive of a EST Node	
4	From the GUS/EST station (where R684 personality is installed) open the LCNP status display and select RESET LCNP . NOTE: Ensure Enable LCNP Restart is selected in the Configuration Utility for a successful GUS/EST station load. However, if it is not selected, you can load the GUS using the HM as the default data source.	
5	Wait until the loading is complete.	

Table 12-12 Migrate USs

Migrate GUS

With TPS 421 the GUS package installer has changed. You must remove the existing GUS package to install the new package. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

ATTENTION

Uninstall the existing personality, before migrating GUS.

Use the following steps to migrate the GUS nodes from R6xx to R684 software.

Step	Action
1	Close all the GUS displays and Native Window.
2	Using the Windows Task Bar, Select Settings→Control Panel→Add/Remove Programs..
3	From the Add/Remove Programs window, Select TPS – UNPW and click Change/Remove button. Click Yes on the Confirm File Deletion window. Verify the uninstallation window.
4	When the uninstallation window appears, select OK . Select OK in the Add/Remove Programs window.
5	To install GUS personality, insert the GUS-TPN Software CD .
5	Use Windows Explorer to navigate to the following directory on the CD: \GUS Double-click the tps - unpw.msi file. Follow the instructions on the installation wizard for installing the package.
7	Using the windows task bar: START → PROGRAMS → HONEYWELL TPS → CONFIGURATION UTILITY → CONFIGURE → BOARD 0 (or the name configured for Board 0). Select Enable LCNP Auto Restart and click OK .

Table 12-13 Migrate GUS

Step	Action
8	Call up the LCNP Status by performing the following: Using the windows task bar: START → PROGRAMS → HONEYWELL TPS → LCNP STATUS Reset the LCNP by performing the following: On LCNP status window, Select Details. Select Reset LCNP. Are you sure you want to RESET the LCNP? Message is displayed. Select Yes. Wait for the status to change to State OK and SubState OK.
9	Log off and log on to the system again.
10	Call up the Native Window and Select Console Status. Ensure the node indicates GUS. Note: Ensure the TPN Server is in the Running or Idle (if no clients are connected) state. If the TPN Server does not enter the state, the node is to be rebooted. TIP: You can ascertain the status of the TPN Server through the System Management Display.

Table 12-13 Migrate GUS (continued)

12.5. Migrate USs/GUS/EST

Migrate ESTs

With TPS 421 the GUS package installer has changed. You must remove the existing GUS package to install the new package. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

Use the following steps to migrate the ESTs from R6xx to R684 software.

Step	Action
1	Close all the Experion Windows, GUS displays and Native Window
2	Using the Windows Task Bar, Select Settings→Control Panel→Add/Remove Programs .
3	From the Add/Remove Programs Window, select TPS – UNPW and click Change/Remove button. Click Yes on the Confirm File Deletion window. Verify the uninstallation window.
4	When the uninstallation window appears, select OK . Select OK from the Add/Remove Programs window.
5	To install GUS personality, insert the GUS-TPN Software CD .
6	Use Windows Explorer to navigate to the following directory on the CD: \GUS Double-click the tps - unpw.msi file. Follow the instructions on the installation wizard for installing the package.
7	Using the windows task bar: START → PROGRAMS → HONEYWELL TPS → CONFIGURATION UTILITY → CONFIGURE → BOARD 0 (or the name configured for Board 0). Select Enable LCNP Auto Restart and click OK .

Step	Action
8	Call up the LCNP Status by performing the following: Using the windows task bar: START → PROGRAMS → HONEYWELL TPS → LCNP STATUS Reset the LCNP by performing the following: On LCNP status window, select Details. Select Reset LCNP. Are you sure you want to RESET the LCNP? Message is displayed. Select Yes. Wait for the status to change to State OK and SubState OK.
9	Log off and log on to the system.
10	Call up the Native Window and Select Console Status. Ensure the node indicates EST. Note: Ensure the TPN Server is in the Running or Idle (if no clients are connected) state. If the TPN Server does not enter the state, the node is to be rebooted. TIP: You can ascertain the status of the TPN Server through the System Management Display.

Table 12-14 Migrate ESTs

12.6 Migrate Redundant HGs

**Migrate HG (or
PLCG/EPLCGs)**

If there are no HGs, PLCGs, or EPLCGs, skip to section 12.9. If you do not have redundant HGs, PLCGs, or EPLCGs, skip to the next section (section 12.7 in this document).

**Load HGs and/or
PLCGs/EPLCGs**

Steps 1 through 8 will be used to load migration software into an HG pair or a PLCG/EPLCG pair.

Step	Action	Done (√)
1	From any US/GUS/EST, at the System Status Display, SHUTDOWN the BACKUP HG. (NOTE: At this point, the primary HG will go to WARNING, due to loss of redundancy.)	
2	SELECT LOAD SELECT and PRESS <ENTER>	
3	WAIT until the "Completed Loading" message is displayed.	
4	After the HG is in the BACKUP state and has been running for at least 2 minutes, NOTE: The time delay allows for verification of compatibility across releases for redundant nodes. SHUTDOWN the primary HG.	
5	WAIT for the backup to go to WARNING, then SELECT LOAD SELECT and PRESS <ENTER>	
6	WAIT until the "Completed Loading" message is displayed.	
7	PERFORM a Demand Checkpoint Save to NET <u>TWICE</u> for ALL BOXES.	
8	REPEAT Steps 1 through 7 for each remaining REDUNDANT HG, PLCG, or EPLCG.	

Table 12-15 Load HGs and/or PLCGs/EPLCGs R6xx to R684

12.7 Migrate Nonredundant HGs

Load Nonredundant HGs and/or PLCGs/EPLCGs

Perform this procedure for nonredundant HGs/PLCGs/EPLCGs.
 Steps 1 through 6 will be used to load migration software into a nonredundant HG or a nonredundant PLCG/EPLCG.

Step	Action	Done (✓)
1	From any US/GUS/EST, SHUTDOWN the HG.	
2	SELECT LOAD SELECT and PRESS <ENTER>	
3	WAIT until the "Completed Loading" message is displayed.	
4	For PLCGs/EPLCGs only: Perform hiway command from Hiway Status Displays. NOTE: All DHP boxes will be in ADDR ERR state. Perform INIT ADDR command to the hiway. The status of the boxes will go to RESET. Load data to all boxes from NET. Status goes to OK.	
5	SET UP the hiway and box control states as you desire.	
6	PERFORM a Demand Checkpoint Save to NET <u>TWICE</u> for ALL BOXES.	
7	REPEAT Steps 1 through 7 for each remaining NONREDUNDANT HG, PLCG, or EPLCG.	

Table 12-16 Load Non-Redundant HGs and/or PLCGs/EPLCGs

12.8 Enable Auto Checkpointing - HGs

Enable Auto Checkpointing for HGs

OPTIONAL STEPS—Perform the following Auto Checkpointing steps **only if you want to enable Auto Checkpointing on the HGs**. Otherwise, continue with section 12.9.

Step	Action	Done (✓)
1	FOR each HG: At any US/GUS/EST, from the PROCESS NETWORKS NODE STATUS menu, SELECT the HG SELECT AUTO SAVE SELECT ENABLE SELECT NTWK STATUS SELECT HIWAY CMND SELECT CKPT ENABLE SELECT EXECUTE COMMAND	
2	FOR HG Boxes: SELECT BOX STATUS ENTER lowest box number, and SELECT CKPT ENABLE SELECT EXECUTE COMMAND	
3	For additional boxes on the hiway, Use the <DISP FWD> key to get to the next box status to enable checkpointing. SELECT CKPT ENABLE SELECT EXECUTE COMMAND	
4	Repeat steps 1 through 3 of this table for each additional hiway and box.	

Table 12-17 Enable Auto-checkpointing HGs R6xx to R684

12.9 Migrate NIMs

ATTENTION

If there are no NIMs, skip to section 12.16.

ATTENTION

Throughout the sections 12.9 ~ 12.11, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms **EHPM**, **ENIM**, and **EUCN**, respectively.

Important!

IMPORTANT NOTE: The UCN servicing these NIMs may include multiple LCNs. If your system has multiple LCNs and any are running a release before R6xx that are serviced by this UCN, you must contact the Honeywell Technical Assistance Center for special procedures. **You cannot follow the procedures in this section.**

CAUTION

NIM/UCN users should review all cautions and notes pertaining to NIM/UCN devices as found in the R684 Software Change Notice.

If External Shutdown timers are in use in conjunction with the NIM, be sure to check the timer function to ensure the timers are set at a minimum of 3 seconds. Performing a swap of NIMs requires that these timers be set to 3 seconds or longer.

12.9. Migrate NIMs

Migrate Redundant NIMs Perform the steps below to migrate your Redundant NIMs.

Step	Action	Done (√)
1	From any US/GUS/EST, SELECT one of a NIM pair.	
2	SHUTDOWN the Backup NIM.	
3	WAIT for the node to go to QUALIF, then SELECT LOAD SELECT, and PRESS <ENTER>	
4	WAIT until the "Completed Loading" message is displayed.	
5	After the NIM is in BACKUP state and has been running for at least two minutes, from any US/GUS/EST, NOTE: The time delay allows for verification of compatibility across releases for redundant nodes. SHUTDOWN the primary NIM.	
6	SELECT LOAD SELECT, and PRESS <ENTER>	
7	WAIT until the "Completed Loading" message is displayed.	
8	VERIFY its status goes to OK BACKUP.	
9	REPEAT Steps 1 through 8 for any remaining REDUNDANT NIMs.	

Table 12-18 Migrate Redundant NIMs R6xx to R684

**Migrate
Nonredundant NIMs**

Perform the steps below to migrate your Nonredundant NIMs

Step	Action	Done (√)
1	From any US/GUS/EST, SELECT a NIM.	
2	SHUTDOWN the node.	
3	WAIT for the node to go to QUALIF, then SELECT LOAD SELECT, and PRESS <ENTER>	
4	WAIT until the "Completed Loading" message is displayed.	
5	VERIFY its status goes to OK.	
6	REPEAT Steps 1 through 5 for any remaining NONREDUNDANT NIMs.	

Table 12-19 Migrate Nonredundant NIMs R6xx to R684

12.10 Migrate Redundant UCN Devices

**Confirm proper
procedure**

If your UCN Devices are NOT redundant, skip forward to section 12.11.

**Migrate Redundant
PM, APM, and HPM
modules**

Within this section, the term “device” is used to refer to any PM, APM or HPM.

NIM/UCN users should review all cautions and notes pertaining to NIM/UCN devices as found in the R684 Software Change Notice.

CAUTION

The following precaution must be followed when migrating redundant APMs and HPMs that use SI Array points which reference external data. Failure to observe this procedure could result in a loss of data from one or more SI Array points.

If you have any questions regarding this migration precaution, please contact your local Honeywell National Response Center (NRC), Technical Assistance Center (TAC), or Customer Response Center (CRC).

If your redundant APM / HPM are currently running below the minimum level shown, you are at risk and must follow this special precaution.

	Minimum R6xx Version
APM-cntrl (umf)	64.2
APM-comm (umc)	63.0
HPM-cntrl (hmf)	63.2
HPM-comm (hpmc)	63.0

Prepare a list of all APM/HPM SI Array points whose “External Data Option” is NOT equal to “NONE”. Use the Data Entity Builder “Print Entities” function to create “exception build” (.EB) files for these points.

After performing an on-line upgrade of the APM/HPM, we recommend that your SI Array points be reloaded from the .EB files you created in

the previous step. Otherwise, it is possible that one or more SI Array points will be displaying invalid data (i.e., NaN). Reloading the SI Array points on the new software, and taking a checkpoint, will correct the problem for future migrations.

Your particular application may require additional precautions to prevent possible plant upset.

This problem will not occur if you perform an off-line upgrade of the APM/HPM. (i.e., shut down the redundant pair and reload with new software.) However, this causes a loss of view and control for the affected points.

NOTE: If External Shutdown timers are in use in conjunction with the devices, check the timer function to ensure the timers are set at a minimum of 3 seconds. Performing a swap of the UCN devices requires that these timers be set to 3 seconds or longer. Perform the following steps to migrate EACH of your REDUNDANT devices.

Step	Action	Done (✓)
1	From any US/GUS/EST, on the UCN Status Display, SELECT the BACKUP device, and perform a SHUTDOWN. The device status will go to ALIVE on the UCN Status Display.	
2	At any US/GUS/EST, LOAD the secondary device manually, using NET as source for the PGM SOURCE and the DATA SOURCE. The UCN status indicates LOADING until the load is complete. The status for this device will then go to BACKUP. The color of the letters goes to cyan when load is completed. WAIT 5 minutes <u>after</u> the color has gone to cyan. (This allows time for background diagnostics to run.) If the Node Type ID letters remain in yellow, <u>do not continue</u>. Call TAC at 1-800-822-7673 as soon as possible.	

Table 12-20 Migrate Redundant UCN Devices R6xx to R684

12.10. Migrate Redundant UCN Devices**Migrate Redundant PM, APM, and HPM modules, continued**

Step	Action	Done (✓/)
3	NOTE: If you are using GUS Graphics during the SWAP PRIMARY procedure below, you may get errors. This is normal and if it occurs, recall the display and everything will be OK. At any US/GUS/EST, SELECT the primary device and perform the SWAP PRIMARY procedure. The device on R684 becomes the primary, and the status of the device on the old release goes to BACKUP. WAIT 7 minutes for the swap to complete. (This allows time for background diagnostics to run.)	
4	Select the Backup device and perform a Shutdown. The device status will go to ALIVE on the UCN Status display.	
5	At any US/GUS/EST, LOAD the secondary device, using NET as the source for the PGM SOURCE and the DATA SOURCE. The UCN status indicates LOADING until the load is complete. The status for this device will then go to BACKUP. If the Node Type ID letters remain in yellow, <u>do not continue</u>. Call TAC at 1-800-822-7673 as soon as possible.	
6	At any US/GUS/EST, VERIFY that the R684 device is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc.	
7	CHECKPOINT the R684 device to the NET <u>two times</u> .	
8	REPEAT Steps 1-7 for all redundant devices being migrated on each UCN.	

Table 12-20 Migrate Redundant UCN Devices R6xx to R684 (continued)

ATTENTION
Once all devices have been transitioned to R684, use the PERFMENU, UCNVERSN schematic to verify that those nodes are running R684 software.

12.11 Migrate Nonredundant UCN Devices

**Confirm proper
procedure**

If your UCN Devices are redundant, go back to section 12.10 Migrate Redundant UCN Devices in this document, or if you finished with that section, continue.

**Migrate
Nonredundant UCN
devices to R684**

Within this section, the term “device” is used to refer to any PM, APM or HPM.

NIM/UCN users should review all cautions and notes pertaining to NIM/UCN devices as found in the R684 Software Change Notice.

Since this procedure is off-line, as each device is shutdown, you lose the view to points, all local control, and all peer-to-peer communication stops. Putting the device in IDLE and restoring a checkpoint places all control points in IDLE and causes outputs for component points to go to the FAILSAFE position.

Perform the following steps to migrate nonredundant devices to R683.

Step	Action	Done (✓)
1	From any US/GUS, on the UCN Status Display SELECT the device to be migrated.	
2	SET the selected device to IDLE state, and SHUTDOWN the node. Its status will go to ALIVE on the UCN Status Display	

Table 12-21 Migrate Non-Redundant UCN Devices R6xx to R684

12.11. Migrate Nonredundant UCN Devices

**Migrate
Nonredundant UCN
devices to R684,**
continued

Step	Action	Done (√)
3	LOAD the selected device, using NET as source for the PGM SOURCE and the DATA SOURCE. The UCN status indicates LOADING until the load is complete. The status for this device will then go to IDLE.	
4	STARTUP the selected device using WARM START (or whatever best suits for your system). Its status will go to OK.	
5	At any US/GUS/EST, VERIFY the R683 device is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing modes, etc.	
6	CHECKPOINT the R684 device to NET <u>two times</u> .	
7	REPEAT Steps 1-6 for all remaining non-redundant devices being migrated.	

Table 12-21 Migrate Non-Redundant UCN Devices R6xx to R684 (continued)**ATTENTION**

Once all the devices have been transitioned to R684, use the PERFMENU, UCNVERSN schematic to verify those nodes are running R684 software.

12.12 Migrate Redundant LM, FSC-SM, Triconex-SMM

Preliminary check	Follow this procedure if you have redundant Logic Managers, FSC Safety Managers, or Triconex SMMs on your system. If not, skip this section and continue with section 12.13, Migrate Nonredundant LM, FSC-SM, Triconex-SMM.
Common instructions	In this section the term “LM” is used for Logic Manager, FSC Safety Manager, and Triconex SMM. The instructions are common for all three device types.
Check LM software version	Use PERFMENU, UCNVERSN schematics to determine the software version of your running LM. Compare this version with the new LM version shown on your R684 Software Change Notice. Continue with the migration procedure in this section <u>only</u> if a newer version of LM software is now available. If a newer version is not available, continue with section 12.14, Additional Safety Manager Instructions.
Triconex (SMM)	If you are currently using Triconex SMM software prior to version 65.0 please be aware of the following anomaly: During failover or swap primary of a Triconex SMM, a peer data request could return incorrect digital state data. This could cause applications to take action on false data. This has been corrected in Triconex Safety Manager version 65.0. Before migrating from a pre-65.0 version of Triconex Safety Manager ensure that your applications cannot or will not take undesired action should an incorrect digital state value be read from the Triconex Safety Manager.
Check timers	If any external shut down timers are in use in conjunction with the LM, check the timer function and make appropriate adjustments. The time will vary as a function of the amount of ladder logic in the LM. As you increase your ladder logic, you should adjust the timer function as necessary. Performing a swap of LM primaries requires the external timers to be set at 6.5 seconds. Set external timers to a time of 6.5 seconds or greater to eliminate problems in this area.

12.12. Migrate Redundant LM, FSC-SM, Triconex-SMM

Check preferred controller

The On-line migration requires that the secondary LM be migrated first while the primary LM continues to control the process. Ensure the primary LM is the preferred controller before proceeding.

Migrate redundant LM to R684

Redundant LMs can be migrated to R684 while on process. All LMs on the physical UCN must be migrated to R684 software. No translators are required for this migration. Therefore, the migration will be performed using R6xx checkpoints.

Step	Action	Done (✓)
1	Determine if the lead (PRIMARY) LMM is the "PREF" as follows: At the UCN Status Display, SELECT the "ODD" (OK & PRIMARY) LMM node. SELECT DETAIL STATUS target. VERIFY P/S is PRIMARY, and the FILE POS is PREF. IF it is PREF and "SYNCHST" is SYNCHED, continue with the remainder of this step. If you find that the "NON - PREF" is the lead (PRIMARY), go directly to Step 2. SELECT the LMM. SELECT RUN STATES target. SELECT SWAP PRIMARY. NOTE: After the swap, the non-preferred LM should assume the lead, and will be performing the ladder logic control. WAIT for 7 minutes.	
2	Shutdown the BACKUP LM (this is the preferred LM), as follows: SELECT the EVEN numbered LM (NOTE: you may have to select it twice) SELECT RUN STATES target SELECT SHUTDOWN target SELECT ENTER target The status of the BACKUP LM goes to OFFNET then to the ALIVE state.	

Table 12-22 Migrate Redundant LM to R684

12.12. Migrate Redundant LM, FSC-SM, Triconex-SMM

Step	Action	Done (√)
3	At a US/GUS/EST, reload with new personality (to the BACKUP LM). SELECT LOAD/SAVE/RESTORE. SELECT PROGRAM LOAD. SELECT ENTER target. SELECT DEFAULT SOURCE for the LMM program source. SELECT DEFAULT SOURCE for the data source. When the load completes, the LMM goes to BACKUP. WAIT 5 minutes to allow synchronization to complete. VERIFY the detail status of the backup LMM shows "PGM MISMATCH." This status is okay.	
4	Swap primaries (preferred LM now becomes the lead again). SELECT RUN STATES target SELECT SWAP PRIMARY SELECT ENTER target Once swap is complete, WAIT for another 7 minutes.	
5	Shutdown BACKUP LM (this is the non-preferred LM) as follows: SELECT the EVEN numbered LM. (NOTE: you may have to select it twice.) SELECT RUN STATES target. SELECT SHUTDOWN target. SELECT ENTER target. The status of the BACKUP LM goes to OFFNET then to ALIVE.	
6	RELOAD with new personality (to the BACKUP LM) SELECT LOAD/SAVE/RESTORE. SELECT PROGRAM LOAD. SELECT ENTER target. SELECT DEFAULT SOURCE for the LMM program source. SELECT DEFAULT SOURCE for the data source. When the load completes, the LMM goes to BACKUP. WAIT for 7 minutes to allow synchronization to complete.	

Table 12-22 Migrate Redundant LM to R684 (continued)

12.12. Migrate Redundant LM, FSC-SM, Triconex-SMM

Step	Action	Done (√)
7	At a US/GUS/EST, VERIFY the R684 LM is capable of full operation by • calling up Group Displays, • calling up schematics, • changing the modes of points, etc. If you detect any errors at this time, call Honeywell TAC. Also, if the LM shows a status of OK/NOSYNC (which means there is no synchronization between the primary and secondary LMs), call Honeywell TAC. The LMM is now on R684 with its database synchronized to its R683 partner.	
8	CHECKPOINT the R684 LM to the NET, <u>twice</u> .	
9	SAVE the Ladder Logic to the HM and to removable media for EACH LM. Note the name of the Ladder Logic Save file.	
10	REPEAT steps 1-9 for all redundant LMs being migrated.	

Table 12-22 Migrate Redundant LM to R684 (continued)**CAUTION**

Once all the LM nodes have been transitioned to R684, use the PERFMENU, UCNVERSN schematics to verify that those nodes are running R684 software.

12.13 Migrate Nonredundant LM, FSC-SM, Triconex-SMM

Preliminary check	Follow this procedure if you have non-redundant Logic Managers, FSC Safety Managers, or Triconex SMMs on your system. If not, skip this section and continue with section 12.14, Additional Safety Manager Instructions.
Common instructions	In this section the term “LM” is used for Logic Manager, FSC Safety Manager, and Triconex SMM. The instructions are common for all three device types.
Check LM software version	Use PERFMENU, UCNVERSN schematics to determine the software version of your running LM. Compare this version with the new LM version shown on your R684 Software Change Notice. Continue with the migration procedure in this section only if a newer version of LM software is now available. If a newer version is not available, continue with section 12.14, Additional Safety Manager Instructions.
Migrate Nonredundant LM to R684	All LMs on the physical UCN must be migrated to R684 software. NOTE: Nonredundant LMs can <u>only</u> be migrated off-process.

Step	Action	Done (✓)
1	At a US/GUS/EST, from the UCN Status Display, idle and shut down the LMM, as follows. At the UCN STATUS display, SELECT the LM to be migrated. SELECT RUN STATES. SELECT IDLE. SELECT ENTER (wait for IDLE to be displayed). SELECT SHUTDOWN. SELECT ENTER (wait for the status to go to ALIVE)	

Table 12-23 Migrate Nonredundant LM to R684

**Migrate
Nonredundant
LM to R684,
continued**

Step	Action	Done (✓)
2	At a US or GUS, load the LM. At the UCN STATUS display, SELECT LOAD/SAVE RESTORE. SELECT PROGRAM LOAD. SELECT ENTER. SELECT NET for the PGM source. SELECT NET for the DATA source. If the load status goes to OK, go to Step 4. If the load status goes to PF_IDLE, which indicates that the ladder logic file is not installed, continue on with Step 3. You must know the file name and location of the ladder logic file. Note that the ladder logic file must have a ".PO" extension.	
3	Reload the Ladder logic file from the NET as follows: Select LOAD/SAVE RESTORE. Select LADDER LOAD. Select ENTER. Enter the name of the ladder logic file (up to 8 characters) in the FILE NAME data input port. Do not enter the file extension. Select ENTER. Select the correct SOURCE location. Select EXECUTE. When the load is complete, go to the LM processor card file and change the keyswitch to RUN if required. The status of the LM goes to IDLE. Continue with Step 4.	
4	Start up the LM as follows: SELECT RUN STATES. SELECT STARTUP. SELECT ENTER. The status of the LM goes to OK.	

Table 12-23 Migrate Nonredundant LM to R684 (continued)

**Migrate
Nonredundant
LM to R684,**
continued

Step	Action	Done (√)
5	At a US or GUS, once the LM status is OK, VERIFY the R684 LM is capable of full operation, by • calling up Group Displays, • calling up schematics, • changing the modes of points, etc. If you detect any errors at this time, call Honeywell TAC.	
6	CHECKPOINT the R684 LMM to NET, <u>twice</u> .	
7	SAVE the Ladder Logic to the HM and to removable media for the LM. Note the name of the Ladder Logic Save file.	
8	REPEAT steps 1-7 for all nonredundant LMs being migrated.	

Table 12-23 Migrate Nonredundant LM to R684 (continued)

CAUTION

Once all the LM nodes have been transitioned to R684, use the PERFMENU, UCNVERSN schematics to verify those nodes are running R684 software.

12.14 Additional Safety Manager Instructions

**Additional SM
upgrades if
needed**

The instructions in Sections 12.12 and 12.13 updated the “TPN portion” of the FSC Safety Manager or Triconex-SMM (i.e., the part that connects directly to the UCN and runs TPN software.) If additional software/firmware/hardware upgrades are planned, please refer to appropriate instructions provided by the vendor.

For FSC Safety Manager, refer to the Honeywell HSMS upgrade document and the FSC Software Manual.

12.15 Enable Cable Swapping/Auto CheckPointing - NIMs/UCN Nodes

Ensure cable swapping is enabled

Ensure that UCN cable swapping is enabled for all UCN nodes.

Enable Auto Checkpointing for NIMs and UCN devices

OPTIONAL STEPS—Perform the following Auto Checkpointing steps only if you want to enable Auto Checkpointing on the NIMs and UCN devices. Otherwise, continue with section Migrate Application Modules in this document. Using the following steps, enable Auto Checkpointing only after the entire logical UCN network has been migrated to R684. Auto checkpointing may be selected for the entire NIM or for individual LMs, PMs, APMs, or HPMs. For each box that should be supported by auto checkpointing, proceed as follows.

Step	Action	Done (✓)
1	FOR each NIM: At any US/GUS/EST, from the PROCESS NETWORKS NODE STATUS menu, SELECT NIM SELECT AUTO SAVE SELECT ENABLE SAVE SELECT NETWORK STATUS SELECT AUTO CHECKPT SELECT UCN CKPT ENABLE SELECT ENTER	
2	FOR UCN DEVICES: On the same screen, SELECT DEV CKPT ENABLE	
3	SELECT a device for which AUTO CHECKPOINTING is desired, SELECT ENTER	
4	REPEAT Step 3 until auto checkpointing is enabled for all desired UCN devices.	

Table 12-24 Enable Cable Swapping/Auto Checkpointing NIM/UCN

12.16 Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)

Migrate Application Modules

Redundant AMs may be migrated on-line using the steps in section 12.17.

Perform the following steps to upgrade hardware, and load the AM with the R684 software.

Note: Use these Application Module (AM) migration steps when:

- The Application Module is an AMW, APP or non-redundant AM node;
- The hardware is upgraded or changed from current hardware configurations;
- The AM has had an NCF change since the last time the AMs were reloaded;
- The steps for on-line migration of Redundant Application Modules fail.
- The AM software version is R611.2 or greater

ATTENTION

For redundant AM pairs, both AMs must be SHUTDOWN, and then each reloaded.

ATTENTION

For each AM, AMW and APP node save a checkpoint of the R6xx database for use if regressing from R684 to R6xx.

Step	Action	Done (✓)
1	CAUTION Choosing cold, warm, or hot restart: If you use totalizers in your AM and you wish to retain those values, screen print each AM totalizer to record the values. If you restart with COLD or WARM, the totalizer values will go to zero. To retain the totalizer values in R684, load the AM using the HOT restart.	
2	SHUTDOWN an AM (both AMs, if redundant). The AM status now goes to QUALIF.	
3	LOAD the AM, via AUTOLOAD NET target. If the AM is redundant, load the backup AM as well.	

Table 12-25 Migrate Application Modules

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)

Perform the following steps to enable AM control, start AM control, take checkpoints to the NET for each AM, and enable Auto Checkpointing (if Auto Checkpointing is desired).

Step	Action	Done (√)
4	SET control status for the hiway and its boxes to FULL CONTROL, to enable AM control (no points in cascade at this time).	
5	SET control from the UCN and its devices to FULL CONTROL to enable AM control.	
6	START AM control by placing Hiway/UCN points in CASCADE mode. Nodes on the UCN and Extended Controllers on Data Hiways retain cascade requests previously issued. Basic Controllers and Multifunction Controllers (MCs and AMCs) do not retain former cascade requests. They resume operation with a cold restart.	
7	REPEAT steps 1 through 6 for each remaining AM.	
8	If desired, ENABLE auto checkpointing for each AM.	
9	CHECKPOINT the R6xx AM to the NET.	

Table 12-25 Migrate Application Modules (continued)

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)**Load the R683 AMW personality**

With TPN 683 the AMW package installer has changed. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

Note: The AMW personality is not part of the TPN media from TPN R684.

Perform the following steps to load the AMW personality.

Step	Action	Done (✓)
1	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs →Command Prompt , Type " phdctl stop " press Enter .	
2	Using the Windows Task Bar: START→ HONEYWELL TPS → LCNP STATUS . Select Reset button, Select Yes , and wait until the Node goes to PWR_ON state.	
3	Using the Windows Task Bar: Select Settings→Control Panel→Add/Remove Programs .	
4	From Add/Remove Programs Window Select TPS AMW Personality , Select Add/Remove... button. Verify Uninstallation Window.	
5	Select AMW Personality , Select OK . When the uninstallation window returns, select OK . Select OK from Add/Remove Programs window.	
6	Insert the TPS – Application Module CD.	
7	Using Windows Explorer, go to the <CD drive>:\AMW and run the amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package.	
8	Using Windows Explorer, check under c:\HWIAC\TPS\PERSONALITIES\AM directory to make sure that the personality files have been loaded.	
9	Do an AUTOLOAD NET on the GANT node. It should come up as an AMW with an OK status.	
10	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs →Command Prompt , Type " phdctl start " then press ENTER .	

Table 12-26 Load the R683 AMW Personality

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)**Load the R684 APP personality**

With TPN 684 the APP package installer has changed. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

The R684 APP personality is part of the R684 TPN media and can be found on the TPN Application Module CD ROM.

Perform the following steps to load the APP personality.

Step	Action	Done (✓)
1	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs → Command Prompt . Type " phdctl stop " then press Enter .	
2	Using the Windows Task Bar: START → HONEYWELL TPS → LCNP STATUS . Select Reset button, Select Yes and wait until the Node goes to PWR_ON state.	
3	Using the Windows Task Bar: Select → Settings → Control Panel → Add/Remove Programs .	
4	From Add/Remove Programs Window Select → TPS AMW Personality (AM) , Select Add/Remove... button. Verify Confirm File Deletion Prompt. "Are you sure you want to completely remove "TPS – amw Personality (AM)" and all of its components?"	
5	Select Yes . When complete, select OK then select OK from Add/Remove Programs window.	
6	Insert the TPN – Application Module CD.	
7	Using Windows Explorer, go to the <CD drive>:\APP and run the tps - amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package.	
8	Do an AUTOLOAD NET on the APP node. It should come up as an APP with an OK status.	
9	If you have a PHD server running on this node: Using the Windows Task Bar: Select → Programs → Command Prompt , Type " phdctl start " then press ENTER .	

Table 12-27 Load the R684 APP Personality

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)**Migrate ESVT**

With TPN 684 the APP package installer has changed. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

Perform the following steps to migrate ESVT.

ATTENTION

Uninstall the existing personality, before migrating ESVT.

Step	Action	Done (✓)
1.	FREEZE the data base, that is, nothing must be loaded or built during the migration on the ESVT nodes.	
Install APP personality on all ESVT nodes.		
2.	To determine which ESVT is running as PRIMARY, call up the station on the ESVT node. Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → SERVER → STATION Select CONFIGURE → SYSTEM HARDWARE → REDUNDANT SERVER	
3.	Verify which ESVT is running as Primary and ensure that the servers are synchronized. If the ESVTs are not synchronized, synchronize them before starting the upgrade.	
4.	Go to the BACKUP ESVT and Call up the Station. Ensure you are accessing the BACKUP ESVT.	
5.	From Add/Remove Programs Window Select→ TPS AMW Personality (AM), Select Add/Remove... button. Verify Confirm File Deletion Prompt . "Are you sure you want to completely remove "TPS – amw Personality (AM)" and all of its components?"	
6.	Select Yes . When complete, select OK from Add/Remove Programs window.	
7.	Insert the TPN– Application Module CD .	
8.	Run the tps - amw personality (am).msi file to copy the R684 personality to the local hard drive following the steps below Install APP LCNP personality. Insert TPN – Application Module CD. Use Windows Explorer to navigate to the following directory on the CD: VAPP and run the tps - amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package.	

Table 12-28 Migrate ESVT

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)

Step	Action	Done (✓)
9.	Check the LCNP Board 0 configuration by performing the following: Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → SYSTEM MANAGEMENT → CONFIGURATION UTILITY → CONFIGURE → BOARD 0 (or the name configured for Board 0) Select Enable LCNP Auto Restart and click OK .	
10.	Call up the LCNP Status by performing the following: Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → TPS APPLICATIONS → LCNP STATUS Select Details . Select Reset LCNP . Are you sure you want to RESET the LCNP? Message is displayed. Select Yes . Wait for the status to change to State OK and SubState OK ..	
11.	Verify the redundant Server status is Synchronized and Running with both LEDs are Green	
To upgrade the Primary Server to R684, perform the following:		
12.	Call up Station. Ensure you are at the PRIMARY ESVT. At the Primary ESVT, select Configure->System Hardware->System Interface->TPS If "Failover to backup server on TPS communications failure if servers are synchronized" is checked, a failover will occur. IF it is NOT checked, failover will not occur.	
13.	Run the tps - amw personality (am).msi file to copy the R684 personality to the local hard drive following the steps below. Install APP LCNP personality. Insert TPN – Application Module CD. Use Windows Explorer to navigate to the following directory on the CD: \APP and run the tps - amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package..	

Table 12-27 Migrate ESVT (continued)

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)

Step	Action	Done (√)
14.	Check the LCNP Board 0 configuration by performing the following: Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → SYSTEM MANAGEMENT → CONFIGURATION UTILITY → CONFIGURE → BOARD 0 (or the name configured for Board 0) Select Enable LCNP Auto Restart and select OK .	
15.	Swap the ESVTs. Therefore, the Primary ESVT becomes the Backup ESVT and the Backup ESVT becomes the Primary ESVT.	
16.	Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → SERVER → STATION	
17.	Select CONFIGURE → SYSTEM HARDWARE → REDUNDANT SERVER.	
18.	Select Manual Failover.	
19.	Select Yes.	
To load the ESVT with the R684 personality, perform the following:		
20.	Call up the LCNP Status by: Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → TPS APPLICATIONS → LCNP STATUS Select Details Select Reset LCNP Are you sure you want to RESET the LCNP? Message is displayed. Select Yes. Wait for the status to change to State OK and SubState OK .	
21.	To synchronize the Servers from the station, Select Synchronize and select Yes.	
22.	Wait for the synchronization to complete with the following status: Synchronized and Running LEDs are both Green.	
When the status is displayed in green, you should get the Experion return the ESVTs to their original PRIMARY/SECONDARY status. To do that perform the following:		

Table 12-27 Migrate ESVT (continued)

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)

Step	Action	Done (√)
23.	From the Redundant Server page in the Station, select Manual Failover . Wait for one minute before continuing. Select Yes to Failover to backup?	
24.	Select Synchronize .	
25.	Select Yes .	
26.	Wait for the status indication to show the original configuration with status as good. Note: Ensure the TPN Server is in the Running or Idle (if no clients are connected) state. If the TPN Server does not enter the state, the node is to be rebooted. TIP: You can ascertain the status of the TPN Server through the System Management Display.	

Table 12-27 Migrate ESVT (continued)

Migrate ACE-T

With TPN 684 the APP package installer has changed. The new personality installation package constitutes a single .msi file as opposed to the setup.exe in the previous package.

Perform the following steps to migrate ACE-T.

ATTENTION

Uninstall the existing personality, before migrating ACE-T.

Step	Action
1	FREEZE the data base, that is, nothing must be loaded or built during the migration on the ACE-T nodes. Save the check points of all the instances of ACE-T using CB.
	Install APP personality on all the ACE-T nodes.
2	Run the Setup.exe to copy the R684 personality to the local hard drive following the steps below. Install APP LCNP personality. Insert TPN – Application Module CD. Use Windows Explorer to navigate to the following directory on the CD: \APP and run the tps - amw personality (am).msi file. Follow the instructions on the installation wizard for installing the package.
3	Reset the LCNP by performing the following: Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → SYSTEM MANAGEMENT → CONFIGURATION UTILITY → CONFIGURE → BOARD 0 (or the name configured for Board 0). Select Enable LCNP Auto Restart and click OK .

Table 12-29 Migrate ACE-T

12.16. Migrate Application Modules (AM, AMW, APP, ESVT, ACE-T)

Step	Action
4	Call up the LCNP Status by: Using the windows task bar: START → PROGRAMS → HONEYWELL EXPERION PKS → TPS APPLICATIONS → LCNP STATUS Select Details Select Reset LCNP Are you sure you want to RESET the LCNP? Message is displayed. Select Yes. Wait for the status to change to State OK and SubState OK Ensure from the system status display (invoked on a US/GUS/ESVT connected to the same LCN) that the personality loaded is APP. Note: While LCNP is reset and reloaded, ACE will not be receiving data from TPN. Switch the control strategies in Experion (which are dependent on TPN data) to Idle state. Reload the strategies after node reload is complete.
	Note: Ensure the TPN Server is in the Running or Idle (if no clients are connected) state. If the TPN Server does not enter the state, the node is to be rebooted. TIP: You can ascertain the status of the TPN Server through the System Management Display.

Table 12-29 Migrate ACE-T (continued)

12.17 On-line Migration of Redundant Application Modules

**Loading the
redundant AM with
the R684 software**

Perform the following steps to load the redundant AM with the R684 software. This allows the installation of functional, maintenance, or point releases into redundant AMs while maintaining the view and control provided by the AM pair.

NOTE: All conditions below must be met for on-line migration of redundant AMs to work properly:

- The current software revision of the redundant AM node is at or above version 62.2 (TPN 611.2);
- The AM hardware configuration stays the same as the last time the AMs were reloaded;
- The AM NCF configuration stays the same as the last time the AMs were reloaded;
- The total size of the configured backplane modules has not grown by more than 8k words.

If these conditions cannot be met, you must use the steps in section 12.16 to update your redundant AMs.

ATTENTION

For each redundant AM pair, save a checkpoint of the R6xx database for use if regressing from R684 to R6xx.

Step	Action	Done (√)
1	To avoid unnecessary reloading of nodes, it is suggested that the primary AM be shutdown first and reloaded with the new release. This will ensure that, once the upgrade is complete, the AM node numbers will be identical to the original LCN primary/secondary node number pair used before the upgrade. From any US/GUS/EST, at the System Status Display, SHUTDOWN the PRIMARY AM. The current secondary becomes the primary. The new primary AM will go to WARNING because of the loss of redundancy.	
2	Select 'LOAD SELECT' and press <ENTER>.	

Table 12-30 On-line Migration of Redundant AMs

**Loading the
redundant AM with
the R684 software,
continued**

Step	Action	Done (√)
3	Wait for the AM to go to OK. (The current primary AM database is transferred to the secondary over the redundancy hardware and this may take several minutes. When completed, the node status will show 'WARNING' and the STATUS DETAIL or Event History - Notification Status will show "AM Redundancy WARNING: SECONDARY IS AN UPGRADE".)	
4	Shutdown the primary AM. (Current upgrade secondary becomes the primary. Its node status will show 'WARNING' and the STATUS DETAIL or Event History - Notification Status will show "WARNING: PRIMARY HAS NO BACKUP"). Select 'LOAD SELECT' and press <ENTER>.	
5	Wait until the "Completed Loading" message is displayed, then select 'RETURN'. (The current primary AM database is transferred to the secondary over the redundancy hardware and this may take several minutes. The node status will show 'OK' and 'BACKUP' and the STATUS DETAIL or Event History - Notification Status will show "AM Redundancy OK: AM redundancy status tracking".)	
6	If the secondary node status remains 'WARNING' and the STATUS DETAIL or Event History - Notification Status shows "AM Redundancy WARNING: SECONDARY IS AN UPGRADE" then the external load modules were upgraded at the same time as the first on-line AM migration was executed and another reload is required. Shutdown the primary AM. (Current upgrade secondary becomes the primary. Its node status will show 'WARNING' and the STATUS DETAIL or Event History - Notification Status will show "WARNING : PRIMARY HAS NO BACKUP"). Select 'LOAD SELECT' and press <ENTER>. Wait until the "Completed Loading" message is displayed, then select 'RETURN'. (The current primary AM database is transferred to the secondary over the redundancy hardware and this may take several minutes. The node status will show 'OK' and 'BACKUP' and the STATUS DETAIL or Event History - Notification Status will show "AM Redundancy OK : AM redundancy status tracking".)	

Table 12-30 On-line Migration of Redundant AMs (continued)

12.18 Migrate Computer Gateways

Migrate Computer Gateways/PLNMs

Perform the following steps to migrate the Computer Gateways/PLNMs

Step	Action	Done (√)
1	From any US/GUS/EST, SHUTDOWN the CG, and wait until its status goes to QUALIF state.	
2	LOAD the CG as follows: SELECT LOAD SELECT PRESS <ENTER> wait for the CG to go to OK.	
3	Checkpoint the CG to the Net.	
4	Auto checkpointing can now be enabled.	

Table 12-31 Migrate the Computer Gateways

12.19 Migrate Network Gateways

Migrate NGs Perform the steps below to migrate the Network Gateway(s).

Step	Action	Done (√)
1	From any US/GUS/EST SHUTDOWN the NG, and wait until it goes to QUALIF state.	
2	LOAD the NG as follows: SELECT LOAD SELECT PRESS <ENTER> Wait for NG to go to OK	
3	REPEAT Steps 1 and 2 for each remaining NG.	

Table 12-32 Migrate Network Gateways

12.20 Migrate X-Layer Nodes

ATTENTION

Refer to the Software Change Notices for TPN and UXS for migration instructions. The UXS was withdrawn from sale in November 1997.

X-layer nodes are different

A^XM nodes cannot be migrated by following the procedures for migrating AM nodes.

The UNIX system must be configured to select the correct personality type to load on the next Auto Load.

Migrate A^XM

Step	Action	Done (√)
1	Following the <i>Application Module^X Customer Release Guide</i> (AX11-490, Section 4), and <i>Application Module^X System Administration</i> , Install the R6xx A ^X M DAT tapes into the UNIX processor on this A ^X M.	
2	IF UPGRADING the processor to a K4LCN, LOG into the A ^X M through a terminal window as ROOT, then INPUT the following two UNIX commands: (if not upgrading processor board, skip to step 3) rm /etc/opt/TDC_Open/common/personality.config Input the next command all in one string, with a <SPACE> after axmk: ln -s /opt/TDC_Open/common/newconfig/personality.config.axmk /etc/opt/TDC_Open/common/personality.config	
3	SHUTDOWN the previous release A ^X M node. Power off the A ^X M.	
4	If you are upgrading from a K2LCN to a K4LCN, (ensure the power is OFF) REPLACE the K2LCN-8 with a K4LCN-8	
5	POWER UP the A ^X M.	
6	When the node goes to PWR ON state, Go to the AM NODE STATUS menu, SELECT this node, SELECT AUTONET LOAD target	
7	When the node status goes to OK, TAKE a demand checkpoint to NET.	
8	SHUTDOWN the AM personality running in this A ^X M.	
9	When the node goes to QUALIF, SELECT Auto Load Net This will cause the A ^X M to boot from its local UNIX drive.	

Table 12-33 Migrate A^XM

13. Backing Out of R684

13.1 Regressing to R6xx Software

For migrations from R6xx to R684, if for any reason you need to back out of R684 software to an earlier R6xx Release, you must understand and execute the following procedure.



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Step	Action
1	Using the media saved in section 12.3, "Save System Data," you must restore each HM on the system, with the data saved on the BACKOUT BACKUP cartridge(s).
2	Every node on the UCN and LCN will have to be SHUTDOWN and reloaded. THIS CANNOT BE PERFORMED ON-PROCESS.
3	All R684 hardware upgrades that will not work with the earlier R6xx release migrated from must be removed, and any previously removed hardware reinstalled. See section 3.3.

Table 13-1 Regressing to R6xx Software from R684

Set up to Load an R6xx Universal Node If you need to reload an R60x, R61x Universal Node on an R684 system you must execute the following procedure. If you don't, the US will fail.

Step	Action	Done (√)
1	Create a removable media which contains the &ASY, &DSY, &CUS and &Dnn directories. CR \$Fd>&ASY> Where d = the removable media cartridge	
2	Create a directory on the removable media cartridge for the &DSY directory. CD \$fd>&ASY> &DSY Where d = the removable media cartridge	
3	Create a directory on the removable media cartridge for the &CUS directory. CD \$fd>&ASY> &CUS Where d = the removable media cartridge	
4	Create a directory on the removable media cartridge for the Area Database, &Dnn. CD \$fd>&Dnn Where d = the removable media cartridge, and nn = the Area number	
5	Copy the &ASY directory from the HM onto a removable media. CPV NET>&ASY> \$Fd>&ASY> -d Where \$Fd = the removable media drive	
6	Copy the &DSY directory from the HM onto a removable media. CPV NET>&DSY> \$Fd>&DSY> -d Where \$Fd = the removable media drive	

Table 13-2 Set Up to Load an R6xx Universal Node

**Set up to Load
an R6xx
Universal Node,**
continued

Step	Action	Done (√)
7	Copy the &CUS directory from the HM onto a removable media. CPV NET>&CUS> \$Fd>&CUS> -d Where \$Fd = the removable media drive	
8	Copy the &Dnn directory from the HM onto a removable media. CPV NET>&Dnn> \$Fd>&Dnn> -d Where \$Fd = the removable media drive, and nn = the Area number	
9	Unprotect the file TOP2LINE.DO on the removable media. UNPT \$Fd>&DSY>TOP2LINE.DO Where \$Fd = the removable media drive	
10	From the &Z1 cartridge for the R6xx release you want to load, load the file TOP2LINE.DO from the &DSY directory. CP \$Fs>&DSY>TOP2LINE.DO \$Fd>&DSY>TOP2LINE.DO Where \$Fs = the drive containing the &Z1 cartridge, and \$Fd = the drive containing the removable media	
11	You can now load the Universal node, using the &Z1 cartridge of the R6xx release you want to load for the Personality Image, and the Removable Media created above for the DATA source.	

Table 13-2 Set Up to Load an R6xx Universal Node (continued)

14. Troubleshooting Migration Problems

14.1 Overview

Introduction	When it comes to troubleshooting and successful diagnosis of a migration problem, understanding and recording what occurred before and during a failure is extremely important. The following procedures should be used for reporting problems encountered during the migration.
If you need assistance	If you need technical assistance, contact your local Honeywell Service Organization, as explained in the following paragraphs. It is a good idea to make specific notes about the problem before making the call. This will help to reduce delays and expedite answers.
International customers	Outside of the United States, contact your local Honeywell Service Organization. If you are not sure of the location or telephone number, call your Honeywell representative for information.
Customers inside the United States and Canada	Within the United States, call the Technical Assistance Center (TAC) at the toll free number 1-800-822-7673.
Arizona customers	Within Arizona, the local number for TAC is 602-313-5558.
Services provided	Calls to TAC are answered by a dispatcher from 6:00 A.M. to 4:00 P.M. Outside of these hours, emergency calls—those which affect your ability to control or view a process—will be received by an answering service, and returned within one hour. TAC maintains its own TPS system, and frequently can duplicate problems on this equipment.

14.2 Collect Failure Data

Overview If a failure has occurred, specific data should be collected that will help in the diagnosis of the problem. Examine the following symptom labels in this section:

- System or module working, but not properly,
- Node not responding, and
- Universal Station node cannot be booted.

Where applicable, use these procedures to collect failure data.

System or module working, but not properly

Perform the following steps as they apply to your situation.

Step	Action	Examples/Hints
1	Record system software release number.	R510.1.2 for example
2	Record all symptoms of failure.	Slow response from node Error indication on node
3	Record all user actions that occurred immediately before the failure.	Type of operation being performed, such as • Area database configuration • Point building • Building trend displays • Building graphic displays
4	Record the personalities being used.	Universal, Operator, History Module, Application Module, Computer Gateway, Network Gateway, NIM, Hiway Gateway
5	Record error messages during the time of the error (in their entirety).	Real Time Journal printouts Event History Retrieval printouts
6	If custom graphic displays or user-written CL programs are involved, copy related information.	Schematic source files Related subpictures Related CL programs
7	If the problem is related to particular points, IDFs, or exception-build files, submit them to TAC.	FAX to: 602 313-5476
8	If you have the TLK1 Tool kit, use NODEPERF to print the screen of data shown for the node.	For use of Tool kit, see the <i>Customer Resource Manual</i> , Section 25.

Table 14-1 System or Module Working, But Not Properly

14.2. Collect Failure Data**Node not responding**

Perform previous Steps 1 through 7 in “System or module working, but not properly,” then perform these steps as they apply to your situation.

Step	Action	Examples/Hints
1	Record state of LEDs on module control and processor boards.	
2	Record value in 3-digit display on processor board.	-190 for example.
3	Use SMCC to display Module Errors, or Detailed Module Errors for the failed node, and print a copy of the display.	Displays last error in the personality.
4	Dump the contents of memory to removable media, using dump function. NOTE: Cartridge preferred. (Replacement cartridge will be sent to customer)	See <i>Engineer's Reference Manual</i> , Section 20.
5	Copy contents of !CSY directory from NET to removable media, using this command. CP NET>!CSY>*.* \$Fd>!CSY>==D where d = destination drive number	Cartridge preferred. Use same cartridge as in Step 4.
6	Obtain a printout of the board type and revision status of the failed node. Reload the node with the same personality it had when it failed, then • Select SMCC (from Eng. Main Menu) • Select REV/CONFIG Status	
7	Print at least the last two pages of the Real Time Journal that showed the failure.	
8	Print Event History Retrieval for all system errors immediately before and during the time of the failure.	
9	If a hardware error is suspected, run the appropriate HVTs software.	See <i>Hardware Verification Test System</i> in <i>LCN Service - 2</i> binder.
10	If failed node is an HG or a NIM, and if 3-digit LED display shows either -189, -190, or -195, display all error blocks in the redundant node, and print this information	Use SMCC to obtain printout.

Table 14-2 Node Not Responding

**Universal Station
node cannot be
booted**

Perform the following steps as they apply to your situation.

If ">" prompt does not appear on the screen after <RESET> button is pressed:

Step	Action	Examples/Hints
1	Record value in 3-digit display on processor board.	-190 for example.
2	Use SMCC to display Module Errors, or Detailed Module Errors for the failed node, and print a copy of the display.	Displays last error in the personality.

If node failed after <LOAD> button was pressed:

Step	Action	Examples/Hints
3	Ensure cartridge is properly seated in drive and HM is OK.	Remove, inspect, re-insert, or re-load.
4	Ensure the medium has the personality image.	
5	Ensure correct data, command, and/or syntax was entered.	

Table 14-3 Universal Station Node Cannot Be Booted

14.3 Status and Error Messages When Translating

User-visible messages (when translating)

The status messages shown in the following table could be seen when performing translations using the command processor.

Message	Explanation
Translating NIM Checkpoint Files, Please Wait...	Signals start of NIM database translation
Translating PM Checkpoint Files to APM, Please Wait...	Signals start of PM to APM database translation
Translating PM Checkpoint Files to HPM, Please Wait...	Signals start of PM to HPM database translation
Translating APM Checkpoint Files to HPM, Please Wait...	Signals start of APM to HPM database translation
Building Process Point Data...	Shows current phase of NIM database translation
Building Pool_pt1...	Shows current phase of NIM database translation
Building Hash Table...	Shows current phase of NIM database translation
Building Data Access Names Table...	Shows current phase of NIM database translation
Building Point Record Section...	Shows current phase of NIM database translation
Copying Unit DB...	Shows current phase of NIM database translation
Building Logic DB Library...	Shows current phase of NIM database translation
Building Sequence Library...	Shows current phase of NIM database translation
Database Translation Complete	Shows translation has completed successfully

Table 14-4 User-Visible Messages (when translating)

**Error messages
when executing
translator overlay**

The error messages shown in the following table are seen in the command processor when executing the translator overlay. Syntax type errors that occur on the command line will allow you to re-enter your command. However, after the translator begins the actual translation, any error here will terminate and abort the translation in progress.

Error Message	Explanation
TR NM <Source Device> <Destination Device> <Network #>	After any error is encountered in the translator command syntax, this is displayed to show correct usage
TR APM HPM <Source Device> <Destination Device> <Network #> <Node #>	After any error is encountered in the translator command syntax, this is displayed to show correct usage
TR PM APM <Source Device> <Destination Device> <Network #> <Node #>	After any error is encountered in the translator command syntax, this is displayed to show correct usage
TR PM HPM <Source Device> <Destination Device> <Network #> <Node #>	After any error is encountered in the translator command syntax, this is displayed to show correct usage
FATAL ERROR : Invalid Destination Device (Range \$F1 - \$F20)	Invalid drive syntax or out of range number entered for destination device in translator command
FATAL ERROR : Invalid UCN Network Number (Range 1 - 20)	Out of range network number entered in translator command
FATAL ERROR : Invalid UCN Node Number. (Range 1- 63, odd only)	Out of range node number entered in translator command

Table 14-5 Error Messages when Executing Translator Overlay

14.3. Status and Error Messages When Translating

**Error messages
when executing
translator overlay,
continued**

Error Message	Explanation
FATAL ERROR : Invalid Source Device. (Range \$F1 - \$F20)	Invalid drive syntax or out of range number entered for source device in translator command
FATAL ERROR : Could Not Create Destination DB File	Destination checkpoint database file could not be created by File Manager
FATAL ERROR : Could Not Close Destination File.	Destination checkpoint database file could not be closed by File Manager
FATAL ERROR : Could Not Close Master File.	Checkpoint master file could not be closed by File Manager
FATAL ERROR : Could Not Close Source File.	Source checkpoint database file could not be closed by File Manager
FATAL ERROR : Could Not Open Destination File.	Destination checkpoint database file could not be opened by File Manager
FATAL ERROR : Bad file gen or revision code on source master file.	Source master file contains bad revision code or bad gen code
FATAL ERROR : Could Not Open Master File	Checkpoint master file could not be opened by File Manager
FATAL ERROR : Could Not Open Source File	Source checkpoint database file could not be opened by File Manager
FATAL ERROR : Could Not Obtain Destination File Memory.	Memory allocation of a File Manager Request Block for the destination checkpoint database file failed
FATAL ERROR : Could Not Obtain Source File Memory.	Memory allocation of a File Manager Request Block for the source checkpoint database file failed

Table 14-5 Error Messages when Executing Translator Overlay (continued)

**Error messages
when executing
translator overlay,
continued**

Error Message	Explanation
FATAL ERROR : Could Not Obtain Master File Memory.	Memory allocation of a File Manager Request Block for the master file failed
FATAL ERROR : Not enough room on dest drive.	New checkpoint files exceed available disk space on destination drive
FATAL ERROR : Could Not Copy Master File	File Manager reported an error while reading or writing master checkpoint file
FATAL ERROR : Could Not Create Master File.	File Manager reported an error while trying to create master checkpoint file
FATAL ERROR : Could Not Read Master File	File Manager reported an error while trying to read master checkpoint file
FATAL ERROR : Could Not Open Source Database File.	File Manager reported an error while trying to create master checkpoint file
FATAL ERROR : Could Not Read Source File	File Manager reported an error while trying to read source checkpoint database file
FATAL ERROR : Could Not Translate Database.	Final error message displayed after any fatal error has occurred
FATAL ERROR : Could Not Write Destination File.	File Manager reported an error while trying to write destination checkpoint database file
FATAL ERROR : Internal Memory Could Not Be Allocated	The translator could not get local memory for its own use
FATAL ERROR : Reported Size of Translated IOP Exceeds Maximum.	The size of the translated IOP exceeds the maximum size allowed for IOP translation
FATAL ERROR : Reported Size of Translated PM Point Exceeds Maximum.	The size of the translated PM point exceeds the maximum size allowed for PM point translation

Table 14-5 Error Messages when Executing Translator Overlay (continued)

14.3. Status and Error Messages When Translating**Error messages
when executing
translator overlay,
continued**

Error Message	Explanation
FATAL ERROR : Illegal PM Checkpoint File Revision Found.	The PM checkpoint file being translated is not a R4xx or R5xx version checkpoint
FATAL ERROR : Illegal APM Checkpoint File Revision Found.	The APM checkpoint file being translated is not a R4xx or R5xx version checkpoint
FATAL ERROR : Could Not Open New Master File.	File manager reported this error when the master file could not be opened after it was created
FATAL ERROR : Illegal NIM Checkpoint File Revision Found.	The NIM checkpoint file being translated is not a R4xx version checkpoint
FATAL ERROR : Checksum Is Not Correct.	A checksum error was detected in a PM or IOP checkpoint when verifying its checksum
FATAL ERROR : Source Data Is Not Correct.	An error was found in the data of a PM or IOP point being translated
FATAL ERROR : Unexpectedly Found a Newly Indexed Checkpoint Type.	During translation, a point was erroneously translated into an indexed-type point
FATAL ERROR : Not Able to Convert Point Type Found in File.	During translation, a non-supported point type was found
FATAL ERROR : Found APM Point Type in The PM Source File!	During translation, an APM point type was found in the PM checkpoint file
FATAL ERROR : Checkpoint Volume &lxx Does Not Exist on Dest Disk.	Occurs when the destination drive do=0 does not have a checkpoint directory created for this UCN network

Table 14-5 Error Messages when Executing Translator Overlay (continued)

**Error messages
when executing
translator overlay,
continued**

Error Message	Explanation
FATAL ERROR : Exceeded Max Files on Destination Disk.	Error occurs when files writes are attempted to a cartridge drive at or near maximum file capacity
FATAL ERROR : Invalid Destination Node Type.	A non-supported node type was specified for "translation to" in the translator command line
FATAL ERROR : Invalid Source Node Type.	A non-supported node type was specified for "translation from" in the translator command line
FATAL ERROR : Destination Device Not a Bernoulli Drive	Destination drive specified in translator command line is not a cartridge drive
FATAL ERROR : Not able to check if enough space on destination disk.	Error occurred while trying to access the destination drive
FATAL ERROR : Invalid Node Type Combination.	The combination of node being translated from and node being translated to is not supported by the translator
FATAL ERROR : Reported size of translated APM point exceeds maximum.	The size of the translated APM point exceeds the maximum size allowed for APM point translation
FATAL ERROR : Invalid or Incorrect Option Switch.	Invalid option or option syntax error entered in translator command
FATAL ERROR : Incomplete or Invalid Command Line	Incomplete command or command containing syntax errors entered
FATAL ERROR : This Package Option Not Valid for -S Option	The -S Option is only applicable to nonredundant PMs and APMs

Table 14-5 Error Messages when Executing Translator Overlay (continued)

14.4 Diagnose a UCN Cable Fault

Overview To avoid node failures during the diagnosis of cable faults, follow the procedures below to diagnose a UCN cable fault that has been reported in one or more nodes.

Inspect cable and connectors Inspect the suspected faulty cable(s). Listed in the following table are some of the more common failures to look for.

Failure	Symptom(s)
Loose or disconnected drop cable on a tap Loose or disconnected drop cable on a modem	If a drop cable is disconnect from a node, the effected node reports detection of silence and noise on the cable. In addition, some other node may report noise on that cable. The reported noise is the result of the center conductor bouncing when the drop cable was being disconnected. Selecting the suspect cable clears the noise report in the nodes reporting noise, and the system switches to that cable, causing a "silence" report for the missing drop cable
Wrong polarity on tap	
Incorrect tap or node grounding	
Loose or disconnected terminator	
Crossed cables (A to B, B to A) on trunk or drop	
Trunk or Drop Cable cut (by floor tile, furniture rearrangement, etc.)	If a trunk cable is cut, some nodes report noise and some report both noise and silence. Because of the nature of the fault, after enabling the Auto_Swap and selecting the suspected cable, the reported cable error may not go away. The cut trunk appears as an un-terminated trunk to some nodes on the network. Usually noise reported in several nodes indicates an un-terminated (cut) trunk.

Table 14-6 Inspect Cable and Connectors

14.5 Metadata File Error

This error occurs because the system recognizes Fieldbus devices or points and no Metadata file have been created. The Metadata file is created and downloaded using the TPS Builder.

14.6 On-line Migration of Redundant AMs Failure

If the secondary AM software cannot map backup memory into locations that match the primary AM, the secondary load will 'FAIL' with a System Menu/Event History Menu/Status Notify of "INCOMPATIBLE PRIM/SEC MEMORY MAP".

Conditions that can cause this error situation include the following:

- the implementation of a major release
- new code which forced the current AM redundant memory boundary to move
- loading an AM version which is older than the current running AM
- any customer initiated NCF change that caused the current AM configuration to change.

The solutions are to shutdown both redundant AMs and follow the original Migrate Application Modules instructions or to restore the differences that caused the secondary load failure.

14.7 US Nodes With No Keyswitch

At certain points in the software migration you are asked to load a US using the &Z6 Startup NCF and Area Database files. In order to perform the subsequent database translation tasks, your US needs to be in engineer keylevel.

But what if your US has no physical keyswitch? How do you change the US to engineer keylevel?

The R640.1 (and newer) &Z6, Startup Area Database (&D01), comes with a pre-built button file that allows you to change keylevel by use of the “PF1 through PF17” buttons.

- Pressing the PF”n” button will place the US in engineer keylevel.
- Pressing SHIFT and PF”n” will place the US in supervision keylevel.
- Pressing CONTROL and PF”n” will place the US in operator or view keylevel.

“n” = 1-17 (i.e., all of the PF1 through PF17 buttons work the same way)

In Section 7, Table 7-21, you must load a US with R53x software and the R53x &Z6 Startup files. The R53x &Z6 may not have the pre-built button file which defines the PF”n” keys. If not, simply copy the R640.1 (or newer) &D01>BUTTON.KO and &D01>BUTTON.KS to the R53x &Z6 &D01 directory.

```
UNPTV $Fx -NQ
```

```
CP $Fn>&D01>BUTTON.* $Fx>D01>= -D
```

```
PROTV $Fx -NQ
```

where:

“n” is the disk drive number holding the R640.1 (or newer) &Z6

“x” is the disk drive number holding the R53x &Z6

The R53x &Z6 will now be ready to use for the steps in Table 7-21.

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